

A Transport  
Health  
Assessment Tool  
for the City of  
Brisbane

# THAT- Brisbane 2023: Modelling Scenarios

## Acknowledgements

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We further acknowledge additional technical contributions from the Public Health Modelling Team at the MRC Epidemiology Unit at the University of Cambridge for the development of THAT-Melbourne as the precursor to THAT-Brisbane. The contributions of Dr Ali Abbas were funded by the European Research Council under the Horizon 2020 research and innovation programme (grant agreement No 817754) under the GLASST: Global and local health impact assessment of transport project. Dr Belen Zapata-Diomedí was funded by a RMIT University Vice-Chancellor's Postdoctoral Fellowship and the JIBE project (APPI192788) with some components of the code development completed during her placement at the MRC Epidemiology Unit, University of Cambridge under the supervision of Dr James Woodcock.

THAT-Brisbane and THAT-Melbourne build on the original model [1] further developed by Zapata-Diomedí et al (2019) [2], with additional development and chronic diseases [3].

**This report provides active transport modelling scenario outcomes produced online in the THAT-Brisbane tool available at: [auo.org.au/that-brisbane](https://auo.org.au/that-brisbane).**

**Additional detailed scenario modelling and visualisations can be accessed through the Australian Urban Observatory ([auo.org.au](https://auo.org.au)).**

## Suggested citation

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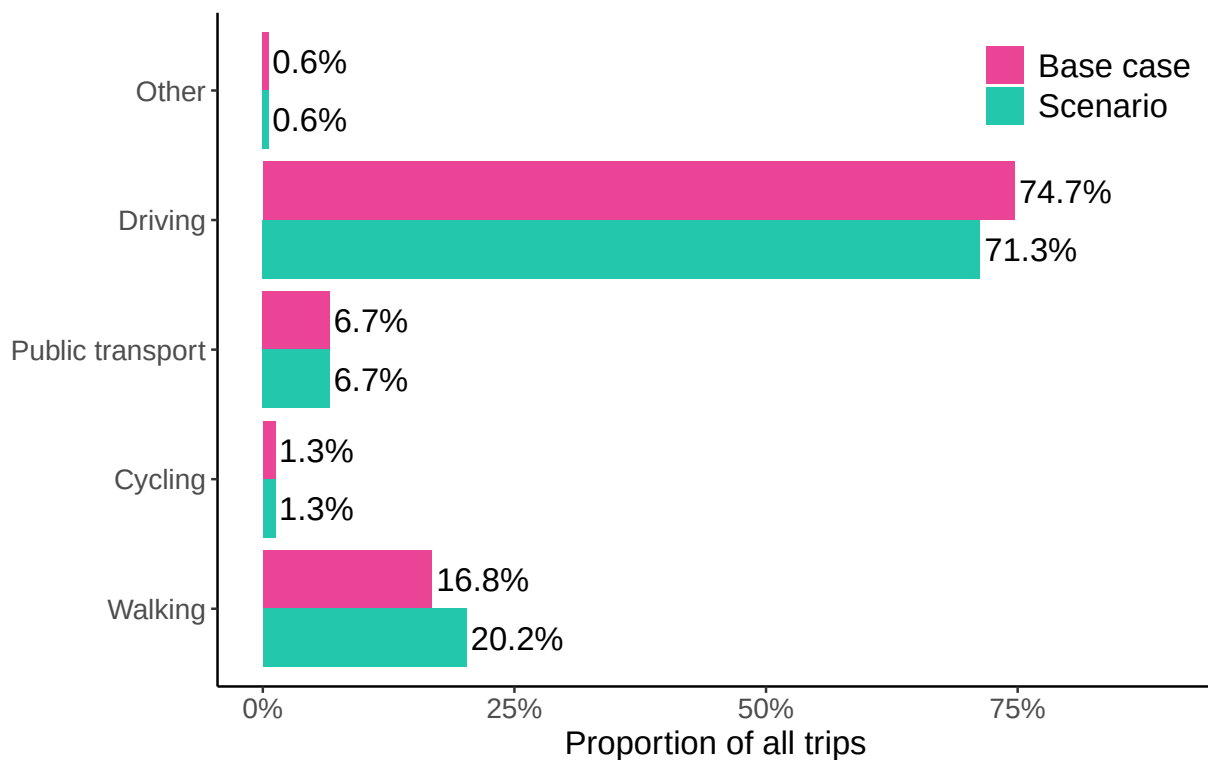
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## Scenario: replacing car trips under 1km with walking for all trip purposes

This scenario shows the results of replacing car trips under 1km for leisure, shopping, work, education or other purposes with walking trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 20.2% and from 74.7% to 71.3% for car trips taken as either a driver or passenger.

Increases in walking translate into a shift from 47.9% to 49.6% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

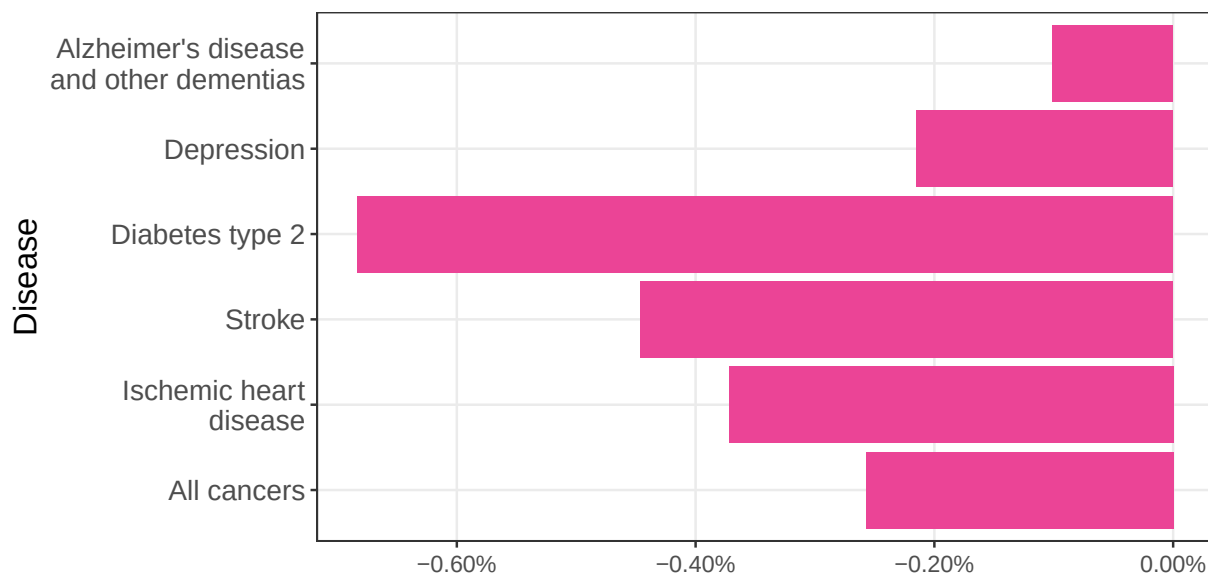
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

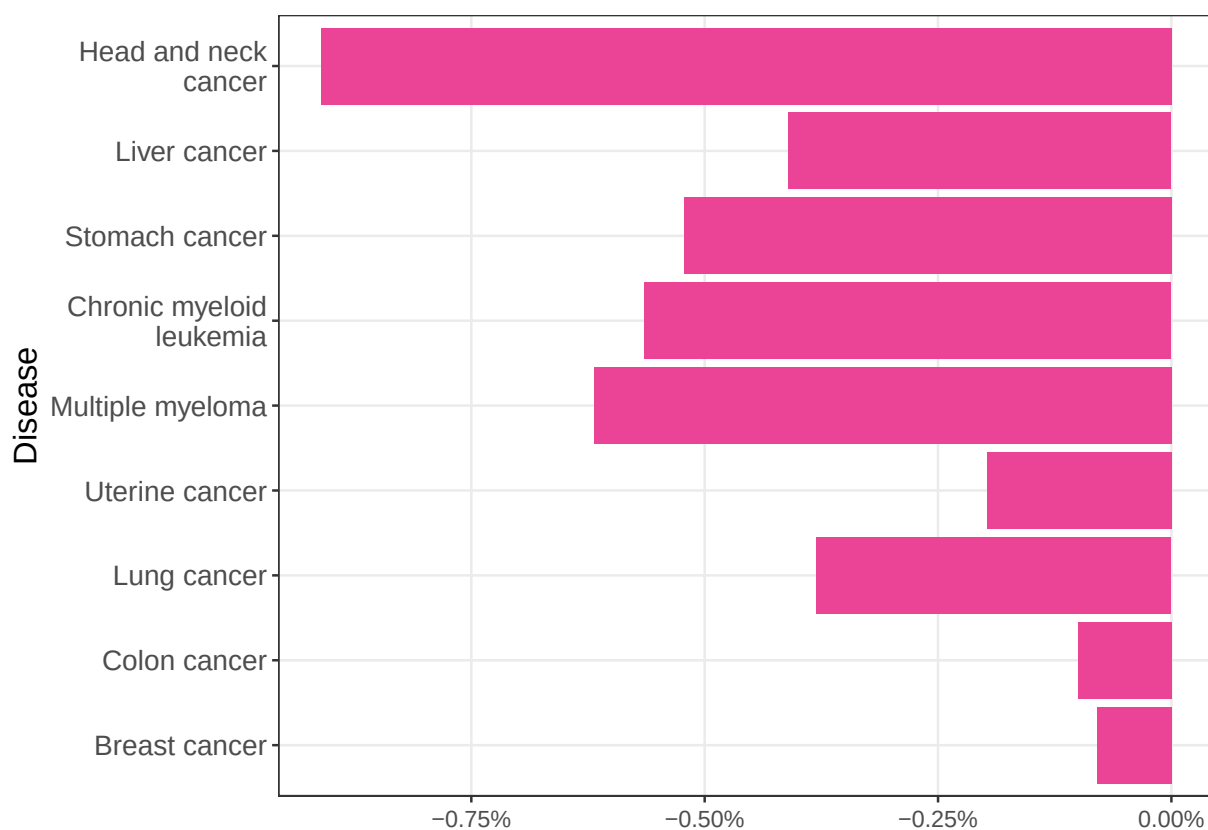
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.10%                              | 644                                                                         |
| Breast cancer                           | 0.08%                              | 58                                                                          |
| All cancers                             | 0.26%                              | 1,014                                                                       |
| Colon cancer                            | 0.10%                              | 130                                                                         |
| Chronic myeloid leukemia                | 0.56%                              | 15                                                                          |
| Diabetes type 2                         | 0.68%                              | 2,255                                                                       |
| Depression                              | 0.21%                              | 3,442                                                                       |
| Head and neck cancer                    | 0.91%                              | 40                                                                          |
| Ischemic heart disease                  | 0.37%                              | 3,275                                                                       |
| Liver cancer                            | 0.41%                              | 72                                                                          |
| Multiple myeloma                        | 0.62%                              | 120                                                                         |
| Stomach cancer                          | 0.52%                              | 125                                                                         |
| Stroke                                  | 0.45%                              | 1,162                                                                       |
| Lung cancer                             | 0.38%                              | 437                                                                         |
| Uterine cancer                          | 0.20%                              | 17                                                                          |

\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.

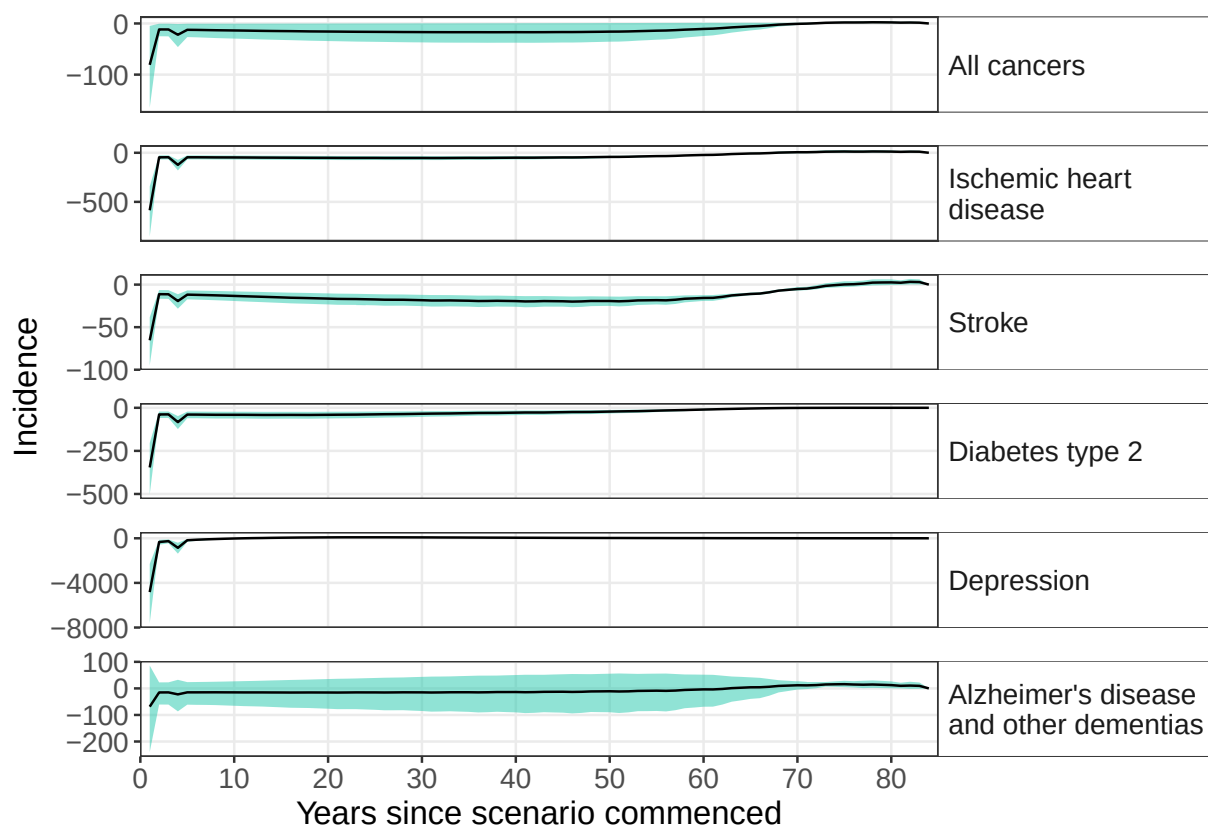


**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



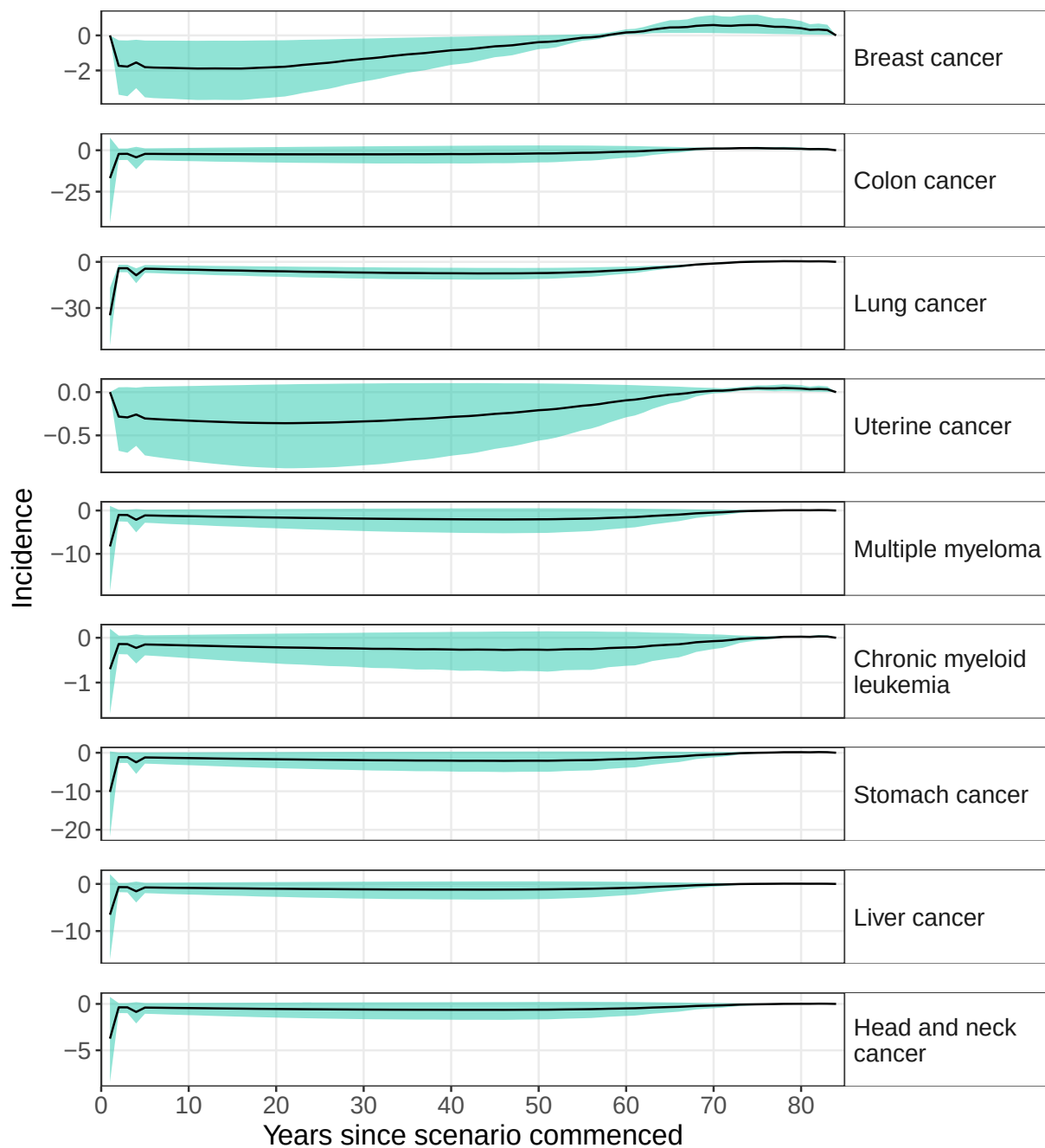
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.





**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

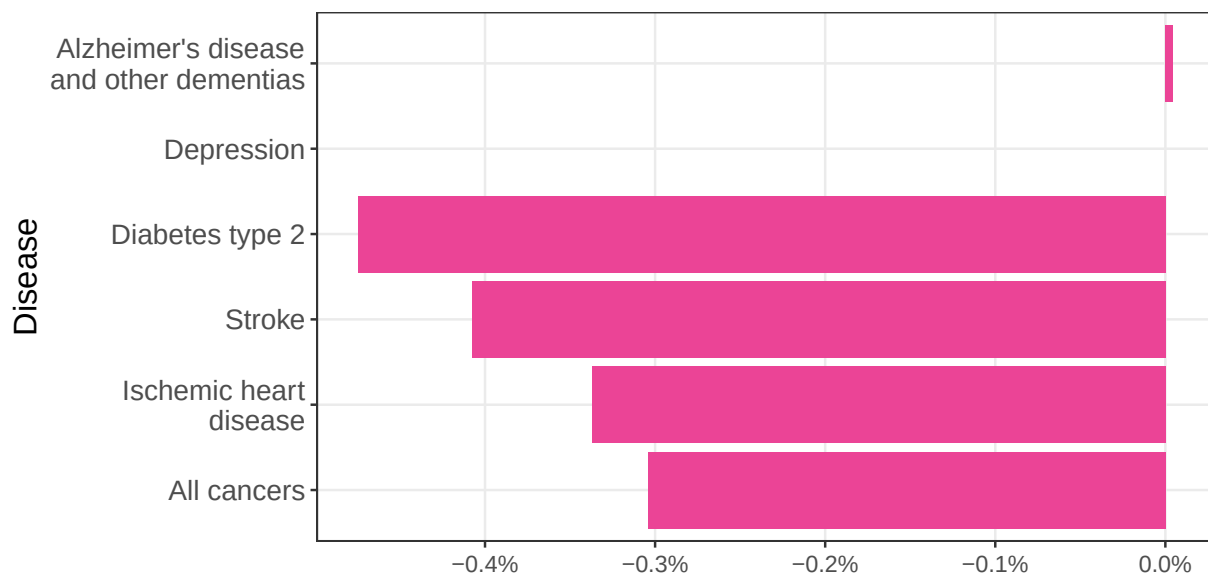
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

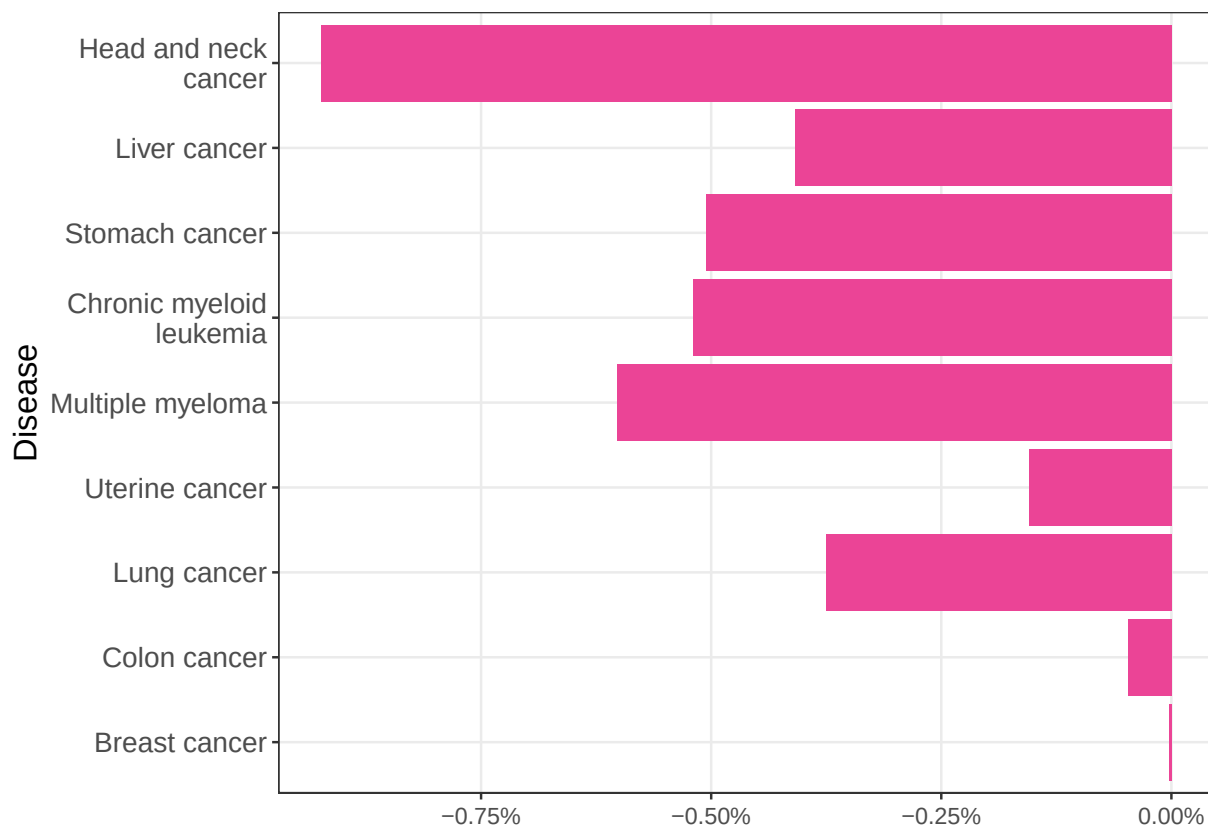
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.00%                   | -11                                                               |
| Breast cancer                           | 0.00%                   | 1                                                                 |
| All cancers                             | 0.30%                   | 771                                                               |
| Colon cancer                            | 0.05%                   | 20                                                                |
| Chronic myeloid leukemia                | 0.52%                   | 11                                                                |
| Diabetes type 2                         | 0.47%                   | 235                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.92%                   | 37                                                                |
| Ischemic heart disease                  | 0.34%                   | 1,265                                                             |
| Liver cancer                            | 0.41%                   | 70                                                                |
| Multiple myeloma                        | 0.60%                   | 105                                                               |
| Stomach cancer                          | 0.51%                   | 107                                                               |
| Stroke                                  | 0.41%                   | 713                                                               |
| Lung cancer                             | 0.38%                   | 409                                                               |
| Uterine cancer                          | 0.15%                   | 11                                                                |

\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.

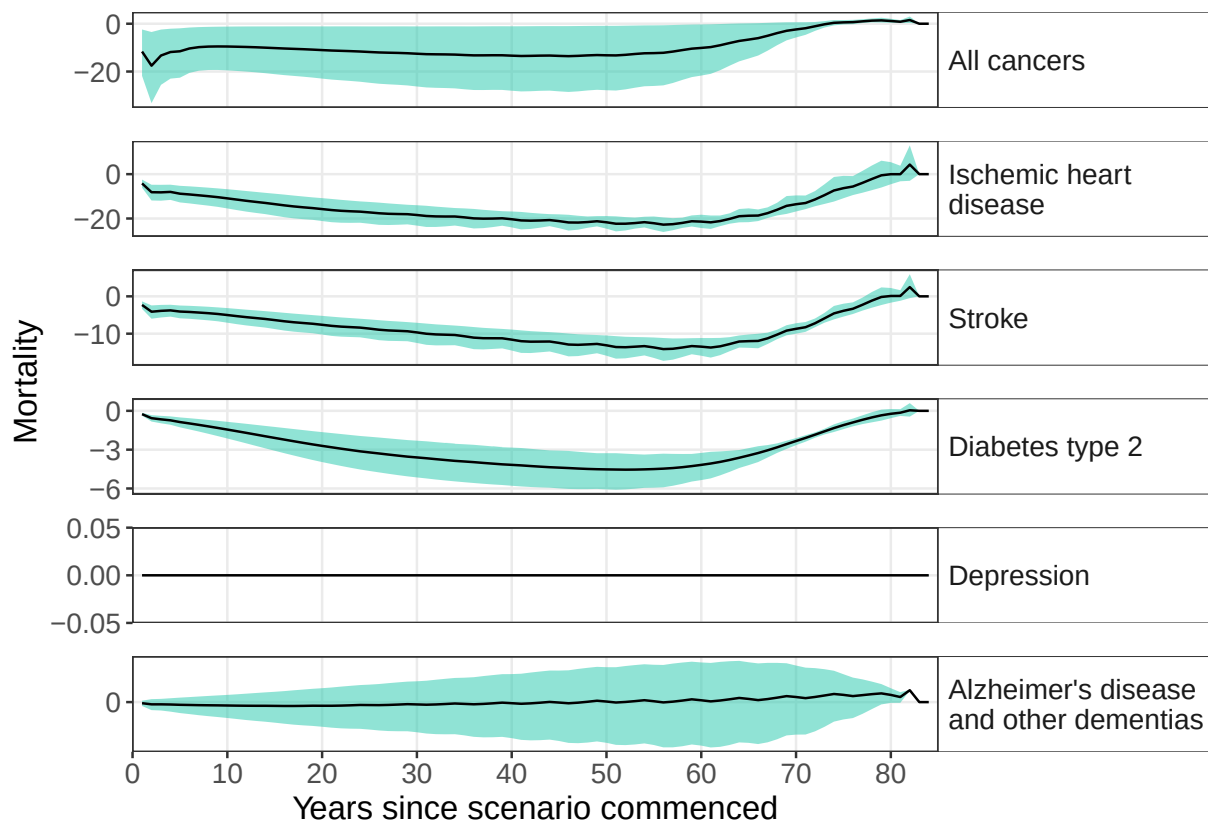


**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



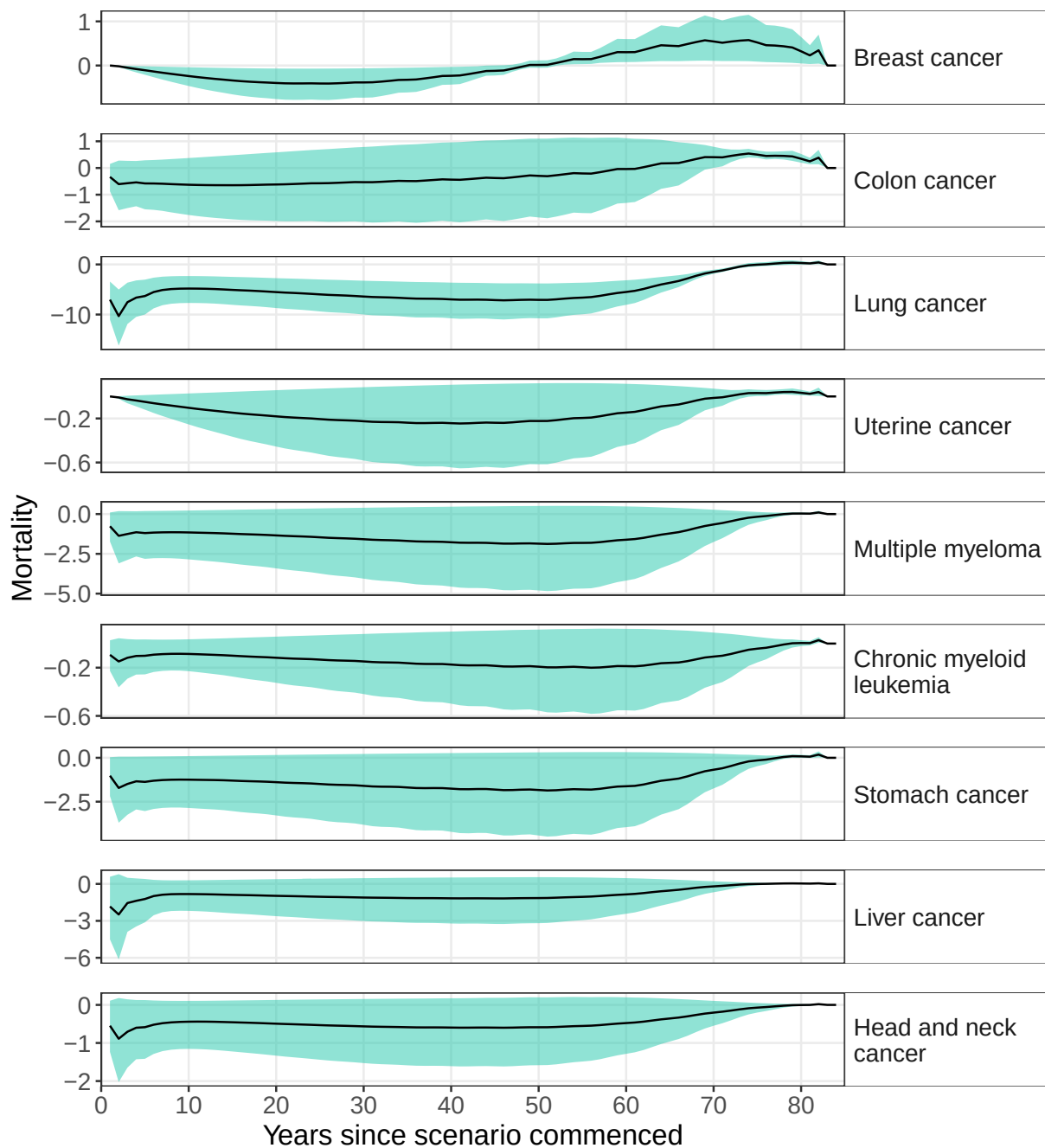
**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.





**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

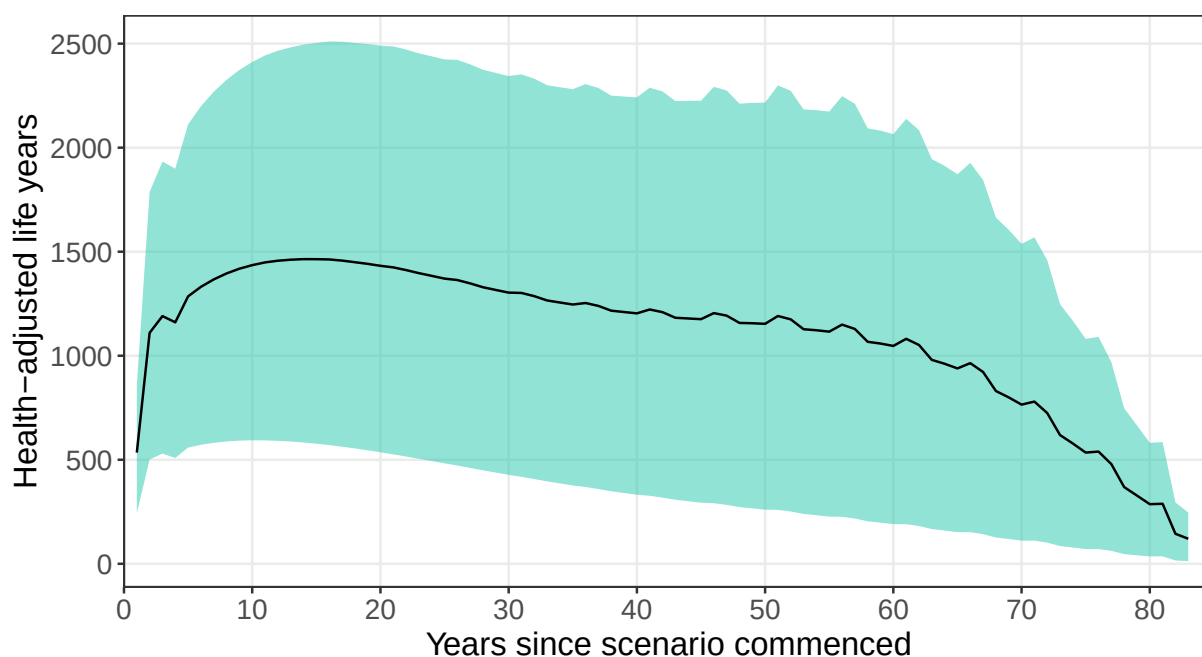
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **91,651** HALYs for the scenario population, which is **46** HALYs per 1,000 members of the population.



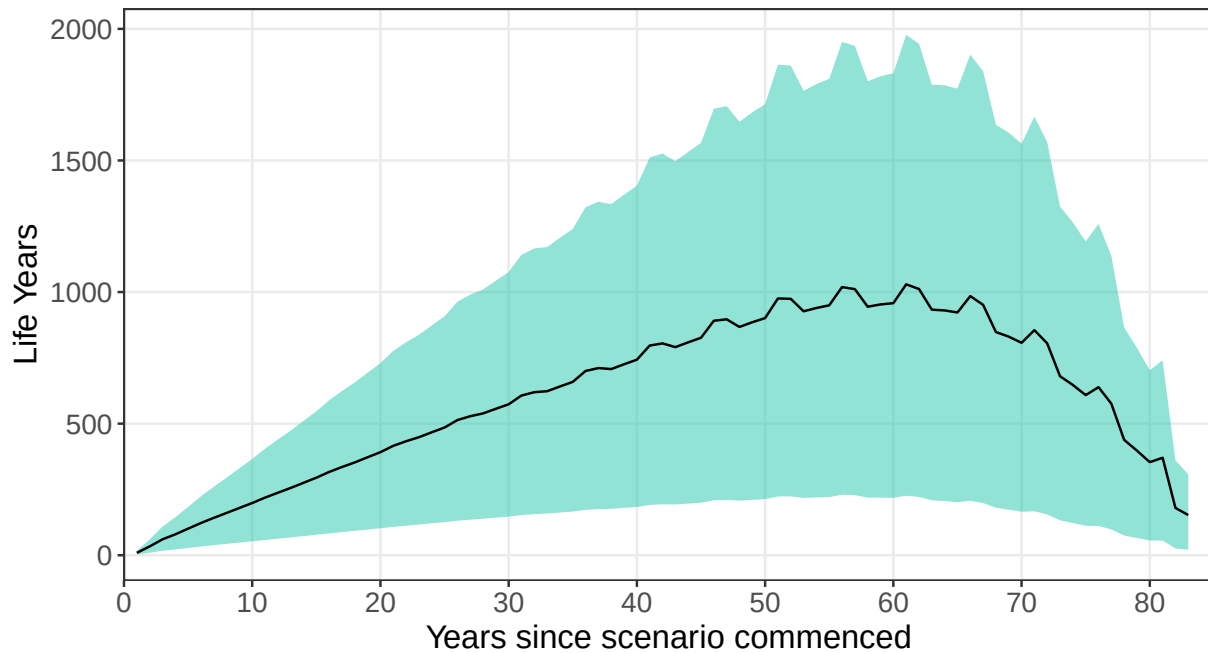
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **50,589** Life Years for the scenario population, which is **25** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **46** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **4,023,467** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **2,667,277** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **1,940,503** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].

## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.



This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$2,378          | \$1,532          | \$1,077          |
| Breast cancer                           | \$12,160         | \$7,949          | \$5,413          |
| All cancers                             | \$47,347         | \$31,797         | \$22,896         |
| Colon cancer                            | \$11,173         | \$8,060          | \$6,079          |
| Chronic myeloid leukemia                | \$3,931          | \$2,353          | \$1,557          |
| Diabetes type 2                         | \$14,963         | \$9,522          | \$6,592          |
| Depression                              | \$114,123        | \$88,757         | \$71,001         |
| Head and neck cancer                    | \$636            | \$438            | \$329            |
| Ischemic heart disease                  | \$38,043         | \$24,967         | \$17,901         |
| Liver cancer                            | \$477            | \$343            | \$269            |
| Multiple myeloma                        | \$9,292          | \$6,166          | \$4,485          |
| Stomach cancer                          | \$3,373          | \$2,253          | \$1,651          |
| Stroke                                  | \$5,389          | \$3,287          | \$2,210          |
| Lung cancer                             | \$5,431          | \$3,705          | \$2,767          |
| Uterine cancer                          | \$876            | \$536            | \$352            |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

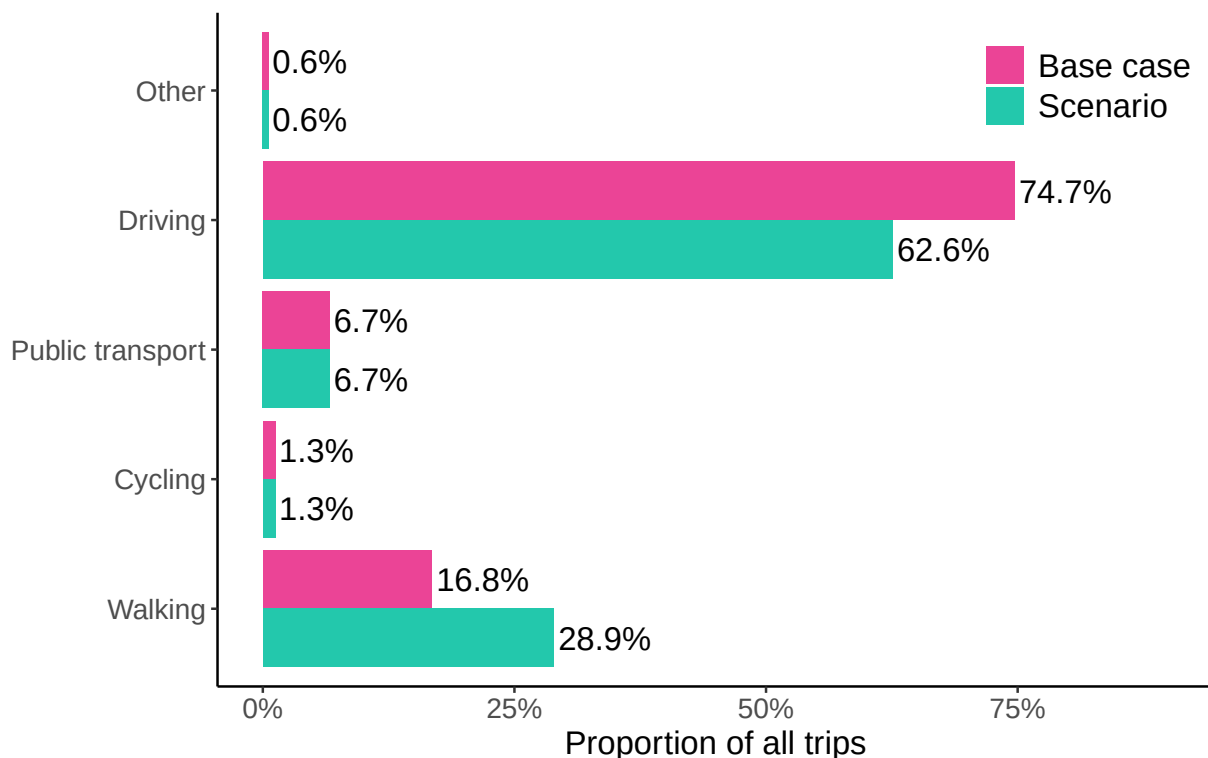
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## Scenario: replacing car trips under 2km with walking for all trip purposes

This scenario shows the results of replacing car trips under 2km for leisure, shopping, work, education or other purposes with walking trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 28.9% and from 74.7% to 62.6% for car trips taken as either a driver or passenger.

Increases in walking translate into a shift from 47.9% to 56.1% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

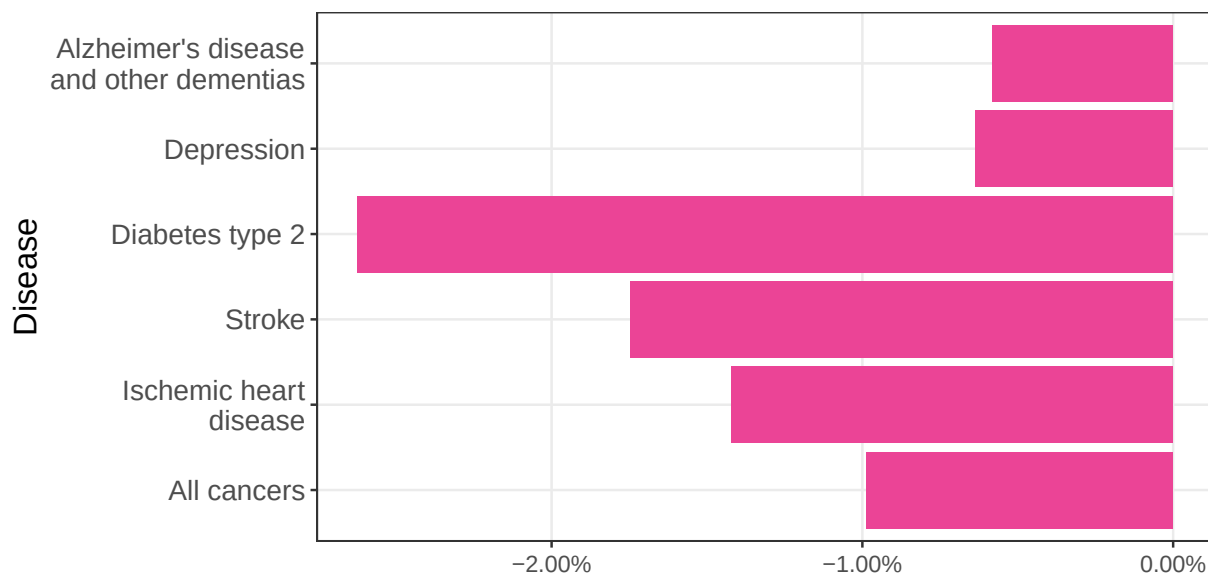
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

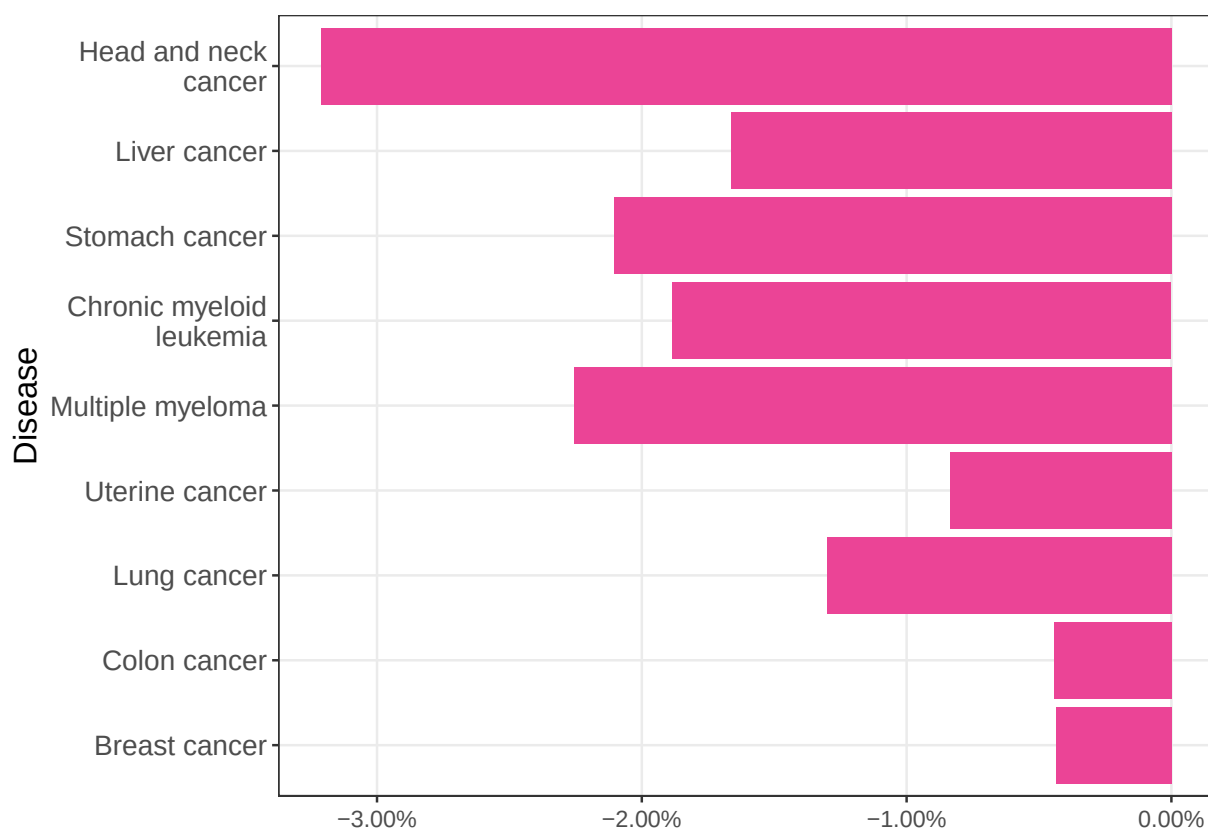
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.58%                              | 3,706                                                                       |
| Breast cancer                           | 0.44%                              | 320                                                                         |
| All cancers                             | 0.99%                              | 3,889                                                                       |
| Colon cancer                            | 0.44%                              | 576                                                                         |
| Chronic myeloid leukemia                | 1.88%                              | 51                                                                          |
| Diabetes type 2                         | 2.62%                              | 8,662                                                                       |
| Depression                              | 0.64%                              | 10,191                                                                      |
| Head and neck cancer                    | 3.21%                              | 142                                                                         |
| Ischemic heart disease                  | 1.42%                              | 12,515                                                                      |
| Liver cancer                            | 1.66%                              | 293                                                                         |
| Multiple myeloma                        | 2.26%                              | 437                                                                         |
| Stomach cancer                          | 2.11%                              | 503                                                                         |
| Stroke                                  | 1.75%                              | 4,546                                                                       |
| Lung cancer                             | 1.30%                              | 1,495                                                                       |
| Uterine cancer                          | 0.83%                              | 71                                                                          |

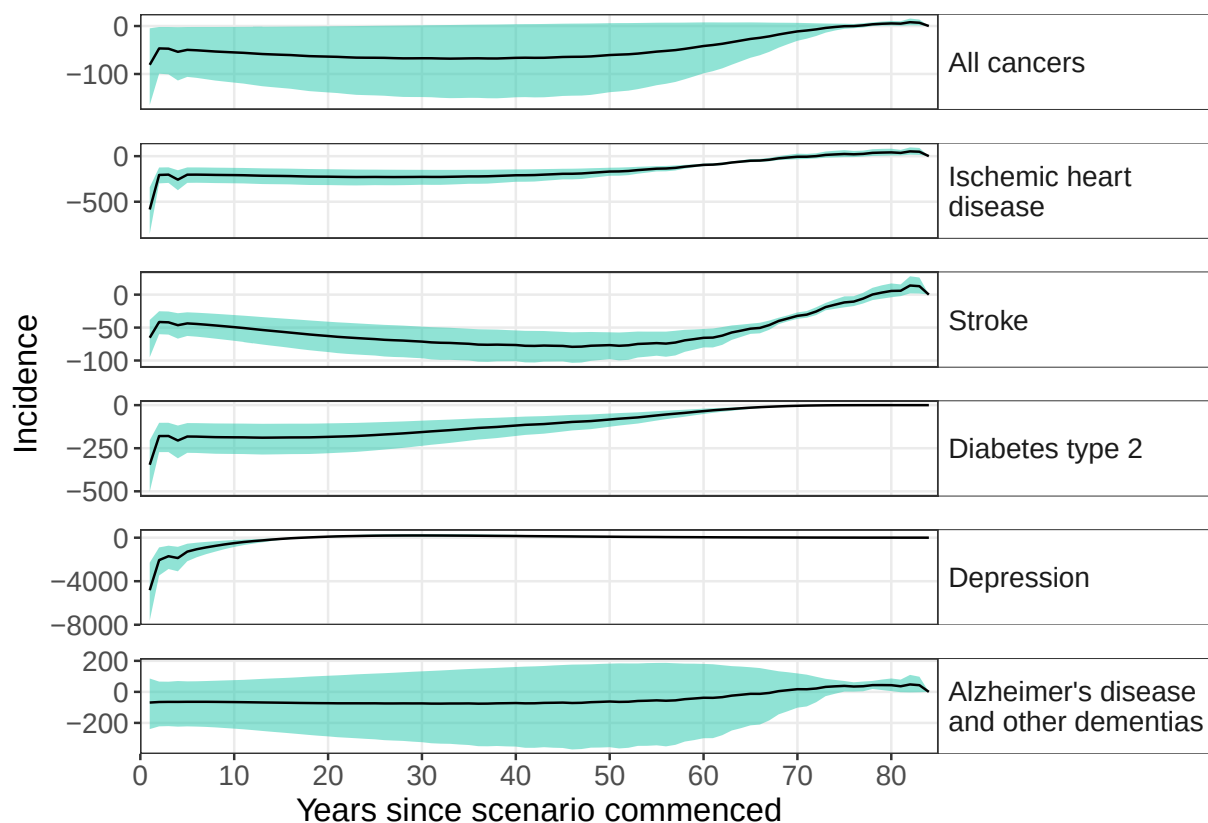
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



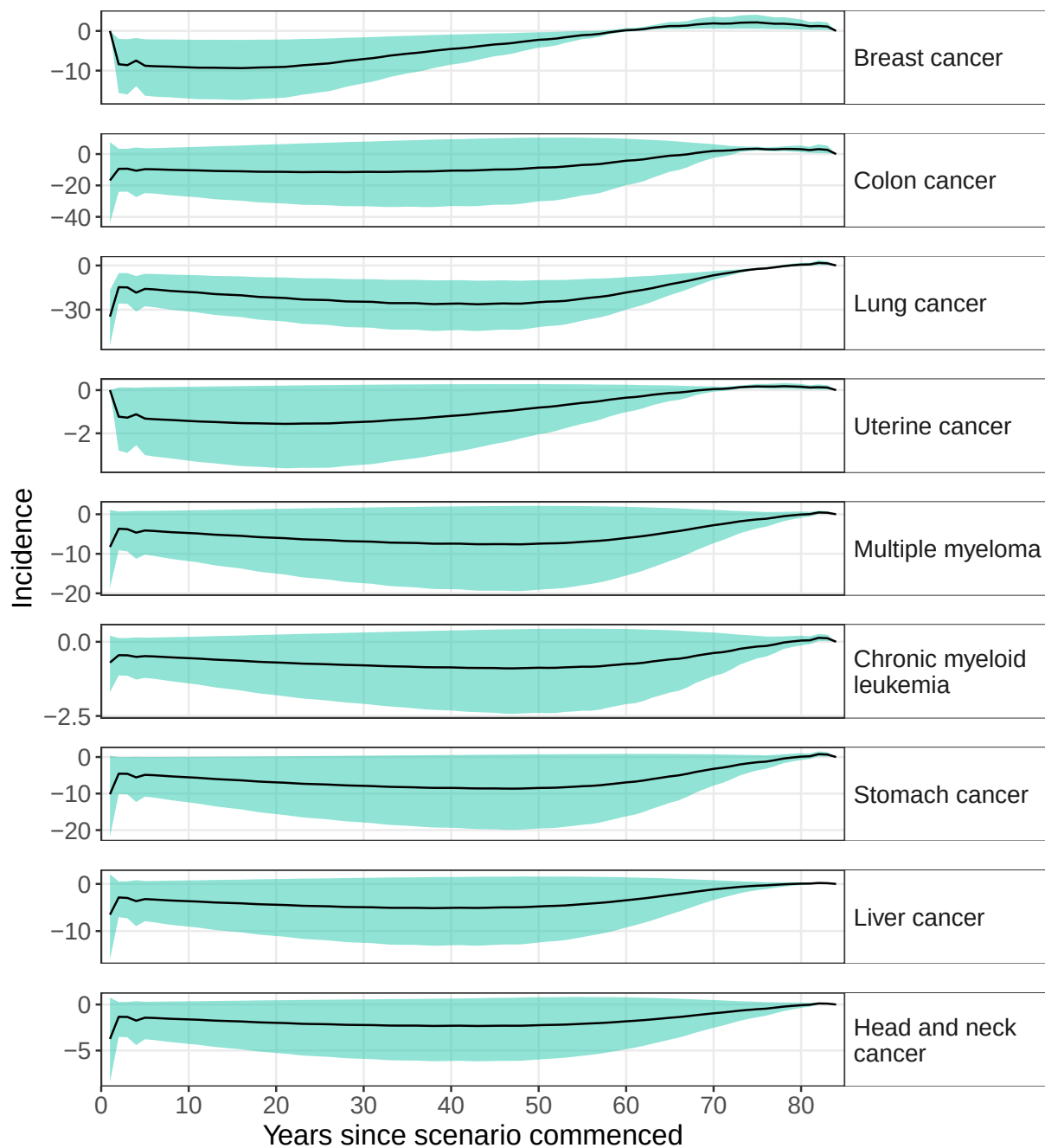
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

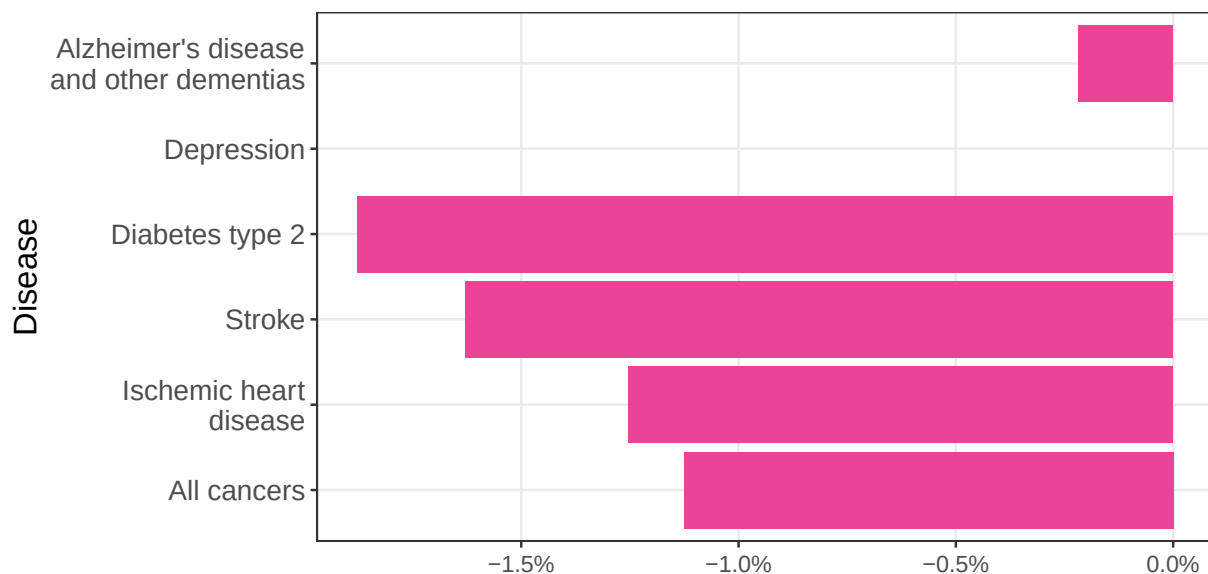
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

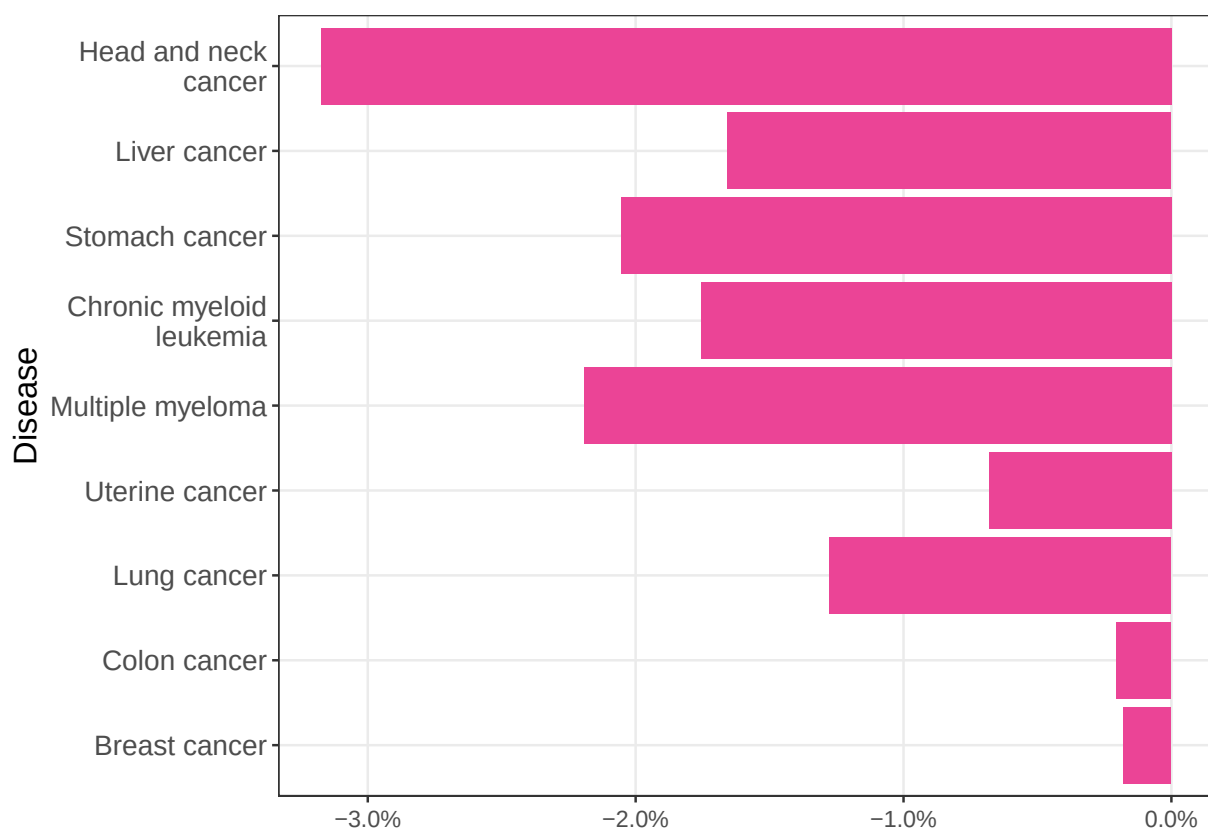
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.22%                   | 521                                                               |
| Breast cancer                           | 0.18%                   | 58                                                                |
| All cancers                             | 1.13%                   | 2,850                                                             |
| Colon cancer                            | 0.21%                   | 88                                                                |
| Chronic myeloid leukemia                | 1.76%                   | 36                                                                |
| Diabetes type 2                         | 1.88%                   | 927                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 3.17%                   | 126                                                               |
| Ischemic heart disease                  | 1.25%                   | 4,700                                                             |
| Liver cancer                            | 1.66%                   | 283                                                               |
| Multiple myeloma                        | 2.19%                   | 382                                                               |
| Stomach cancer                          | 2.05%                   | 434                                                               |
| Stroke                                  | 1.63%                   | 2,848                                                             |
| Lung cancer                             | 1.28%                   | 1,392                                                             |
| Uterine cancer                          | 0.68%                   | 49                                                                |

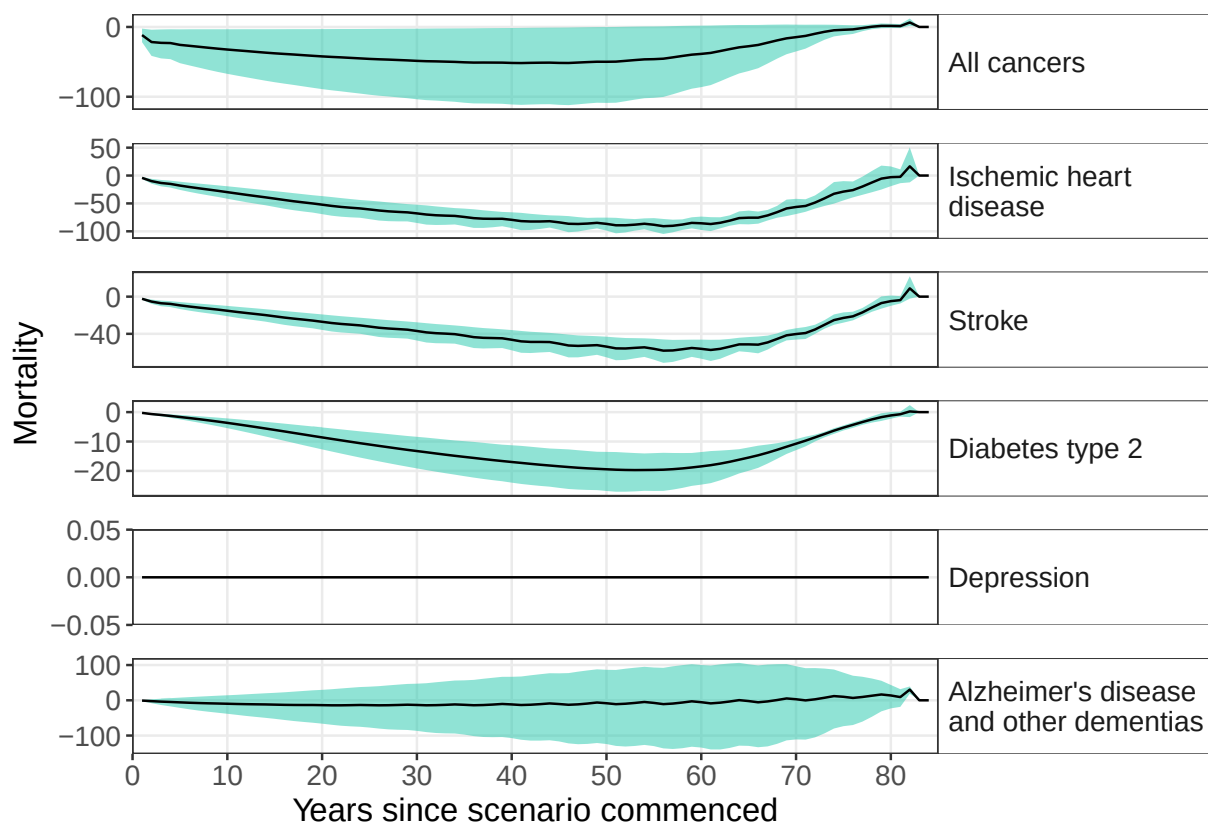
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

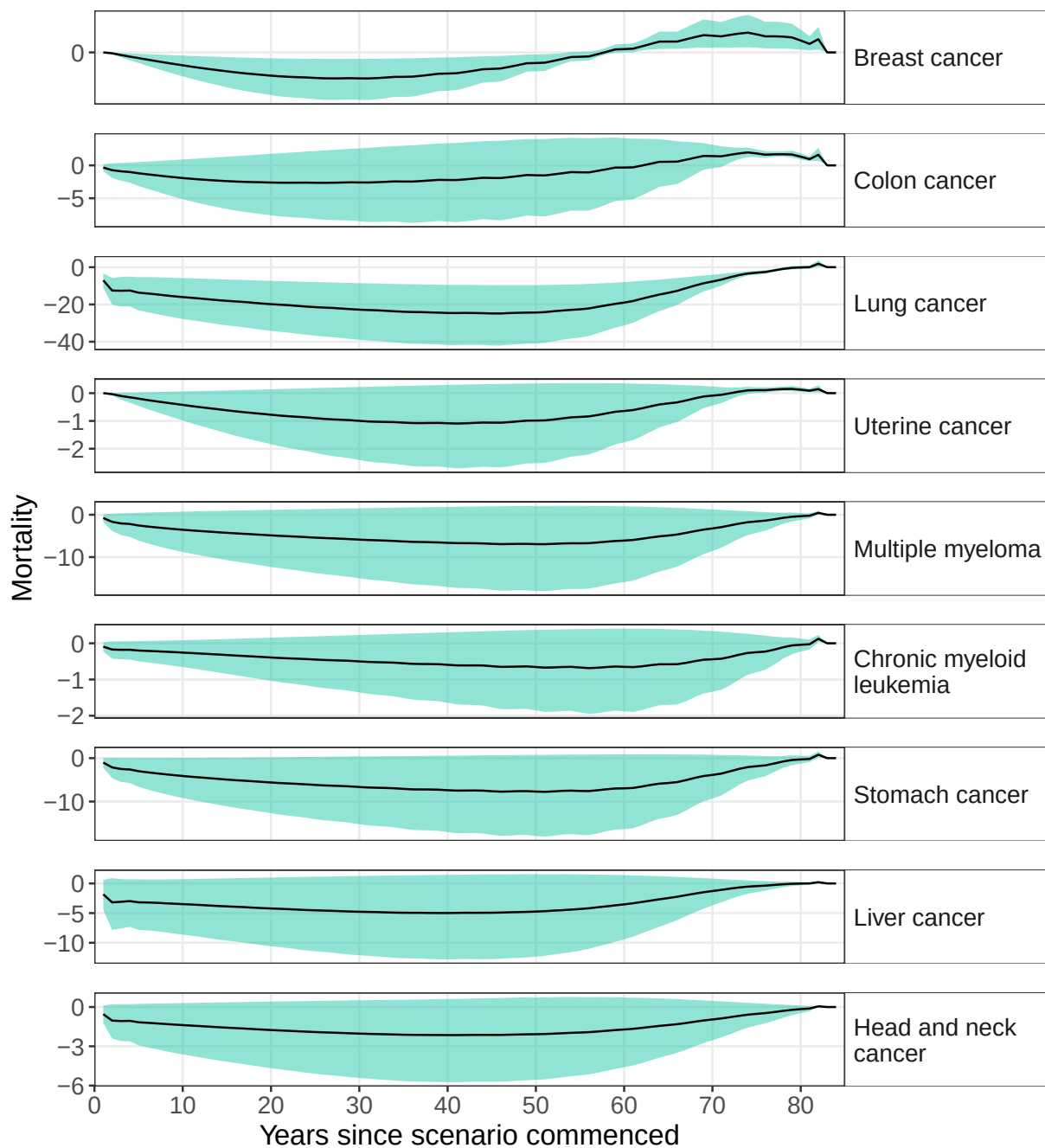


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

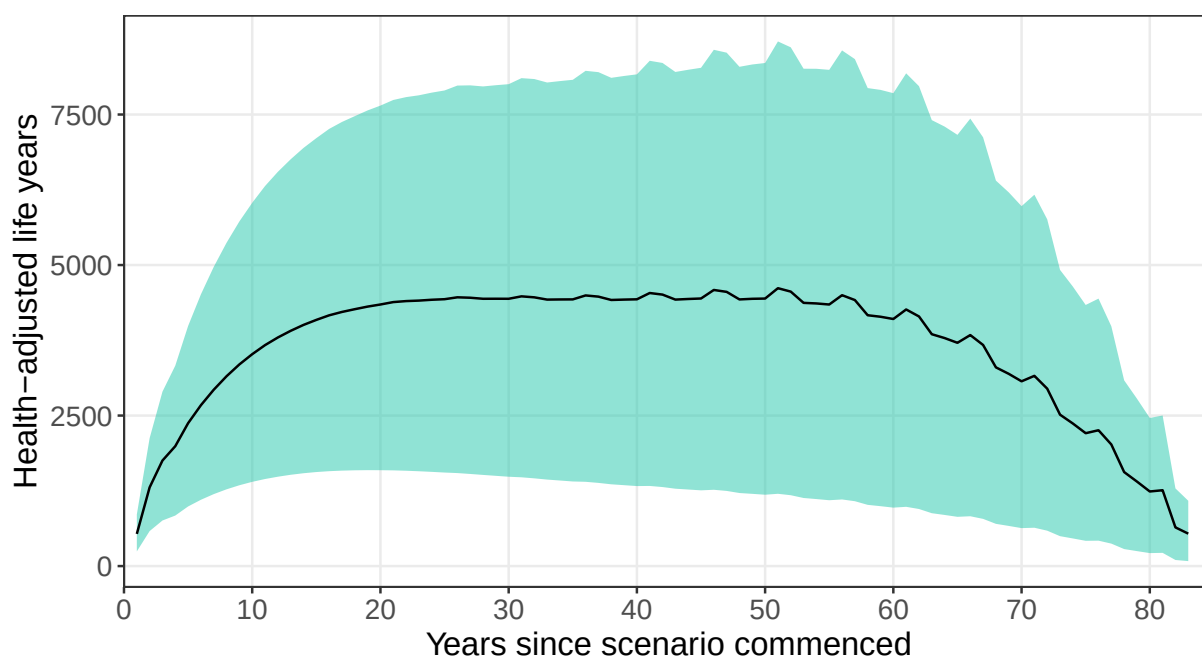
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **305,291** HALYs for the scenario population, which is **152** HALYs per 1,000 members of the population.



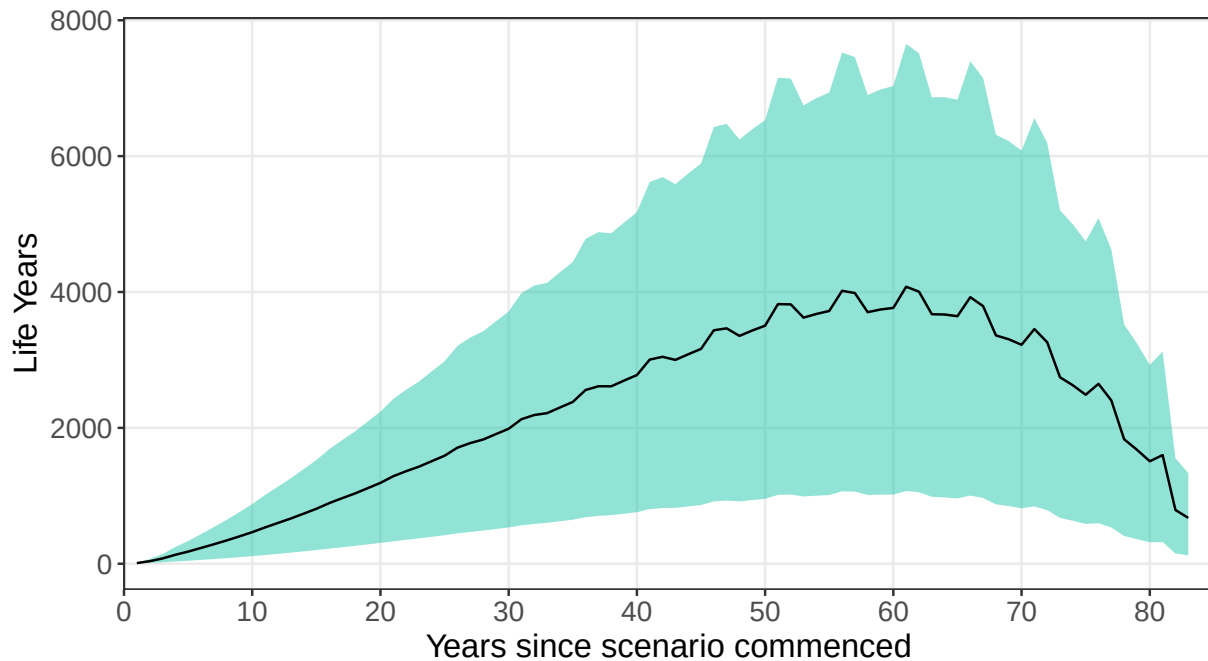
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **189,216** Life Years for the scenario population, which is **94** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **152** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **11,777,115** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **7,210,756** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **4,898,813** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$9,308          | \$5,609          | \$3,704          |
| Breast cancer                           | \$67,653         | \$42,574         | \$28,318         |
| All cancers                             | \$184,699        | \$117,571        | \$80,153         |
| Colon cancer                            | \$41,129         | \$27,592         | \$19,414         |
| Chronic myeloid leukemia                | \$11,734         | \$6,889          | \$4,445          |
| Diabetes type 2                         | \$53,469         | \$31,851         | \$20,651         |
| Depression                              | \$286,291        | \$216,104        | \$167,268        |
| Head and neck cancer                    | \$1,940          | \$1,254          | \$884            |
| Ischemic heart disease                  | \$118,670        | \$71,560         | \$47,331         |
| Liver cancer                            | \$1,676          | \$1,114          | \$806            |
| Multiple myeloma                        | \$28,899         | \$18,085         | \$12,378         |
| Stomach cancer                          | \$11,207         | \$7,009          | \$4,802          |
| Stroke                                  | \$18,017         | \$10,483         | \$6,713          |
| Lung cancer                             | \$16,569         | \$10,703         | \$7,544          |
| Uterine cancer                          | \$3,951          | \$2,415          | \$1,583          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

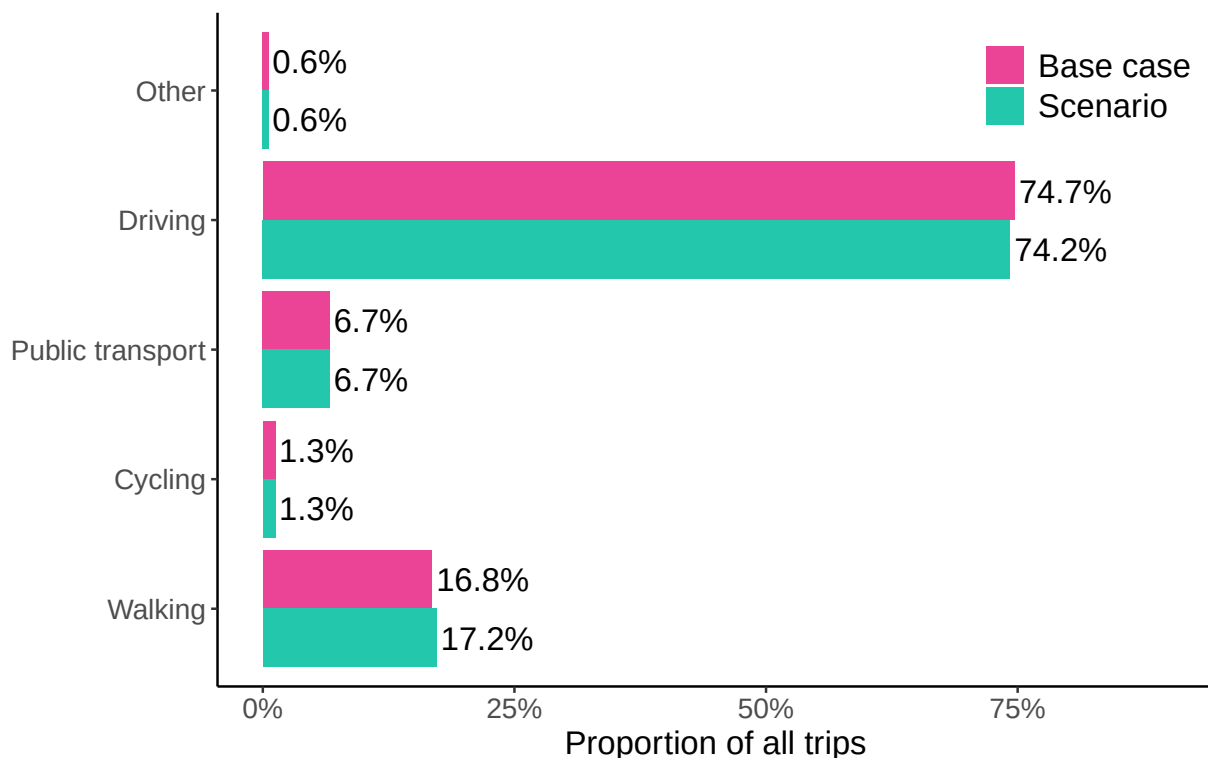
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
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4. Terrill, M. & Batrouney, H. (2018). *Unfreezing discount rates: Transport infrastructure for tomorrow*. Grattan Institute.
5. Abelson, P. (2008). *Establishing a monetary value for lives saved: issues and controversies*. Canberra: Office of Best Practice Regulation, Department of Finance and Deregulation.
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7. Haacker, M., Hallett, T.B. & Atun, R. (2020). *On discount rates for economic evaluations in global health*. Health Policy and Planning, 2020. 35(1): p. 107-114.
8. Australian Institute of Health and Welfare (2023). Technical Notes: Estimating Spending per prevalent case of disease. Health system spending per case of disease and for certain risk factors, Estimating the spending per prevalent case of disease - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
11. Zapata-Diomedí, B., Boulangé, C., Giles-Corti, B., Phelan, K., Washington, S., Veerman, L.J., & Gunn, L. (2019). Physical activity-related health and economic benefits of building walkable neighbourhoods: A modelled comparison between brownfield and greenfield developments. International Journal of Behavioural Nutrition and Physical Activity.
12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577-1590.

## Scenario: replacing car trips under 1km with walking for commuting trip purposes

This scenario shows the results of replacing car trips under 1km for work related or education purposes with walking trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 17.2% and from 74.7% to 74.2% for car trips taken as either a driver or passenger.

Increases in walking translate into a shift from 47.9% to 48.1% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

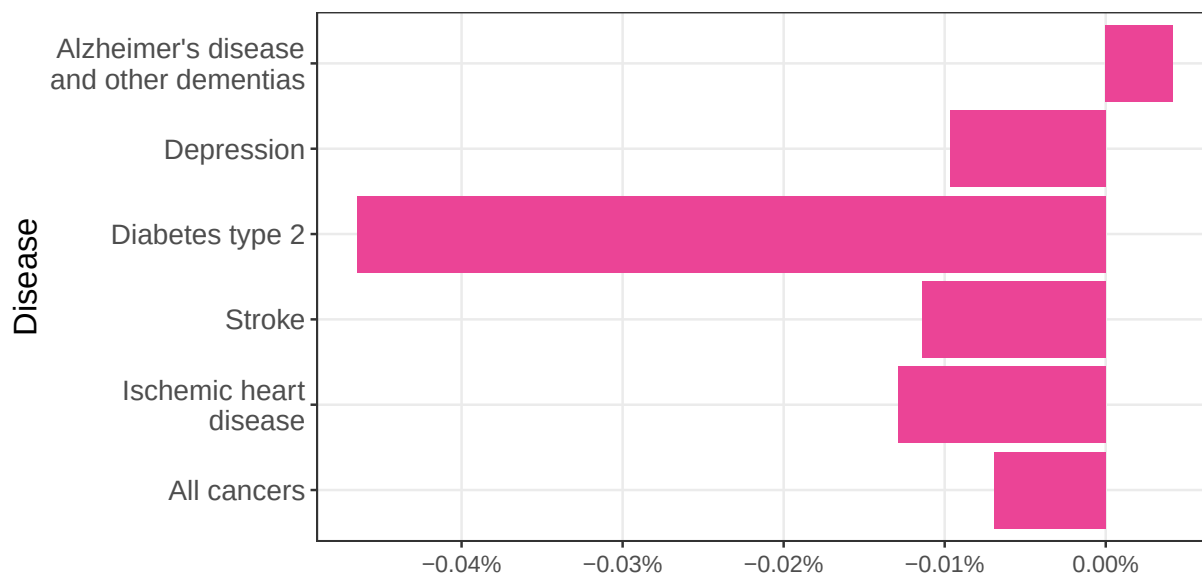
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

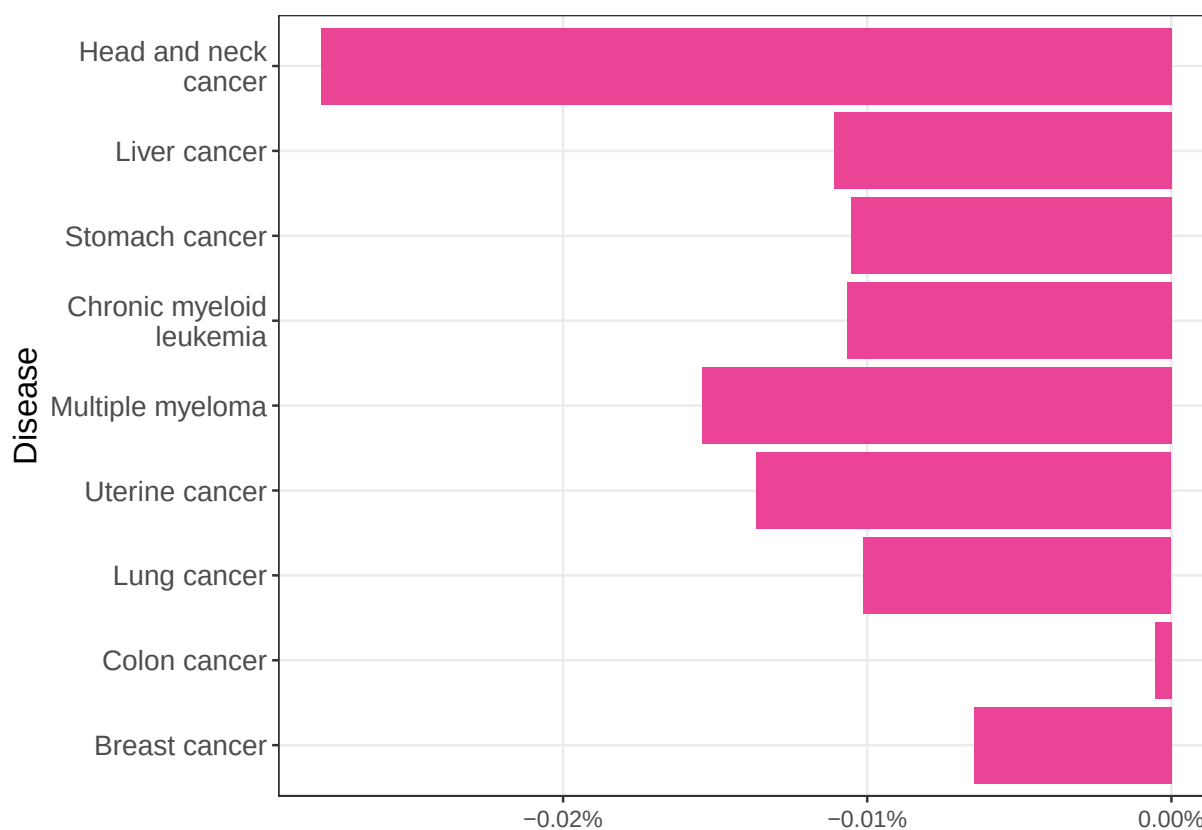
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.00%                              | -27                                                                         |
| Breast cancer                           | 0.01%                              | 5                                                                           |
| All cancers                             | 0.01%                              | 27                                                                          |
| Colon cancer                            | 0.00%                              | 1                                                                           |
| Chronic myeloid leukemia                | 0.01%                              | 0                                                                           |
| Diabetes type 2                         | 0.05%                              | 153                                                                         |
| Depression                              | 0.01%                              | 154                                                                         |
| Head and neck cancer                    | 0.03%                              | 1                                                                           |
| Ischemic heart disease                  | 0.01%                              | 114                                                                         |
| Liver cancer                            | 0.01%                              | 2                                                                           |
| Multiple myeloma                        | 0.02%                              | 3                                                                           |
| Stomach cancer                          | 0.01%                              | 3                                                                           |
| Stroke                                  | 0.01%                              | 30                                                                          |
| Lung cancer                             | 0.01%                              | 12                                                                          |
| Uterine cancer                          | 0.01%                              | 1                                                                           |

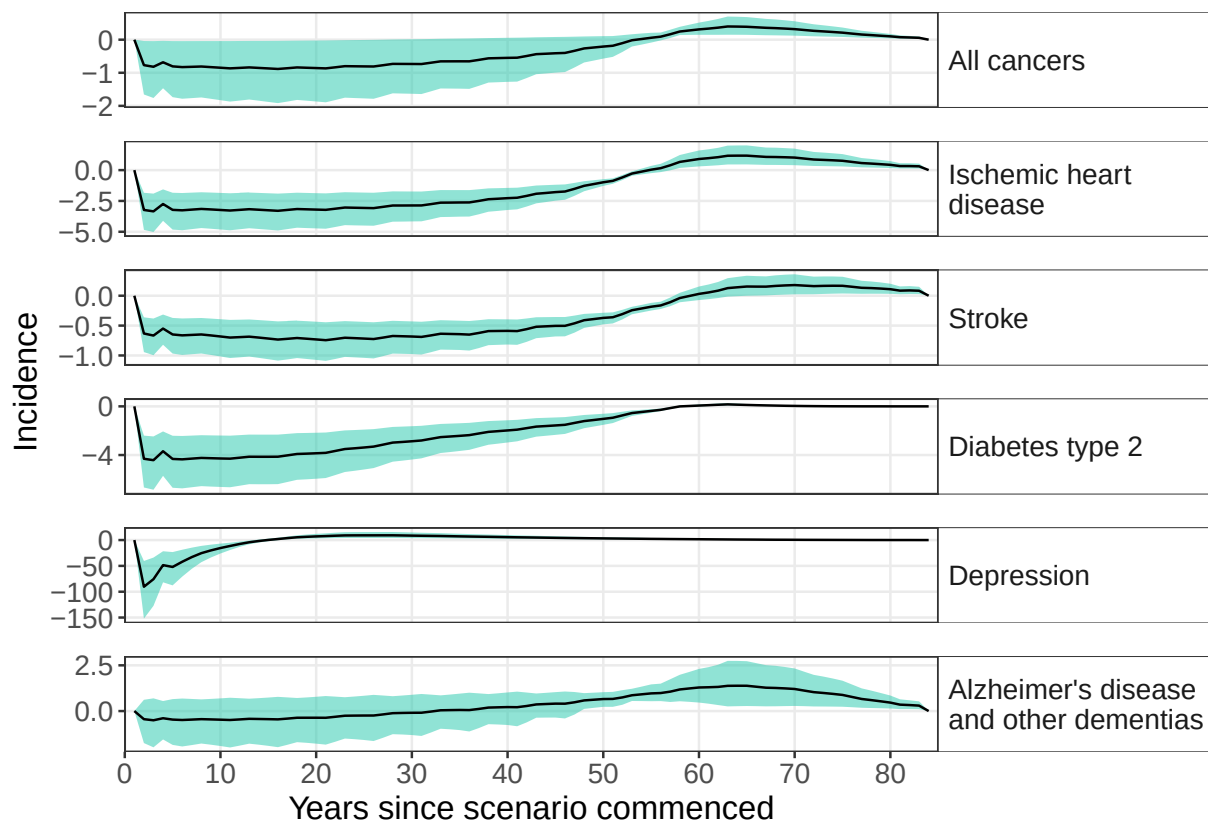
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



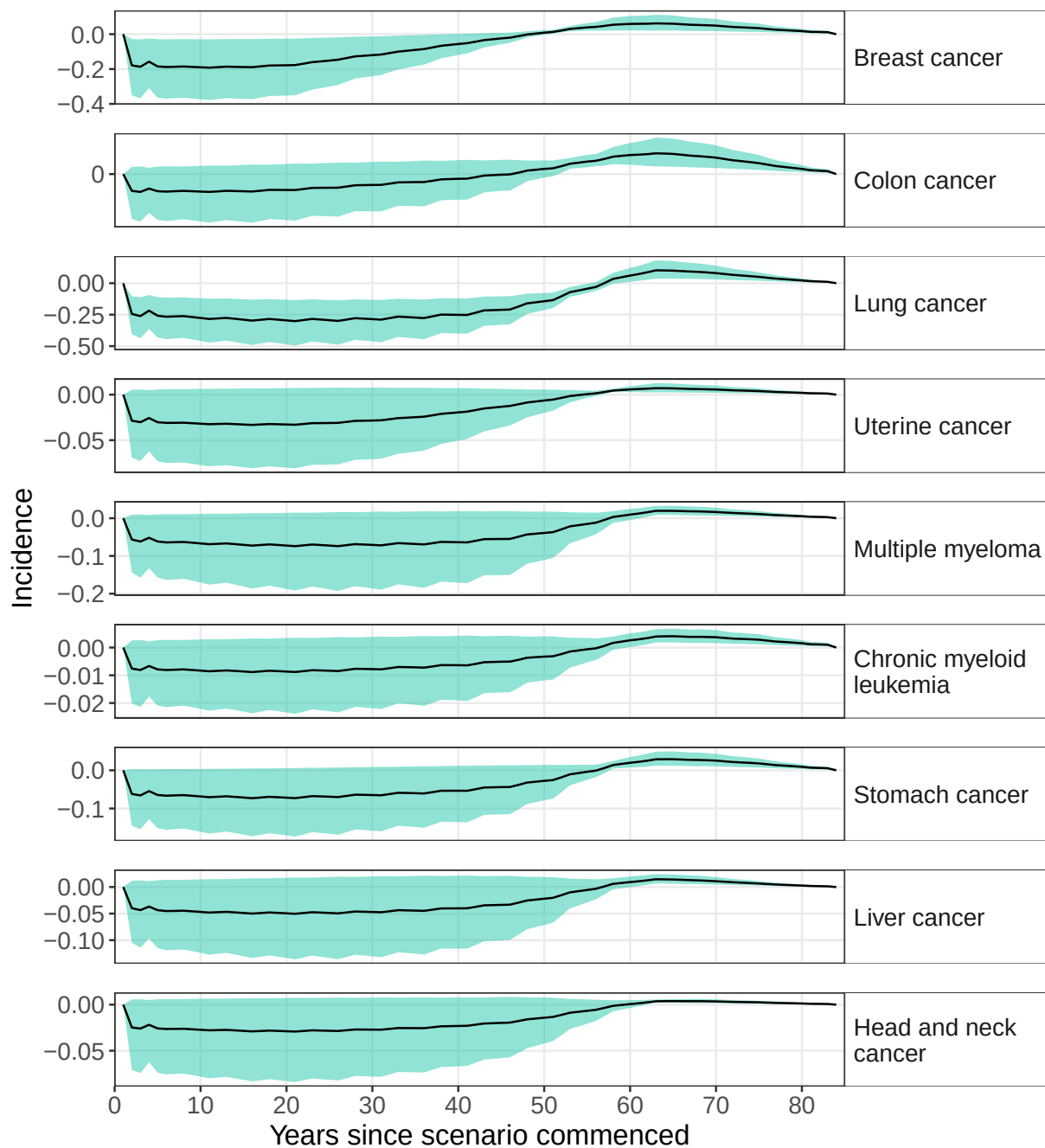
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

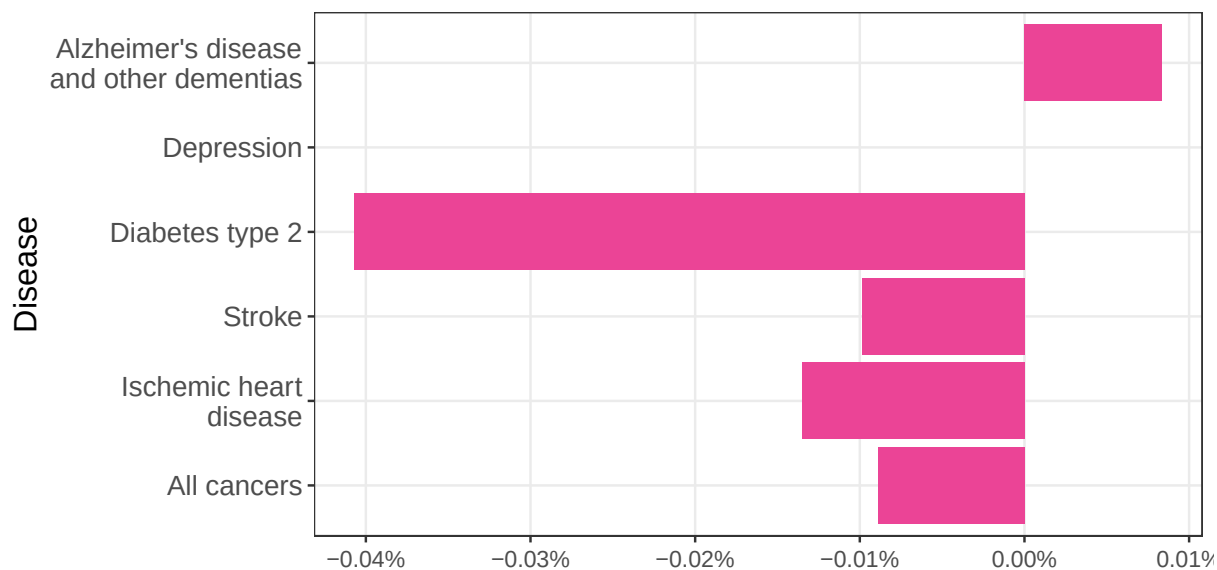
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

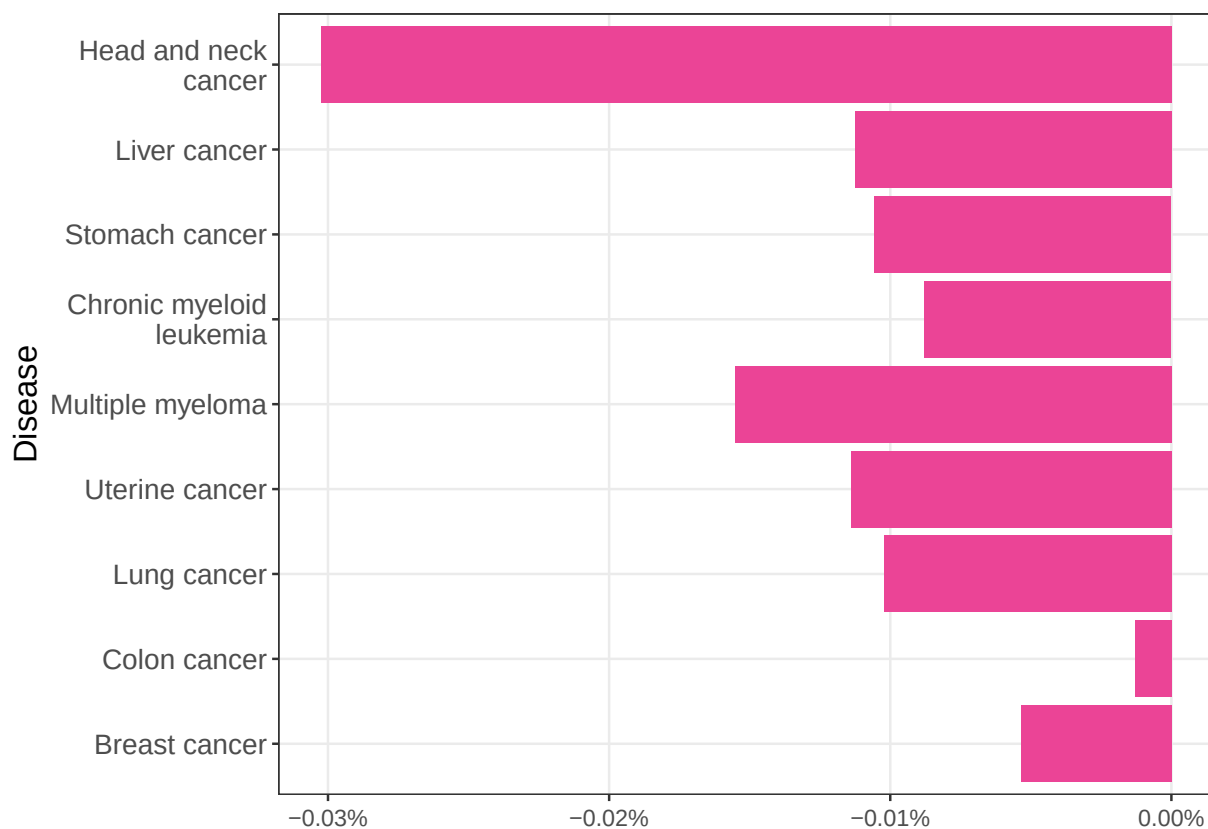
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.01%                  | -20                                                               |
| Breast cancer                           | 0.01%                   | 2                                                                 |
| All cancers                             | 0.01%                   | 22                                                                |
| Colon cancer                            | 0.00%                   | 1                                                                 |
| Chronic myeloid leukemia                | 0.01%                   | 0                                                                 |
| Diabetes type 2                         | 0.04%                   | 20                                                                |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.03%                   | 1                                                                 |
| Ischemic heart disease                  | 0.01%                   | 51                                                                |
| Liver cancer                            | 0.01%                   | 2                                                                 |
| Multiple myeloma                        | 0.02%                   | 3                                                                 |
| Stomach cancer                          | 0.01%                   | 2                                                                 |
| Stroke                                  | 0.01%                   | 17                                                                |
| Lung cancer                             | 0.01%                   | 11                                                                |
| Uterine cancer                          | 0.01%                   | 1                                                                 |

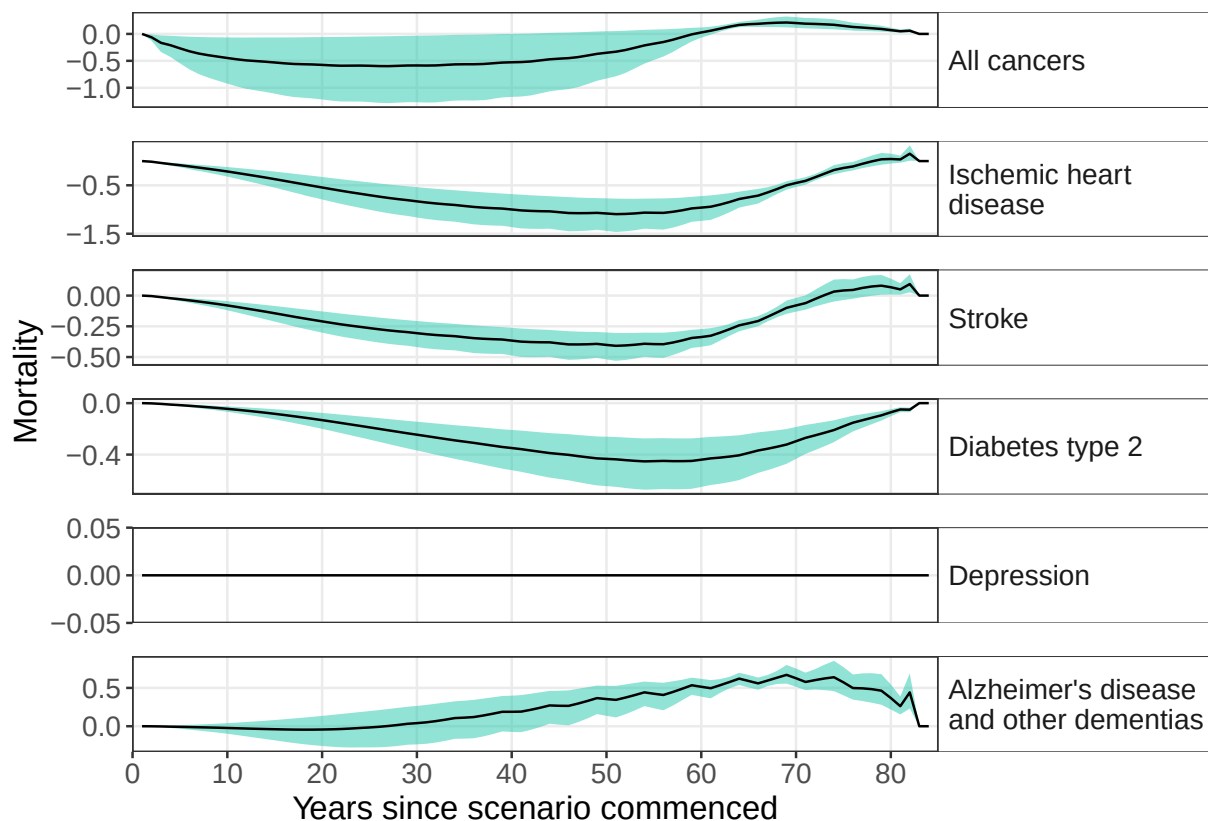
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

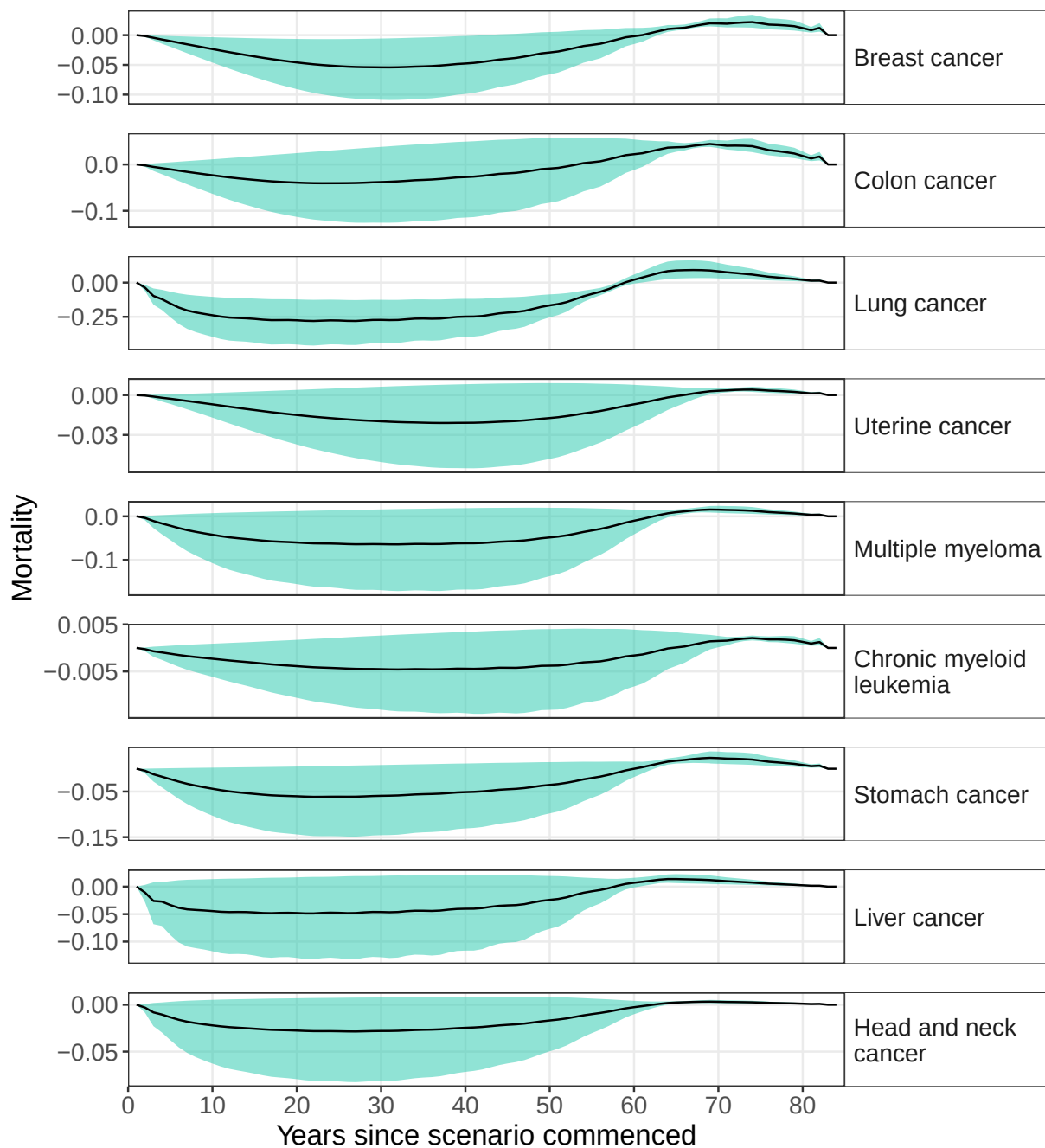


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

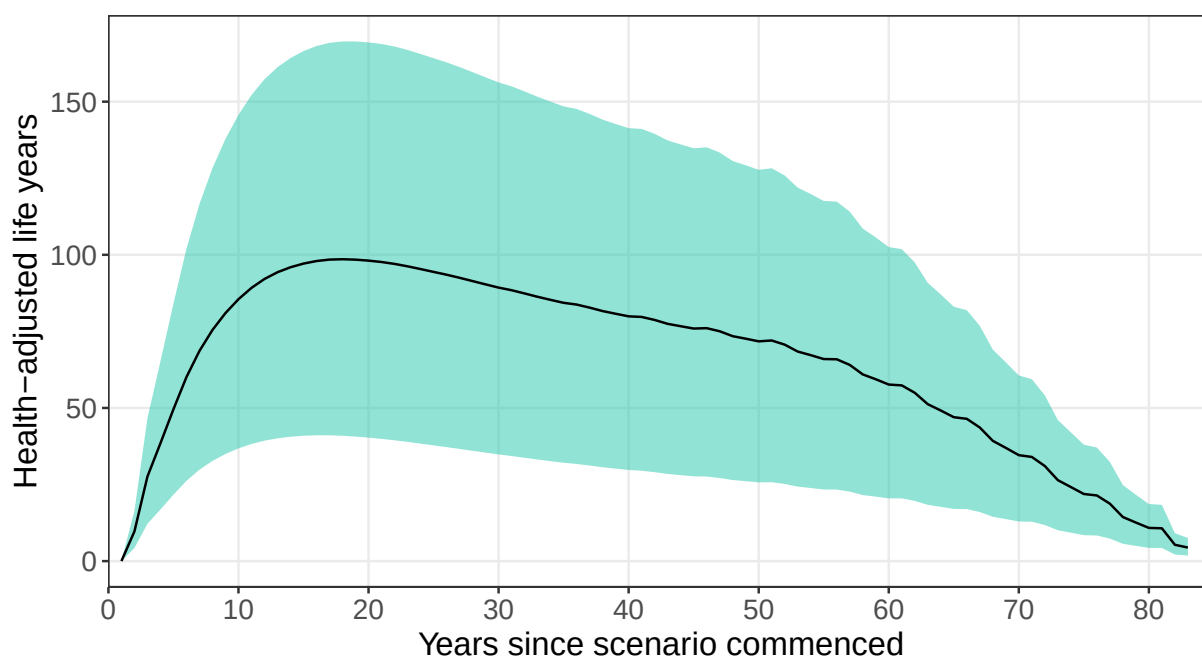
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **5,388** HALYs for the scenario population, which is **2.7** HALYs per 1,000 members of the population.



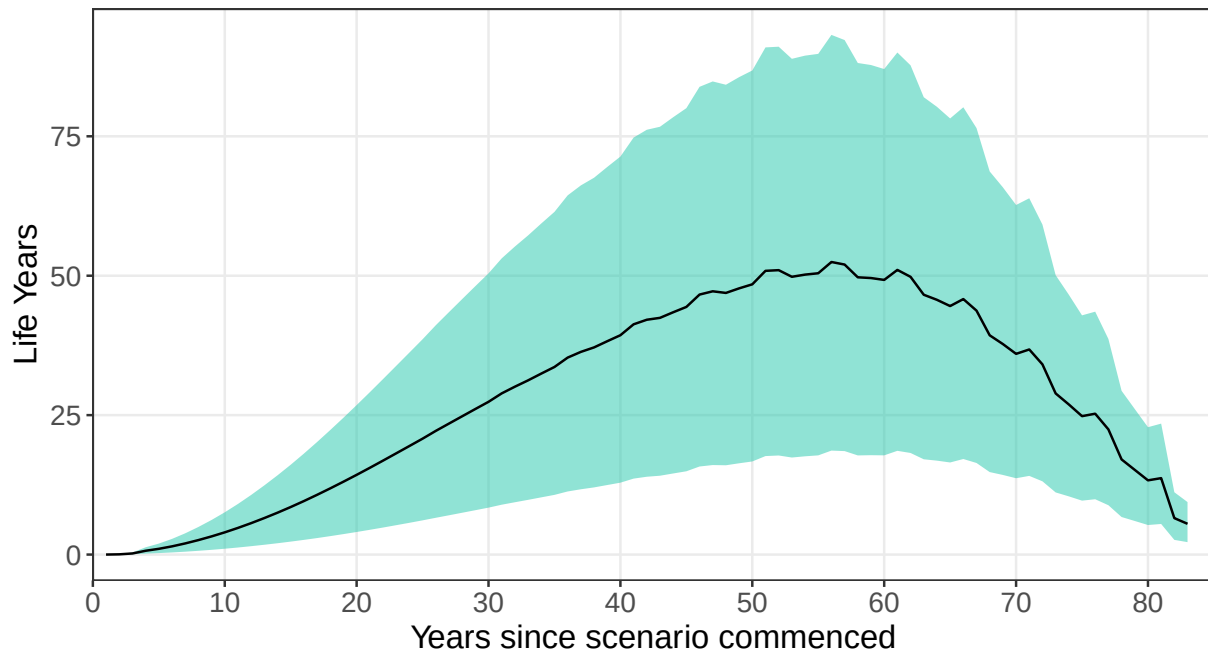
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **2,356** Life Years for the scenario population, which is **1.2** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **2.7** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **233,423** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **149,311** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **103,775** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$58             | \$42             | \$30             |
| Breast cancer                           | \$1,563          | \$974            | \$644            |
| All cancers                             | \$3,382          | \$2,162          | \$1,461          |
| Colon cancer                            | \$557            | \$384            | \$269            |
| Chronic myeloid leukemia                | \$246            | \$147            | \$95             |
| Diabetes type 2                         | \$1,160          | \$674            | \$426            |
| Depression                              | \$8,123          | \$5,931          | \$4,484          |
| Head and neck cancer                    | \$32             | \$21             | \$15             |
| Ischemic heart disease                  | \$1,902          | \$1,109          | \$701            |
| Liver cancer                            | \$20             | \$14             | \$10             |
| Multiple myeloma                        | \$455            | \$290            | \$198            |
| Stomach cancer                          | \$149            | \$96             | \$66             |
| Stroke                                  | \$325            | \$191            | \$121            |
| Lung cancer                             | \$262            | \$174            | \$123            |
| Uterine cancer                          | \$100            | \$61             | \$40             |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

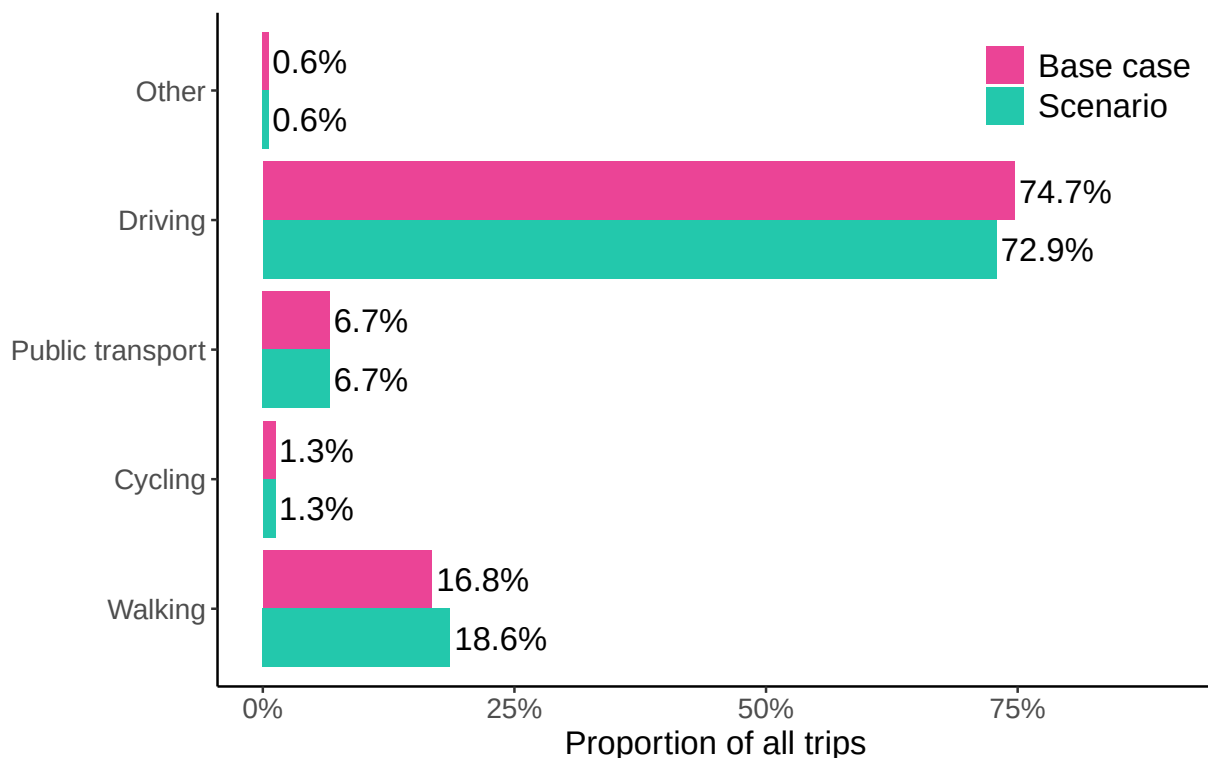
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
3. Ananthapavan, J., Moodie, M., Milat, A.J., & Carter, R. (2021). *Systematic review to update 'value of a statistical life' estimates for Australia*. International journal of environmental research and public health, 2021. 18(11): p. 6168.
4. Terrill, M. & Batrouney, H. (2018). *Unfreezing discount rates: Transport infrastructure for tomorrow*. Grattan Institute.
5. Abelson, P. (2008). *Establishing a monetary value for lives saved: issues and controversies*. Canberra: Office of Best Practice Regulation, Department of Finance and Deregulation.
6. Department of the Prime Minister and Cabinet. (2022). *Best practice regulation guidance note: Value of statistical life*. Australian Government.
7. Haacker, M., Hallett, T.B. & Atun, R. (2020). *On discount rates for economic evaluations in global health*. Health Policy and Planning, 2020. 35(1): p. 107-114.
8. Australian Institute of Health and Welfare (2023). Technical Notes: Estimating Spending per prevalent case of disease. Health system spending per case of disease and for certain risk factors, Estimating the spending per prevalent case of disease – Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data – Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
11. Zapata-Diomedí, B., Boulangé, C., Giles-Corti, B., Phelan, K., Washington, S., Veerman, L.J., & Gunn, L. (2019). Physical activity-related health and economic benefits of building walkable neighbourhoods: A modelled comparison between brownfield and greenfield developments. International Journal of Behavioural Nutrition and Physical Activity.
12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577-1590.

## Scenario: replacing car trips under 2km with walking for commuting trip purposes

This scenario shows the results of replacing car trips under 2km for work related or education purposes with walking trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 18.6% and from 74.7% to 72.9% for car trips taken as either a driver or passenger.

Increases in walking translate into a shift from 47.9% to 49.2% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

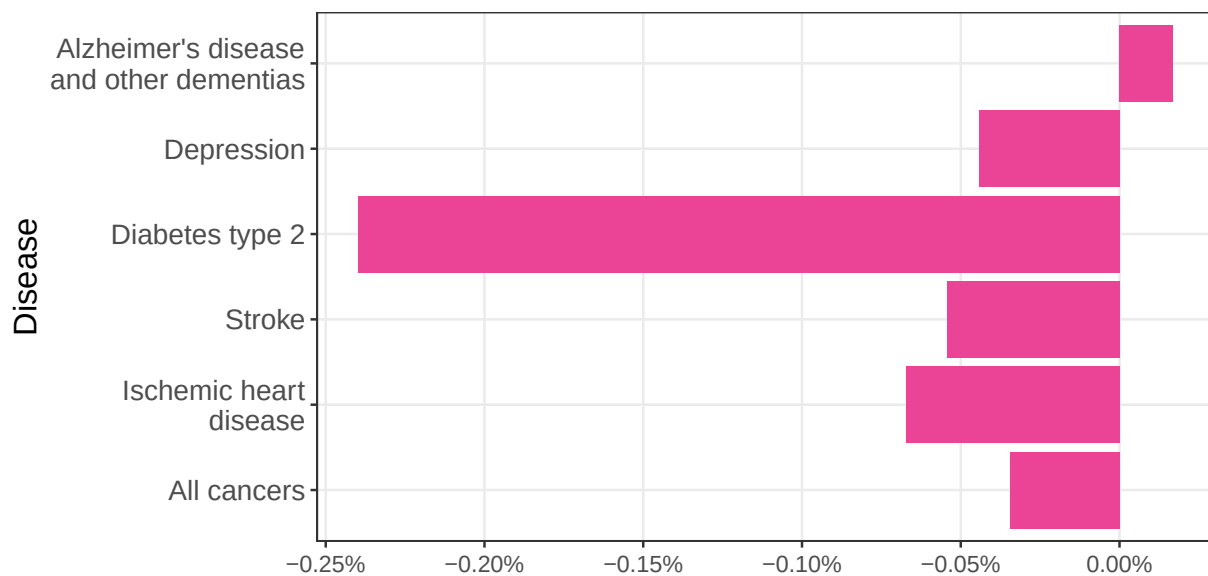
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

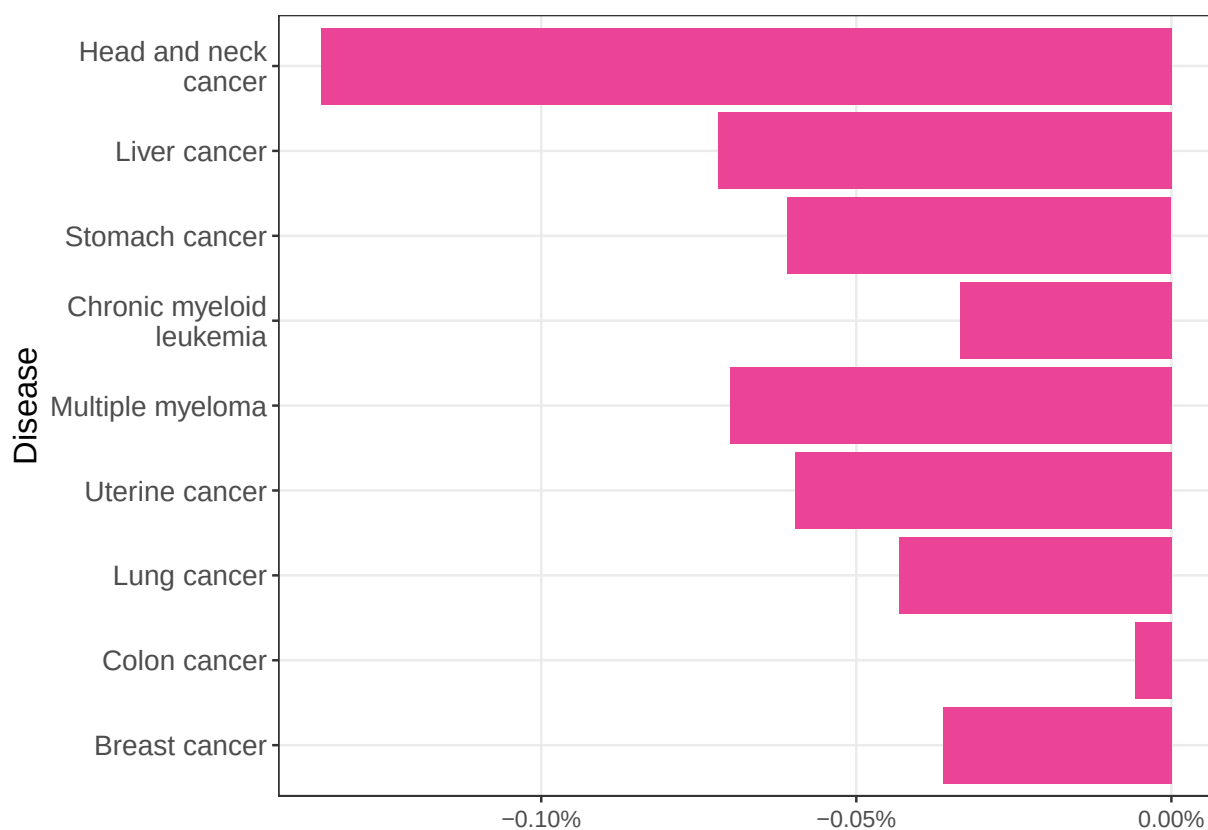
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.02%                             | -108                                                                        |
| Breast cancer                           | 0.04%                              | 27                                                                          |
| All cancers                             | 0.03%                              | 136                                                                         |
| Colon cancer                            | 0.01%                              | 7                                                                           |
| Chronic myeloid leukemia                | 0.03%                              | 1                                                                           |
| Diabetes type 2                         | 0.24%                              | 792                                                                         |
| Depression                              | 0.04%                              | 708                                                                         |
| Head and neck cancer                    | 0.13%                              | 6                                                                           |
| Ischemic heart disease                  | 0.07%                              | 593                                                                         |
| Liver cancer                            | 0.07%                              | 13                                                                          |
| Multiple myeloma                        | 0.07%                              | 14                                                                          |
| Stomach cancer                          | 0.06%                              | 15                                                                          |
| Stroke                                  | 0.05%                              | 142                                                                         |
| Lung cancer                             | 0.04%                              | 50                                                                          |
| Uterine cancer                          | 0.06%                              | 5                                                                           |

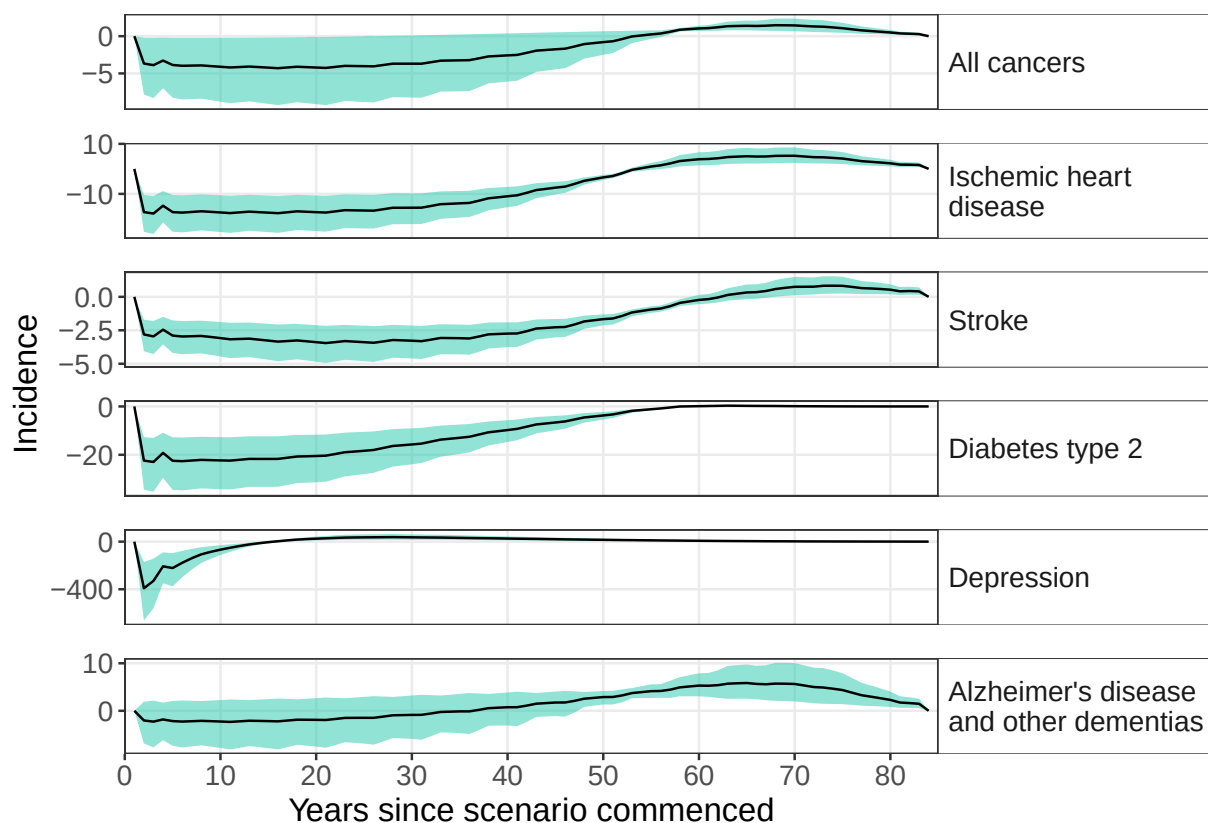
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



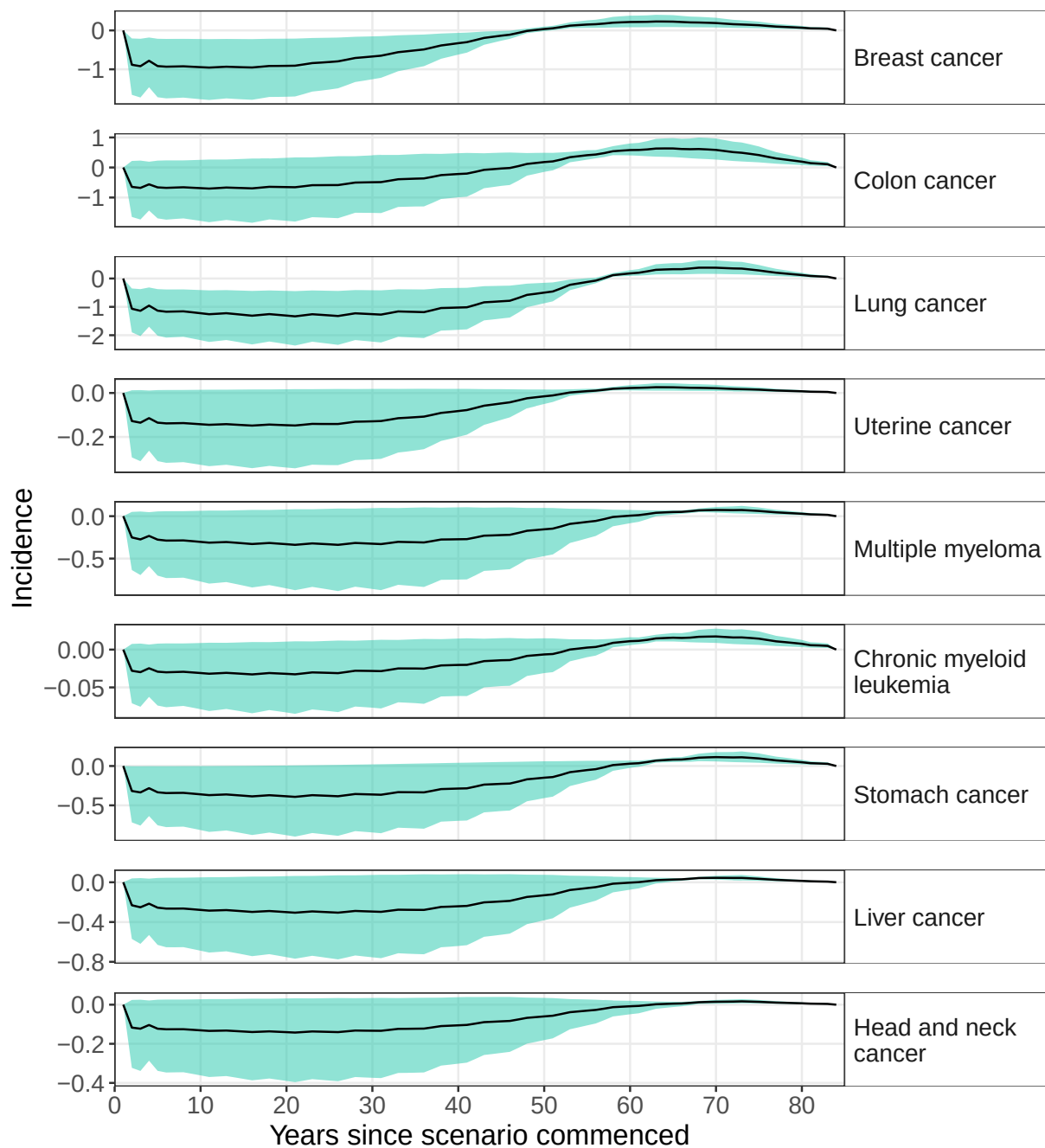
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

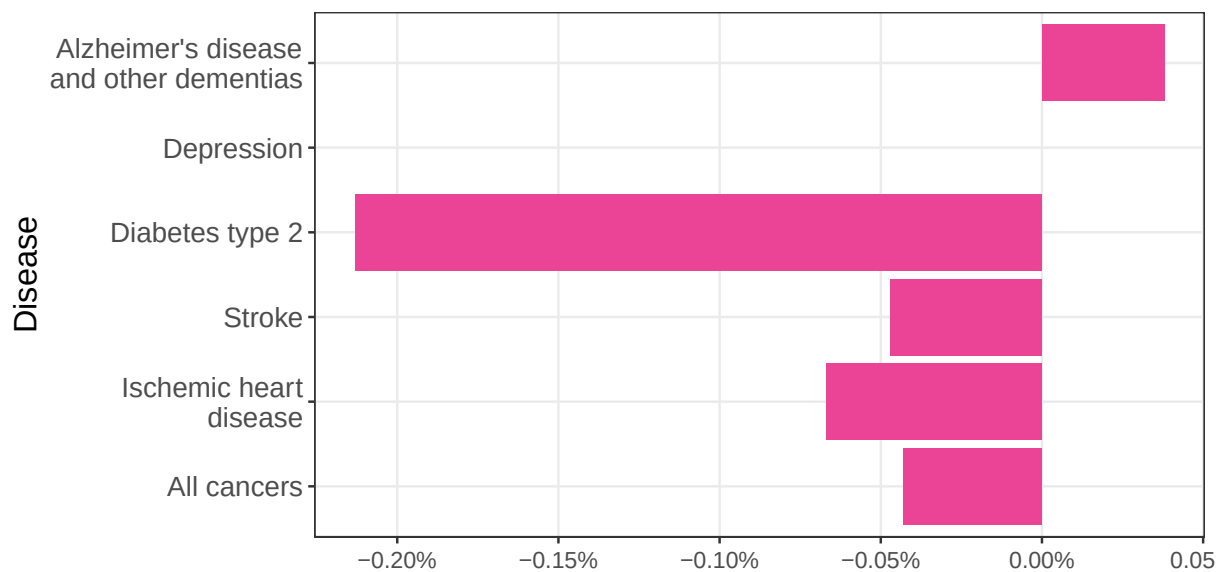
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

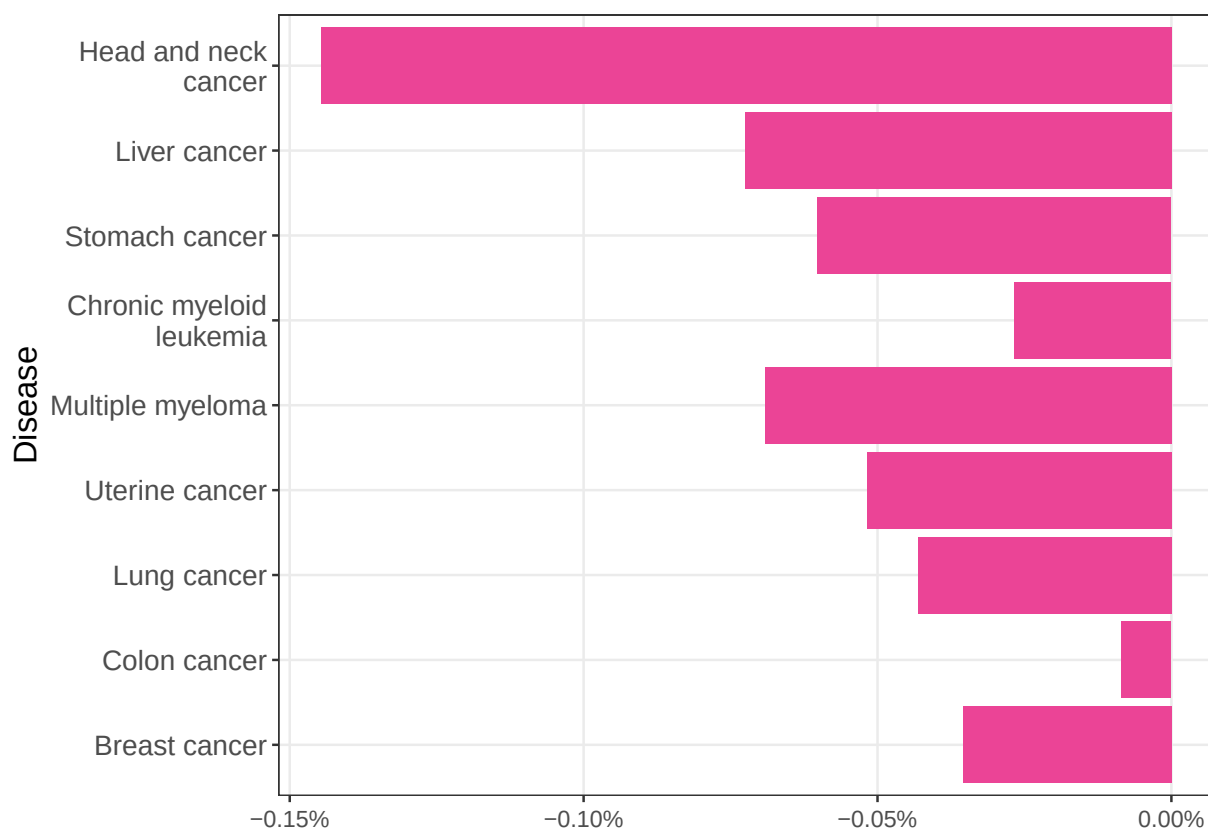
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.04%                  | -91                                                               |
| Breast cancer                           | 0.04%                   | 11                                                                |
| All cancers                             | 0.04%                   | 109                                                               |
| Colon cancer                            | 0.01%                   | 4                                                                 |
| Chronic myeloid leukemia                | 0.03%                   | 1                                                                 |
| Diabetes type 2                         | 0.21%                   | 105                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.14%                   | 6                                                                 |
| Ischemic heart disease                  | 0.07%                   | 251                                                               |
| Liver cancer                            | 0.07%                   | 12                                                                |
| Multiple myeloma                        | 0.07%                   | 12                                                                |
| Stomach cancer                          | 0.06%                   | 13                                                                |
| Stroke                                  | 0.05%                   | 82                                                                |
| Lung cancer                             | 0.04%                   | 47                                                                |
| Uterine cancer                          | 0.05%                   | 4                                                                 |

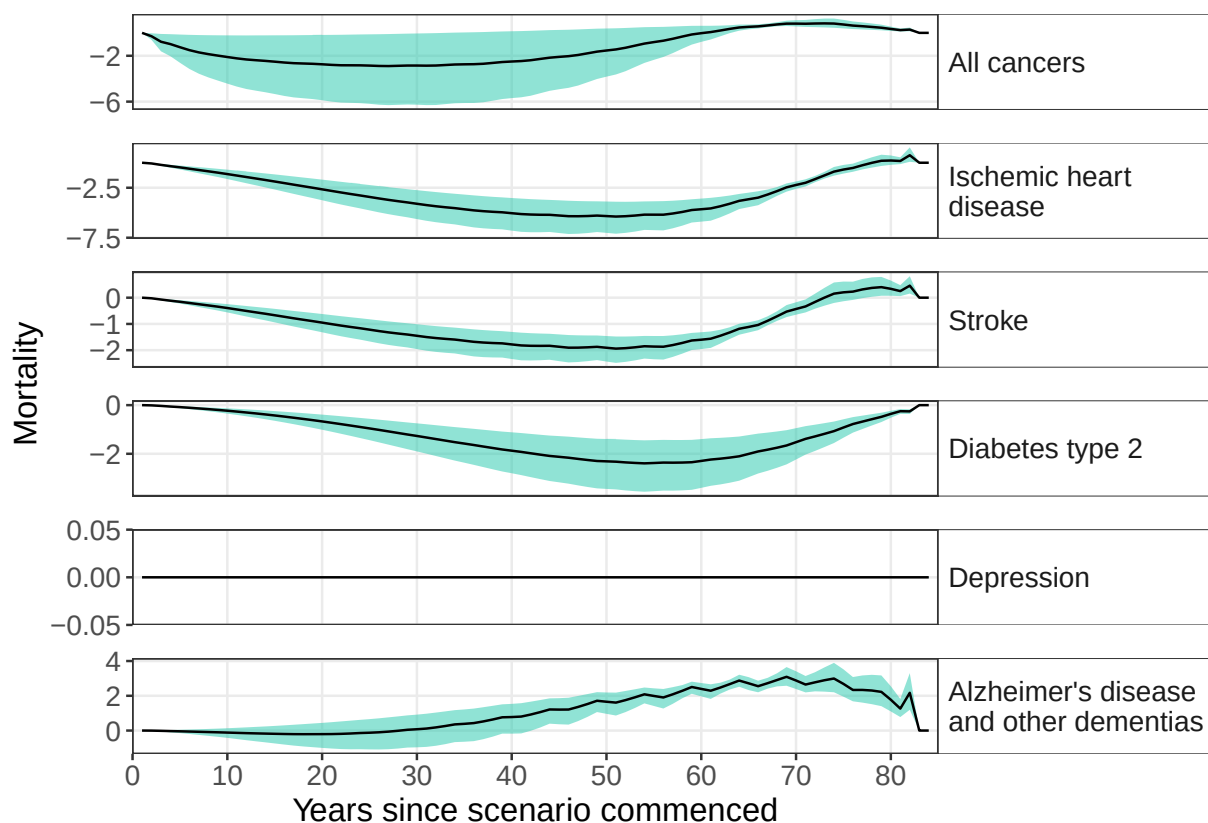
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

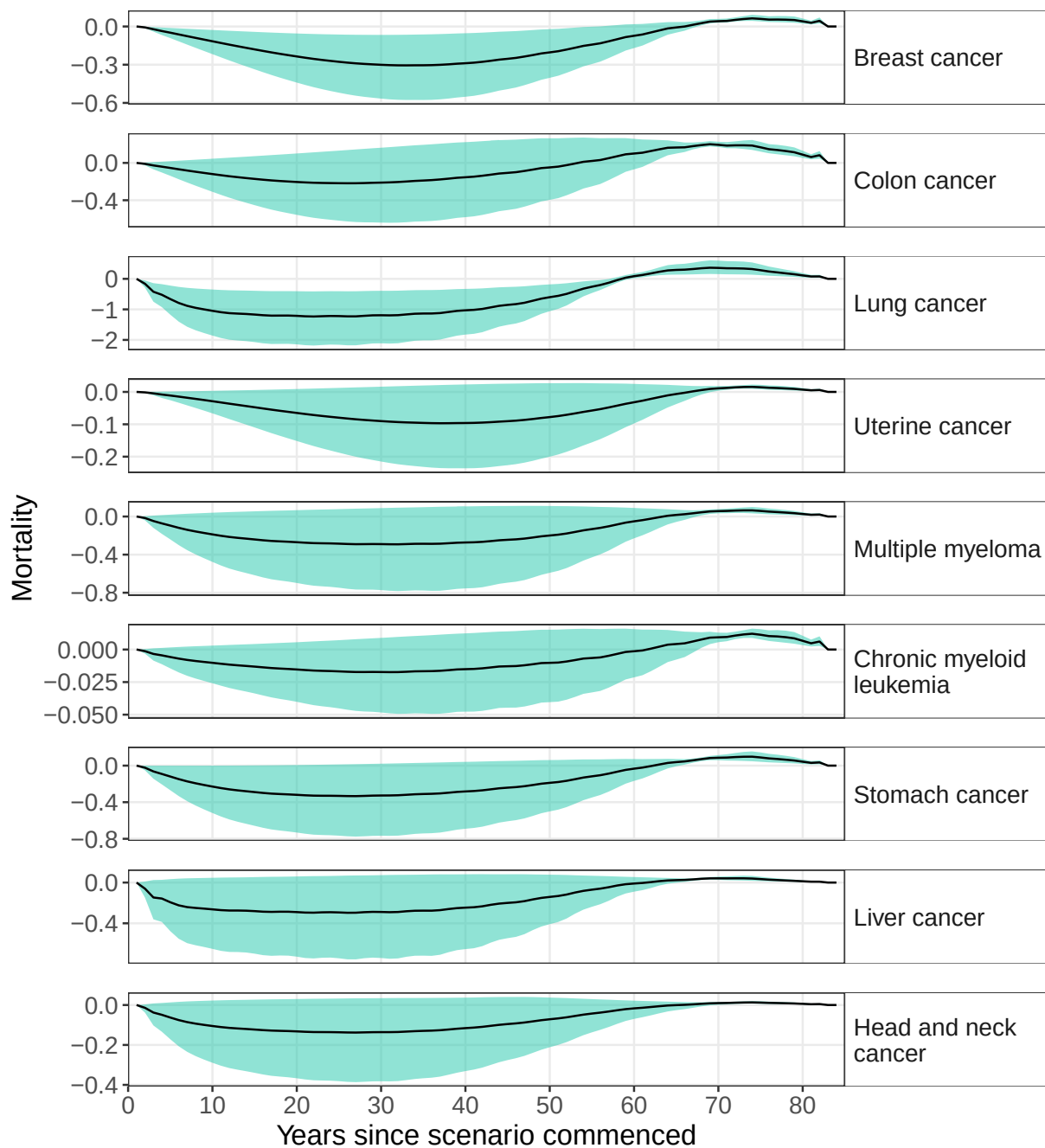


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

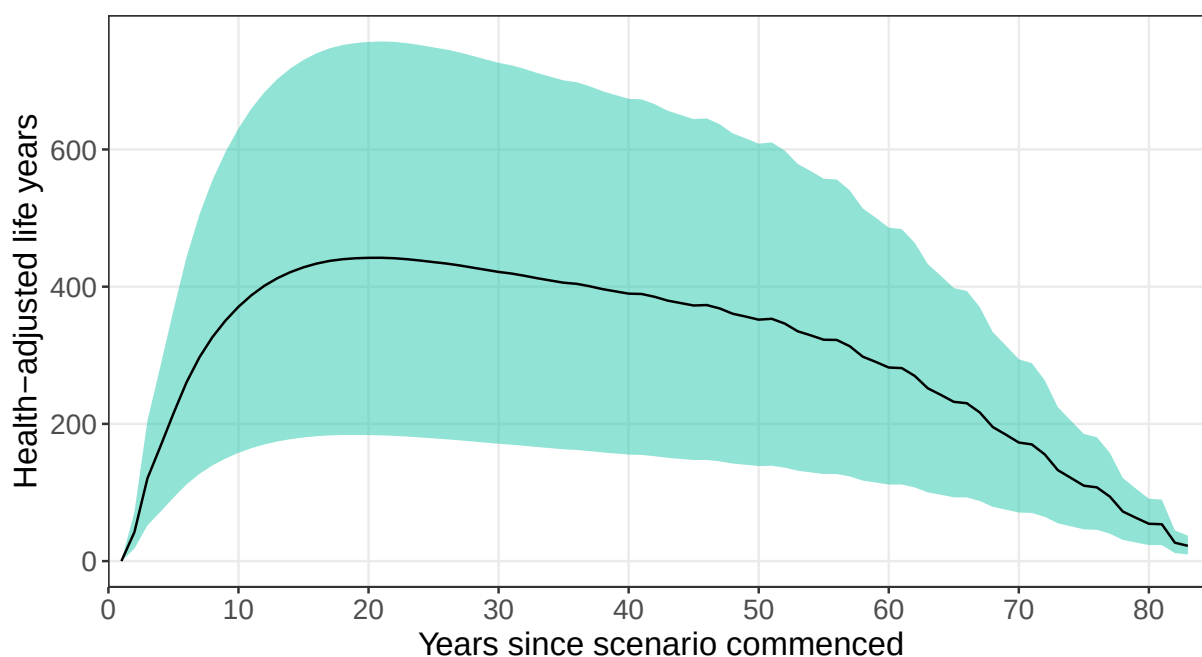
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **25,360** HALYs for the scenario population, which is **13** HALYs per 1,000 members of the population.



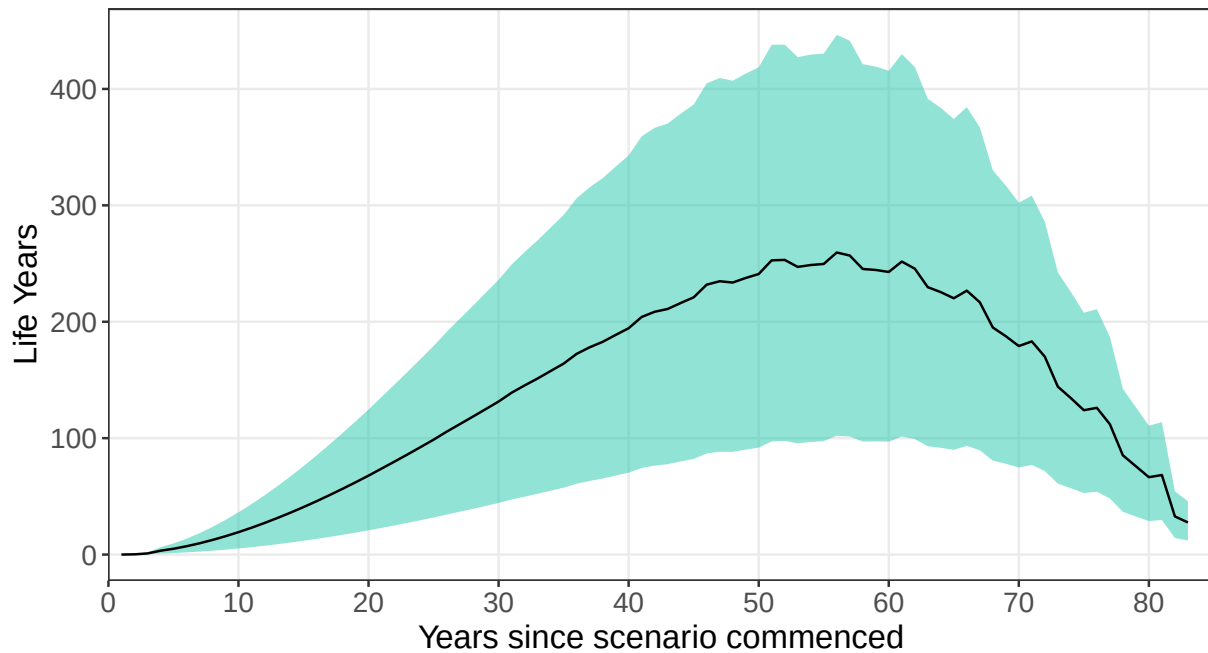
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **11,597** Life Years for the scenario population, which is **5.8** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated<sup>[a]</sup> by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year<sup>[b]</sup>. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting<sup>[c]</sup> to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).<sup>[2]</sup> Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations<sup>[3, 4]</sup> and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **13** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes<sup>[d]</sup>.

The HALYs gained in this scenario have a statistical value of:

- **1,071,442** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **676,202** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **465,123** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year<sup>[b]</sup>.



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$294            | \$206            | \$145            |
| Breast cancer                           | \$8,322          | \$5,097          | \$3,329          |
| All cancers                             | \$16,794         | \$10,649         | \$7,153          |
| Colon cancer                            | \$2,950          | \$2,012          | \$1,402          |
| Chronic myeloid leukemia                | \$835            | \$505            | \$328            |
| Diabetes type 2                         | \$6,216          | \$3,608          | \$2,273          |
| Depression                              | \$34,996         | \$25,488         | \$19,225         |
| Head and neck cancer                    | \$146            | \$99             | \$71             |
| Ischemic heart disease                  | \$10,551         | \$6,164          | \$3,897          |
| Liver cancer                            | \$127            | \$87             | \$63             |
| Multiple myeloma                        | \$2,047          | \$1,310          | \$895            |
| Stomach cancer                          | \$764            | \$493            | \$339            |
| Stroke                                  | \$1,442          | \$843            | \$534            |
| Lung cancer                             | \$1,143          | \$765            | \$542            |
| Uterine cancer                          | \$459            | \$282            | \$185            |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## **b. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **References**

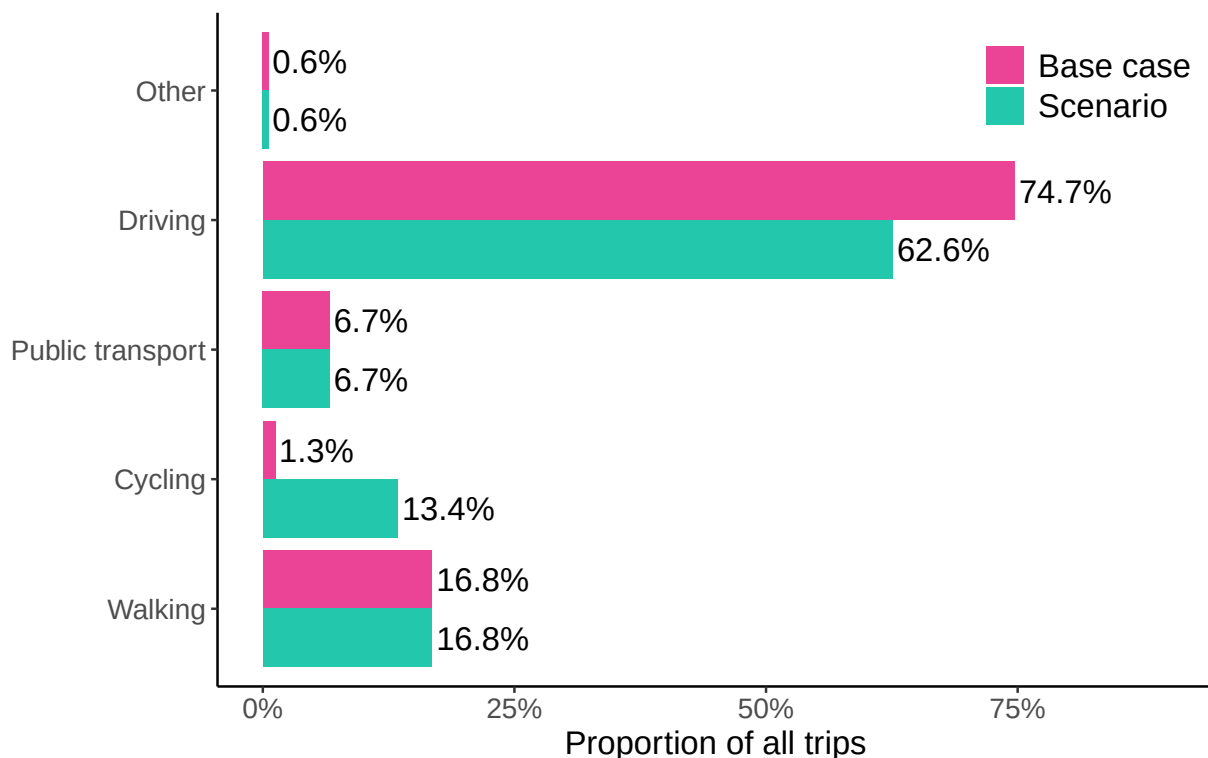
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## Scenario: replacing car trips under 2km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 2km for leisure, shopping, work, education or other purposes with cycling trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in cycling from 1.3% to 13.4% and from 74.7% to 62.6% for car trips taken as either a driver or passenger.

Increases in cycling translate into a shift from 47.9% to 54.7% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

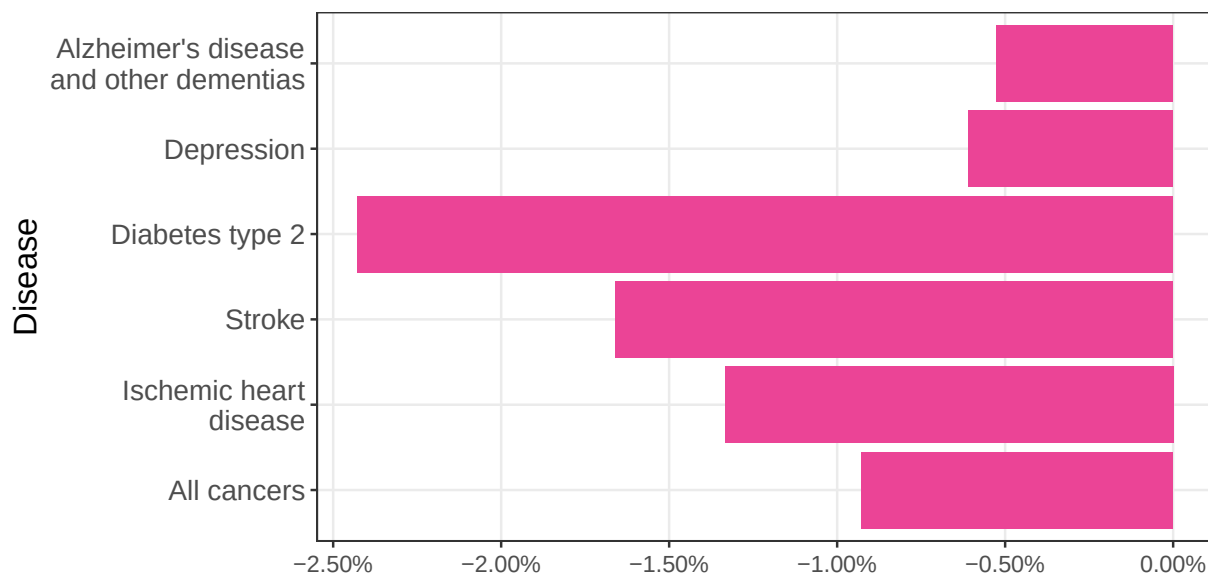
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

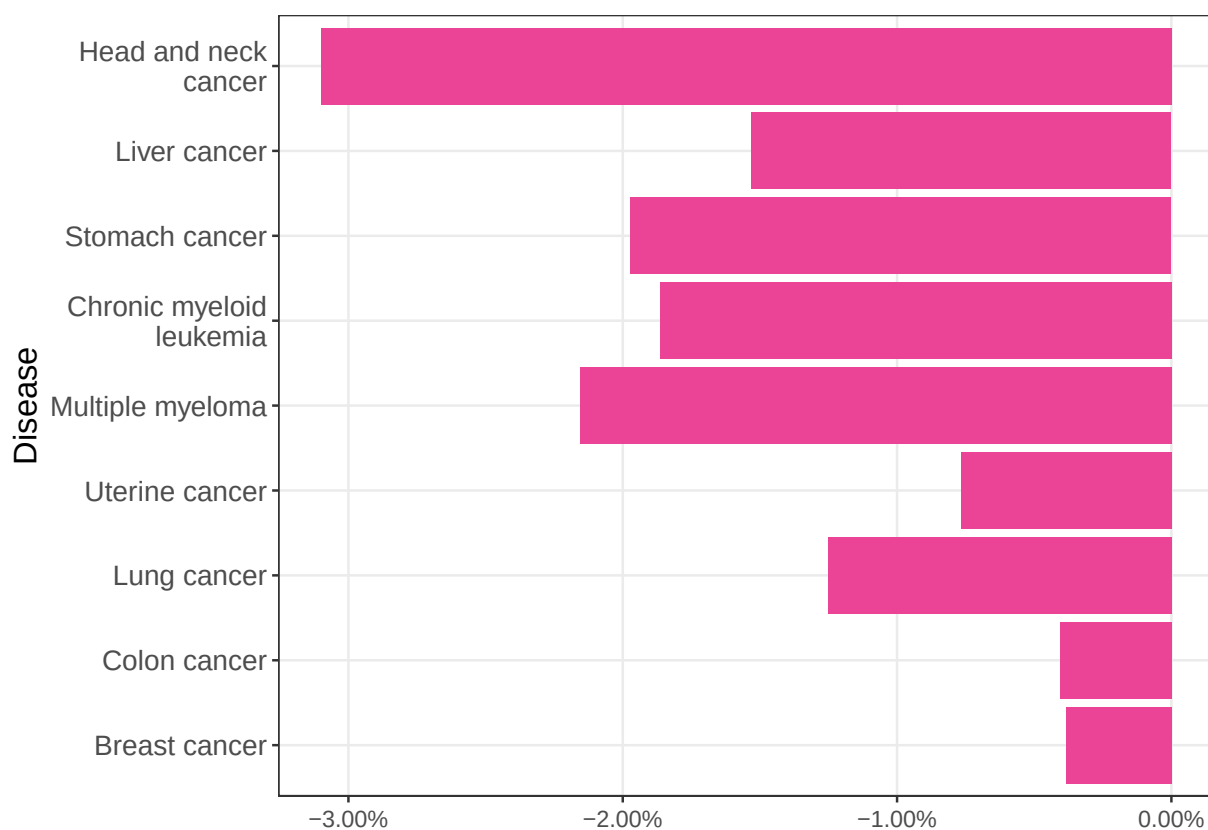
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.53%                              | 3,351                                                                       |
| Breast cancer                           | 0.38%                              | 281                                                                         |
| All cancers                             | 0.93%                              | 3,657                                                                       |
| Colon cancer                            | 0.40%                              | 524                                                                         |
| Chronic myeloid leukemia                | 1.86%                              | 51                                                                          |
| Diabetes type 2                         | 2.43%                              | 8,011                                                                       |
| Depression                              | 0.61%                              | 9,751                                                                       |
| Head and neck cancer                    | 3.10%                              | 137                                                                         |
| Ischemic heart disease                  | 1.33%                              | 11,741                                                                      |
| Liver cancer                            | 1.53%                              | 269                                                                         |
| Multiple myeloma                        | 2.16%                              | 418                                                                         |
| Stomach cancer                          | 1.97%                              | 471                                                                         |
| Stroke                                  | 1.66%                              | 4,317                                                                       |
| Lung cancer                             | 1.25%                              | 1,440                                                                       |
| Uterine cancer                          | 0.77%                              | 65                                                                          |

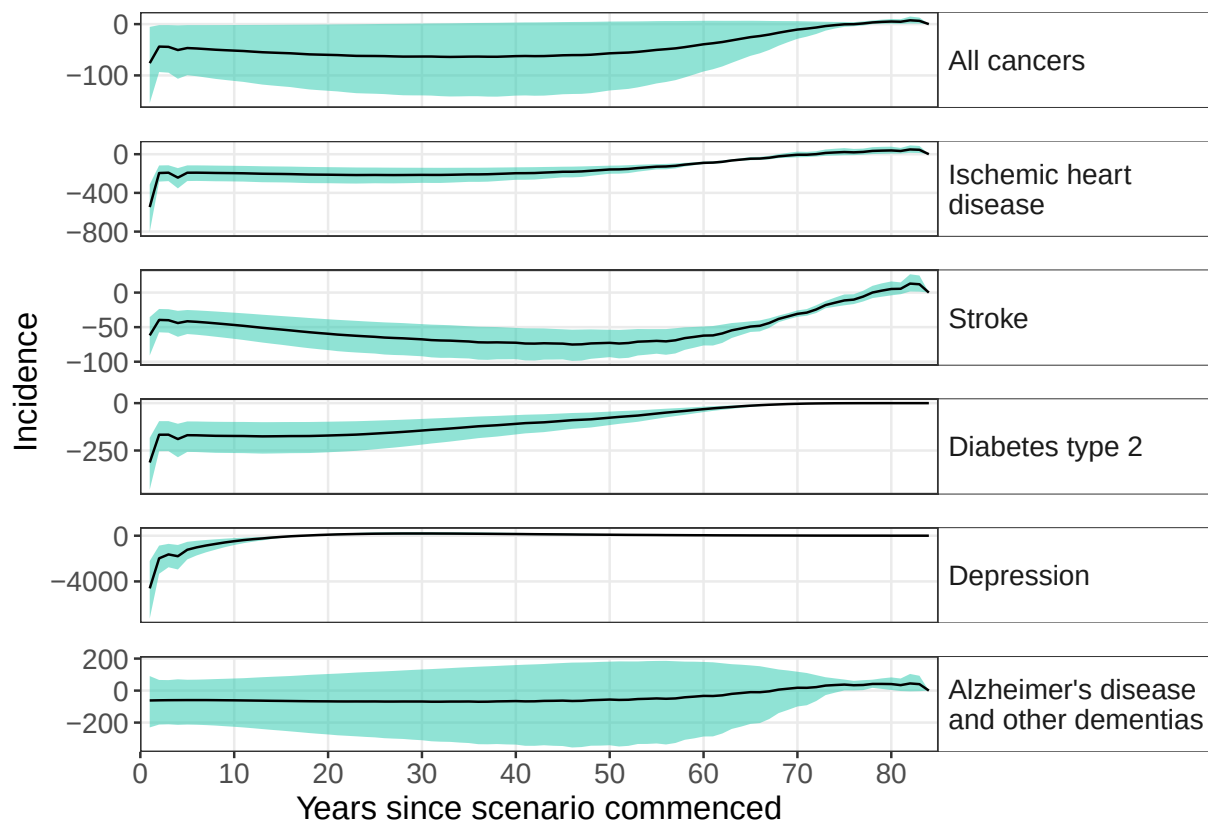
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



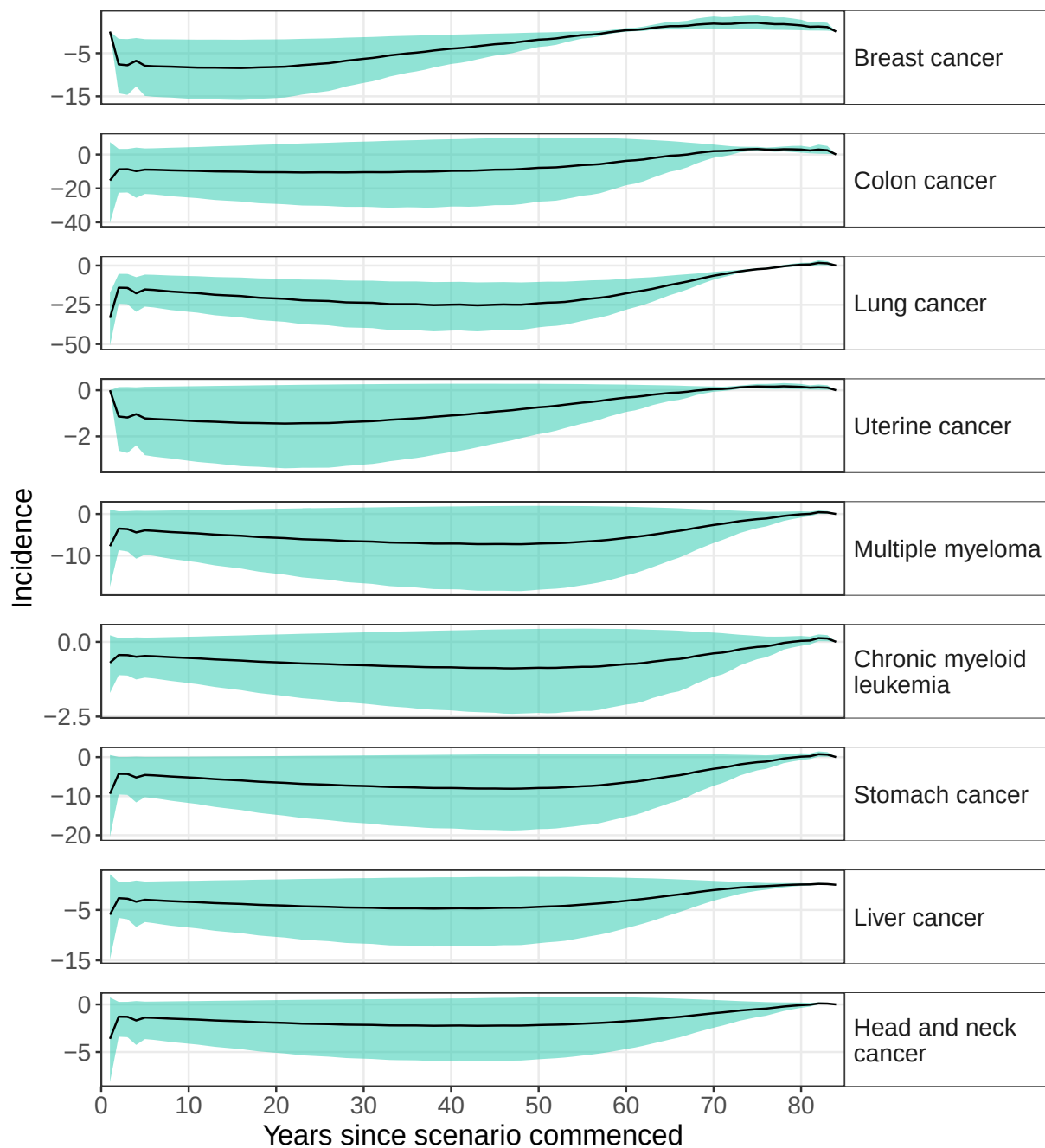
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

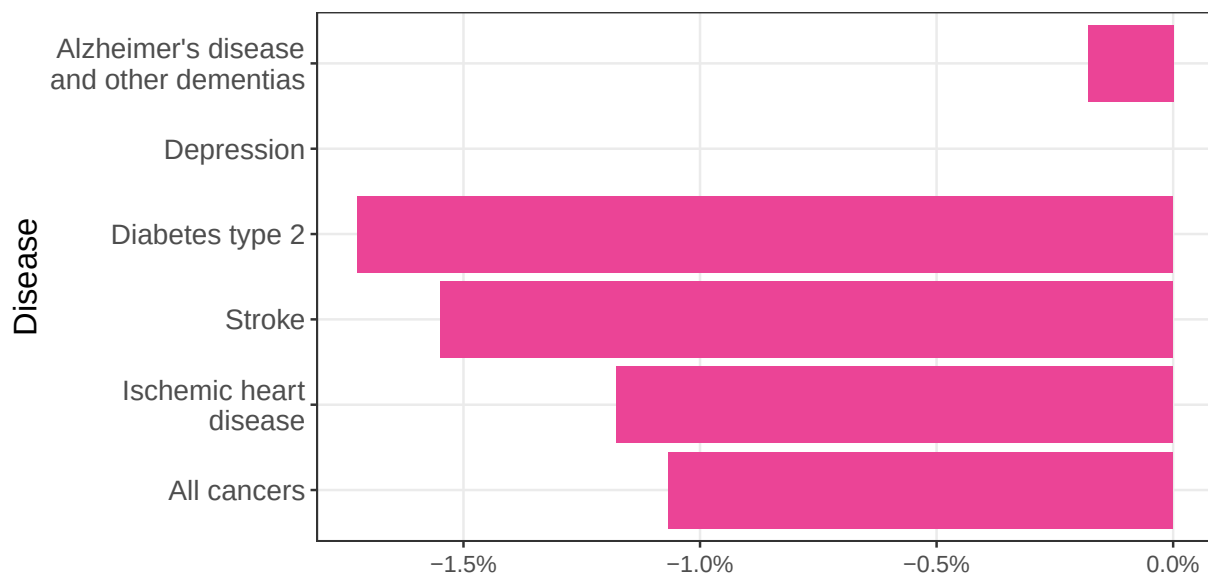
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

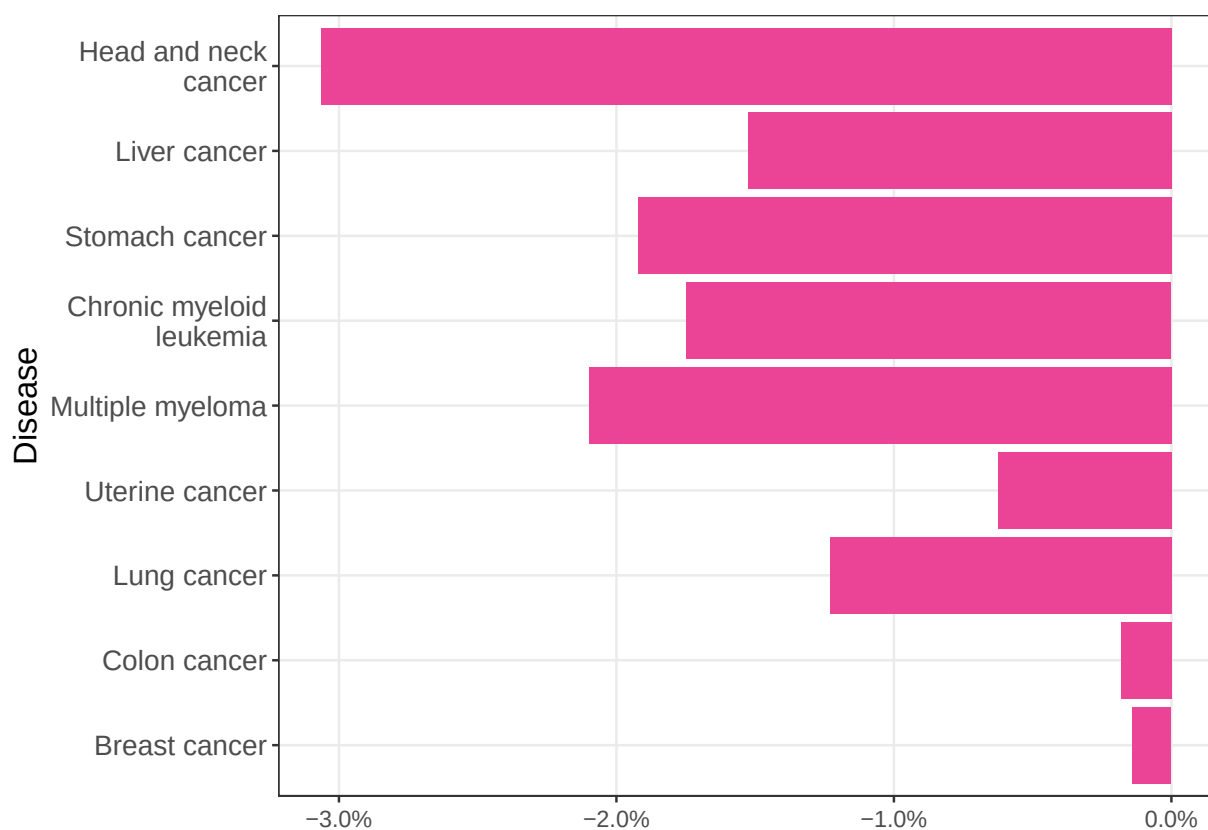
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.18%                   | 430                                                               |
| Breast cancer                           | 0.14%                   | 45                                                                |
| All cancers                             | 1.07%                   | 2,701                                                             |
| Colon cancer                            | 0.18%                   | 78                                                                |
| Chronic myeloid leukemia                | 1.75%                   | 36                                                                |
| Diabetes type 2                         | 1.72%                   | 852                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 3.06%                   | 122                                                               |
| Ischemic heart disease                  | 1.18%                   | 4,414                                                             |
| Liver cancer                            | 1.52%                   | 260                                                               |
| Multiple myeloma                        | 2.10%                   | 366                                                               |
| Stomach cancer                          | 1.92%                   | 407                                                               |
| Stroke                                  | 1.55%                   | 2,709                                                             |
| Lung cancer                             | 1.23%                   | 1,341                                                             |
| Uterine cancer                          | 0.62%                   | 45                                                                |

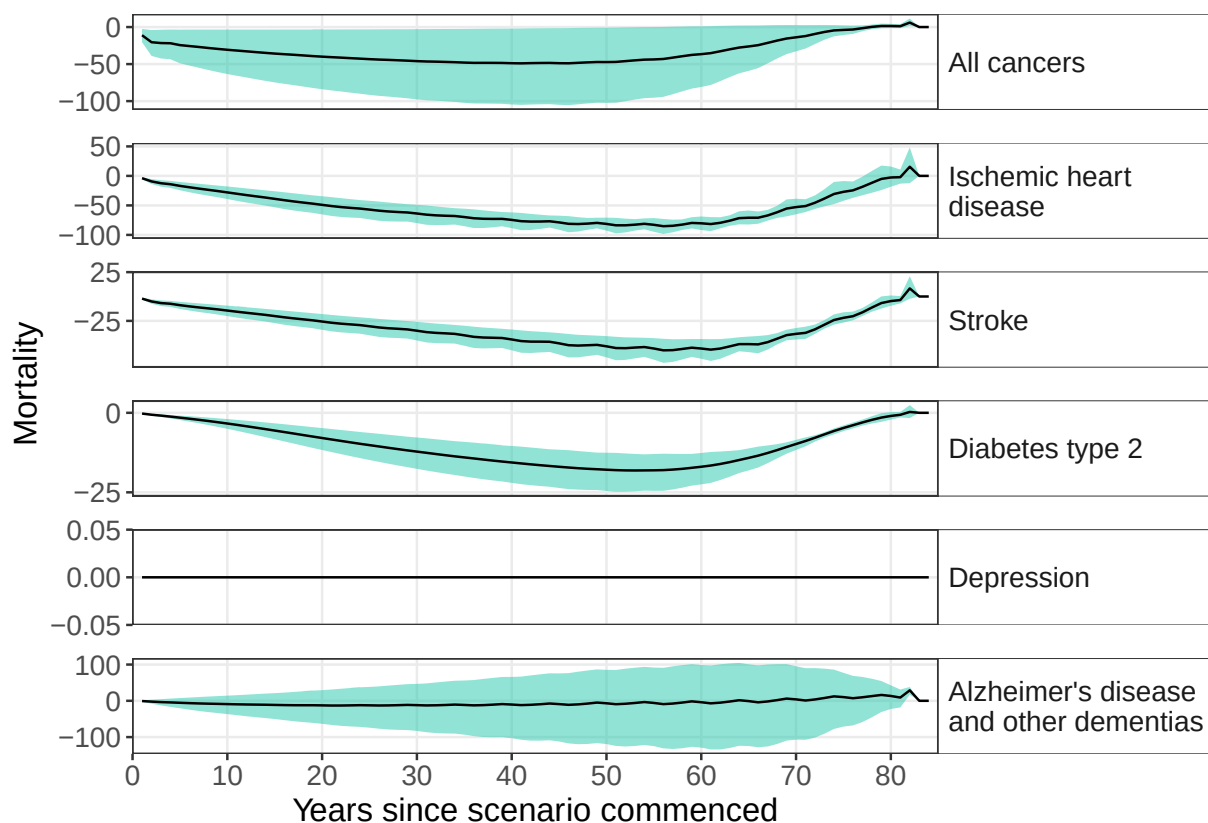
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

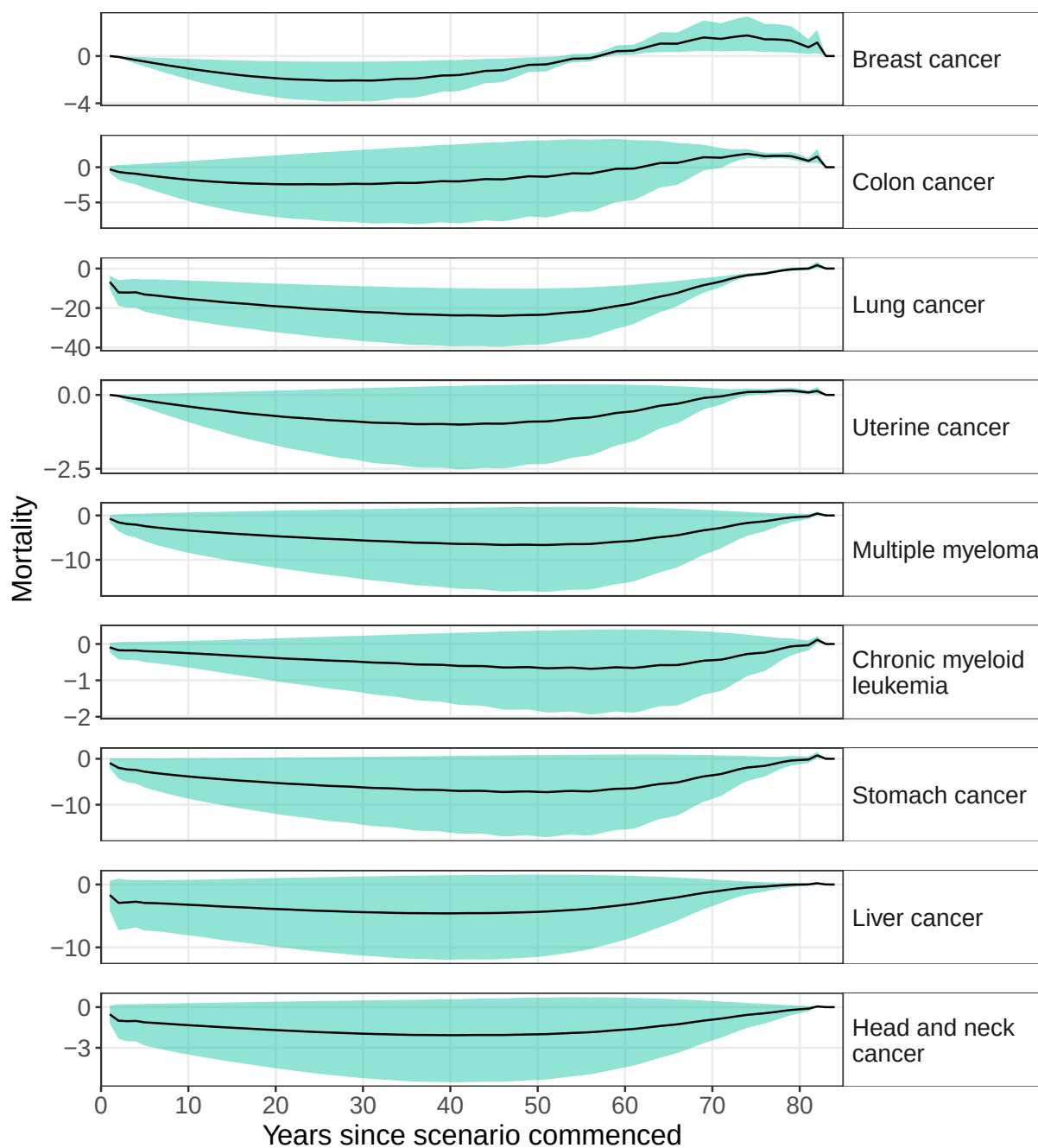


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

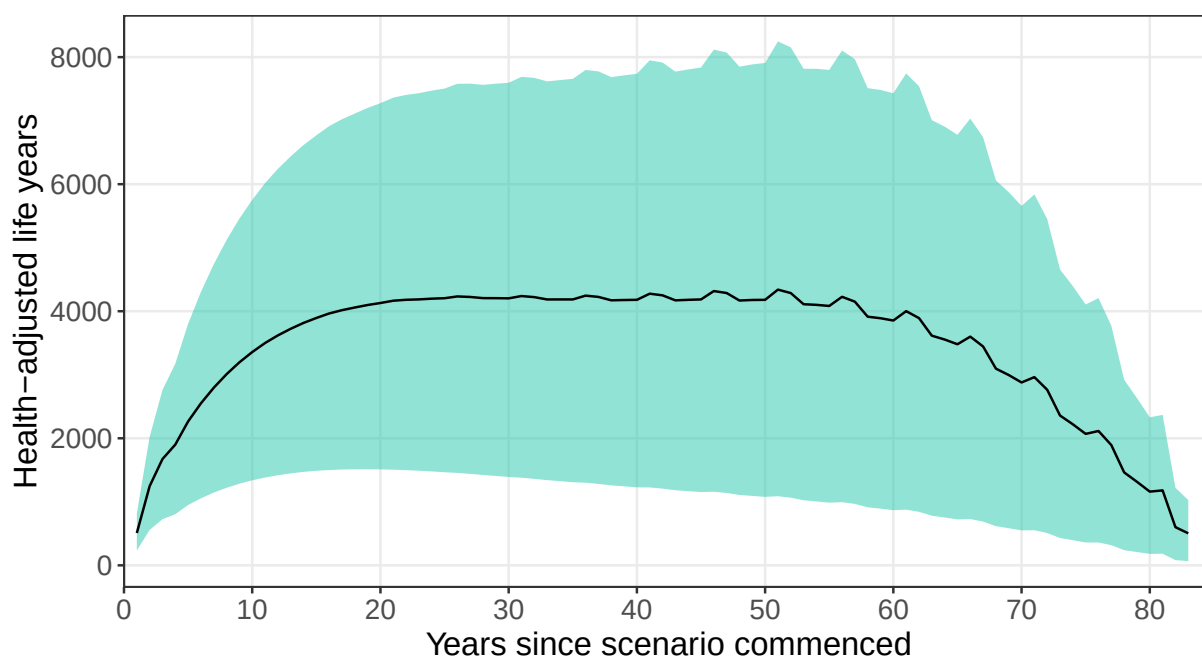
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **288,320** HALYs for the scenario population, which is **143** HALYs per 1,000 members of the population.



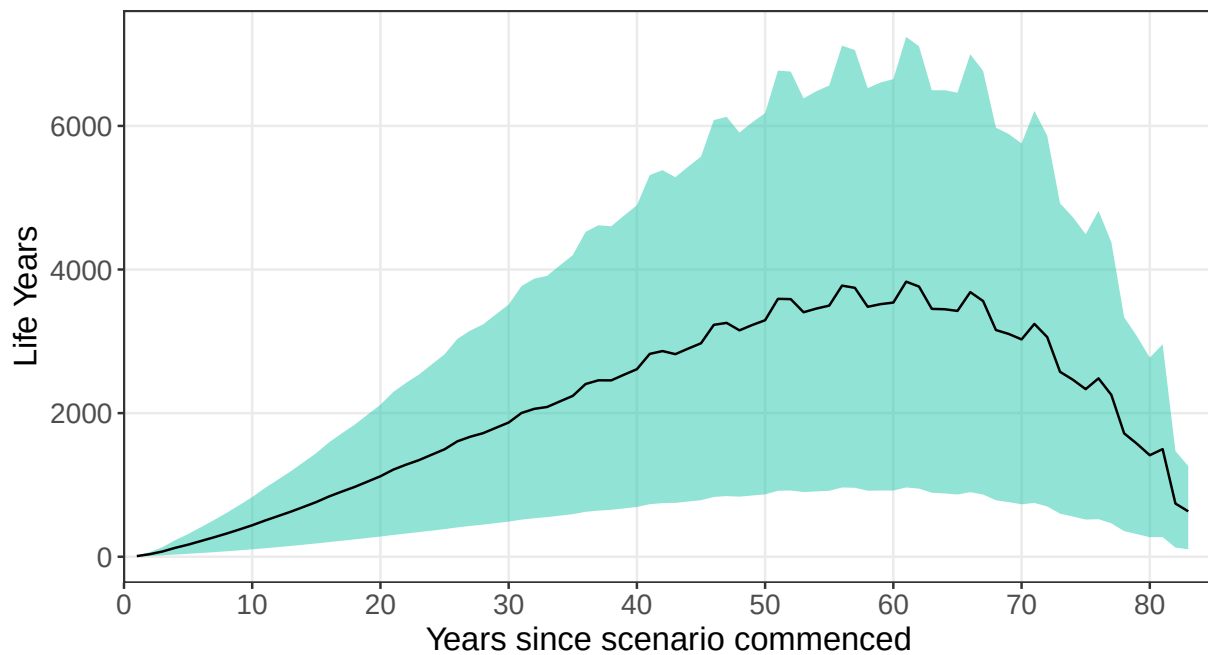
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **177,785** Life Years for the scenario population, which is **88** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated<sup>[a]</sup> by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year<sup>[b]</sup>. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting<sup>[c]</sup> to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).<sup>[2]</sup> Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations<sup>[3, 4]</sup> and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **143** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes<sup>[d]</sup>.

The HALYs gained in this scenario have a statistical value of:

- **11,161,726** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **6,846,716** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **4,657,764** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year<sup>[b]</sup>.



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$8,538          | \$5,156          | \$3,409          |
| Breast cancer                           | \$60,179         | \$38,061         | \$25,394         |
| All cancers                             | \$170,411        | \$108,734        | \$74,248         |
| Colon cancer                            | \$37,542         | \$25,297         | \$17,848         |
| Chronic myeloid leukemia                | \$11,497         | \$6,745          | \$4,350          |
| Diabetes type 2                         | \$49,486         | \$29,473         | \$19,103         |
| Depression                              | \$275,052        | \$207,405        | \$160,448        |
| Head and neck cancer                    | \$1,871          | \$1,209          | \$853            |
| Ischemic heart disease                  | \$111,628        | \$67,309         | \$44,517         |
| Liver cancer                            | \$1,547          | \$1,029          | \$744            |
| Multiple myeloma                        | \$27,689         | \$17,318         | \$11,845         |
| Stomach cancer                          | \$10,522         | \$6,580          | \$4,506          |
| Stroke                                  | \$17,110         | \$9,960          | \$6,377          |
| Lung cancer                             | \$15,924         | \$10,284         | \$7,247          |
| Uterine cancer                          | \$3,657          | \$2,236          | \$1,466          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

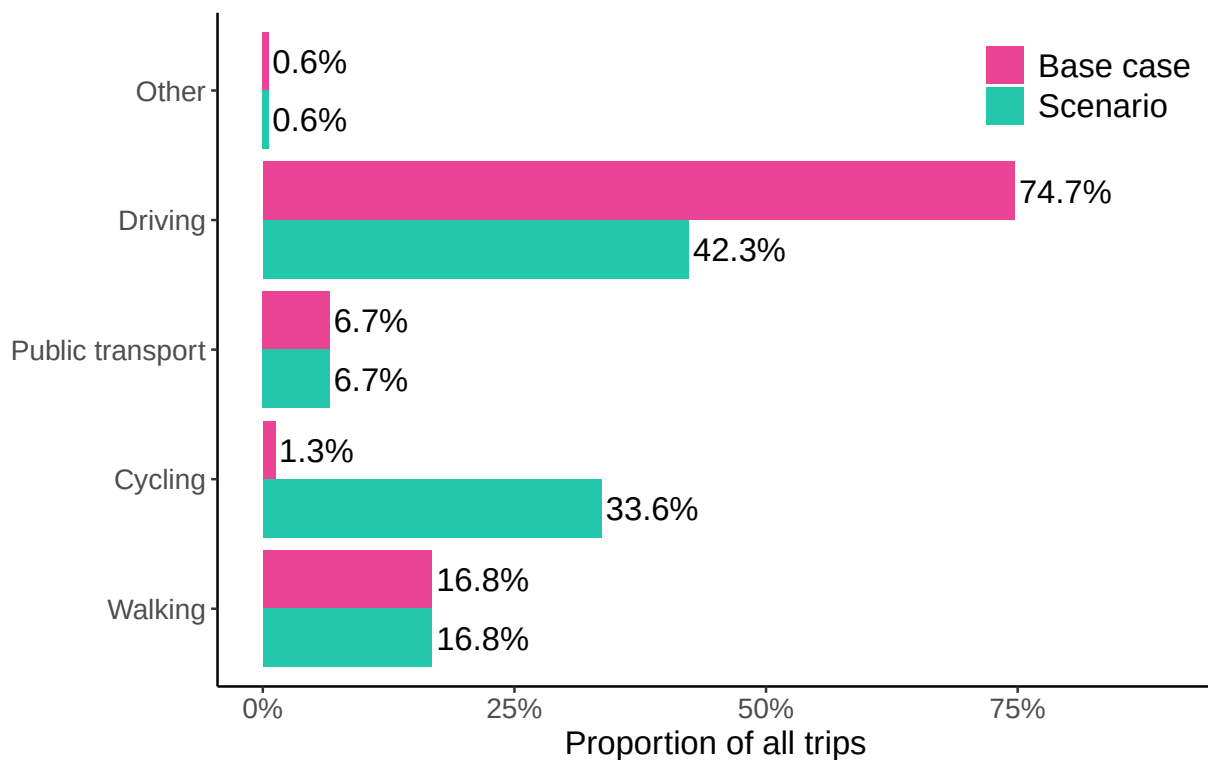
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
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## Scenario: replacing car trips under 5km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 5km for leisure, shopping, work, education or other purposes with cycling trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in cycling from 1.3% to 33.6% and from 74.7% to 42.3% for car trips taken as either a driver or passenger.

Increases in cycling translate into a shift from 47.9% to 67.3% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

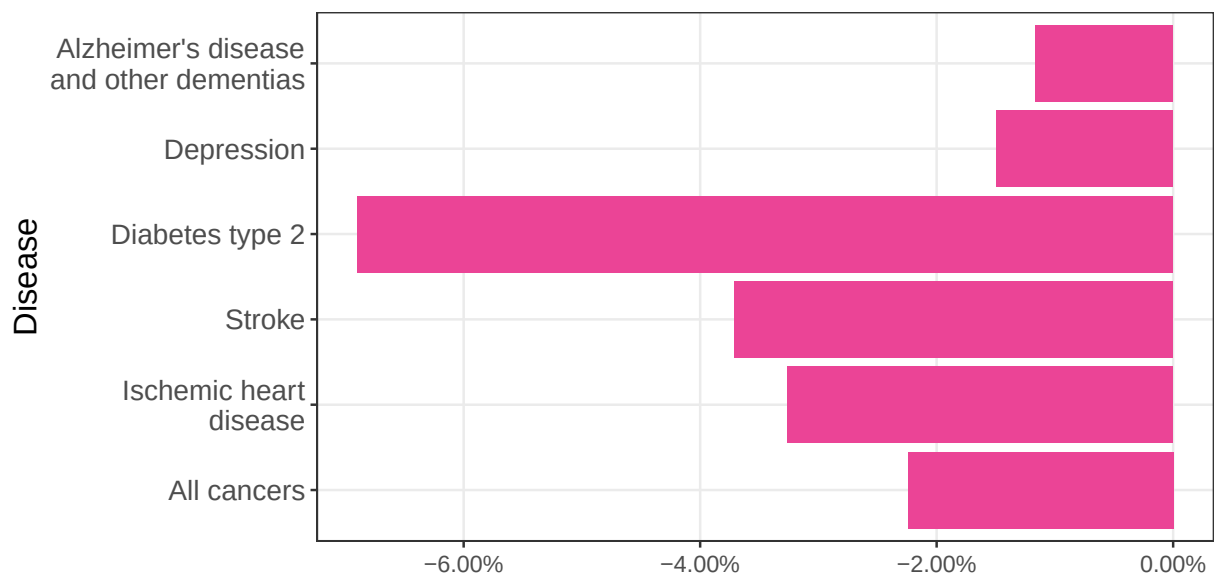
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

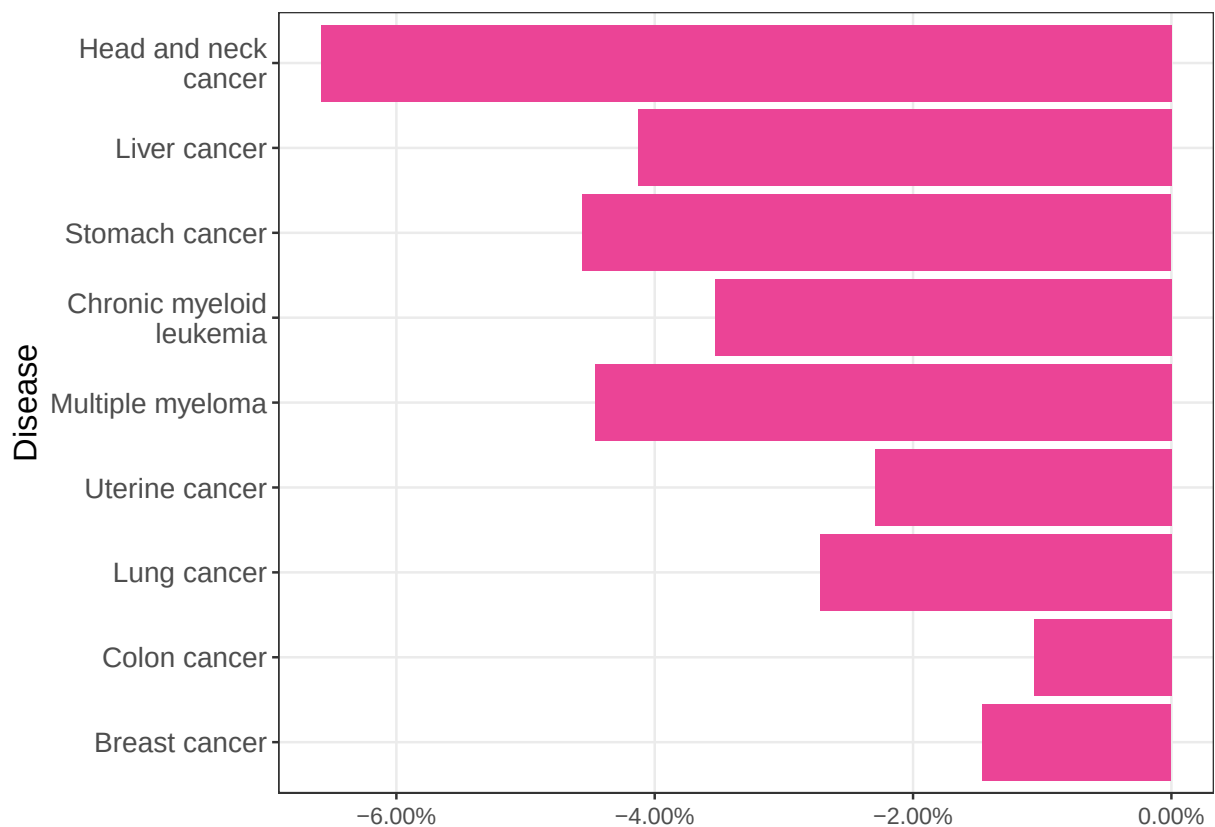
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 1.17%                              | 7,432                                                                       |
| Breast cancer                           | 1.46%                              | 1,071                                                                       |
| All cancers                             | 2.24%                              | 8,838                                                                       |
| Colon cancer                            | 1.06%                              | 1,377                                                                       |
| Chronic myeloid leukemia                | 3.53%                              | 97                                                                          |
| Diabetes type 2                         | 6.90%                              | 22,764                                                                      |
| Depression                              | 1.50%                              | 23,975                                                                      |
| Head and neck cancer                    | 6.58%                              | 290                                                                         |
| Ischemic heart disease                  | 3.26%                              | 28,706                                                                      |
| Liver cancer                            | 4.13%                              | 727                                                                         |
| Multiple myeloma                        | 4.46%                              | 865                                                                         |
| Stomach cancer                          | 4.56%                              | 1,089                                                                       |
| Stroke                                  | 3.71%                              | 9,647                                                                       |
| Lung cancer                             | 2.72%                              | 3,125                                                                       |
| Uterine cancer                          | 2.30%                              | 195                                                                         |

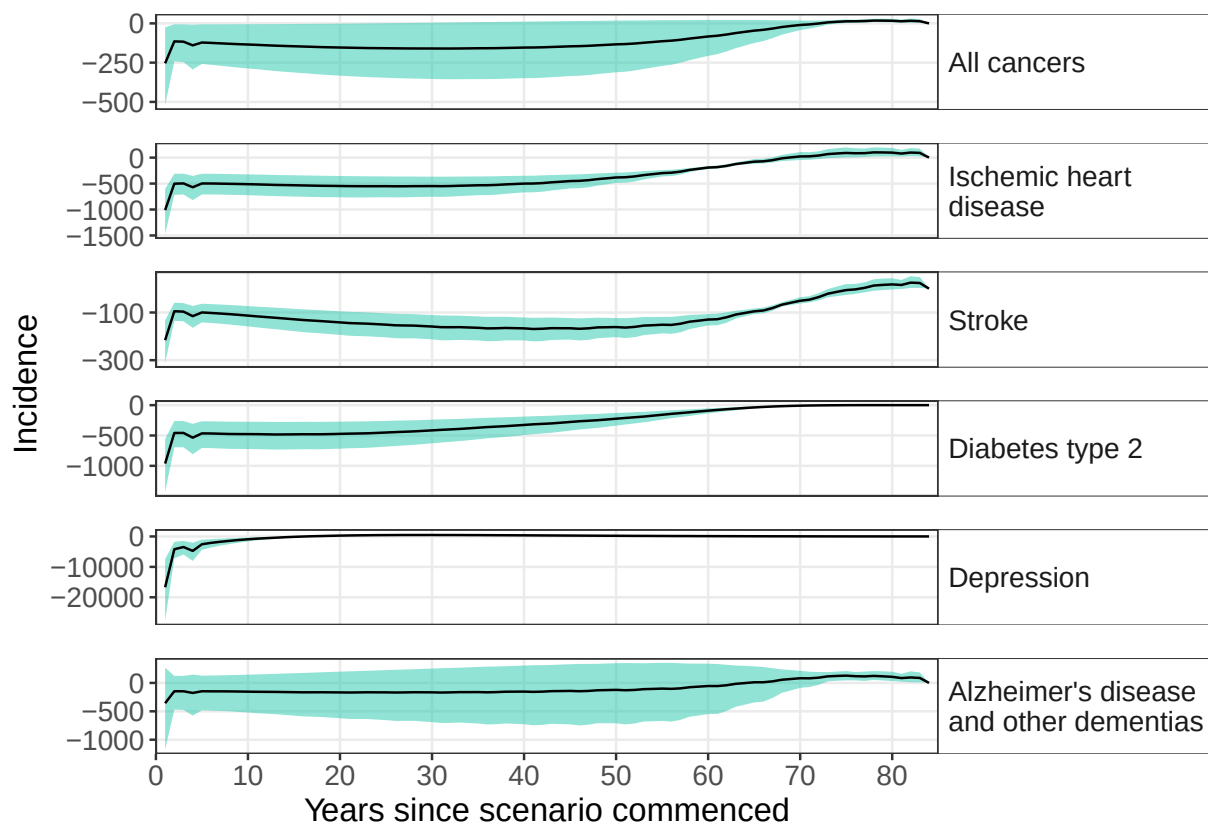
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



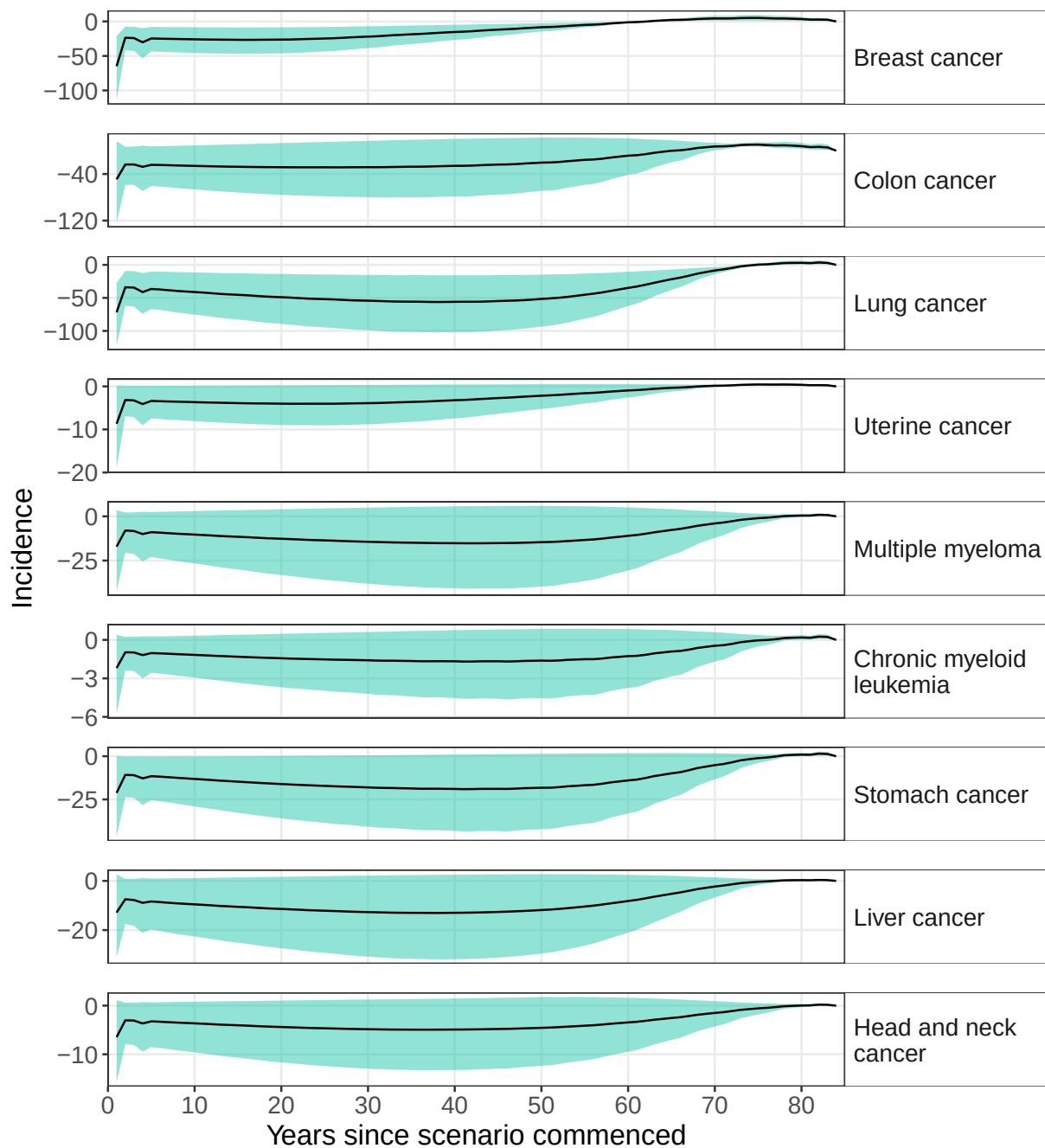
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

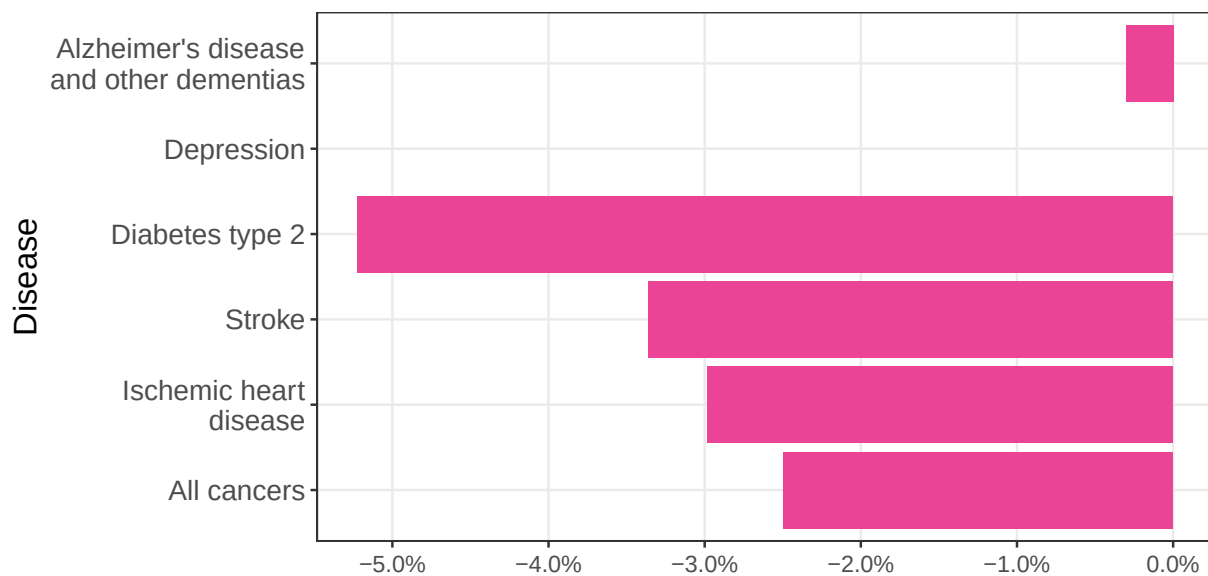
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

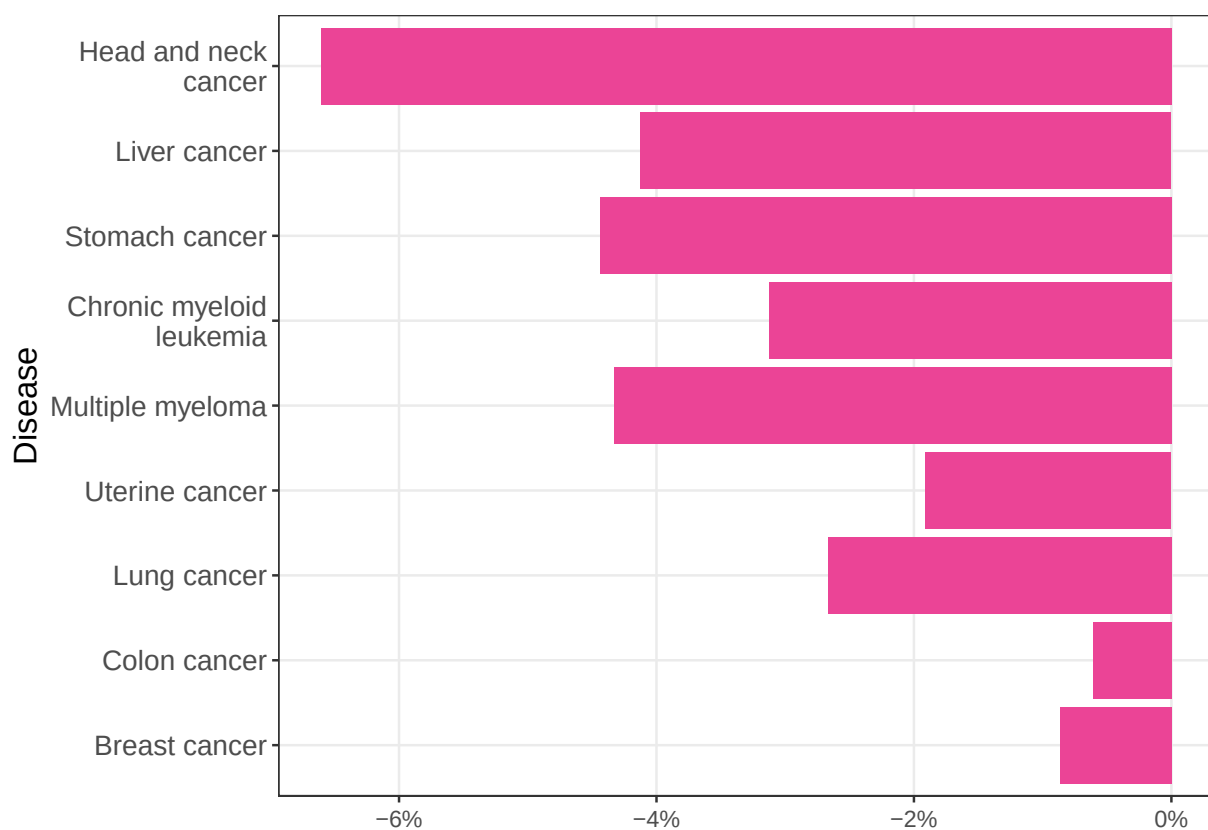
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.30%                   | 720                                                               |
| Breast cancer                           | 0.87%                   | 280                                                               |
| All cancers                             | 2.49%                   | 6,317                                                             |
| Colon cancer                            | 0.61%                   | 263                                                               |
| Chronic myeloid leukemia                | 3.12%                   | 64                                                                |
| Diabetes type 2                         | 5.22%                   | 2,581                                                             |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 6.61%                   | 263                                                               |
| Ischemic heart disease                  | 2.98%                   | 11,181                                                            |
| Liver cancer                            | 4.12%                   | 704                                                               |
| Multiple myeloma                        | 4.33%                   | 754                                                               |
| Stomach cancer                          | 4.44%                   | 939                                                               |
| Stroke                                  | 3.36%                   | 5,875                                                             |
| Lung cancer                             | 2.67%                   | 2,912                                                             |
| Uterine cancer                          | 1.91%                   | 138                                                               |

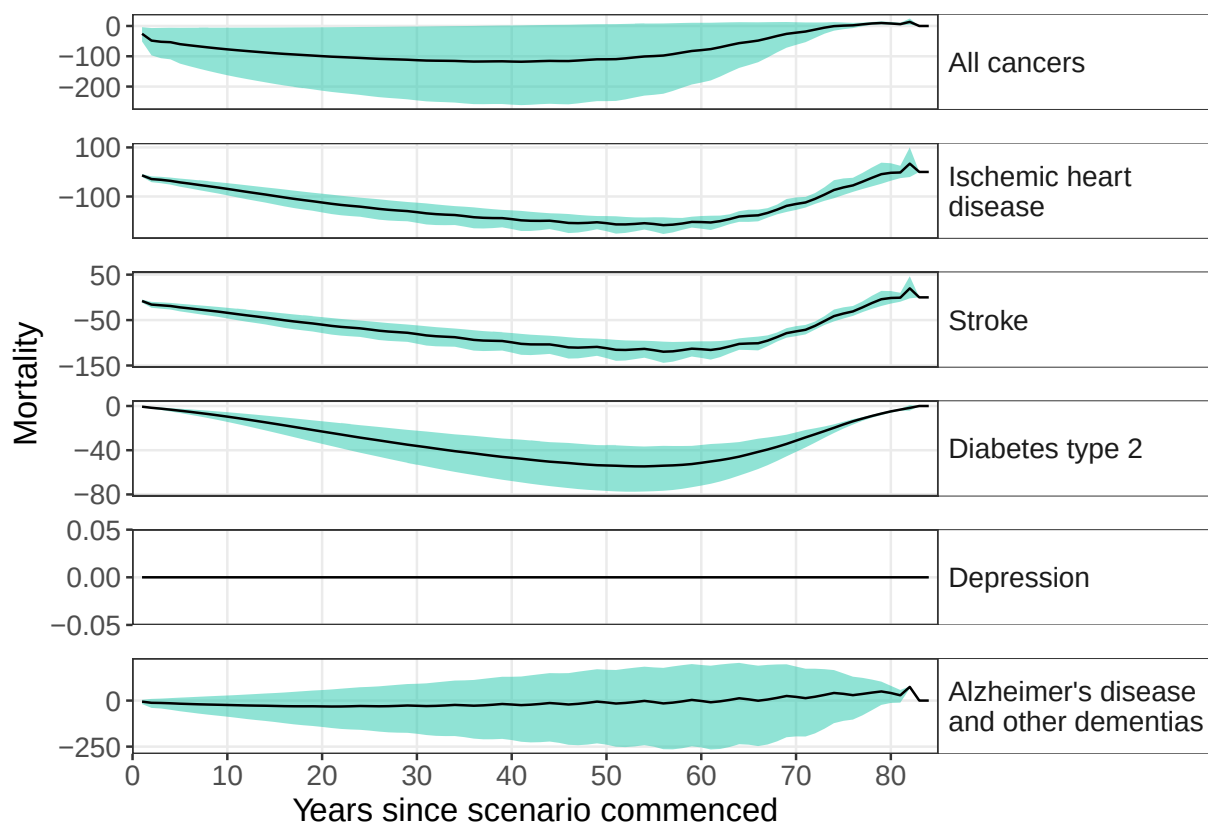
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

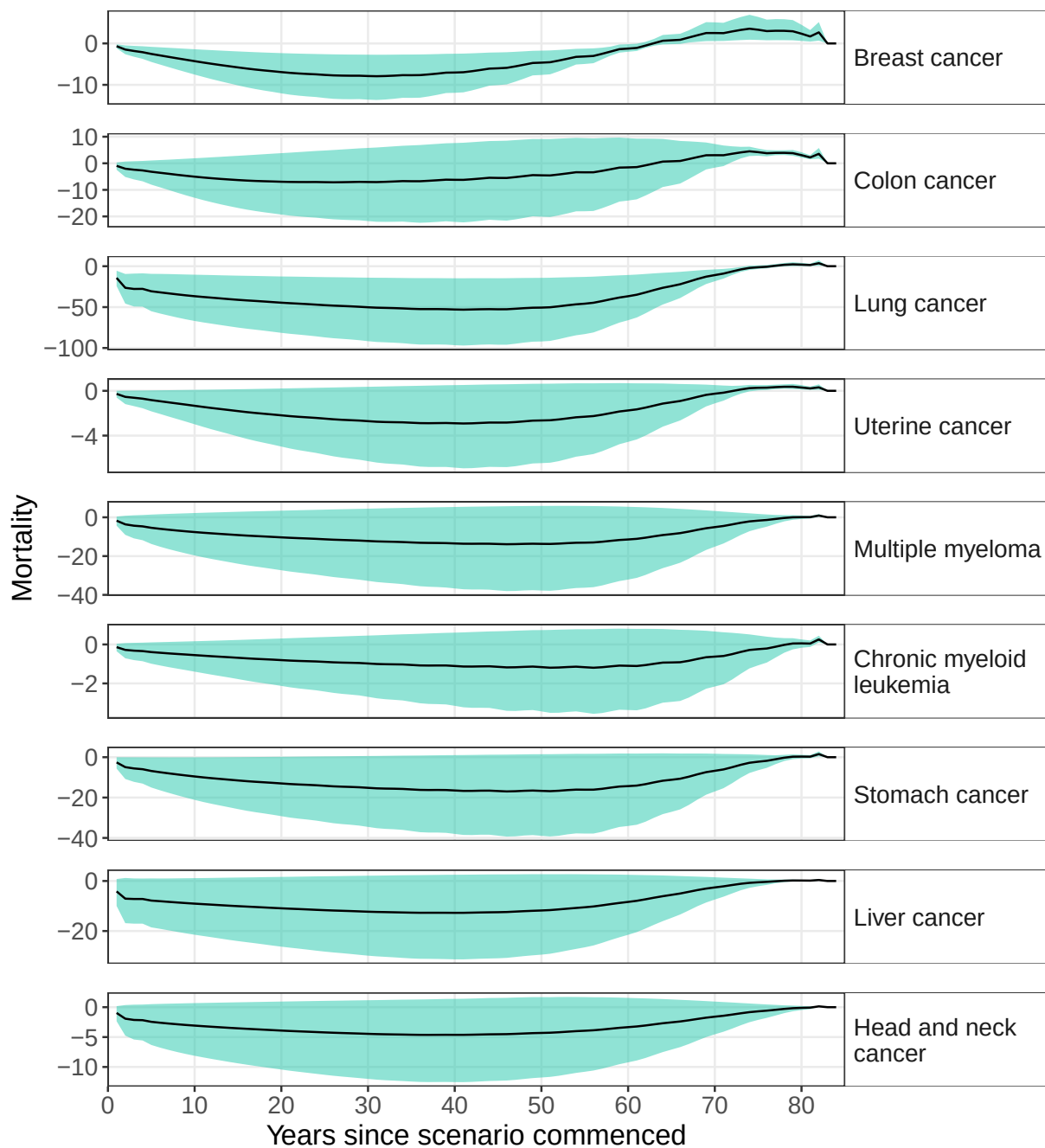


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

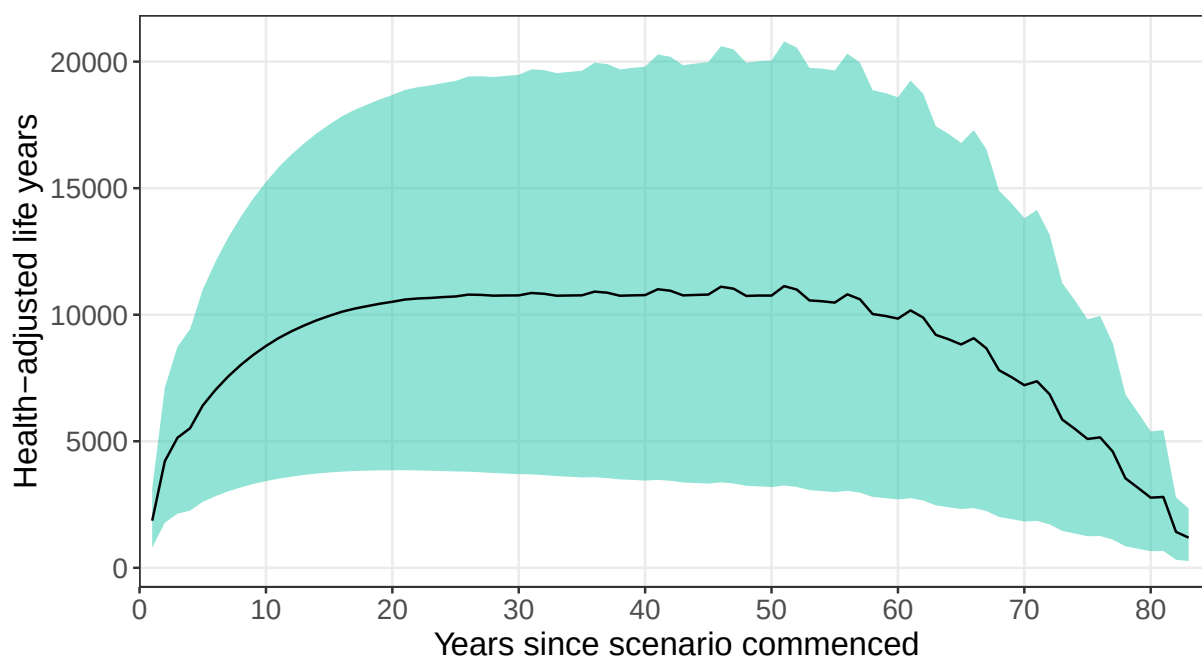
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **740,099** HALYs for the scenario population, which is **368** HALYs per 1,000 members of the population.



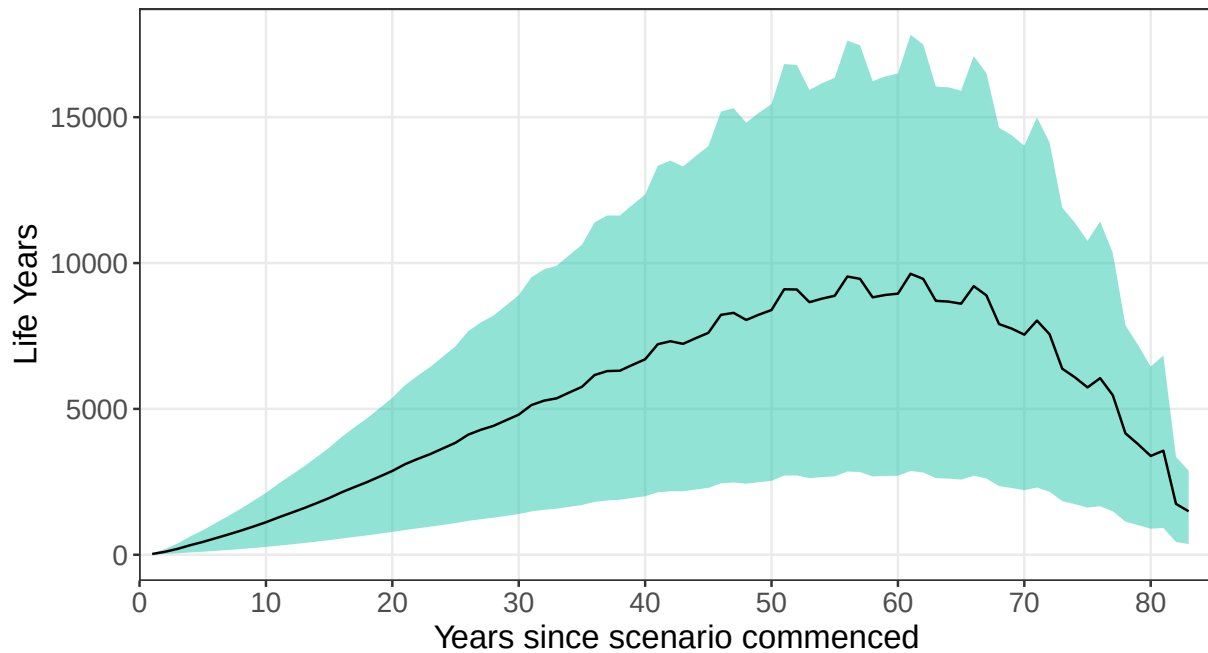
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **448,787** Life Years for the scenario population, which is **223** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **368** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **29,030,974** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **17,972,591** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **12,341,492** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$23,105         | \$14,138         | \$9,458          |
| Breast cancer                           | \$229,566        | \$146,147        | \$99,375         |
| All cancers                             | \$515,299        | \$329,472        | \$226,129        |
| Colon cancer                            | \$112,334        | \$74,271         | \$51,882         |
| Chronic myeloid leukemia                | \$26,482         | \$15,935         | \$10,523         |
| Diabetes type 2                         | \$138,700        | \$82,298         | \$53,222         |
| Depression                              | \$740,456        | \$565,952        | \$443,897        |
| Head and neck cancer                    | \$4,274          | \$2,778          | \$1,964          |
| Ischemic heart disease                  | \$281,483        | \$167,055        | \$108,613        |
| Liver cancer                            | \$4,222          | \$2,777          | \$1,982          |
| Multiple myeloma                        | \$62,783         | \$39,385         | \$26,960         |
| Stomach cancer                          | \$26,673         | \$16,661         | \$11,375         |
| Stroke                                  | \$43,306         | \$25,516         | \$16,537         |
| Lung cancer                             | \$37,711         | \$24,456         | \$17,254         |
| Uterine cancer                          | \$11,292         | \$7,115          | \$4,826          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

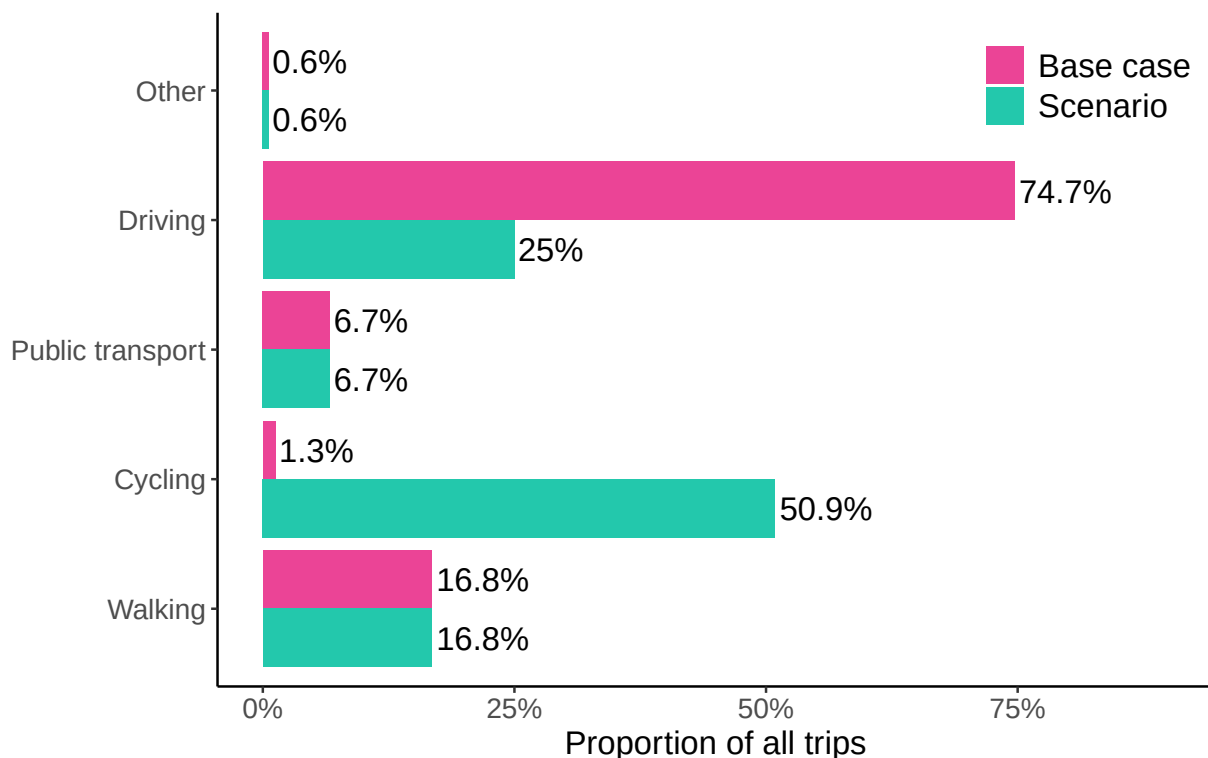
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
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## Scenario: replacing car trips under 10km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 10km for leisure, shopping, work, education or other purposes with cycling trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in cycling from 1.3% to 50.9% and from 74.7% to 25.0% for car trips taken as either a driver or passenger.

Increases in cycling translate into a shift from 47.9% to 74.9% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

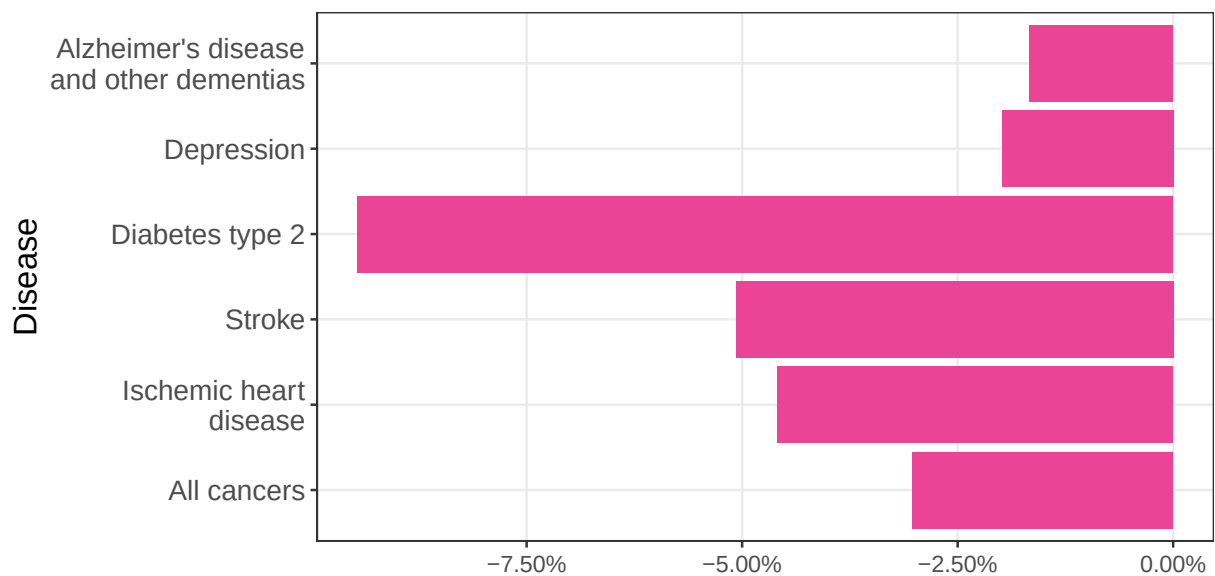
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

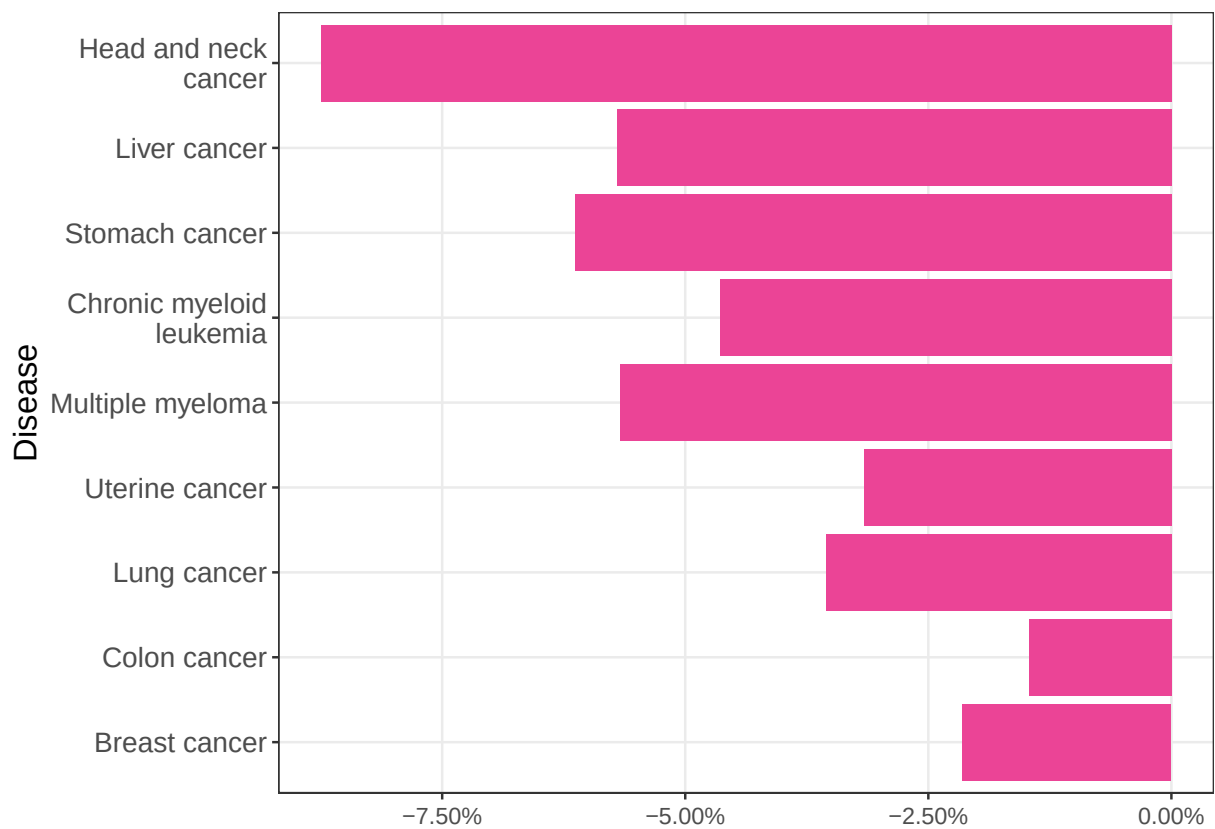
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 1.67%                              | 10,613                                                                      |
| Breast cancer                           | 2.15%                              | 1,573                                                                       |
| All cancers                             | 3.02%                              | 11,906                                                                      |
| Colon cancer                            | 1.46%                              | 1,896                                                                       |
| Chronic myeloid leukemia                | 4.64%                              | 127                                                                         |
| Diabetes type 2                         | 9.46%                              | 31,227                                                                      |
| Depression                              | 1.98%                              | 31,801                                                                      |
| Head and neck cancer                    | 8.75%                              | 385                                                                         |
| Ischemic heart disease                  | 4.59%                              | 40,426                                                                      |
| Liver cancer                            | 5.70%                              | 1,003                                                                       |
| Multiple myeloma                        | 5.68%                              | 1,100                                                                       |
| Stomach cancer                          | 6.14%                              | 1,465                                                                       |
| Stroke                                  | 5.07%                              | 13,201                                                                      |
| Lung cancer                             | 3.55%                              | 4,085                                                                       |
| Uterine cancer                          | 3.16%                              | 268                                                                         |

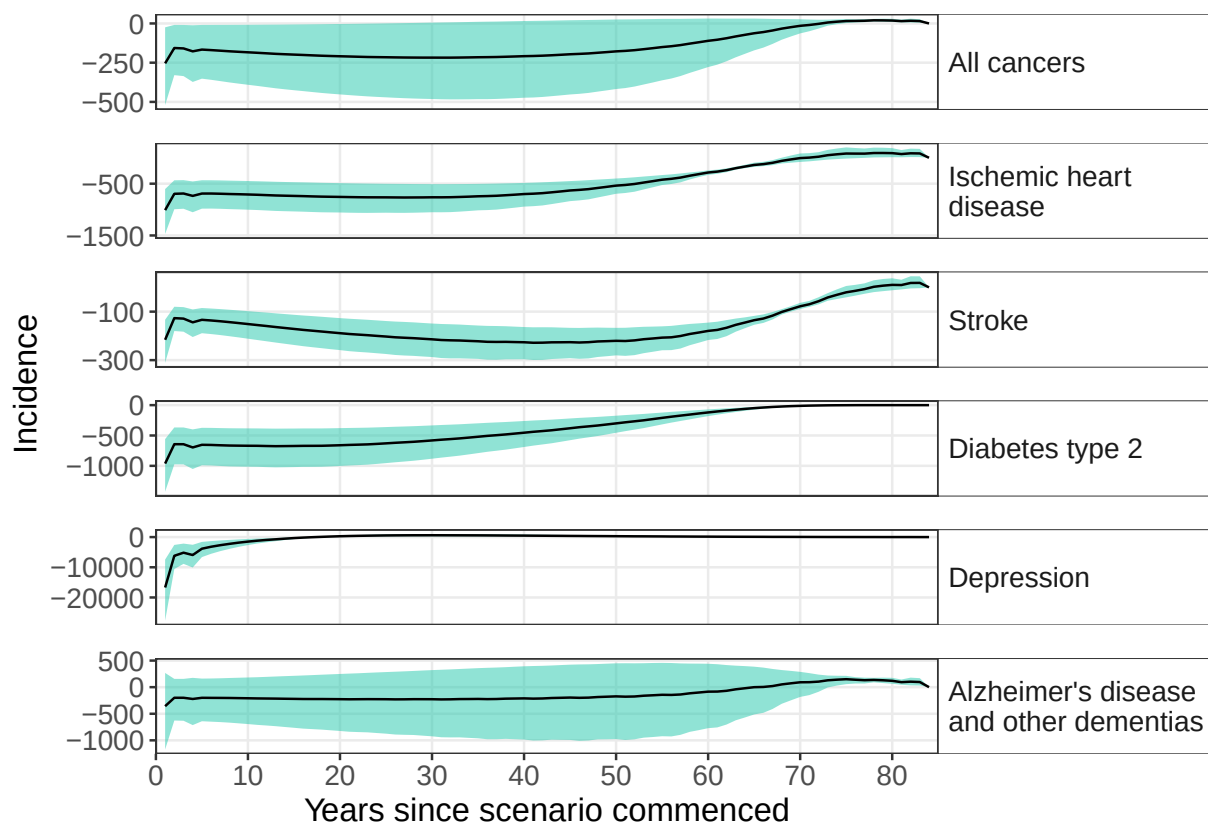
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



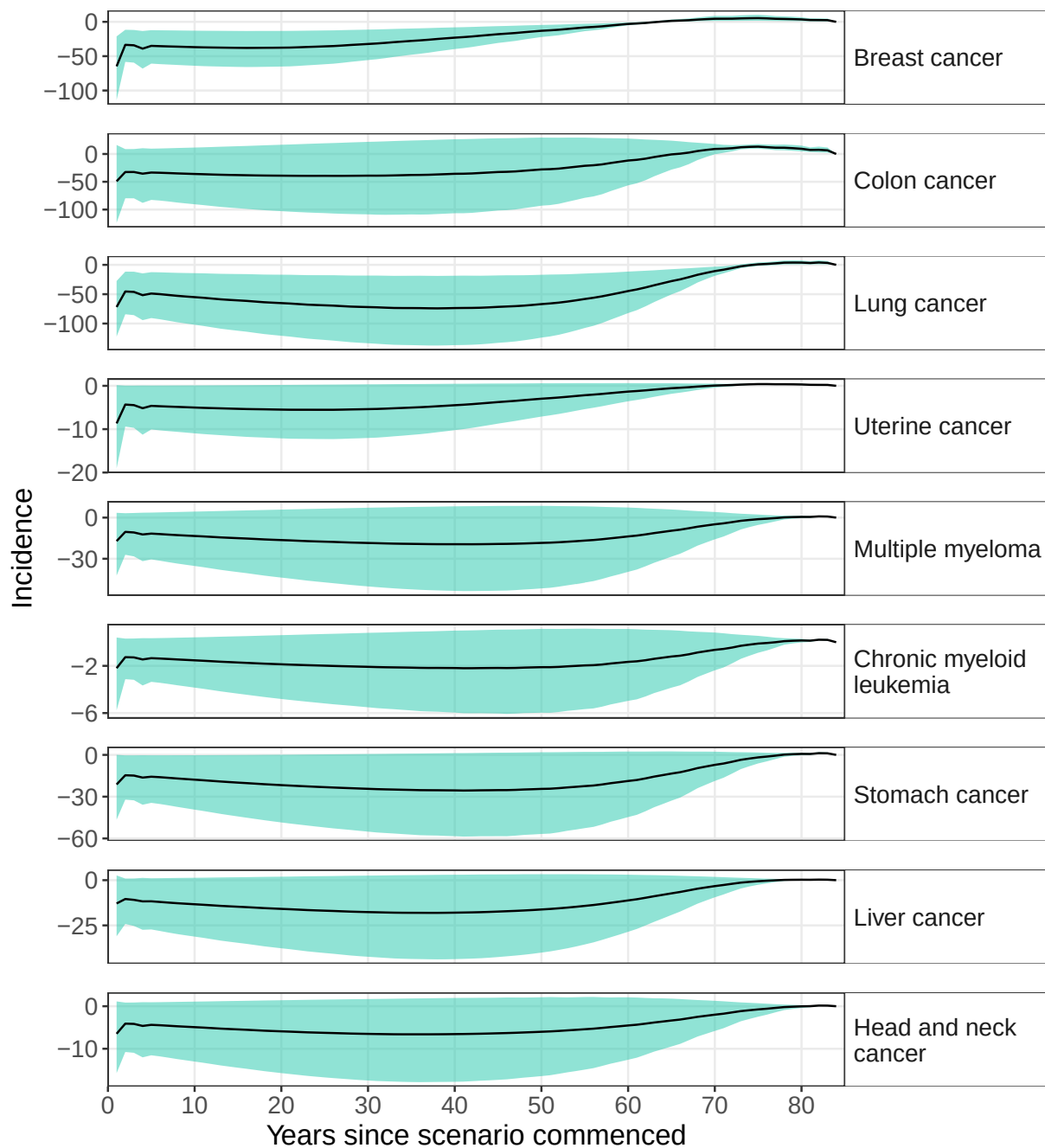
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

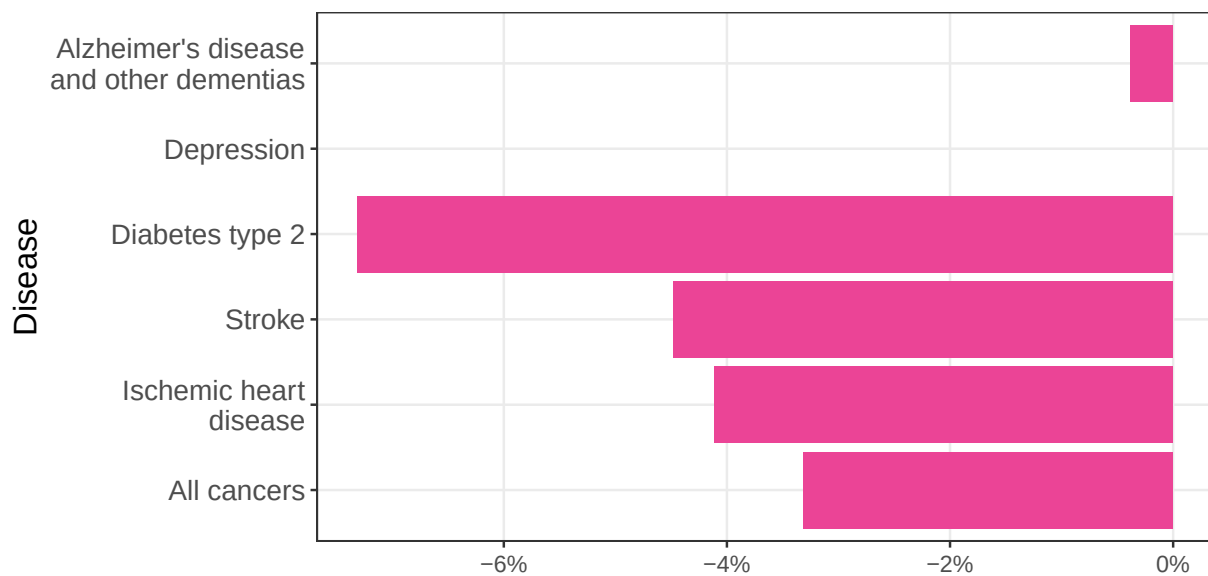
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

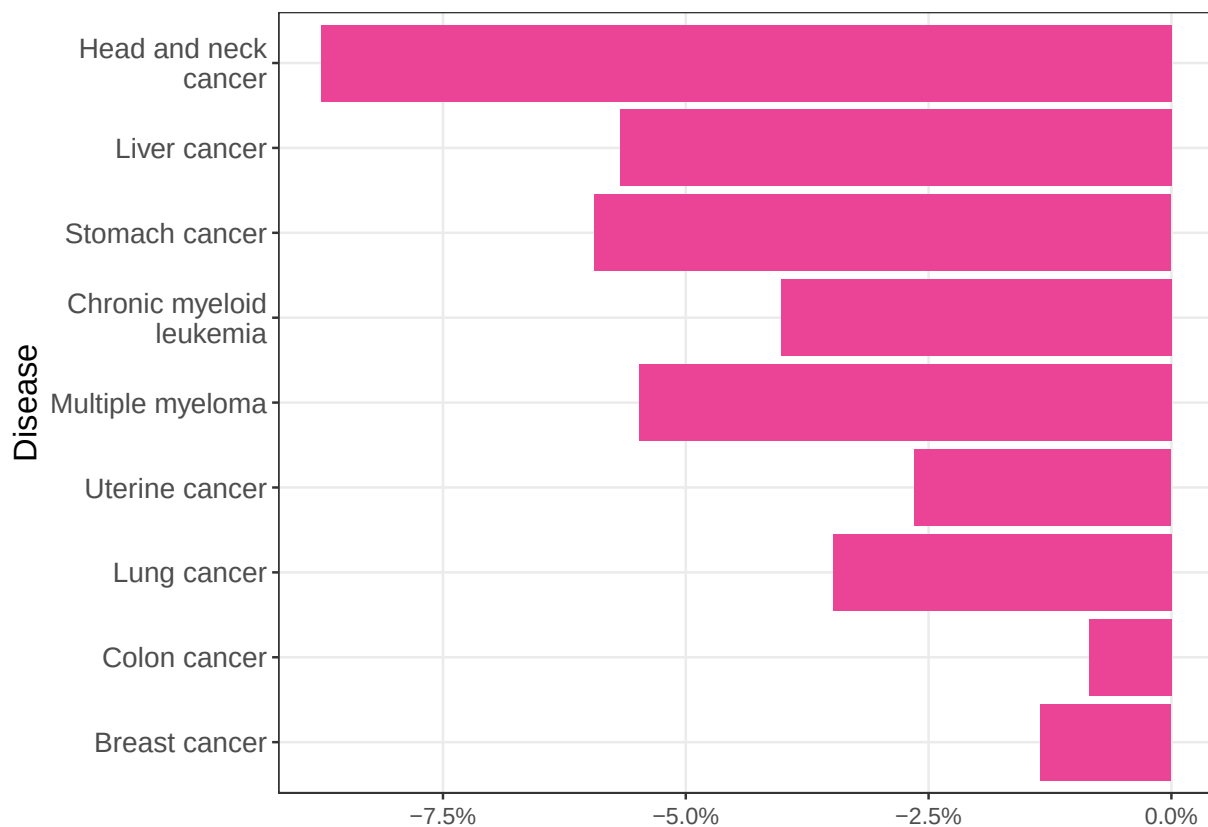
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.38%                   | 907                                                               |
| Breast cancer                           | 1.35%                   | 435                                                               |
| All cancers                             | 3.32%                   | 8,400                                                             |
| Colon cancer                            | 0.84%                   | 364                                                               |
| Chronic myeloid leukemia                | 4.02%                   | 83                                                                |
| Diabetes type 2                         | 7.31%                   | 3,612                                                             |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 8.76%                   | 348                                                               |
| Ischemic heart disease                  | 4.11%                   | 15,427                                                            |
| Liver cancer                            | 5.68%                   | 970                                                               |
| Multiple myeloma                        | 5.48%                   | 955                                                               |
| Stomach cancer                          | 5.94%                   | 1,256                                                             |
| Stroke                                  | 4.48%                   | 7,839                                                             |
| Lung cancer                             | 3.48%                   | 3,798                                                             |
| Uterine cancer                          | 2.64%                   | 191                                                               |

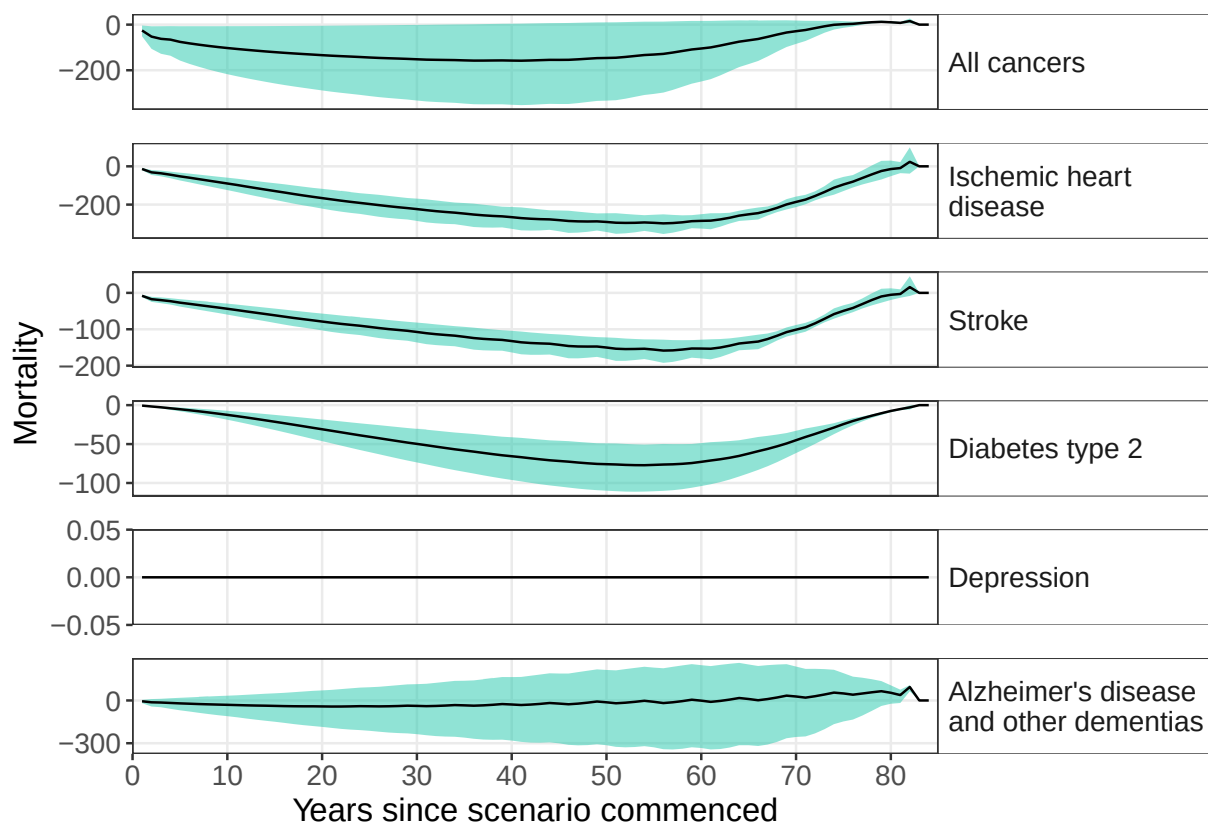
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

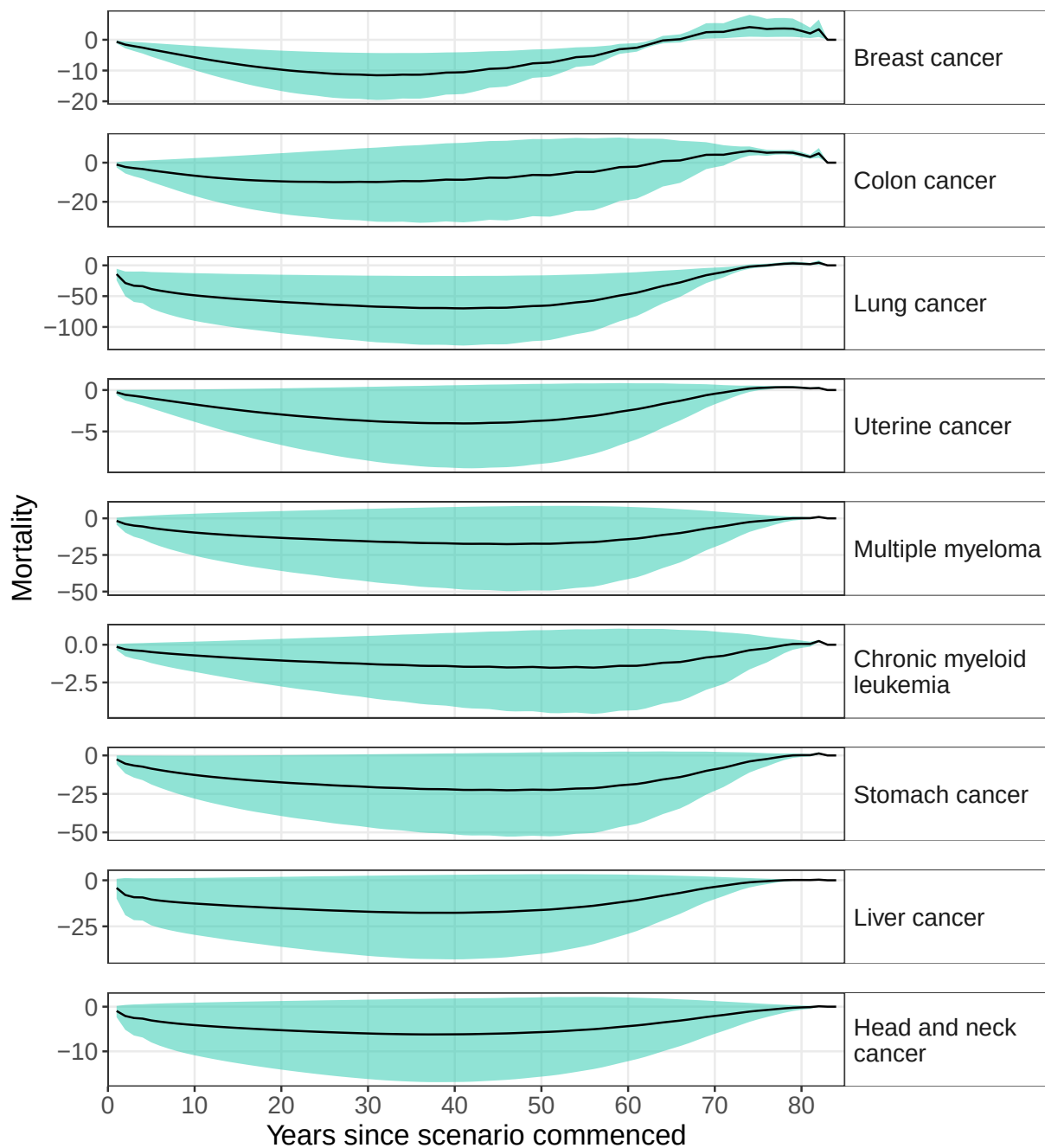


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

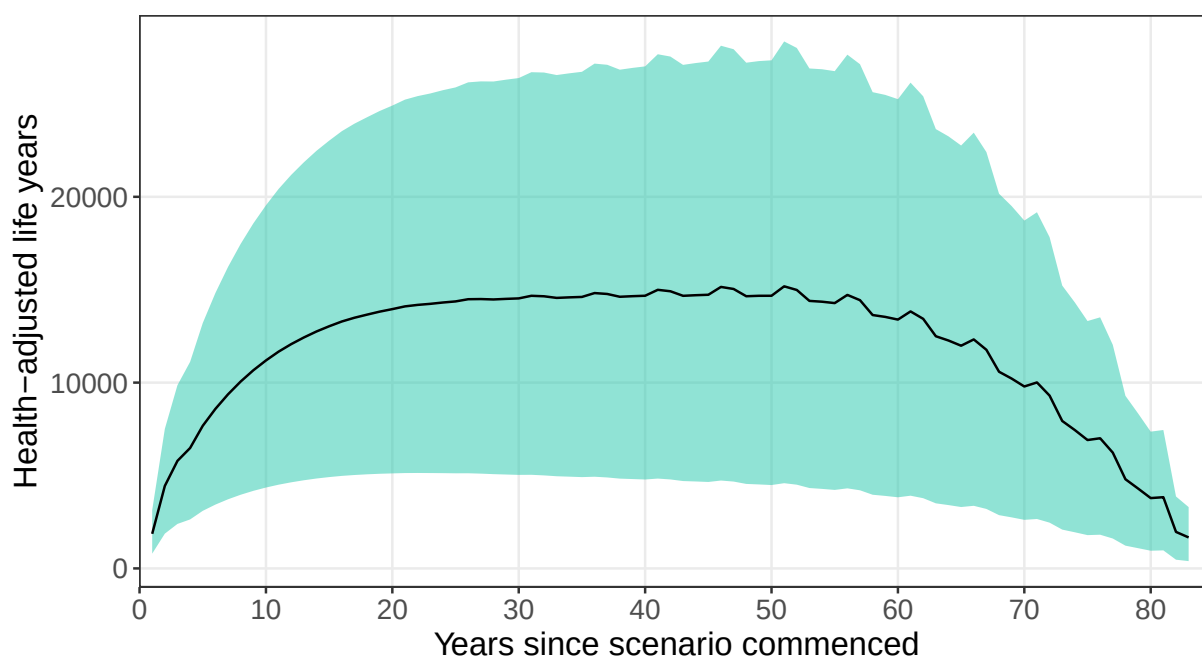
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **990,320** HALYs for the scenario population, which is **492** HALYs per 1,000 members of the population.



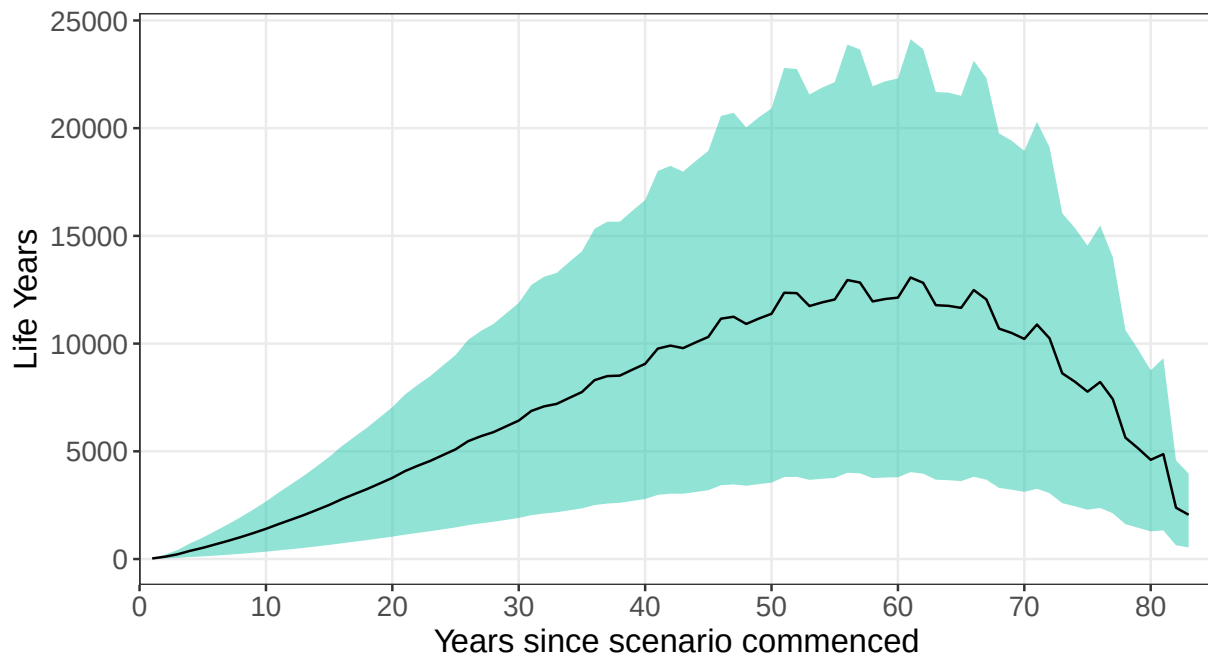
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **604,840** Life Years for the scenario population, which is **300** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **492** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **38,176,100** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **23,333,670** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **15,827,582** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$30,556         | \$18,585         | \$12,343         |
| Breast cancer                           | \$322,060        | \$201,883        | \$135,422        |
| All cancers                             | \$701,618        | \$444,021        | \$301,698        |
| Colon cancer                            | \$152,971        | \$100,246        | \$69,403         |
| Chronic myeloid leukemia                | \$33,787         | \$20,228         | \$13,277         |
| Diabetes type 2                         | \$191,048        | \$112,459        | \$72,126         |
| Depression                              | \$944,468        | \$715,205        | \$555,962        |
| Head and neck cancer                    | \$5,642          | \$3,651          | \$2,565          |
| Ischemic heart disease                  | \$385,529        | \$227,151        | \$146,482        |
| Liver cancer                            | \$5,871          | \$3,846          | \$2,731          |
| Multiple myeloma                        | \$80,787         | \$50,494         | \$34,385         |
| Stomach cancer                          | \$35,677         | \$22,181         | \$15,053         |
| Stroke                                  | \$56,985         | \$33,334         | \$21,435         |
| Lung cancer                             | \$49,740         | \$32,122         | \$22,532         |
| Uterine cancer                          | \$15,129         | \$9,450          | \$6,350          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

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## References

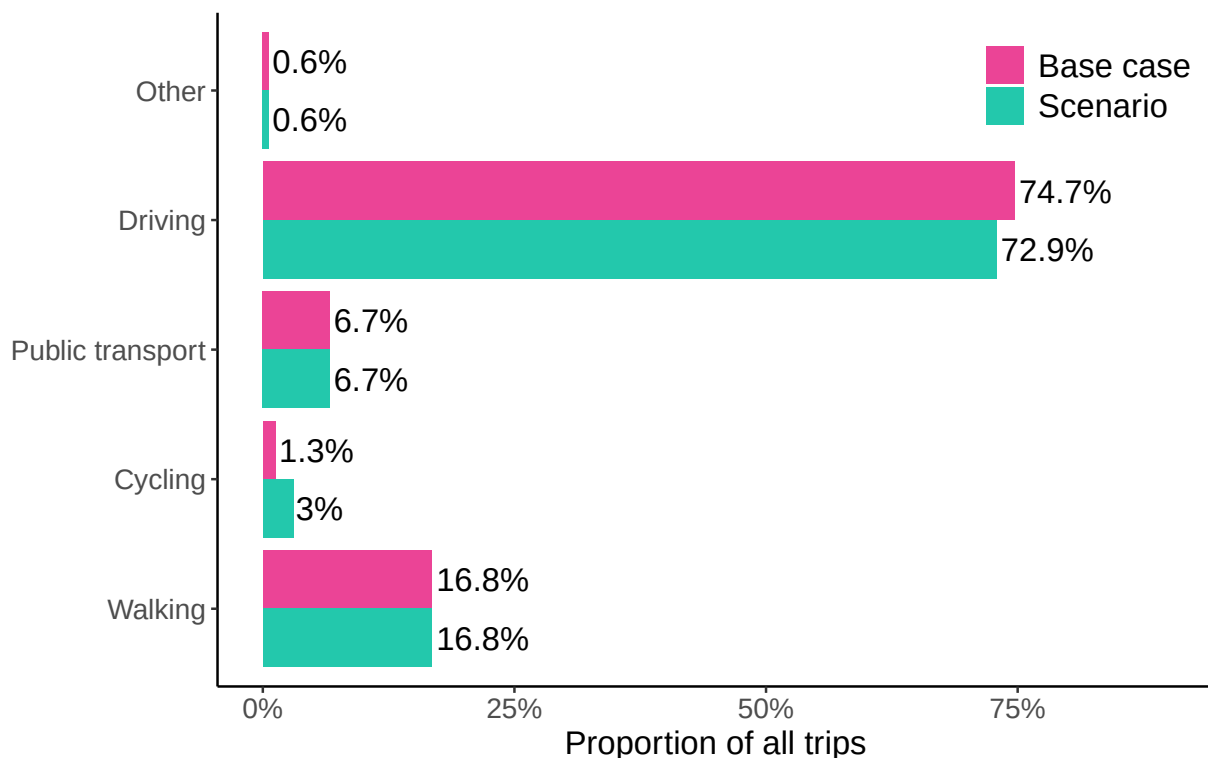
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## Scenario: replacing car trips under 2km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 2km for work related or education purposes with cycling trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in cycling from 1.3% to 3.0% and from 74.7% to 72.9% for car trips taken as either a driver or passenger.

Increases in cycling translate into a shift from 47.9% to 48.9% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

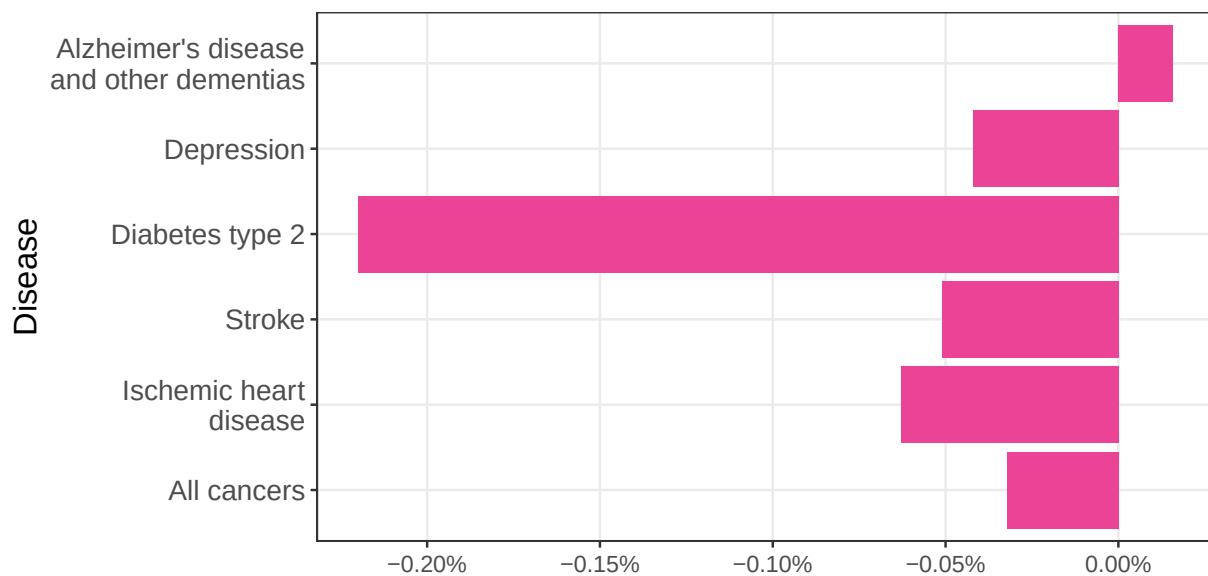
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

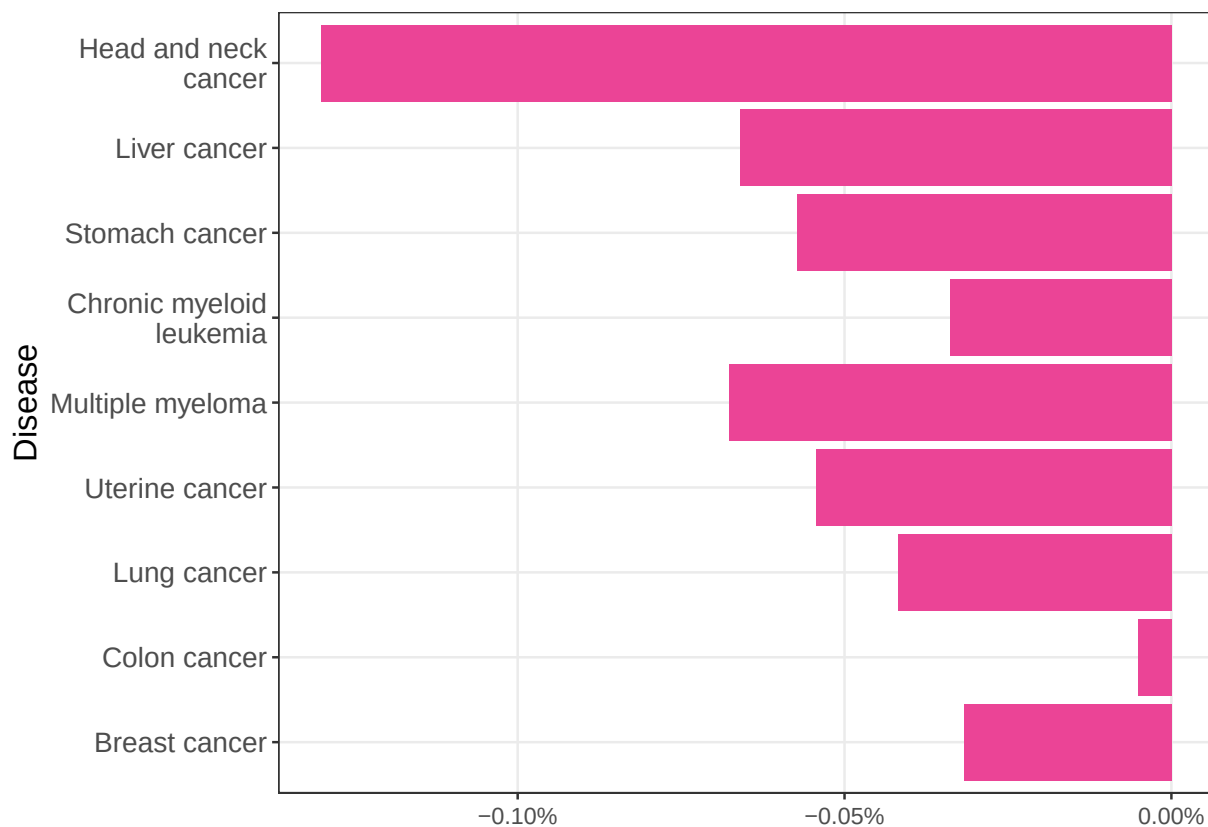
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.02%                             | -101                                                                        |
| Breast cancer                           | 0.03%                              | 23                                                                          |
| All cancers                             | 0.03%                              | 127                                                                         |
| Colon cancer                            | 0.01%                              | 7                                                                           |
| Chronic myeloid leukemia                | 0.03%                              | 1                                                                           |
| Diabetes type 2                         | 0.22%                              | 727                                                                         |
| Depression                              | 0.04%                              | 672                                                                         |
| Head and neck cancer                    | 0.13%                              | 6                                                                           |
| Ischemic heart disease                  | 0.06%                              | 553                                                                         |
| Liver cancer                            | 0.07%                              | 12                                                                          |
| Multiple myeloma                        | 0.07%                              | 13                                                                          |
| Stomach cancer                          | 0.06%                              | 14                                                                          |
| Stroke                                  | 0.05%                              | 133                                                                         |
| Lung cancer                             | 0.04%                              | 48                                                                          |
| Uterine cancer                          | 0.05%                              | 5                                                                           |

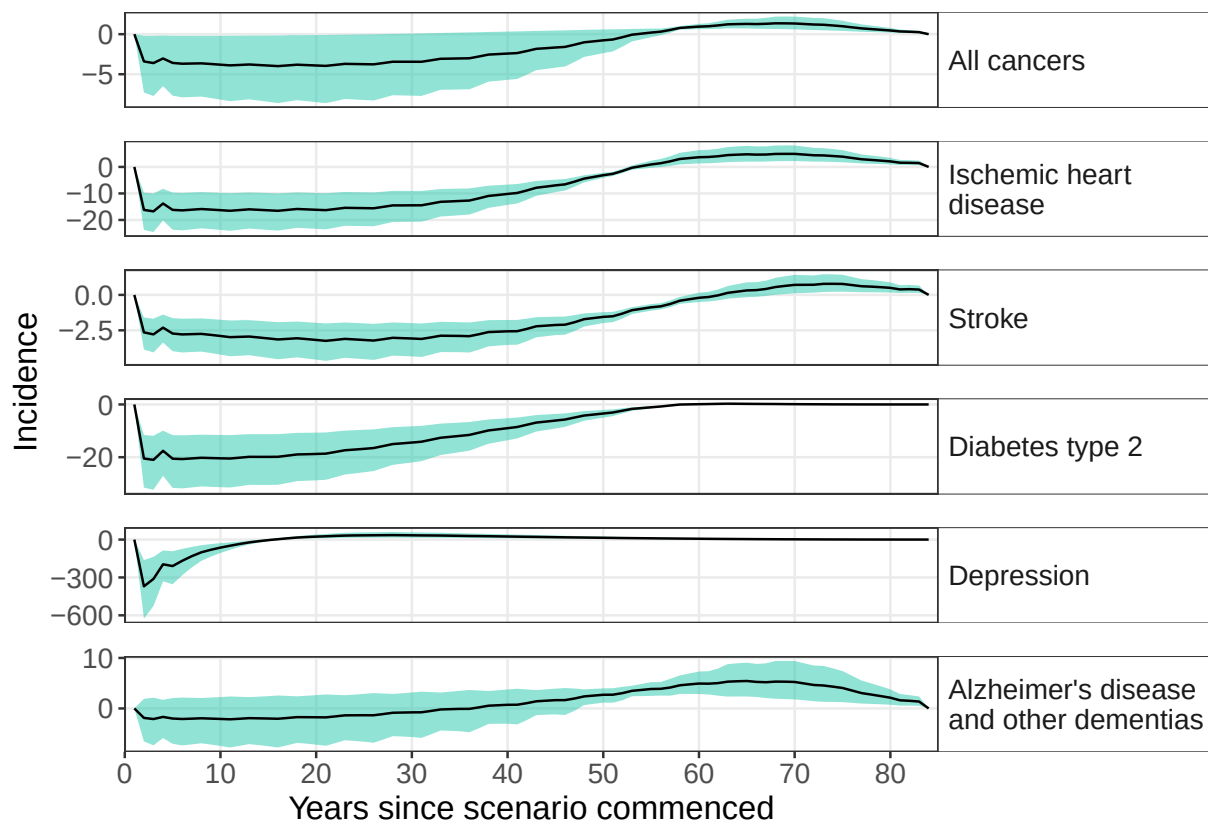
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



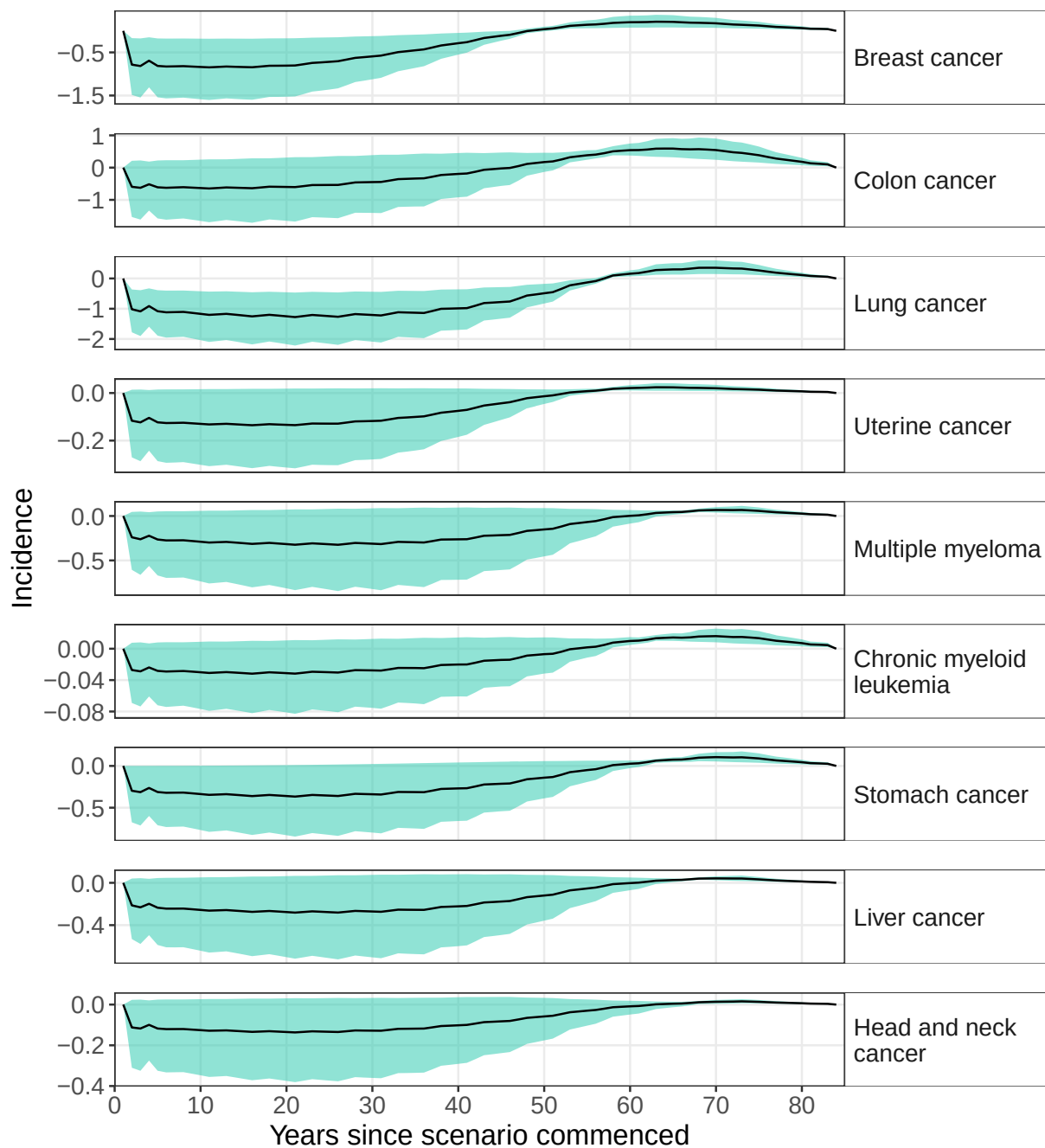
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

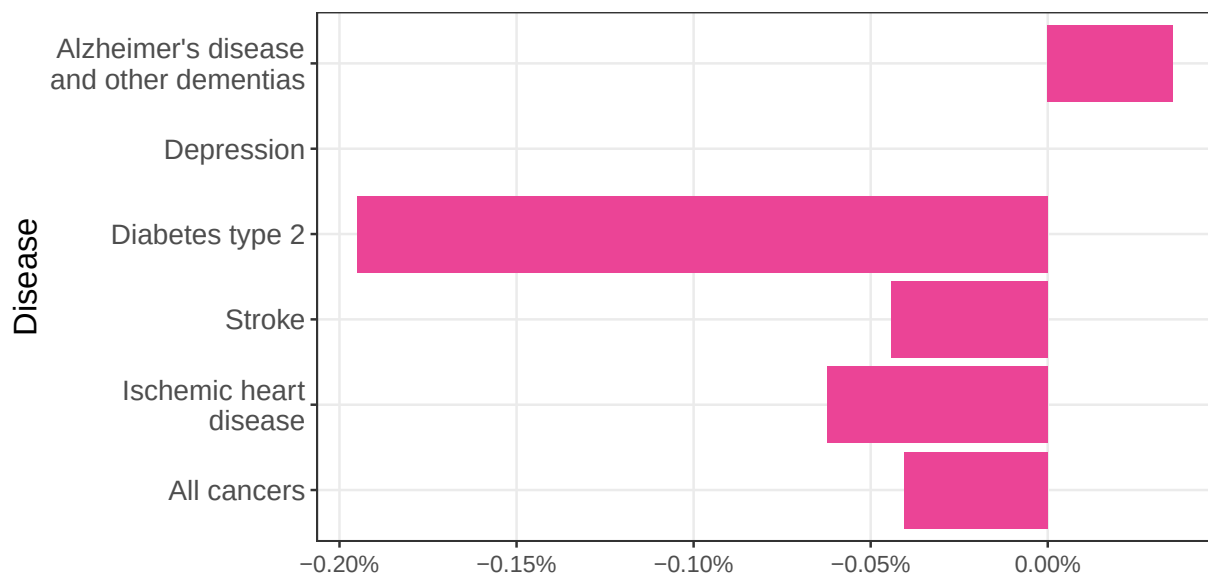
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

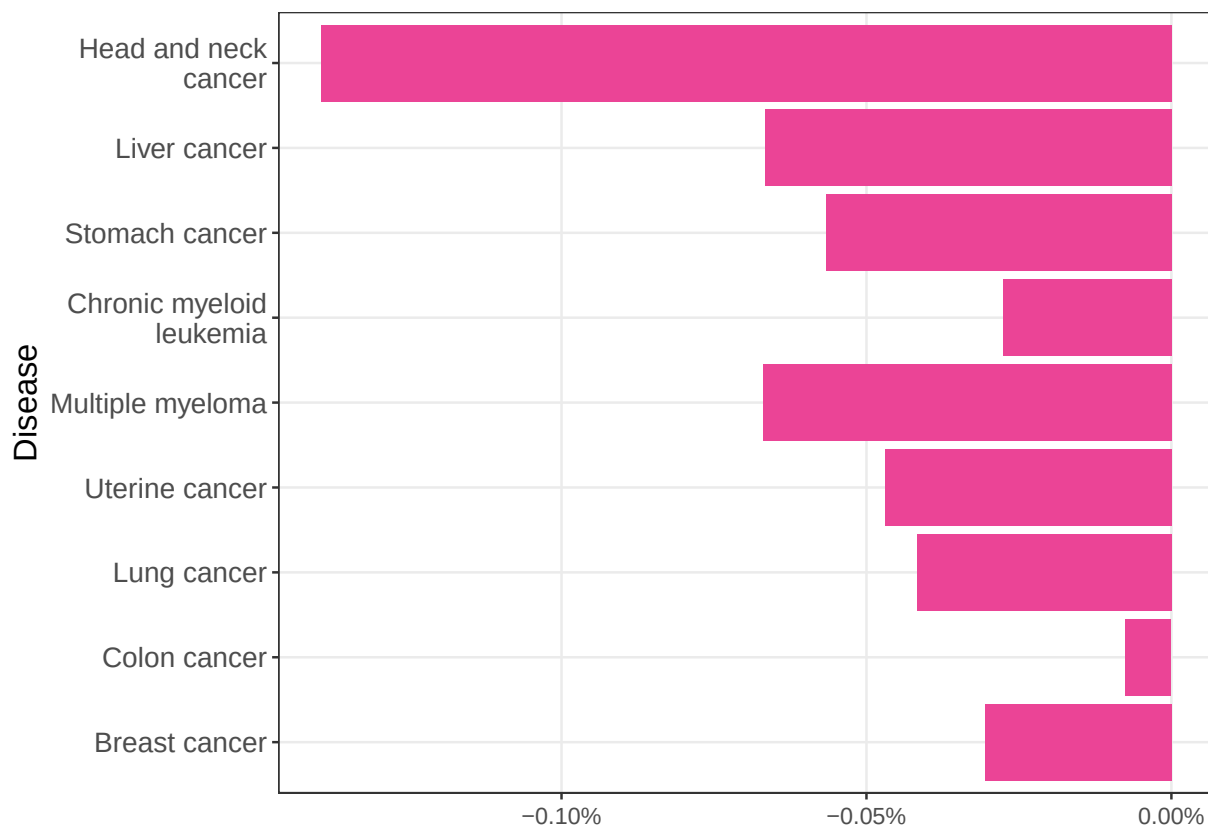
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.04%                  | -85                                                               |
| Breast cancer                           | 0.03%                   | 10                                                                |
| All cancers                             | 0.04%                   | 103                                                               |
| Colon cancer                            | 0.01%                   | 3                                                                 |
| Chronic myeloid leukemia                | 0.03%                   | 1                                                                 |
| Diabetes type 2                         | 0.19%                   | 96                                                                |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.14%                   | 6                                                                 |
| Ischemic heart disease                  | 0.06%                   | 234                                                               |
| Liver cancer                            | 0.07%                   | 11                                                                |
| Multiple myeloma                        | 0.07%                   | 12                                                                |
| Stomach cancer                          | 0.06%                   | 12                                                                |
| Stroke                                  | 0.04%                   | 77                                                                |
| Lung cancer                             | 0.04%                   | 45                                                                |
| Uterine cancer                          | 0.05%                   | 3                                                                 |

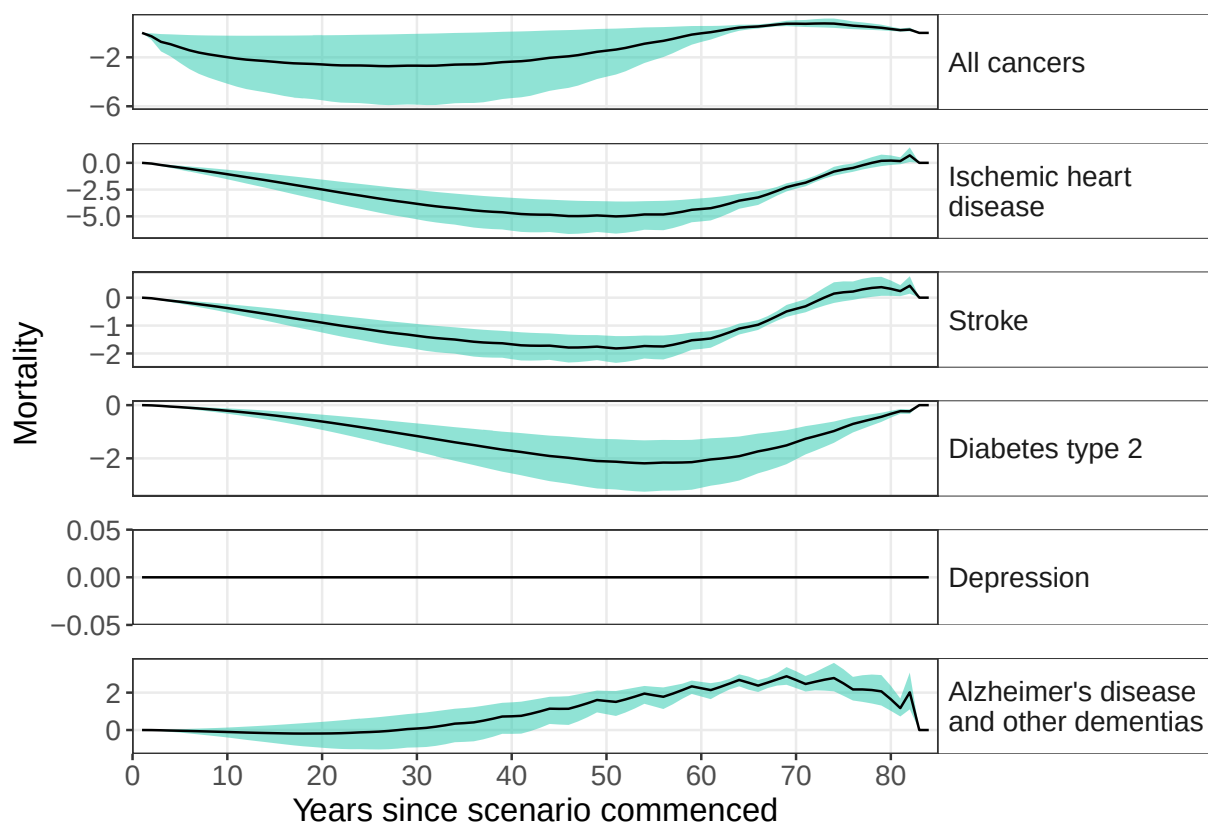
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

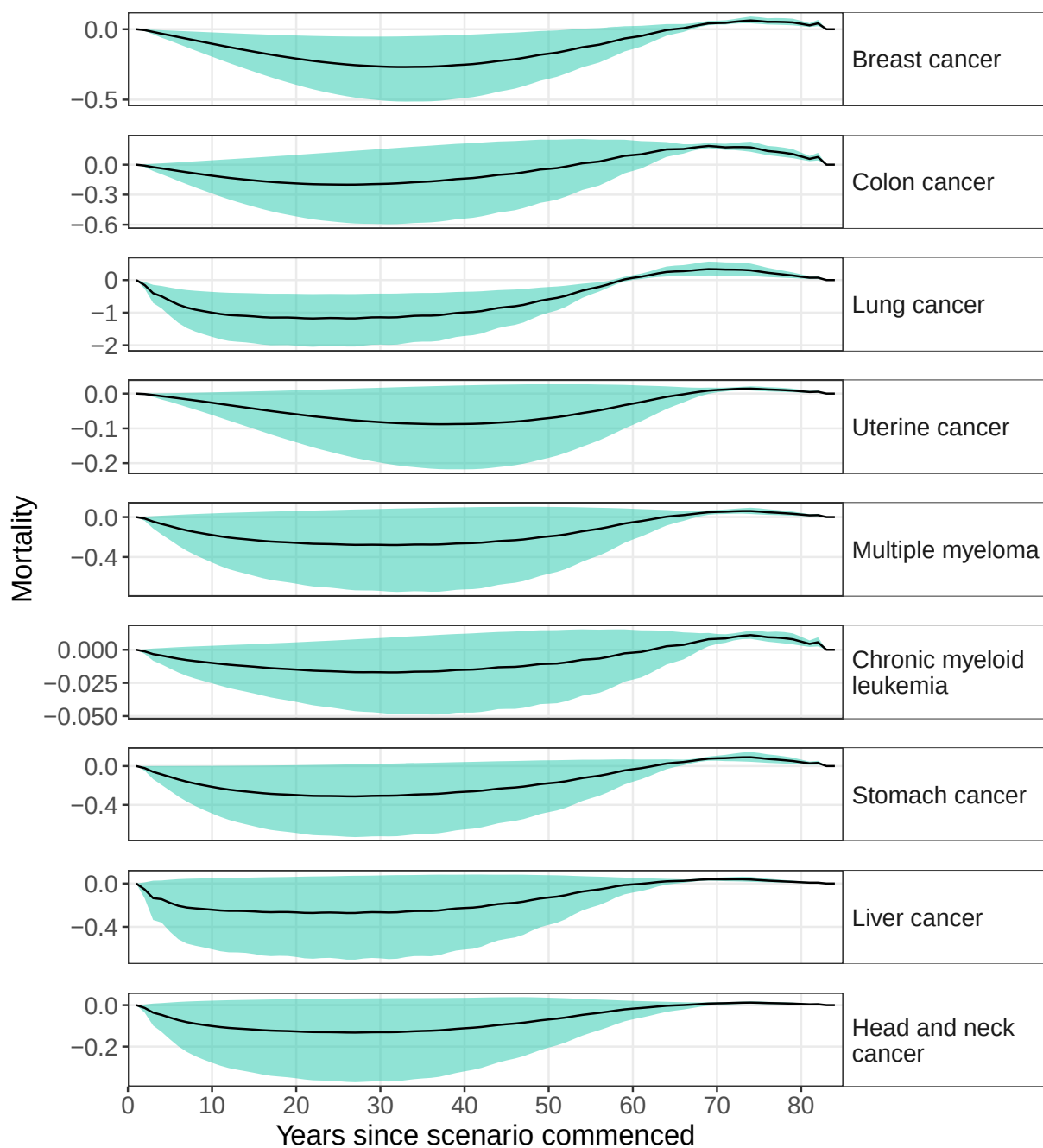


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **23,751** HALYs for the scenario population, which is **12** HALYs per 1,000 members of the population.



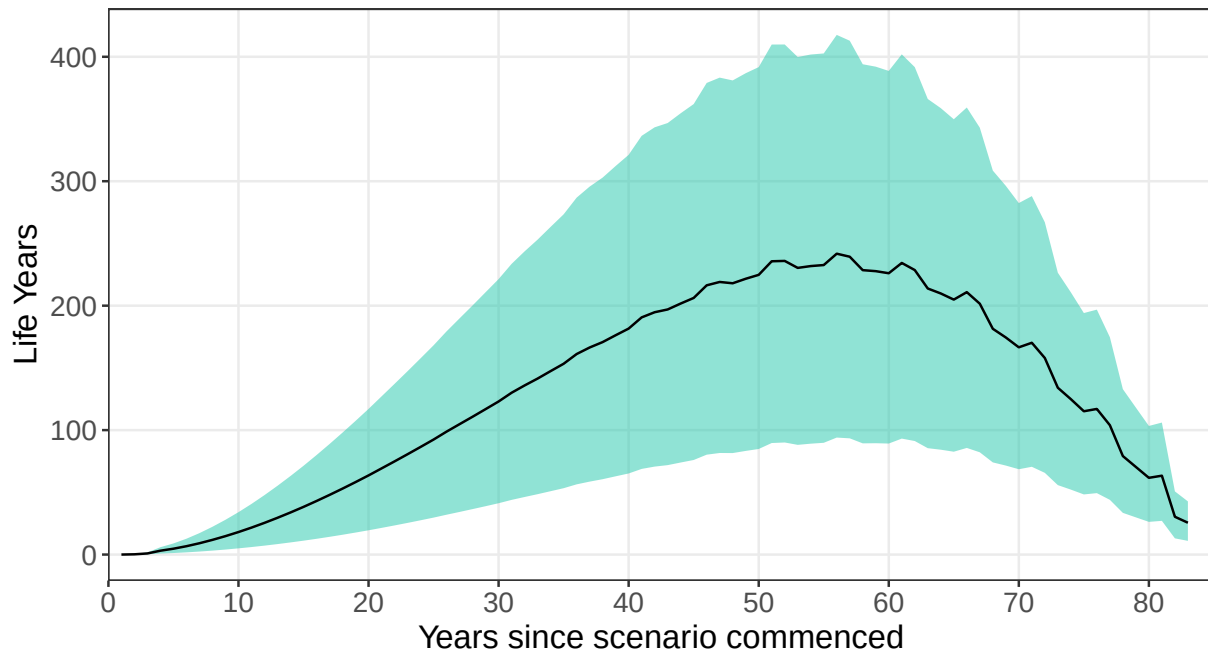
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **10,814** Life Years for the scenario population, which is **5.4** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **12** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **1,006,058** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **635,746** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **437,711** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## b. What is the Value of a Statistical Life and Value of a Statistical Life Year?

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## c. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## d. Application in advocacy and reporting

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$270            | \$190            | \$133            |
| Breast cancer                           | \$7,330          | \$4,498          | \$2,942          |
| All cancers                             | \$15,278         | \$9,707          | \$6,530          |
| Colon cancer                            | \$2,695          | \$1,844          | \$1,287          |
| Chronic myeloid leukemia                | \$810            | \$489            | \$318            |
| Diabetes type 2                         | \$5,686          | \$3,301          | \$2,079          |
| Depression                              | \$33,095         | \$24,094         | \$18,170         |
| Head and neck cancer                    | \$140            | \$95             | \$68             |
| Ischemic heart disease                  | \$9,864          | \$5,765          | \$3,647          |
| Liver cancer                            | \$117            | \$80             | \$58             |
| Multiple myeloma                        | \$1,958          | \$1,253          | \$855            |
| Stomach cancer                          | \$715            | \$461            | \$317            |
| Stroke                                  | \$1,354          | \$792            | \$501            |
| Lung cancer                             | \$1,094          | \$731            | \$518            |
| Uterine cancer                          | \$418            | \$257            | \$169            |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

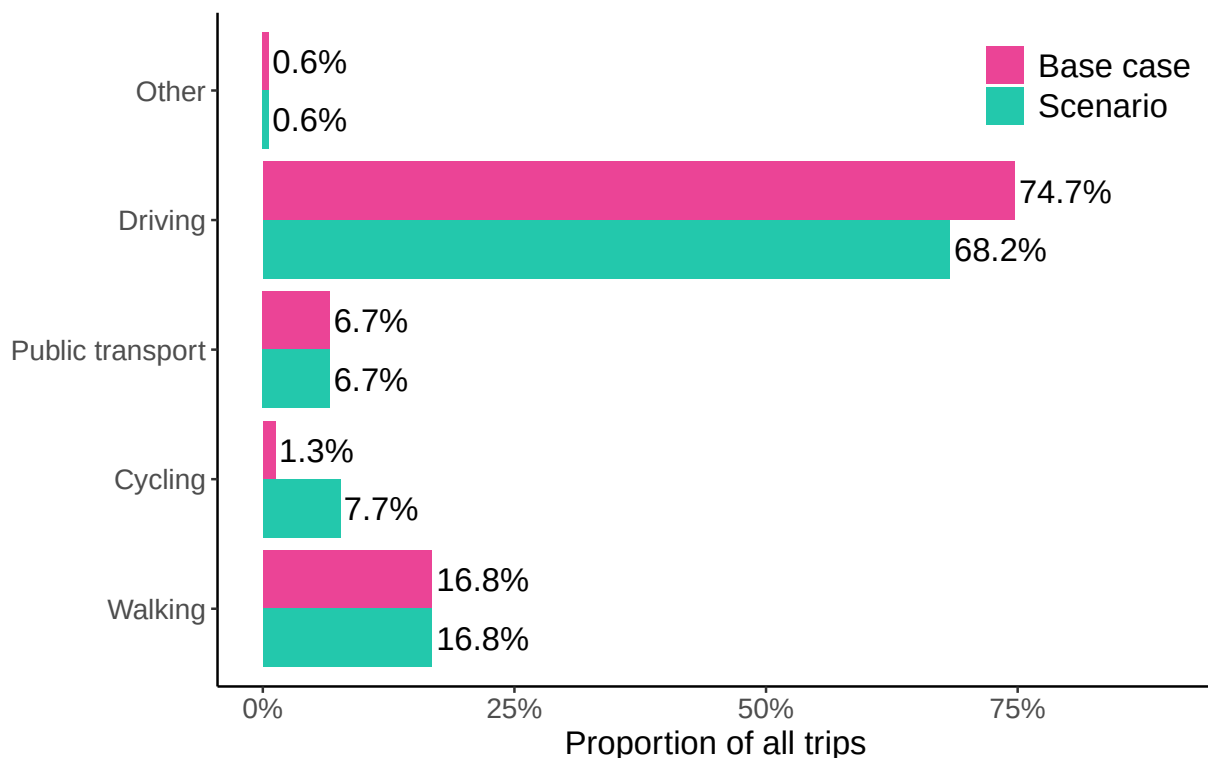
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
3. Ananthapavan, J., Moodie, M., Milat, A.J., & Carter, R. (2021). *Systematic review to update 'value of a statistical life' estimates for Australia*. International journal of environmental research and public health, 2021. 18(11): p. 6168.
4. Terrill, M. & Batrouney, H. (2018). *Unfreezing discount rates: Transport infrastructure for tomorrow*. Grattan Institute.
5. Abelson, P. (2008). *Establishing a monetary value for lives saved: issues and controversies*. Canberra: Office of Best Practice Regulation, Department of Finance and Deregulation.
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## Scenario: replacing car trips under 5km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 5km for work related or education purposes with cycling trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in cycling from 1.3% to 7.7% and from 74.7% to 68.2% for car trips taken as either a driver or passenger.

Increases in cycling translate into a shift from 47.9% to 53.2% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

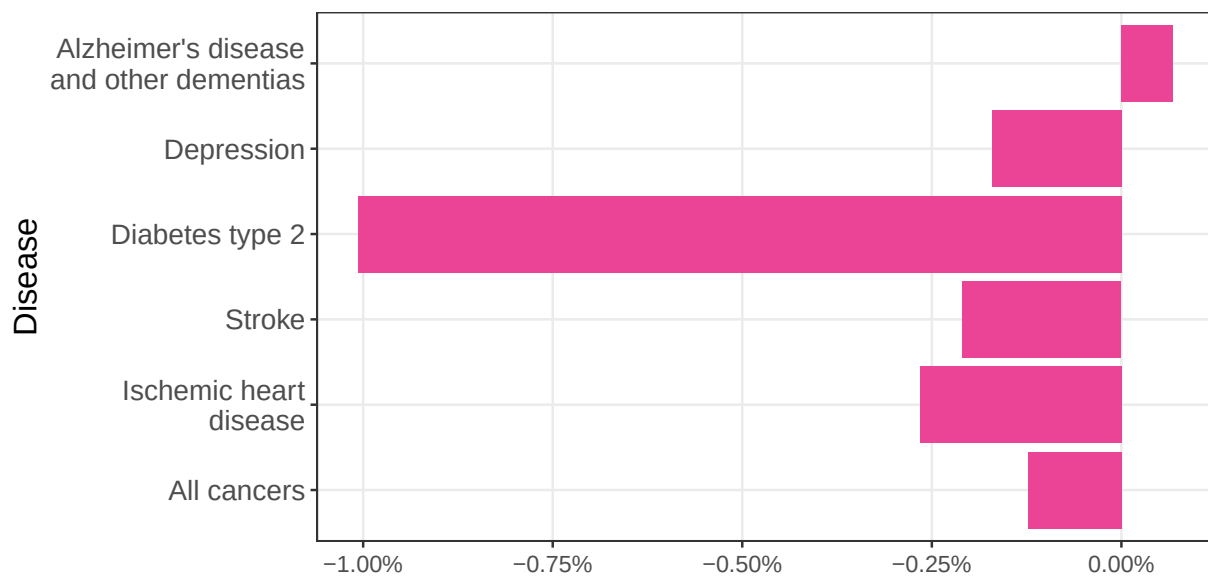
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

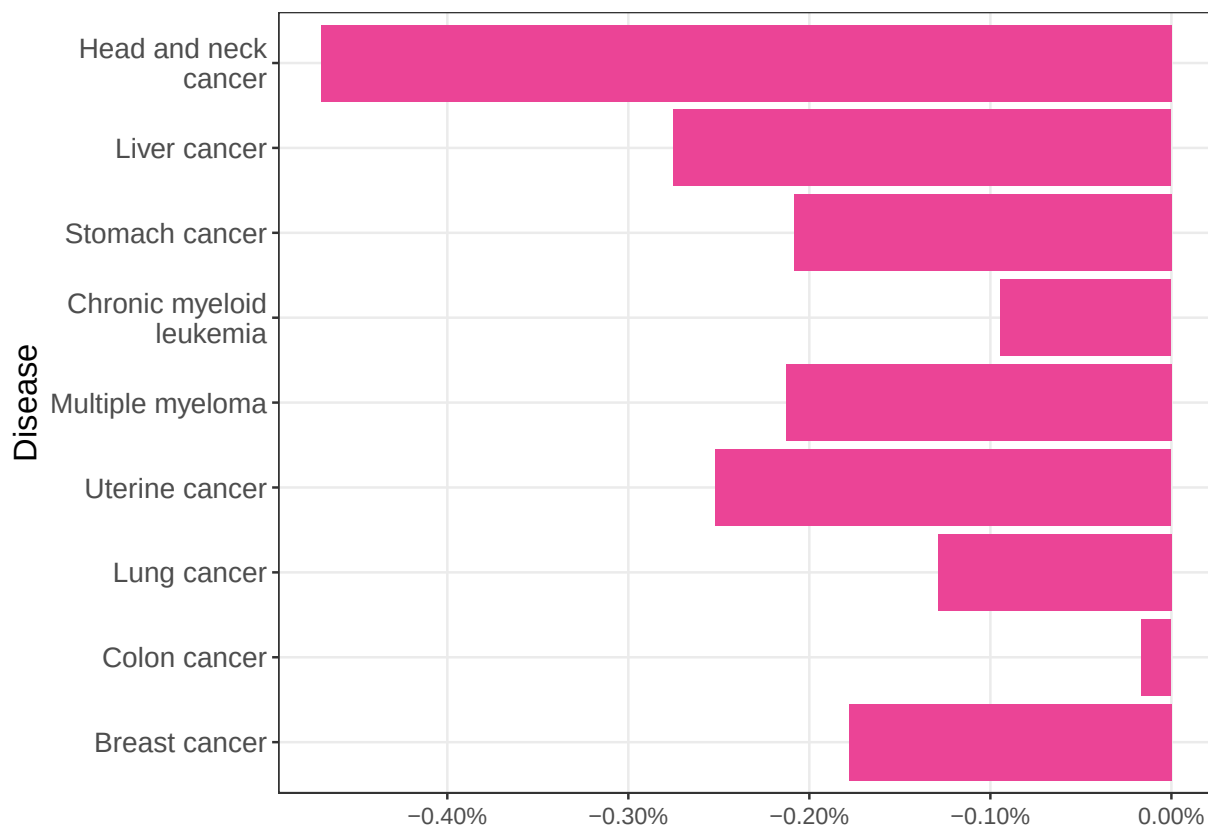
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.07%                             | -435                                                                        |
| Breast cancer                           | 0.18%                              | 130                                                                         |
| All cancers                             | 0.12%                              | 484                                                                         |
| Colon cancer                            | 0.02%                              | 21                                                                          |
| Chronic myeloid leukemia                | 0.09%                              | 3                                                                           |
| Diabetes type 2                         | 1.01%                              | 3,325                                                                       |
| Depression                              | 0.17%                              | 2,735                                                                       |
| Head and neck cancer                    | 0.47%                              | 21                                                                          |
| Ischemic heart disease                  | 0.27%                              | 2,337                                                                       |
| Liver cancer                            | 0.28%                              | 48                                                                          |
| Multiple myeloma                        | 0.21%                              | 41                                                                          |
| Stomach cancer                          | 0.21%                              | 50                                                                          |
| Stroke                                  | 0.21%                              | 546                                                                         |
| Lung cancer                             | 0.13%                              | 148                                                                         |
| Uterine cancer                          | 0.25%                              | 21                                                                          |

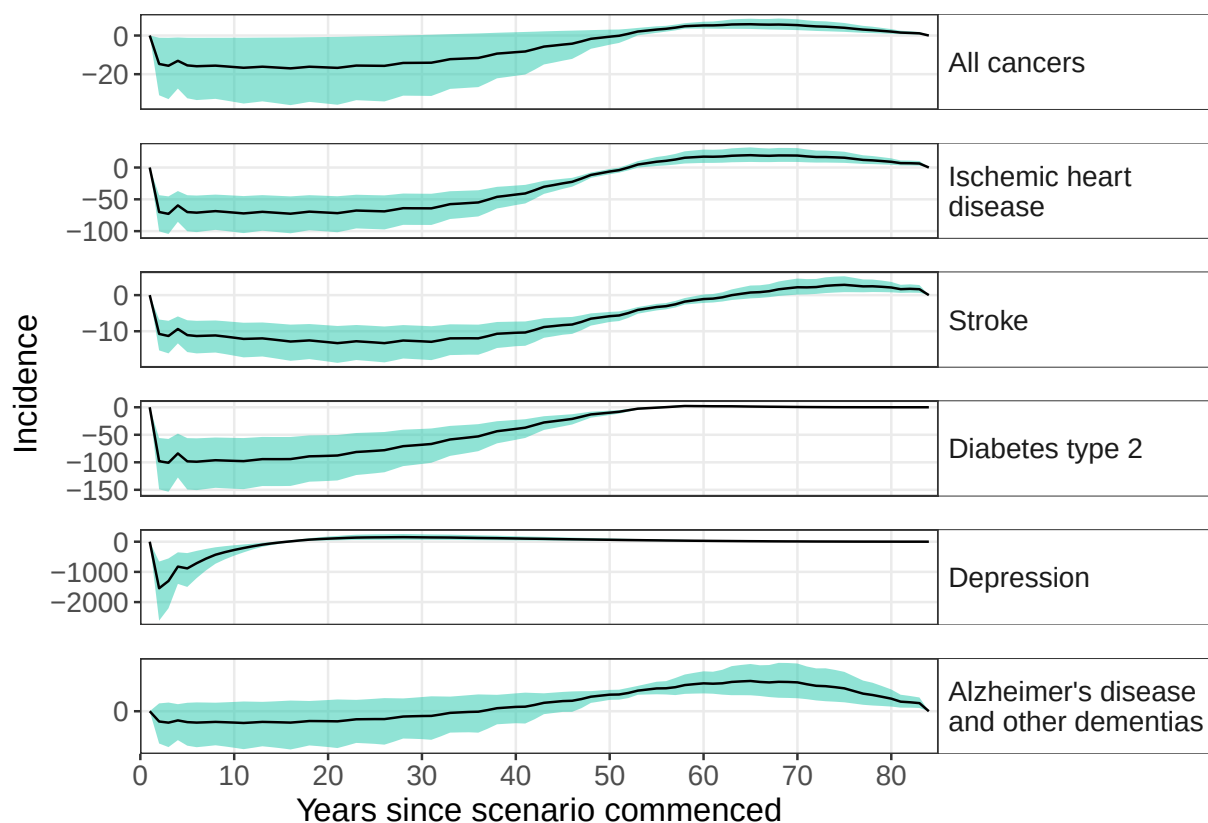
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



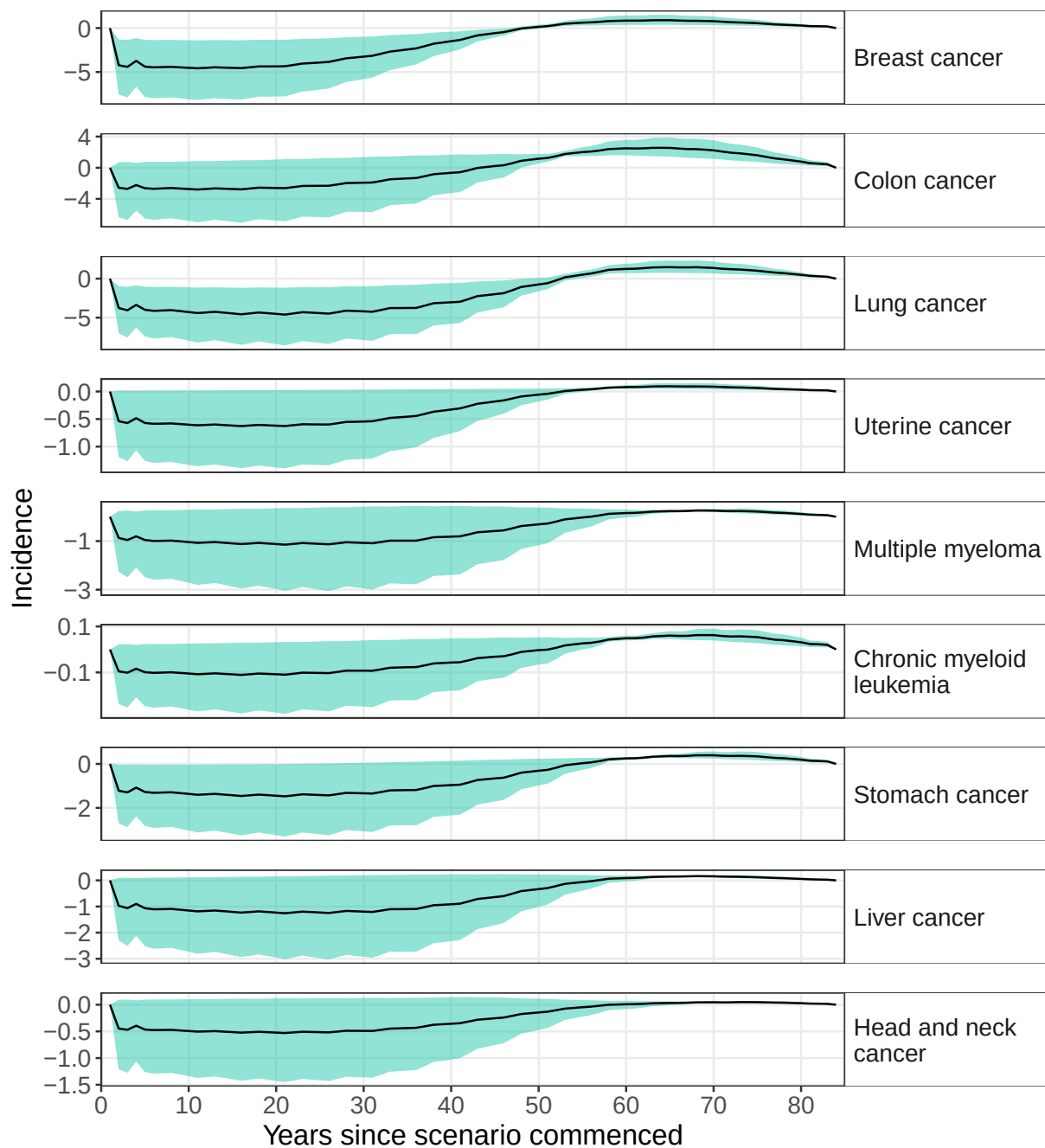
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

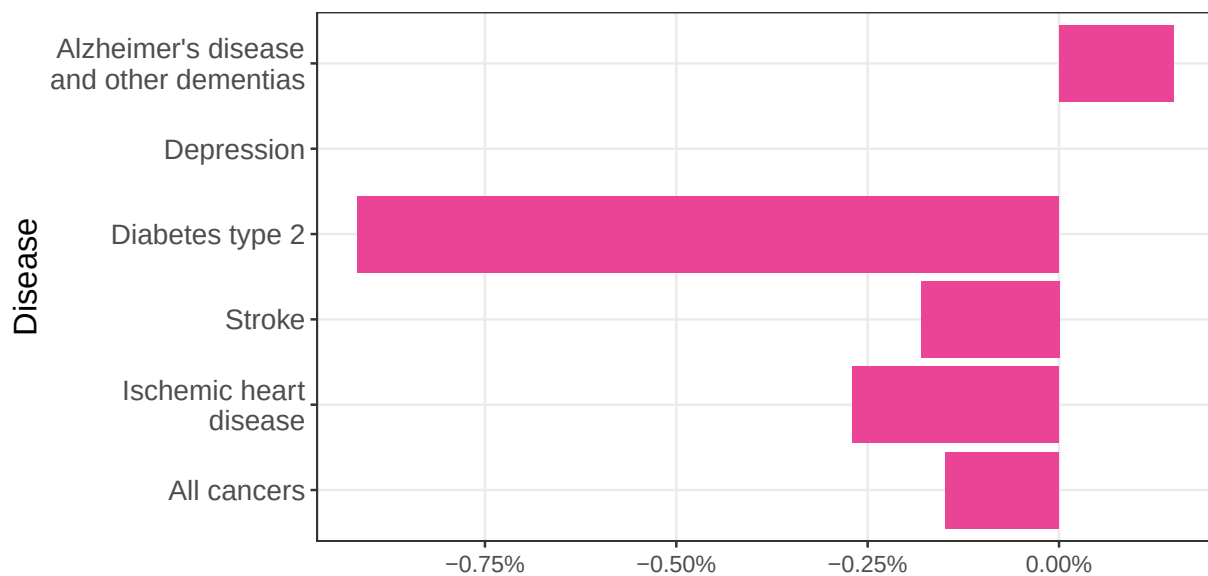
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

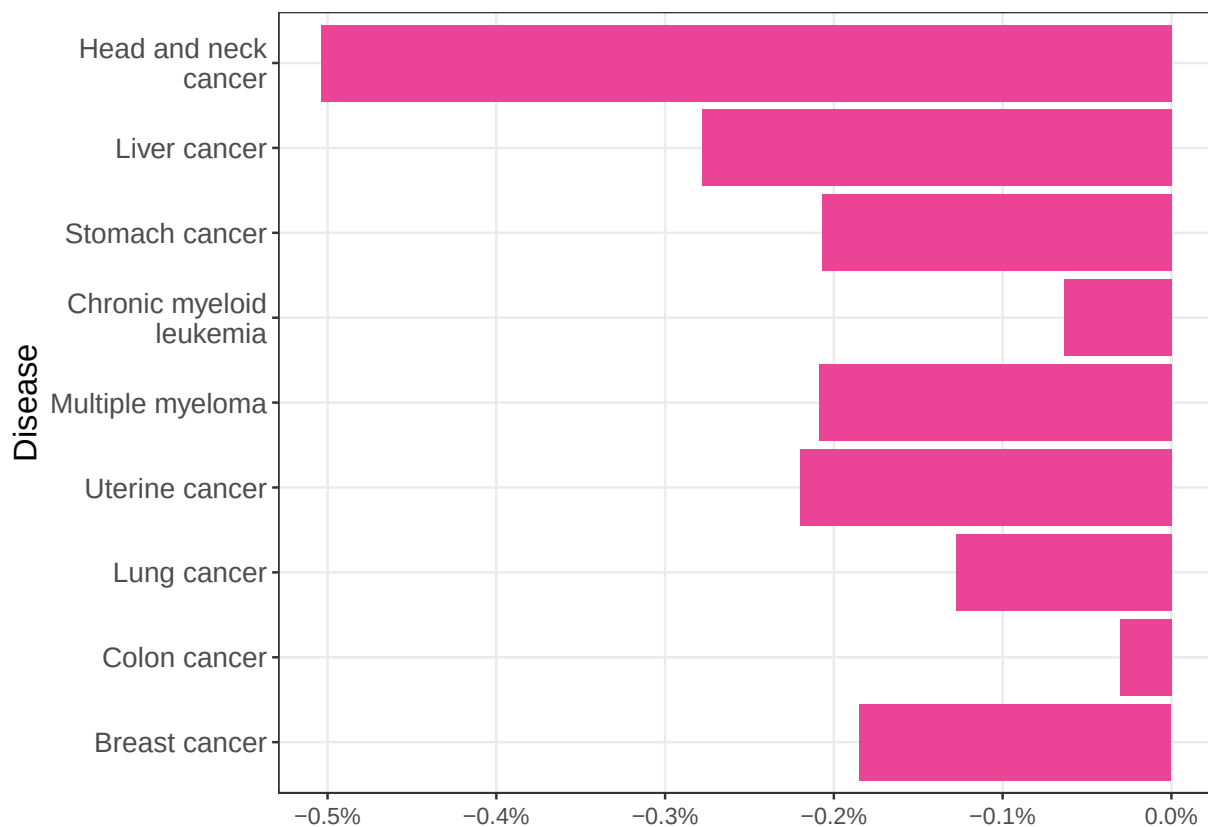
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.15%                  | -357                                                              |
| Breast cancer                           | 0.18%                   | 60                                                                |
| All cancers                             | 0.15%                   | 377                                                               |
| Colon cancer                            | 0.03%                   | 13                                                                |
| Chronic myeloid leukemia                | 0.06%                   | 1                                                                 |
| Diabetes type 2                         | 0.92%                   | 453                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.50%                   | 20                                                                |
| Ischemic heart disease                  | 0.27%                   | 1,012                                                             |
| Liver cancer                            | 0.28%                   | 47                                                                |
| Multiple myeloma                        | 0.21%                   | 36                                                                |
| Stomach cancer                          | 0.21%                   | 44                                                                |
| Stroke                                  | 0.18%                   | 315                                                               |
| Lung cancer                             | 0.13%                   | 139                                                               |
| Uterine cancer                          | 0.22%                   | 16                                                                |

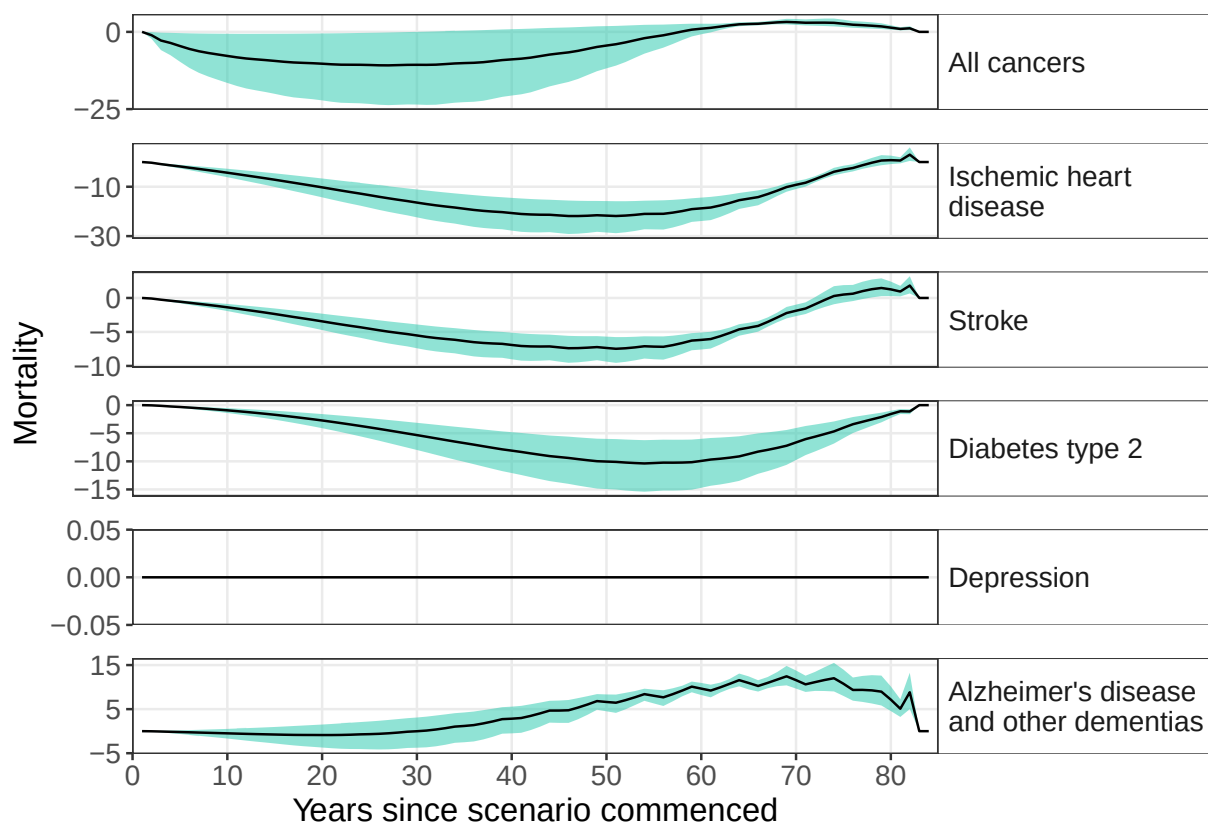
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

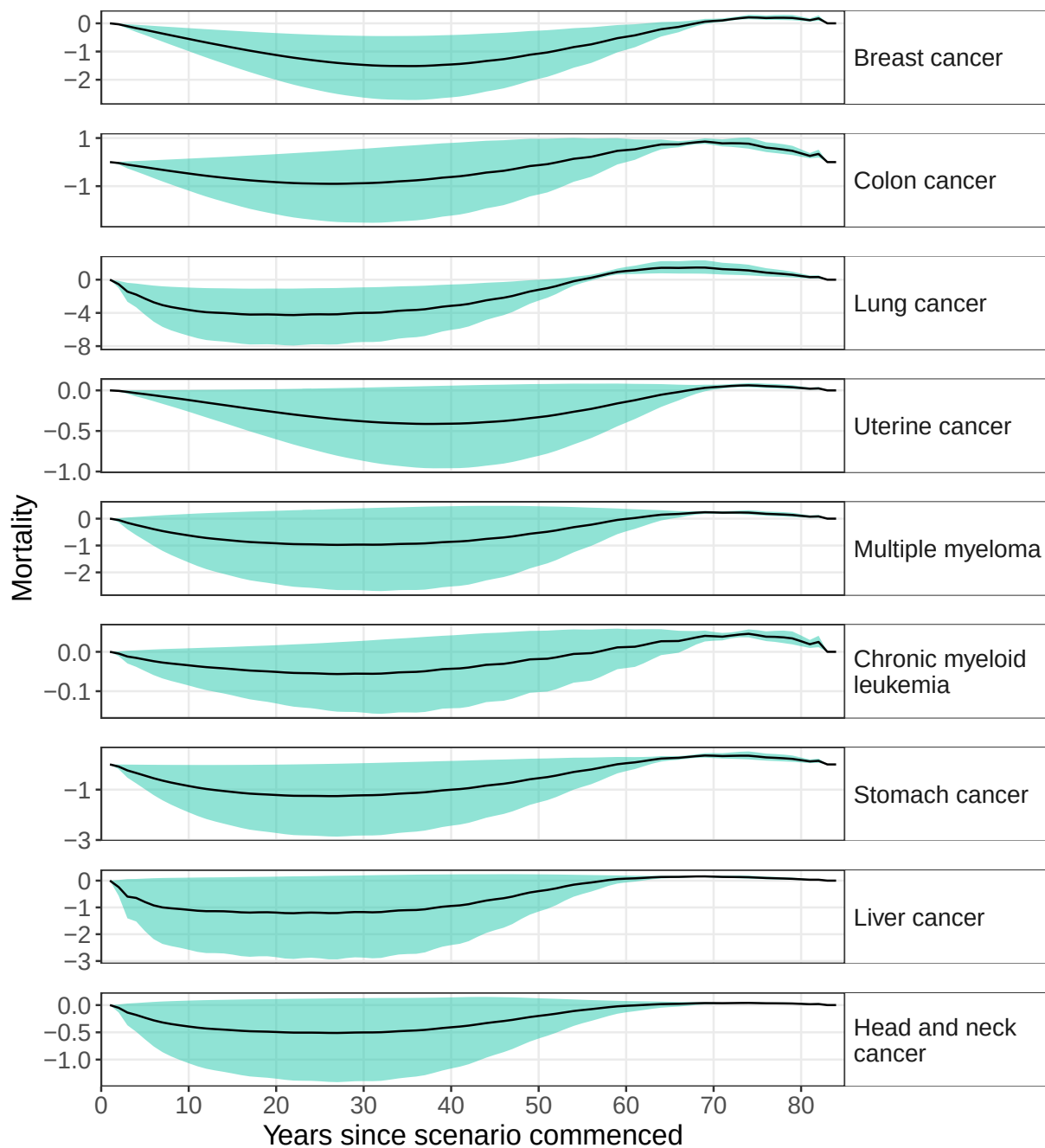


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

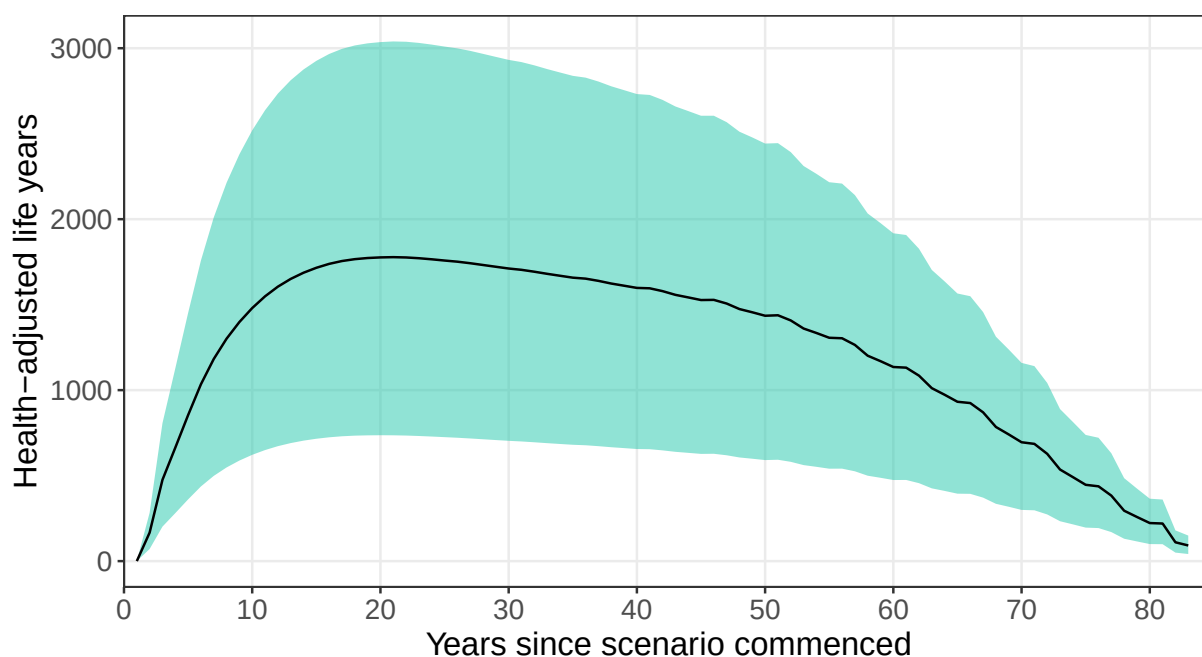
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **102,552** HALYs for the scenario population, which is **51** HALYs per 1,000 members of the population.



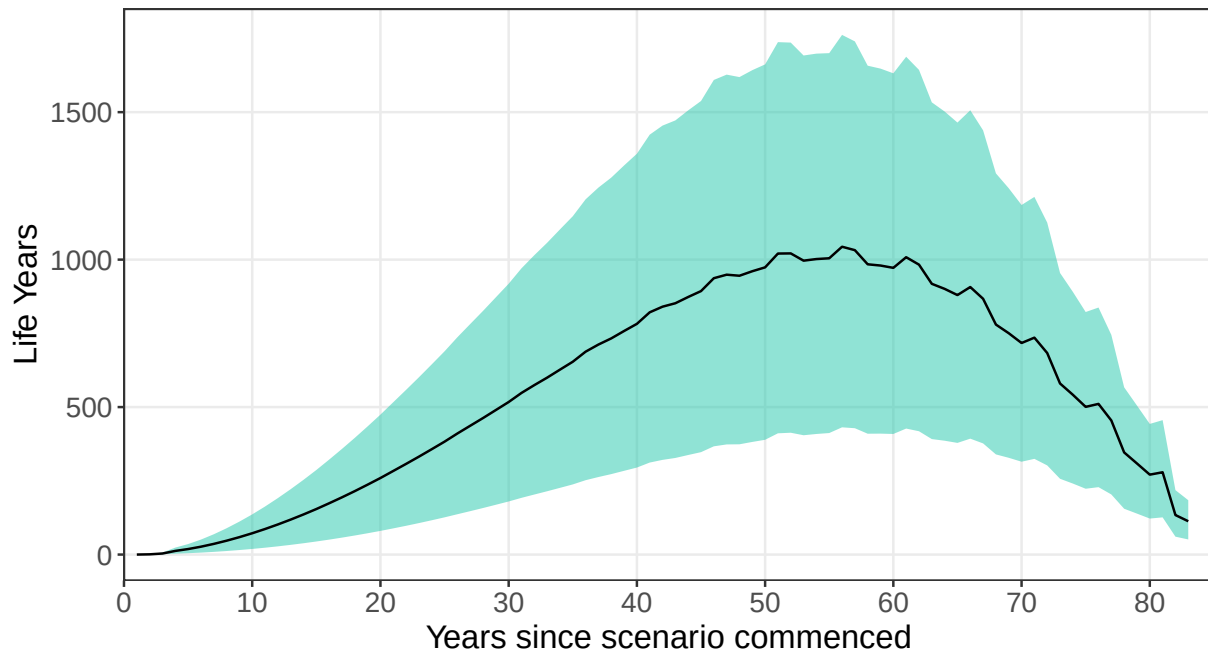
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **46,351** Life Years for the scenario population, which is **23** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **51** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **4,318,397** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **2,718,972** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **1,866,171** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$1,039          | \$750            | \$534            |
| Breast cancer                           | \$40,858         | \$24,875         | \$16,179         |
| All cancers                             | \$72,678         | \$45,965         | \$30,786         |
| Colon cancer                            | \$11,880         | \$8,172          | \$5,712          |
| Chronic myeloid leukemia                | \$2,785          | \$1,695          | \$1,107          |
| Diabetes type 2                         | \$27,451         | \$15,935         | \$10,035         |
| Depression                              | \$139,915        | \$101,720        | \$76,609         |
| Head and neck cancer                    | \$553            | \$378            | \$272            |
| Ischemic heart disease                  | \$44,377         | \$25,905         | \$16,373         |
| Liver cancer                            | \$541            | \$375            | \$274            |
| Multiple myeloma                        | \$7,170          | \$4,640          | \$3,188          |
| Stomach cancer                          | \$2,921          | \$1,903          | \$1,316          |
| Stroke                                  | \$5,657          | \$3,307          | \$2,090          |
| Lung cancer                             | \$4,005          | \$2,723          | \$1,951          |
| Uterine cancer                          | \$1,957          | \$1,205          | \$792            |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

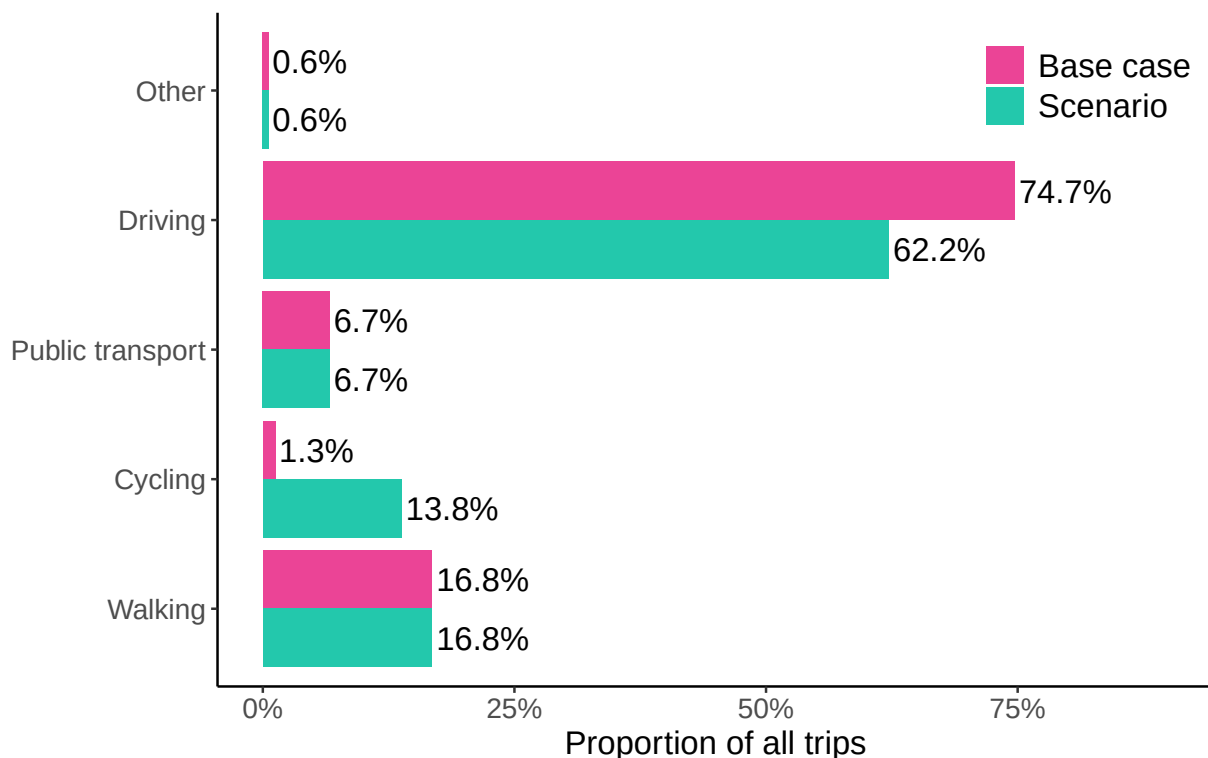
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
3. Ananthapavan, J., Moodie, M., Milat, A.J., & Carter, R. (2021). *Systematic review to update 'value of a statistical life' estimates for Australia*. International journal of environmental research and public health, 2021. 18(11): p. 6168.
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5. Abelson, P. (2008). *Establishing a monetary value for lives saved: issues and controversies*. Canberra: Office of Best Practice Regulation, Department of Finance and Deregulation.
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7. Haacker, M., Hallett, T.B. & Atun, R. (2020). *On discount rates for economic evaluations in global health*. Health Policy and Planning, 2020. 35(1): p. 107-114.
8. Australian Institute of Health and Welfare (2023). Technical Notes: Estimating Spending per prevalent case of disease. Health system spending per case of disease and for certain risk factors, Estimating the spending per prevalent case of disease - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
11. Zapata-Diomedí, B., Boulangé, C., Giles-Corti, B., Phelan, K., Washington, S., Veerman, L.J., & Gunn, L. (2019). Physical activity-related health and economic benefits of building walkable neighbourhoods: A modelled comparison between brownfield and greenfield developments. International Journal of Behavioural Nutrition and Physical Activity.
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## Scenario: replacing car trips under 10km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 10km for work related or education purposes with cycling trips for all adults of all ages.

This implies that the selected scenario results in a mode shift in cycling from 1.3% to 13.8% and from 74.7% to 62.2% for car trips taken as either a driver or passenger.

Increases in cycling translate into a shift from 47.9% to 58.0% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

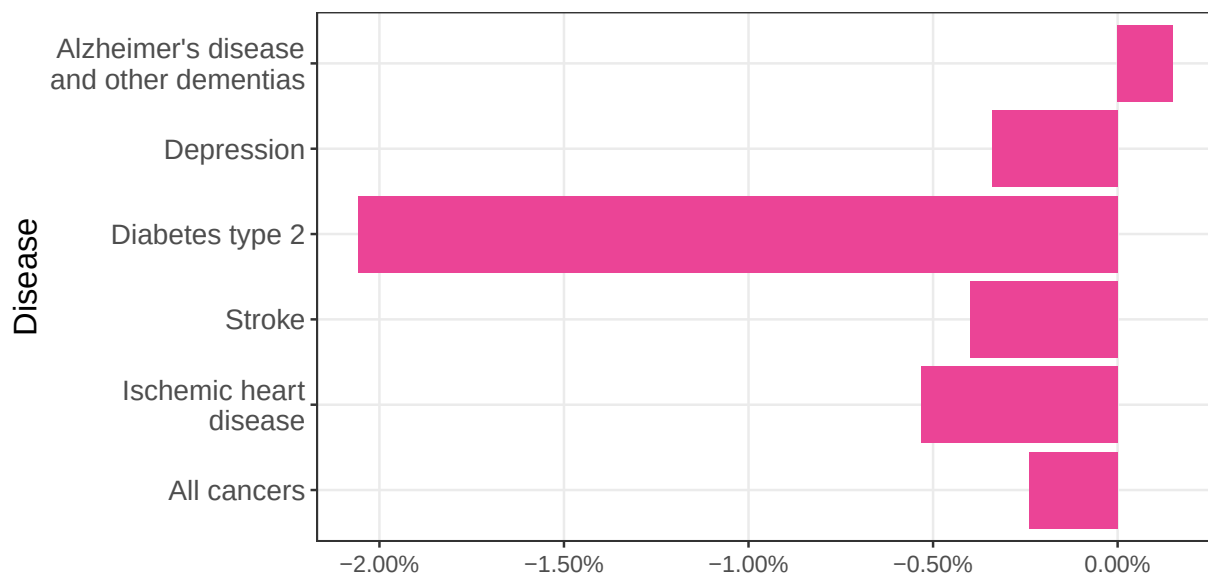
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

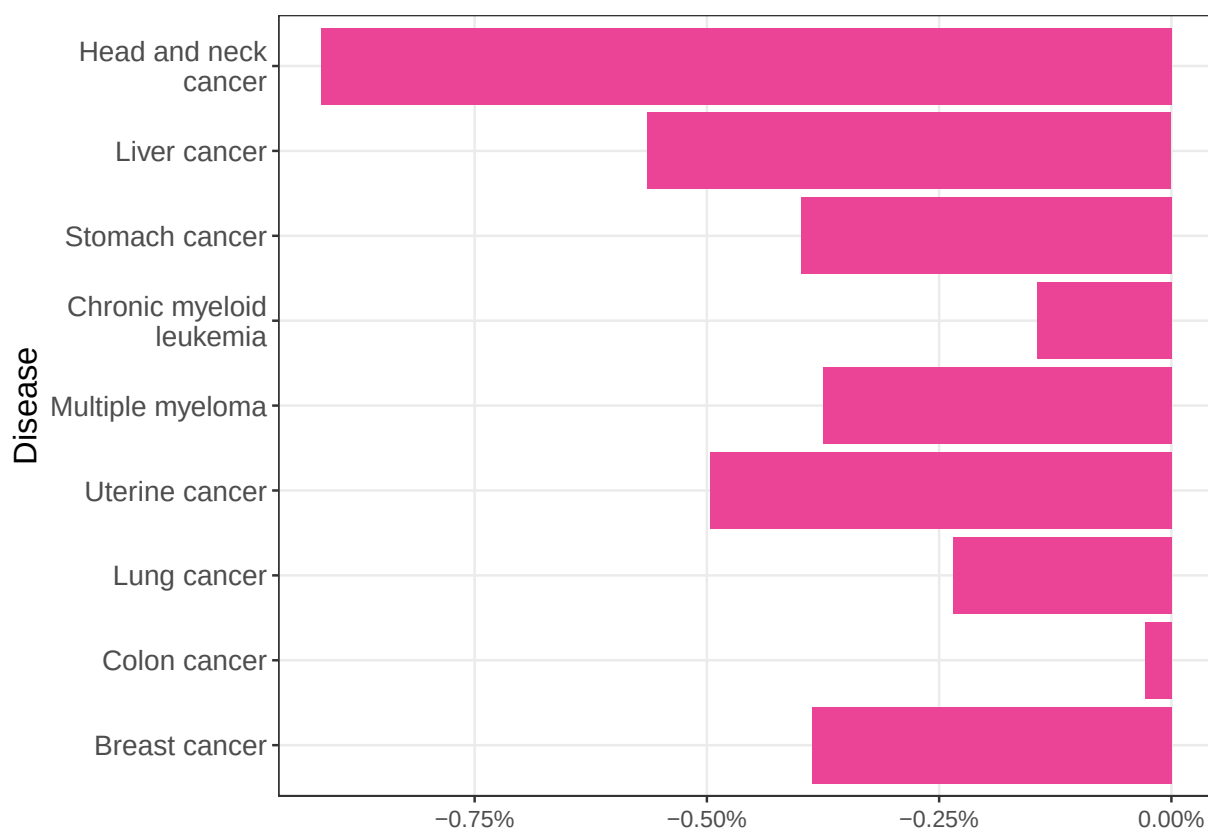
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.15%                             | -958                                                                        |
| Breast cancer                           | 0.39%                              | 283                                                                         |
| All cancers                             | 0.24%                              | 943                                                                         |
| Colon cancer                            | 0.03%                              | 37                                                                          |
| Chronic myeloid leukemia                | 0.14%                              | 4                                                                           |
| Diabetes type 2                         | 2.06%                              | 6,796                                                                       |
| Depression                              | 0.34%                              | 5,468                                                                       |
| Head and neck cancer                    | 0.92%                              | 40                                                                          |
| Ischemic heart disease                  | 0.53%                              | 4,677                                                                       |
| Liver cancer                            | 0.56%                              | 99                                                                          |
| Multiple myeloma                        | 0.37%                              | 73                                                                          |
| Stomach cancer                          | 0.40%                              | 95                                                                          |
| Stroke                                  | 0.40%                              | 1,040                                                                       |
| Lung cancer                             | 0.24%                              | 271                                                                         |
| Uterine cancer                          | 0.50%                              | 42                                                                          |

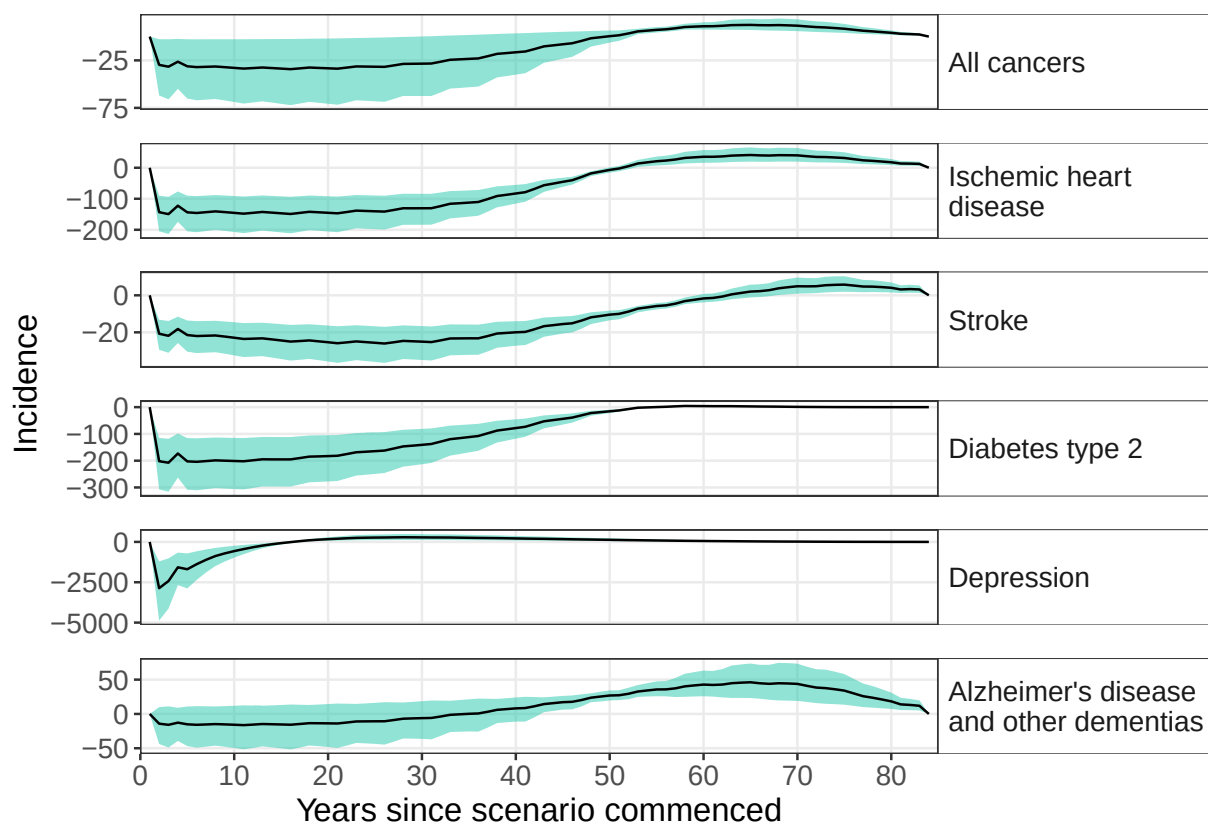
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



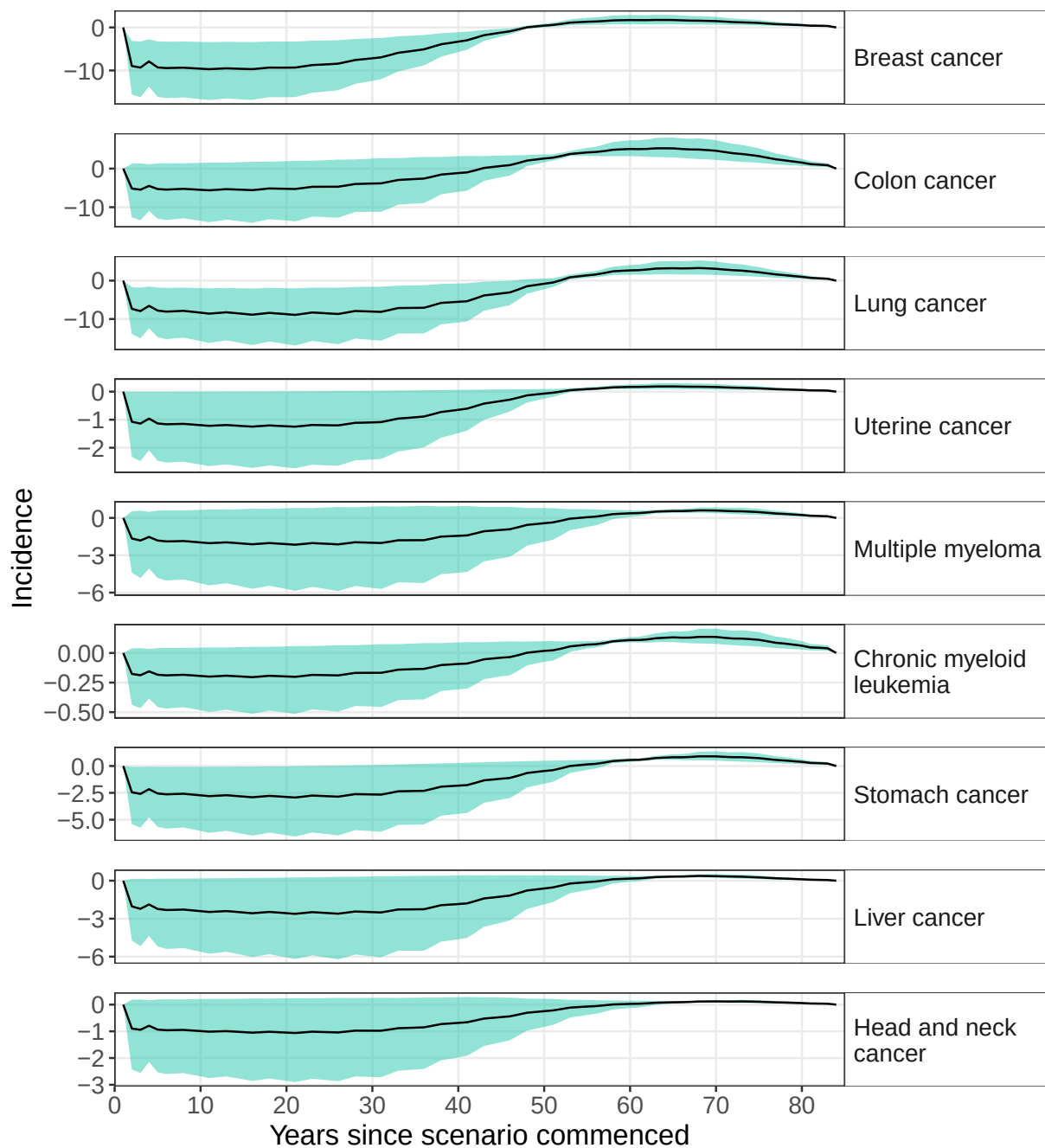
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

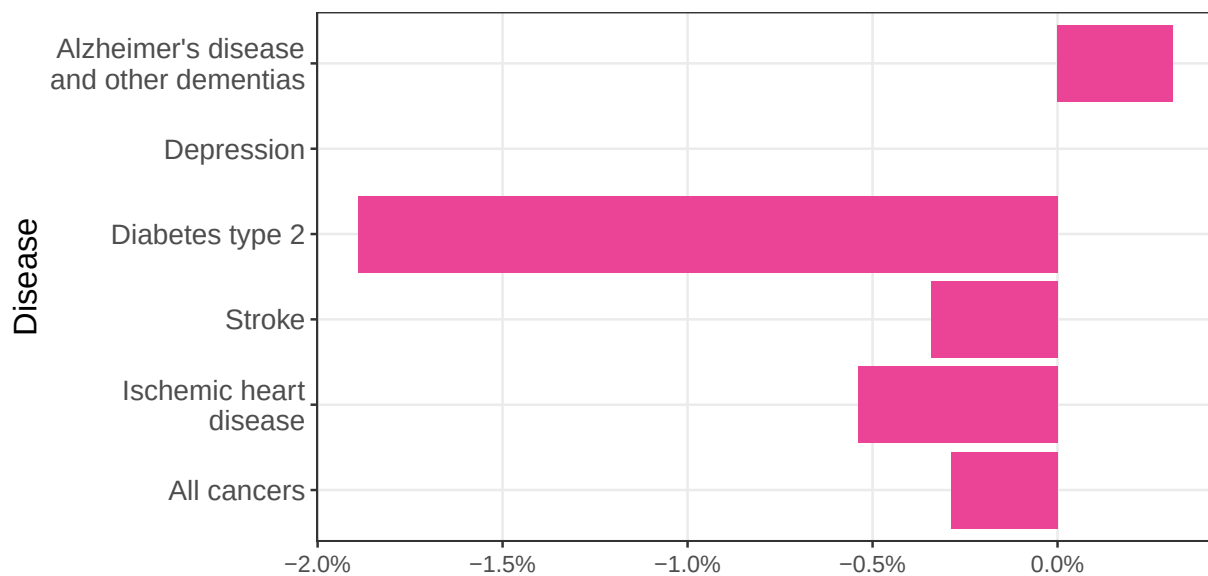
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

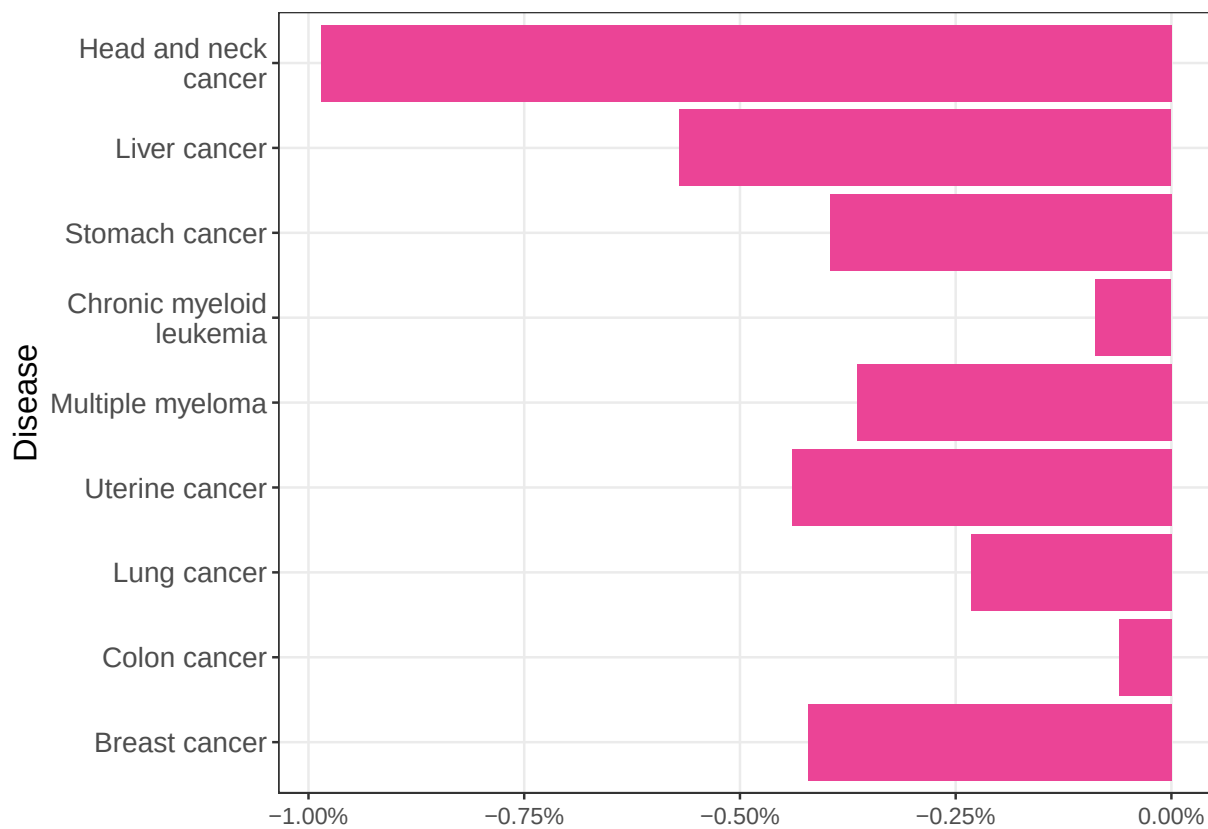
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.31%                  | -748                                                              |
| Breast cancer                           | 0.42%                   | 136                                                               |
| All cancers                             | 0.29%                   | 732                                                               |
| Colon cancer                            | 0.06%                   | 26                                                                |
| Chronic myeloid leukemia                | 0.09%                   | 2                                                                 |
| Diabetes type 2                         | 1.89%                   | 935                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.99%                   | 39                                                                |
| Ischemic heart disease                  | 0.54%                   | 2,023                                                             |
| Liver cancer                            | 0.57%                   | 97                                                                |
| Multiple myeloma                        | 0.36%                   | 64                                                                |
| Stomach cancer                          | 0.40%                   | 84                                                                |
| Stroke                                  | 0.34%                   | 599                                                               |
| Lung cancer                             | 0.23%                   | 253                                                               |
| Uterine cancer                          | 0.44%                   | 32                                                                |

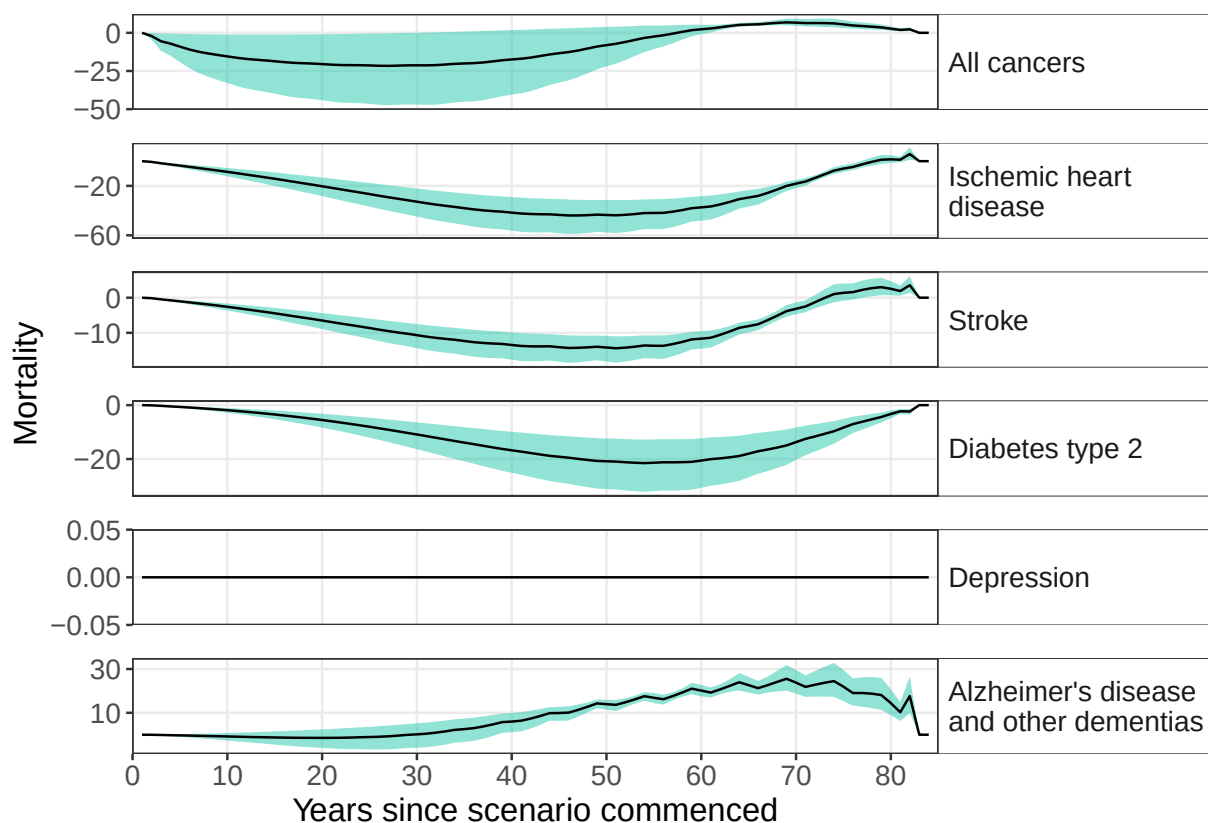
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

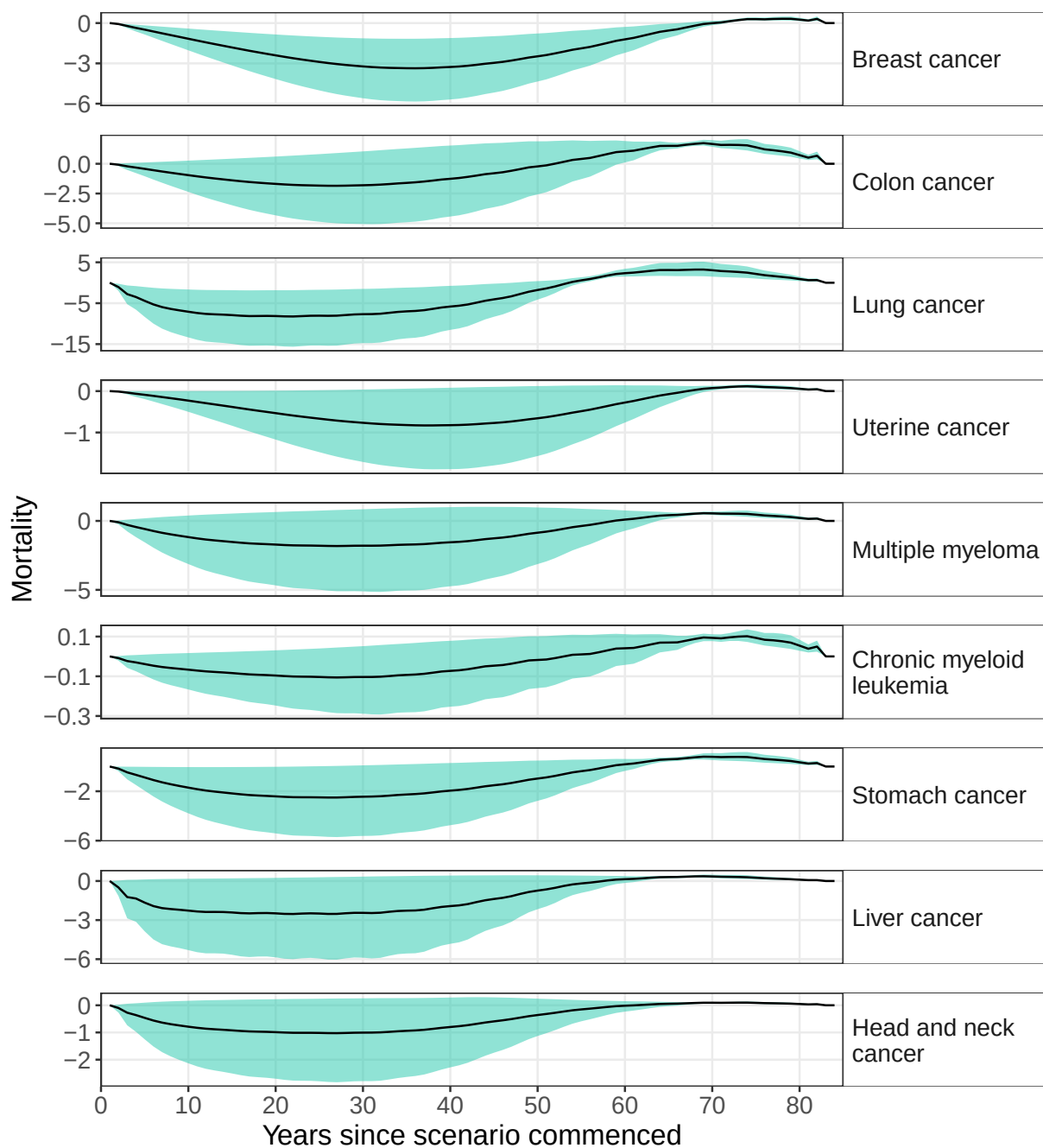


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

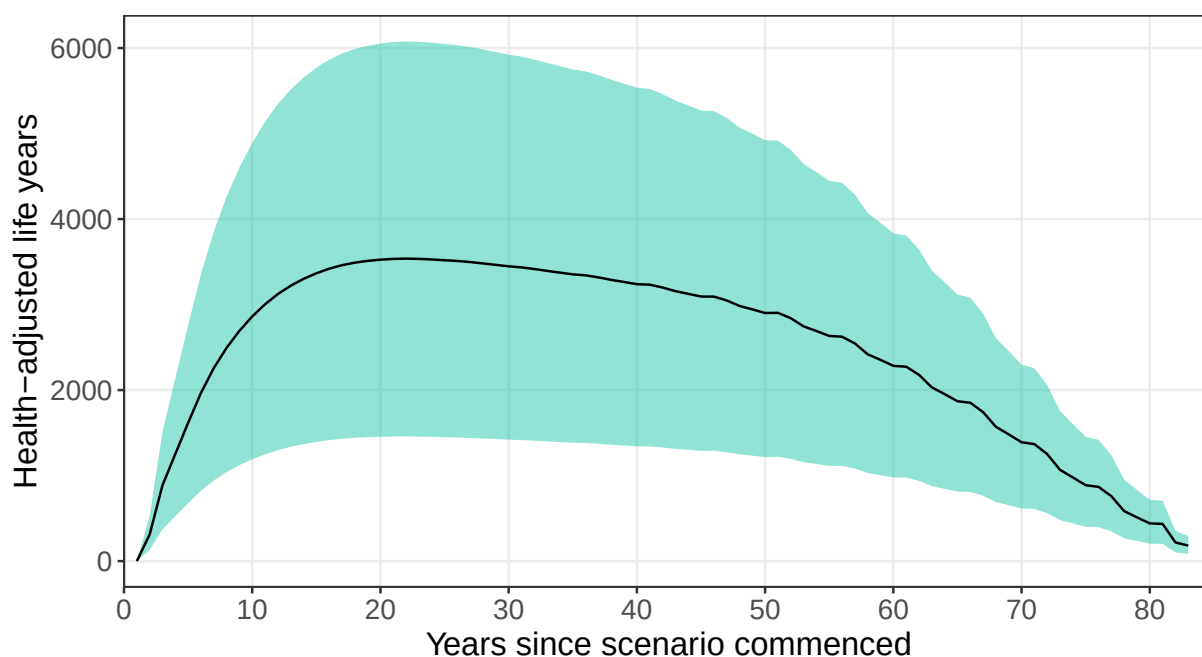
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **204,618** HALYs for the scenario population, which is **102** HALYs per 1,000 members of the population.



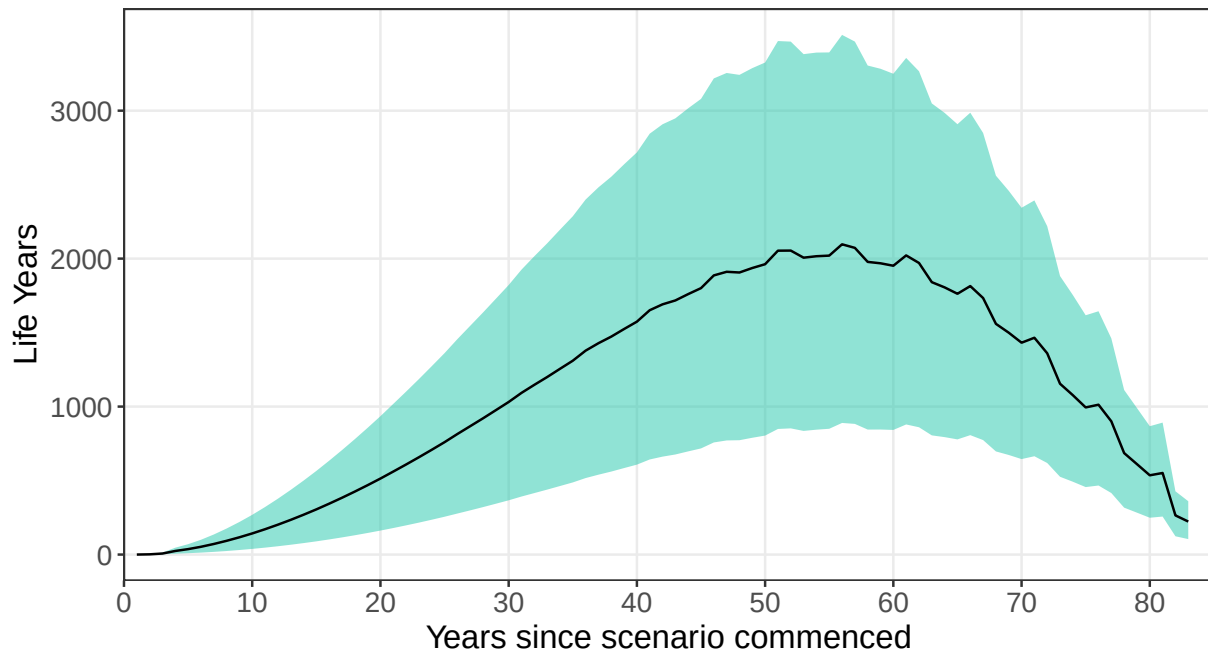
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **92,773** Life Years for the scenario population, which is **46** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **102** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **8,550,235** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **5,353,639** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **3,655,495** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$1,914          | \$1,413          | \$1,015          |
| Breast cancer                           | \$88,805         | \$53,767         | \$34,825         |
| All cancers                             | \$151,320        | \$95,349         | \$63,676         |
| Colon cancer                            | \$24,089         | \$16,588         | \$11,597         |
| Chronic myeloid leukemia                | \$5,010          | \$3,068          | \$2,010          |
| Diabetes type 2                         | \$57,097         | \$33,138         | \$20,859         |
| Depression                              | \$271,164        | \$196,247        | \$147,215        |
| Head and neck cancer                    | \$1,100          | \$755            | \$543            |
| Ischemic heart disease                  | \$92,041         | \$53,770         | \$33,985         |
| Liver cancer                            | \$1,146          | \$797            | \$583            |
| Multiple myeloma                        | \$13,572         | \$8,816          | \$6,072          |
| Stomach cancer                          | \$5,851          | \$3,822          | \$2,648          |
| Stroke                                  | \$11,047         | \$6,459          | \$4,078          |
| Lung cancer                             | \$7,791          | \$5,321          | \$3,821          |
| Uterine cancer                          | \$3,932          | \$2,420          | \$1,588          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

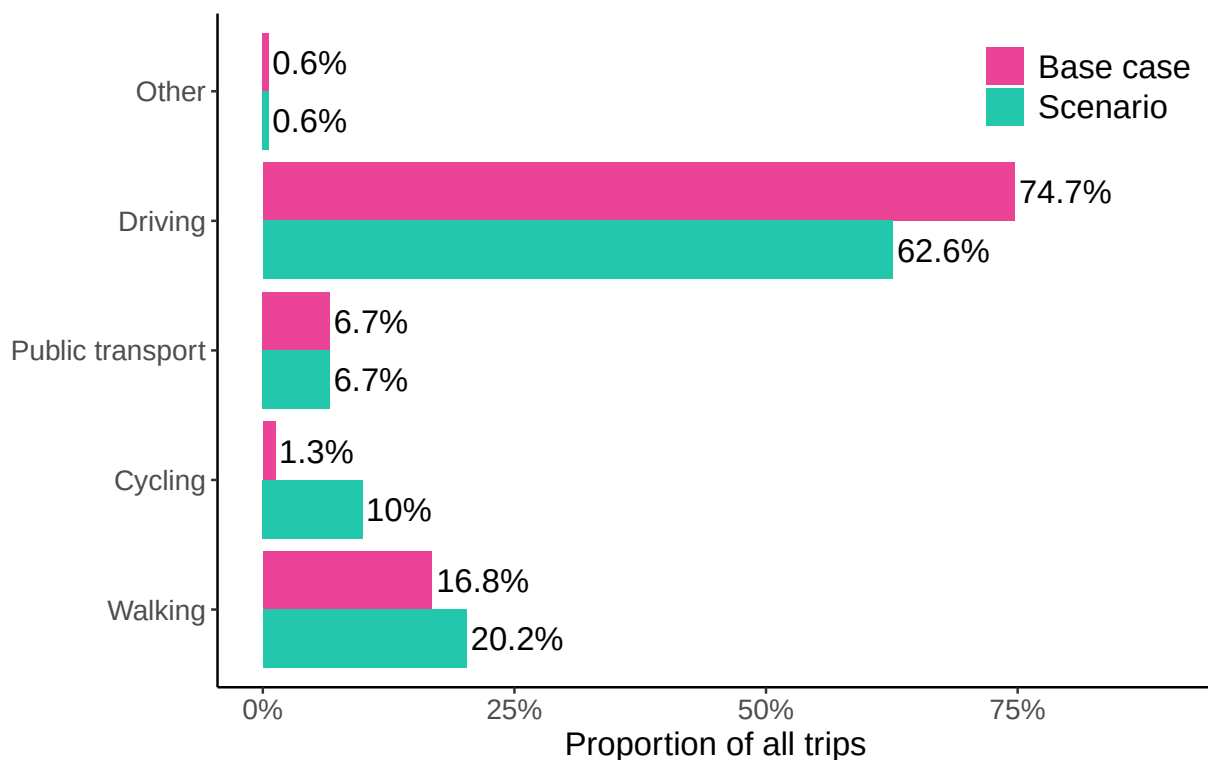
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
3. Ananthapavan, J., Moodie, M., Milat, A.J., & Carter, R. (2021). *Systematic review to update 'value of a statistical life' estimates for Australia*. International journal of environmental research and public health, 2021. 18(11): p. 6168.
4. Terrill, M. & Batrouney, H. (2018). *Unfreezing discount rates: Transport infrastructure for tomorrow*. Grattan Institute.
5. Abelson, P. (2008). *Establishing a monetary value for lives saved: issues and controversies*. Canberra: Office of Best Practice Regulation, Department of Finance and Deregulation.
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7. Haacker, M., Hallett, T.B. & Atun, R. (2020). *On discount rates for economic evaluations in global health*. Health Policy and Planning, 2020. 35(1): p. 107-114.
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9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
11. Zapata-Diomedí, B., Boulangé, C., Giles-Corti, B., Phelan, K., Washington, S., Veerman, L.J., & Gunn, L. (2019). Physical activity-related health and economic benefits of building walkable neighbourhoods: A modelled comparison between brownfield and greenfield developments. International Journal of Behavioural Nutrition and Physical Activity.
12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577-1590.

## Scenario: replacing car trips under 1km with walking, and car trips between 1 and 2km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 1km with walking and replacing car trips between 1km and 2km with cycling for leisure, shopping, work, education or other purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 20.2%; cycling from 1.3% to 10.0%; and, from 74.7% to 62.6% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 55.3% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

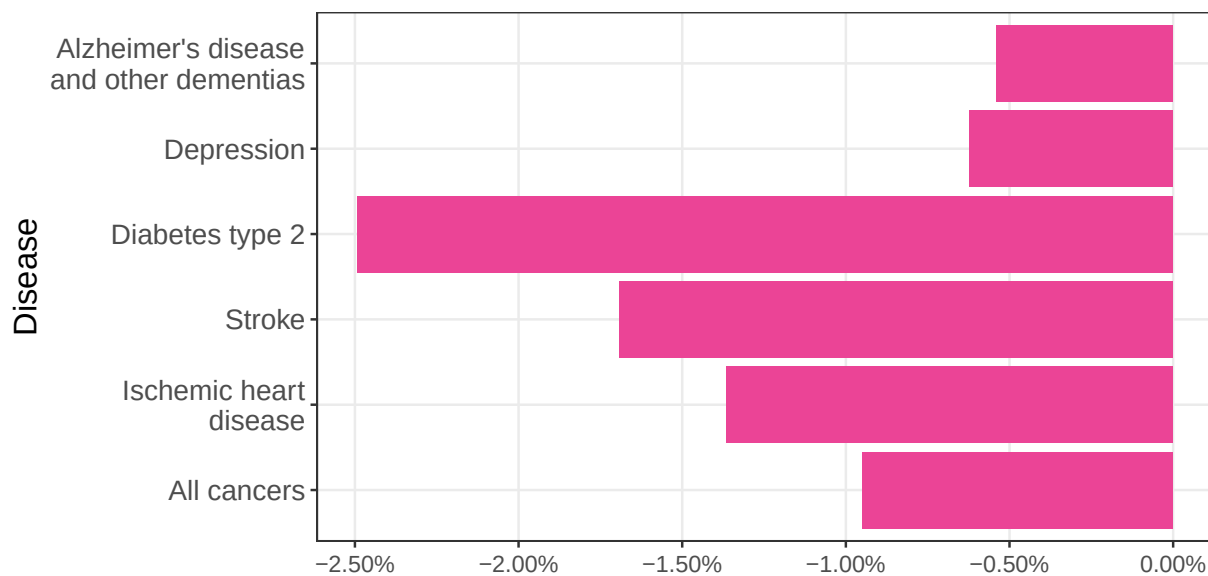
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

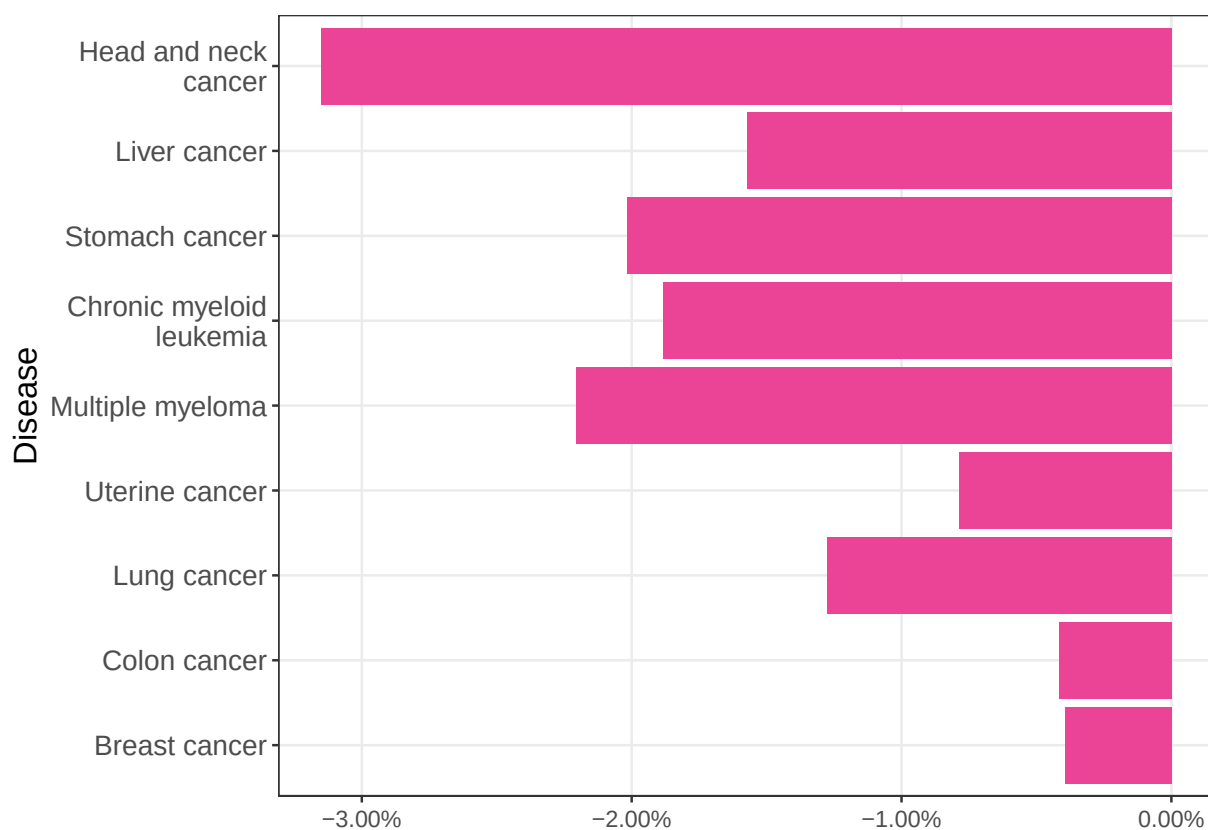
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.54%                              | 3,437                                                                       |
| Breast cancer                           | 0.40%                              | 290                                                                         |
| All cancers                             | 0.95%                              | 3,740                                                                       |
| Colon cancer                            | 0.42%                              | 539                                                                         |
| Chronic myeloid leukemia                | 1.88%                              | 51                                                                          |
| Diabetes type 2                         | 2.49%                              | 8,224                                                                       |
| Depression                              | 0.62%                              | 9,964                                                                       |
| Head and neck cancer                    | 3.15%                              | 139                                                                         |
| Ischemic heart disease                  | 1.36%                              | 12,012                                                                      |
| Liver cancer                            | 1.57%                              | 277                                                                         |
| Multiple myeloma                        | 2.21%                              | 427                                                                         |
| Stomach cancer                          | 2.02%                              | 482                                                                         |
| Stroke                                  | 1.69%                              | 4,400                                                                       |
| Lung cancer                             | 1.28%                              | 1,467                                                                       |
| Uterine cancer                          | 0.79%                              | 67                                                                          |

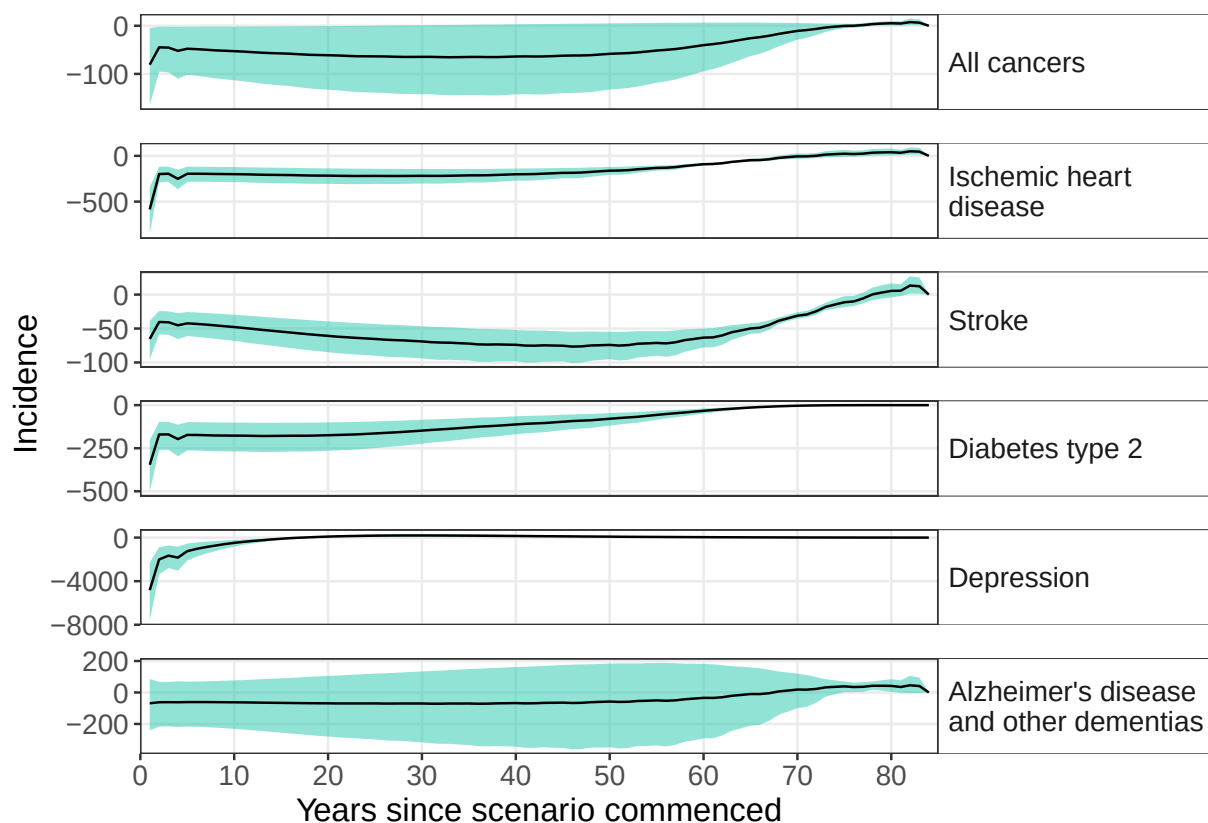
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



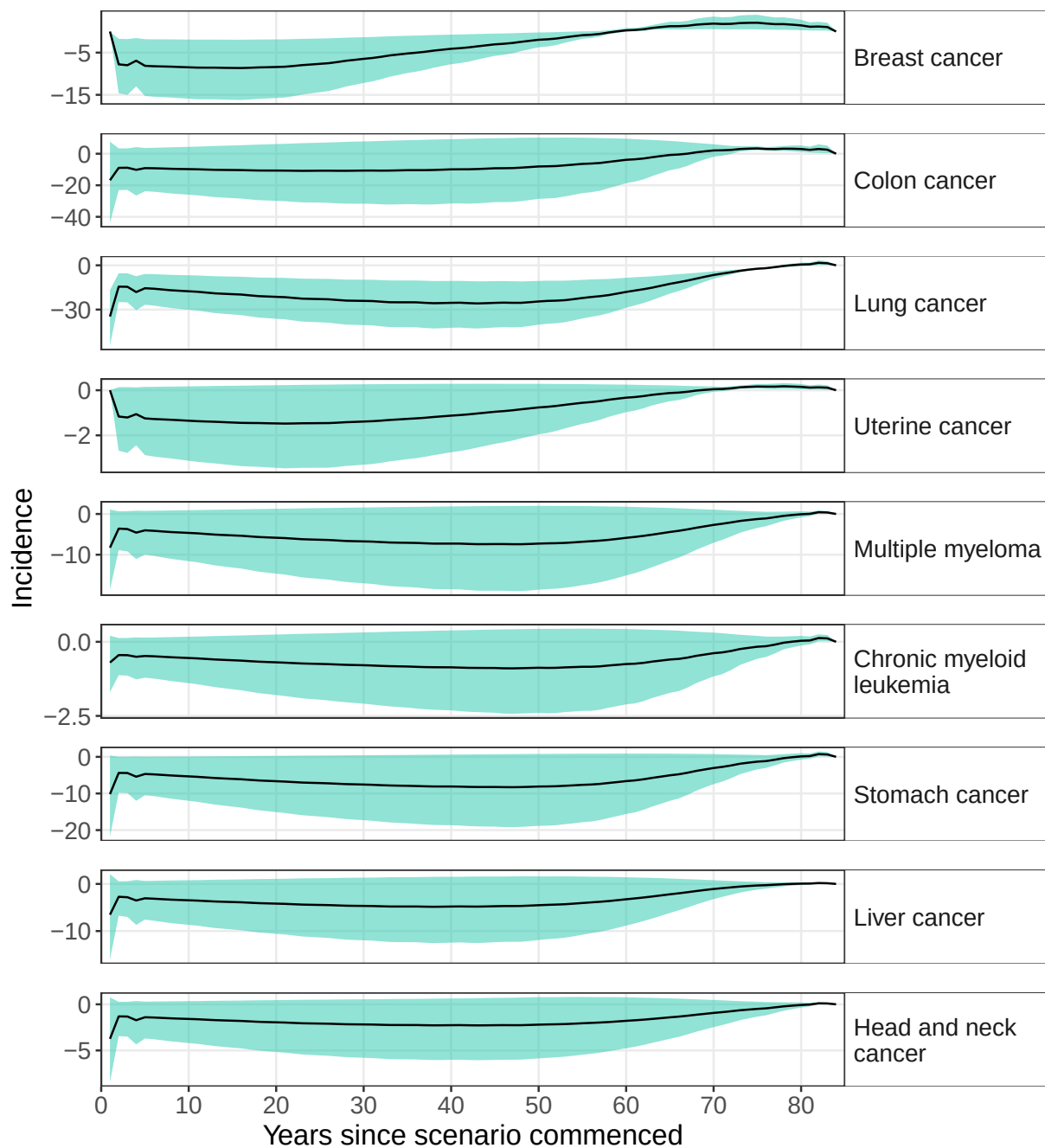
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

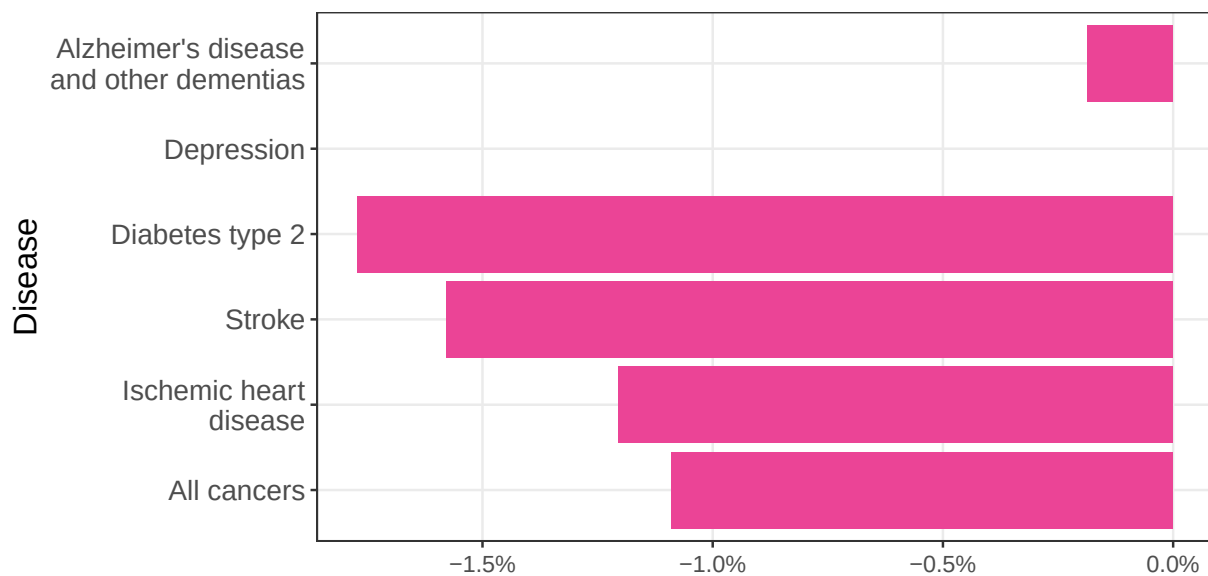
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

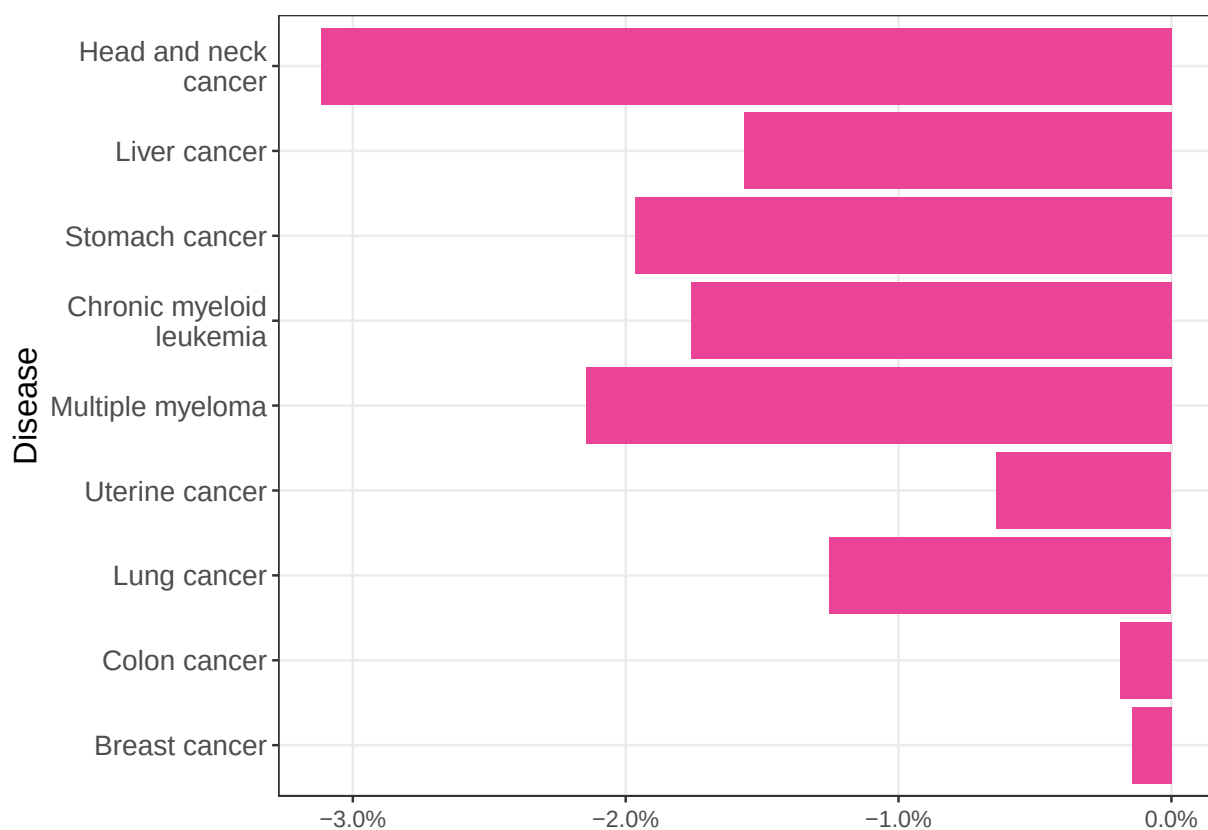
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.19%                   | 444                                                               |
| Breast cancer                           | 0.14%                   | 47                                                                |
| All cancers                             | 1.09%                   | 2,759                                                             |
| Colon cancer                            | 0.19%                   | 81                                                                |
| Chronic myeloid leukemia                | 1.76%                   | 36                                                                |
| Diabetes type 2                         | 1.77%                   | 875                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 3.12%                   | 124                                                               |
| Ischemic heart disease                  | 1.20%                   | 4,519                                                             |
| Liver cancer                            | 1.57%                   | 267                                                               |
| Multiple myeloma                        | 2.15%                   | 374                                                               |
| Stomach cancer                          | 1.97%                   | 416                                                               |
| Stroke                                  | 1.58%                   | 2,759                                                             |
| Lung cancer                             | 1.25%                   | 1,367                                                             |
| Uterine cancer                          | 0.64%                   | 46                                                                |

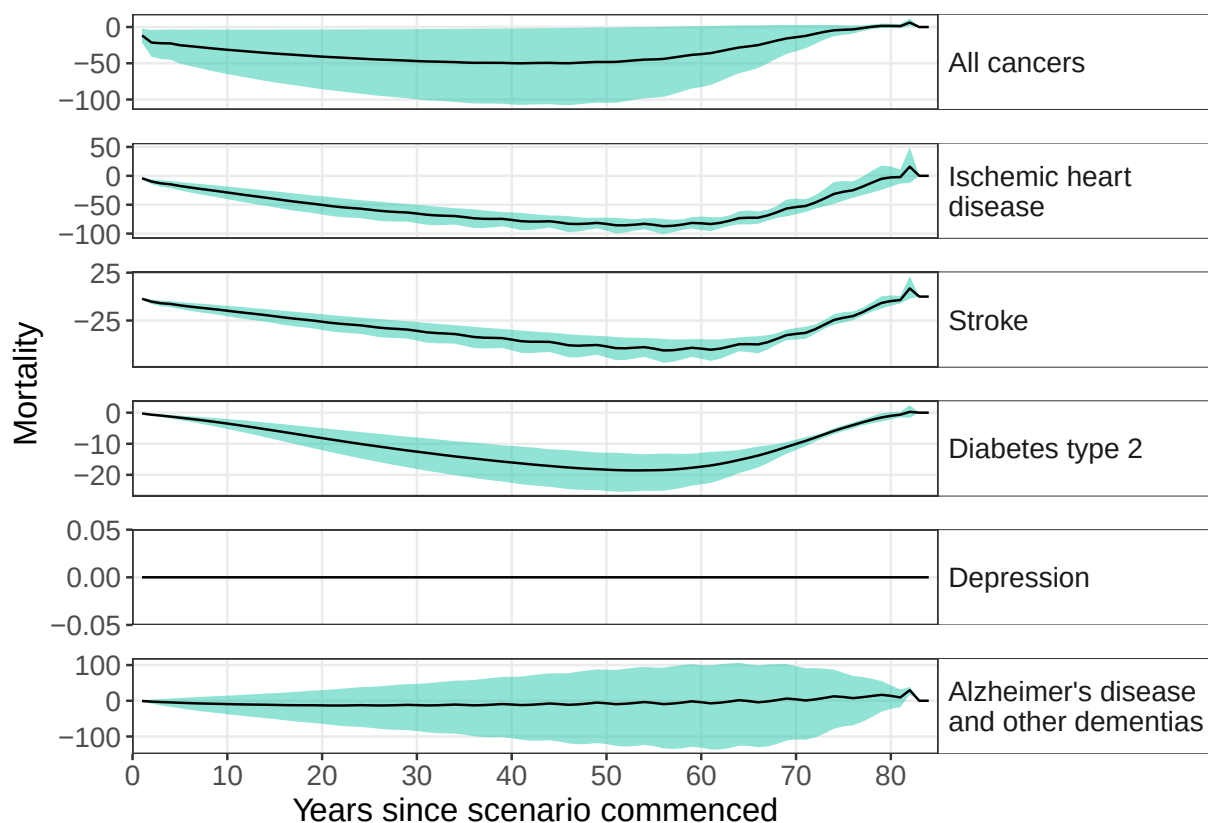
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

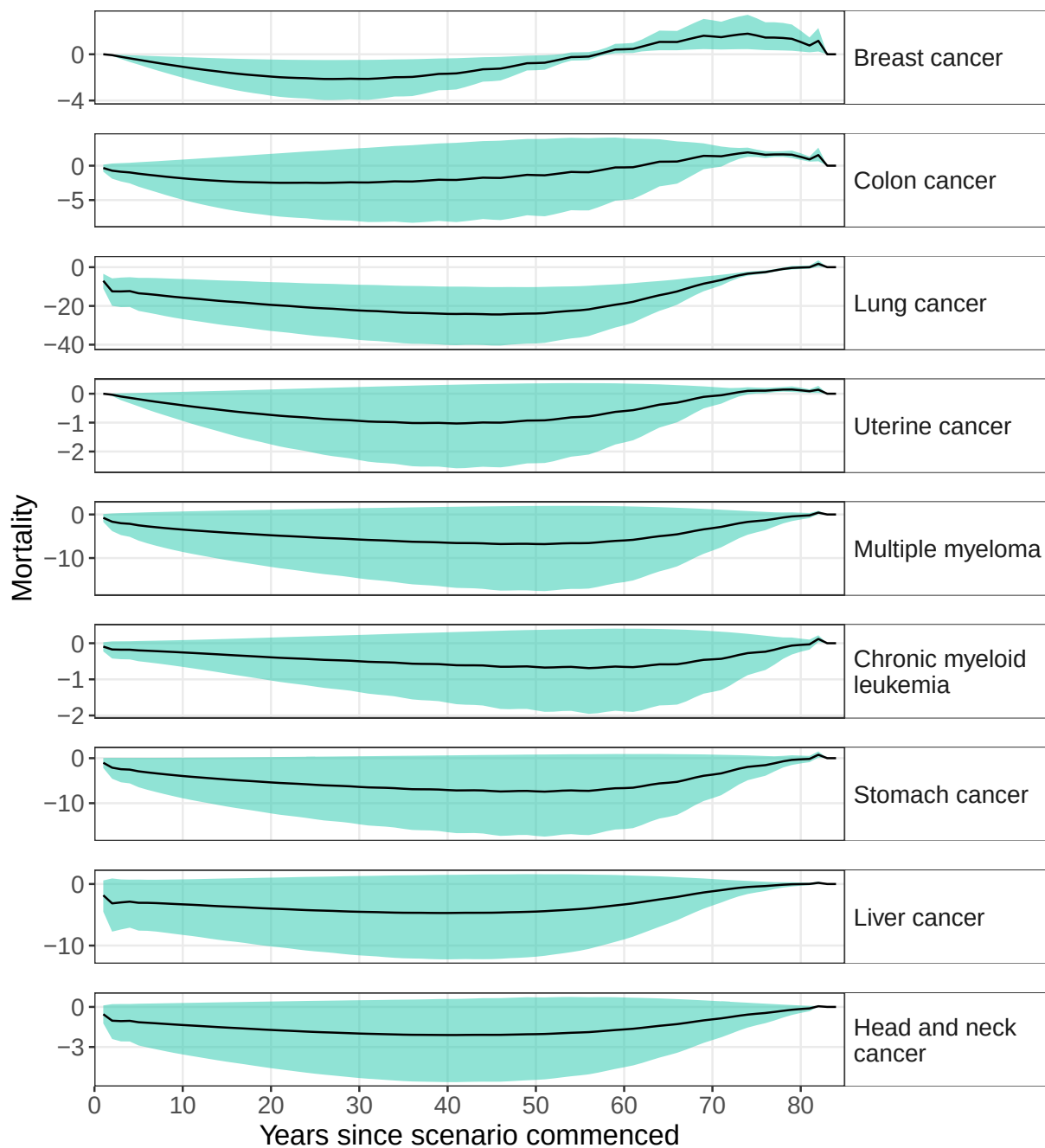


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

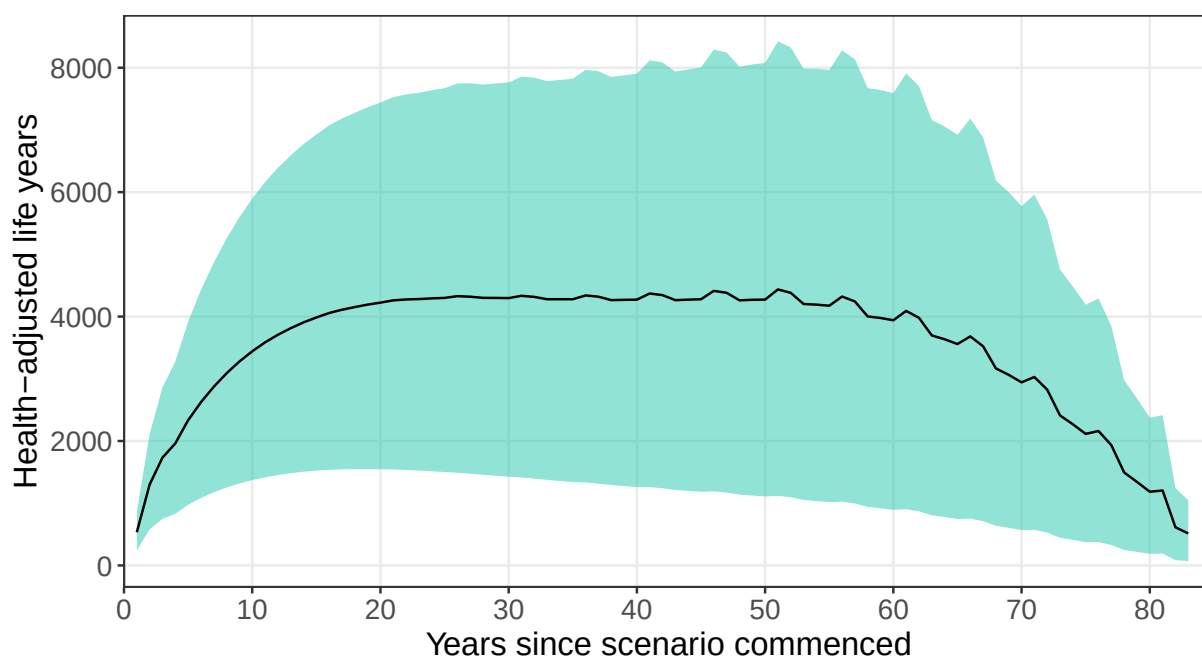
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **294,976** HALYs for the scenario population, which is **146** HALYs per 1,000 members of the population.



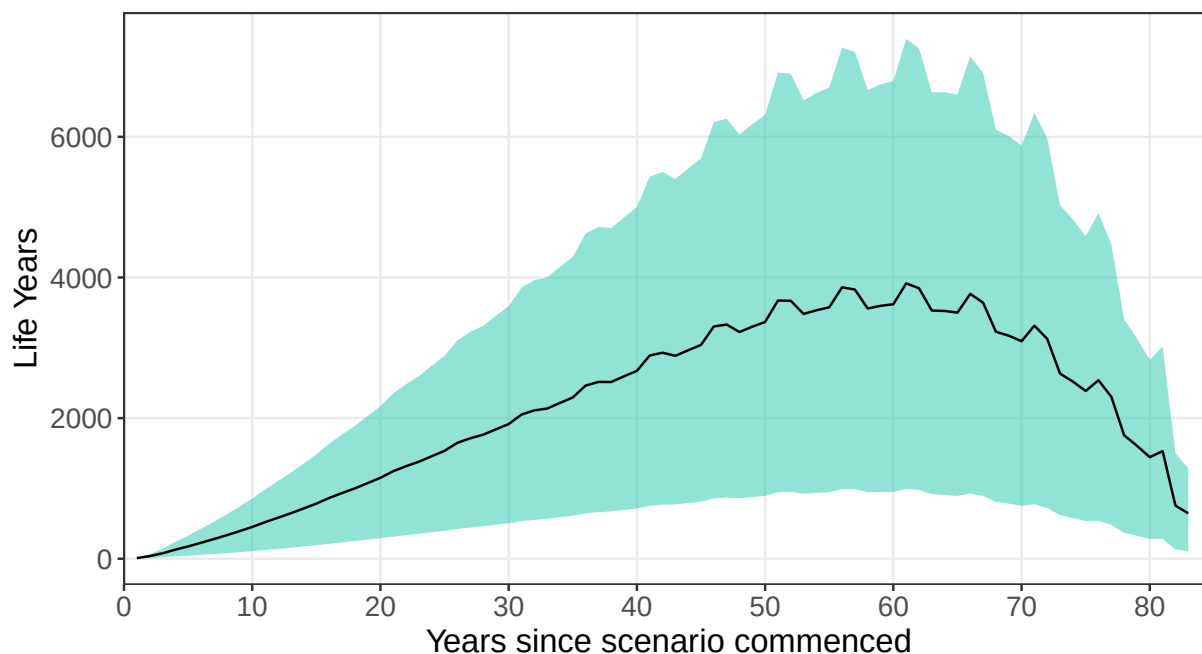
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **181,924** Life Years for the scenario population, which is **90** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated<sup>[a]</sup> by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year<sup>[b]</sup>. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting<sup>[c]</sup> to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).<sup>[2]</sup> Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations<sup>[3, 4]</sup> and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **146** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes<sup>[d]</sup>.

The HALYs gained in this scenario have a statistical value of:

- **11,429,791** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **7,016,152** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **4,776,488** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year<sup>[b]</sup>.



## b. What is the Value of a Statistical Life and Value of a Statistical Life Year?

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## c. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## d. Application in advocacy and reporting

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$8,806          | \$5,324          | \$3,524          |
| Breast cancer                           | \$61,672         | \$38,995         | \$26,015         |
| All cancers                             | \$174,737        | \$111,551        | \$76,225         |
| Colon cancer                            | \$38,760         | \$26,131         | \$18,457         |
| Chronic myeloid leukemia                | \$11,659         | \$6,840          | \$4,411          |
| Diabetes type 2                         | \$50,861         | \$30,347         | \$19,709         |
| Depression                              | \$281,088        | \$212,107        | \$164,192        |
| Head and neck cancer                    | \$1,905          | \$1,232          | \$869            |
| Ischemic heart disease                  | \$114,573        | \$69,204         | \$45,858         |
| Liver cancer                            | \$1,593          | \$1,061          | \$769            |
| Multiple myeloma                        | \$28,371         | \$17,765         | \$12,166         |
| Stomach cancer                          | \$10,805         | \$6,766          | \$4,641          |
| Stroke                                  | \$17,486         | \$10,185         | \$6,527          |
| Lung cancer                             | \$16,253         | \$10,501         | \$7,404          |
| Uterine cancer                          | \$3,738          | \$2,285          | \$1,498          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

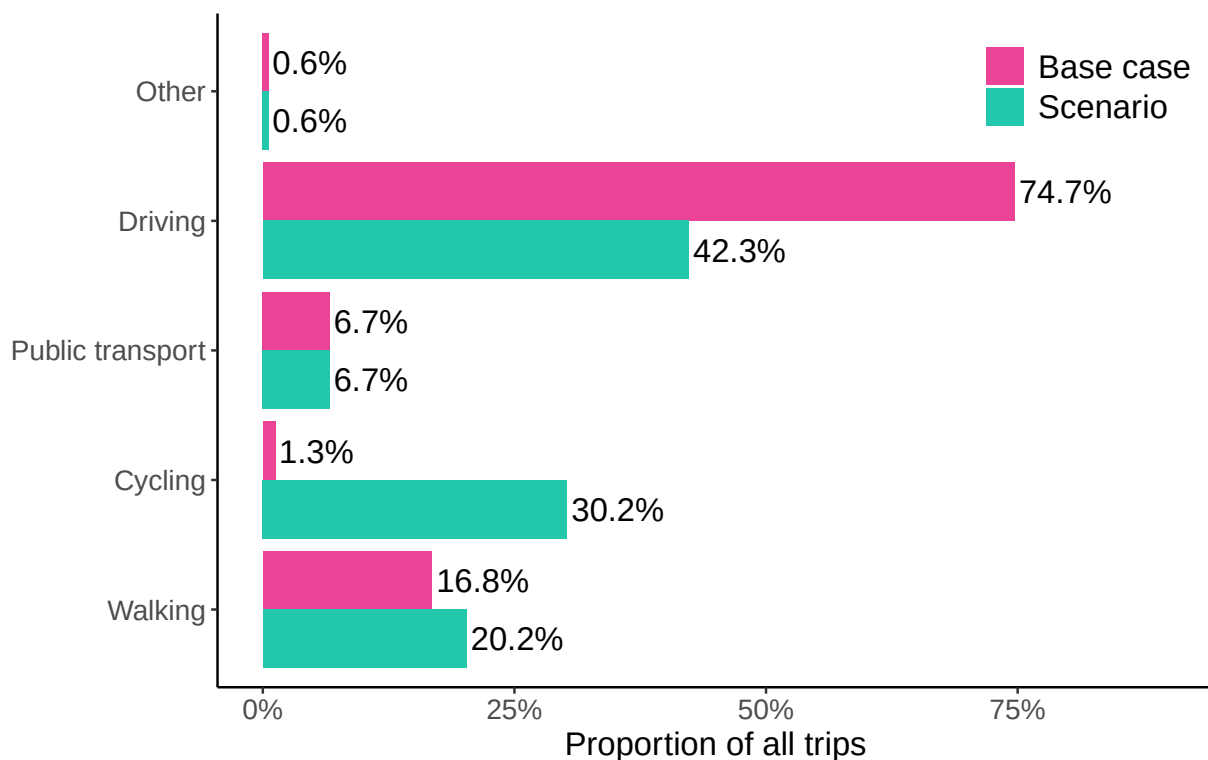
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115–134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745–758.
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12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577–1590.

## Scenario: replacing car trips under 1km with walking, and car trips between 1 and 5km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 1km with walking and replacing car trips between 1km and 5km with cycling for leisure, shopping, work, education or other purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 20.2%; cycling from 1.3% to 30.2%; and, from 74.7% to 42.3% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 67.7% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

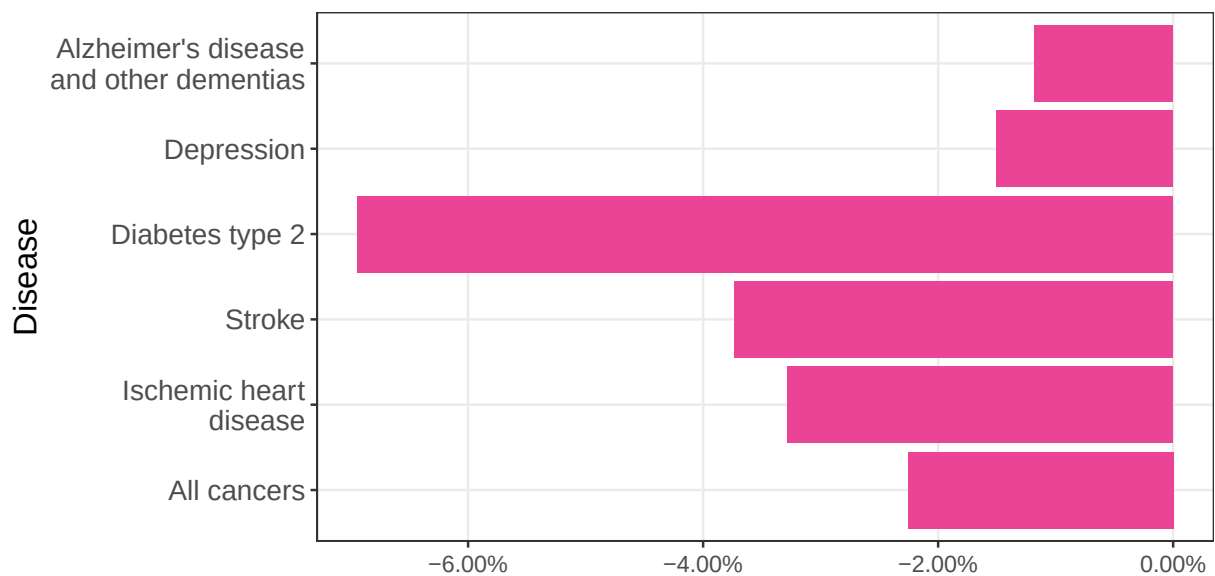
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

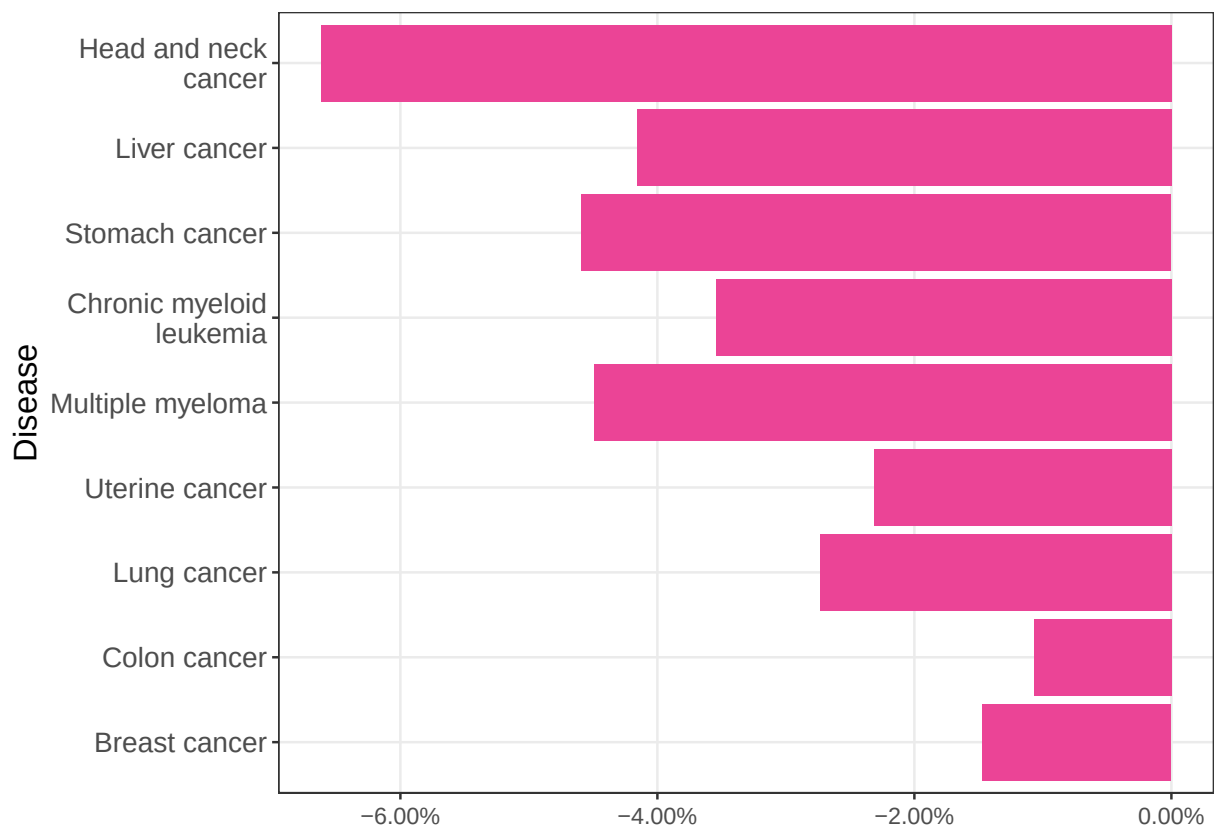
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 1.18%                              | 7,506                                                                       |
| Breast cancer                           | 1.47%                              | 1,077                                                                       |
| All cancers                             | 2.26%                              | 8,895                                                                       |
| Colon cancer                            | 1.07%                              | 1,388                                                                       |
| Chronic myeloid leukemia                | 3.54%                              | 97                                                                          |
| Diabetes type 2                         | 6.94%                              | 22,907                                                                      |
| Depression                              | 1.51%                              | 24,129                                                                      |
| Head and neck cancer                    | 6.62%                              | 292                                                                         |
| Ischemic heart disease                  | 3.28%                              | 28,904                                                                      |
| Liver cancer                            | 4.16%                              | 732                                                                         |
| Multiple myeloma                        | 4.50%                              | 872                                                                         |
| Stomach cancer                          | 4.59%                              | 1,096                                                                       |
| Stroke                                  | 3.73%                              | 9,708                                                                       |
| Lung cancer                             | 2.73%                              | 3,143                                                                       |
| Uterine cancer                          | 2.31%                              | 196                                                                         |

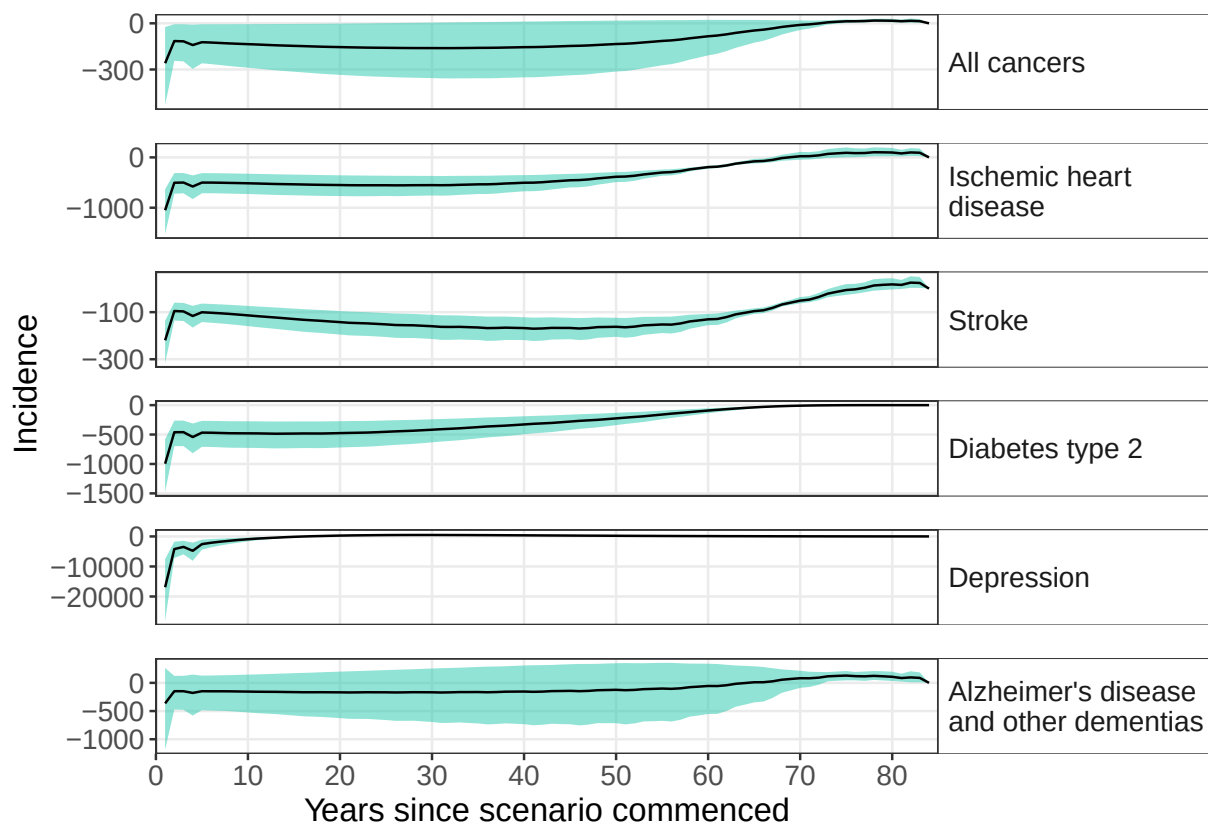
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



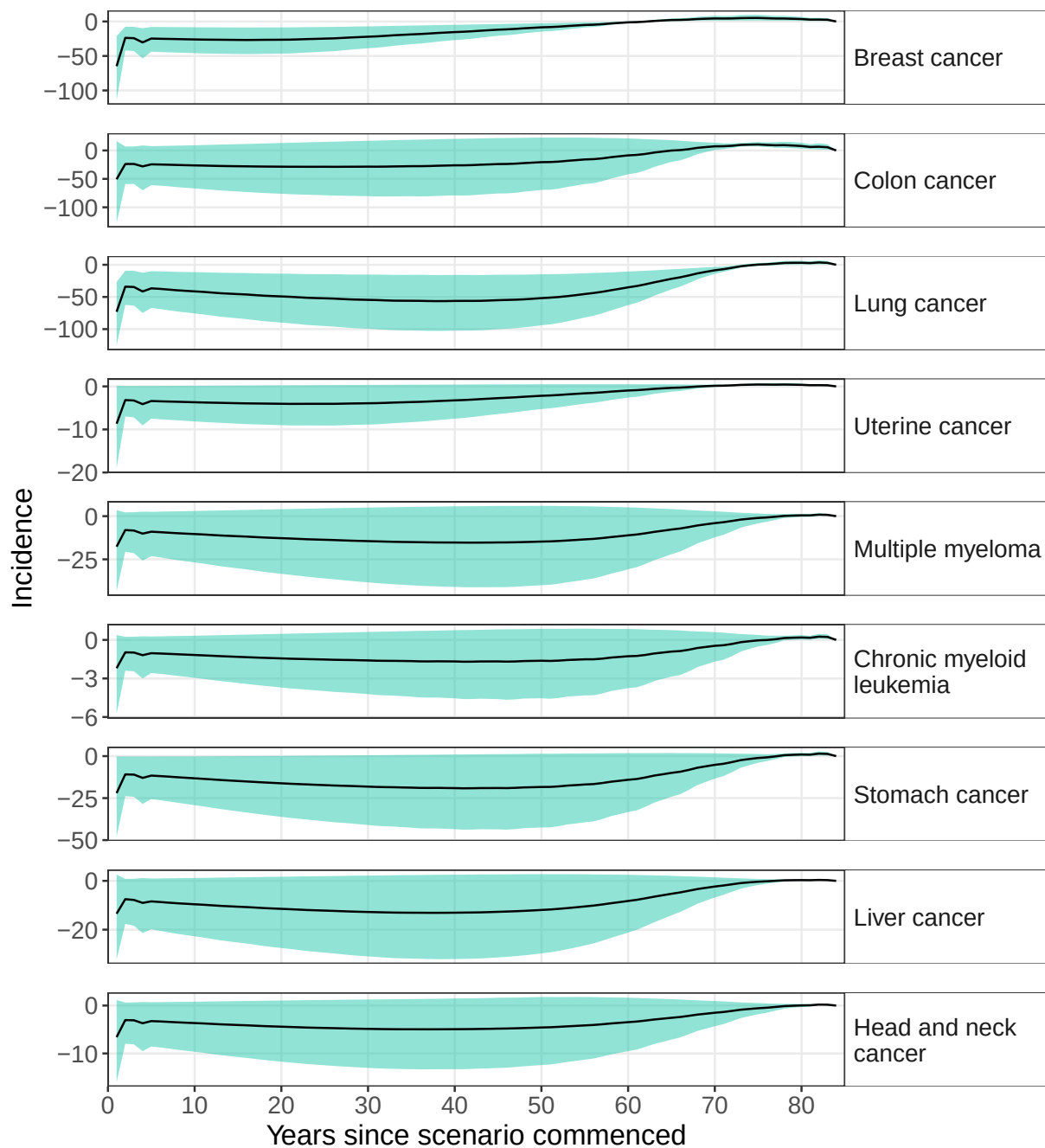
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

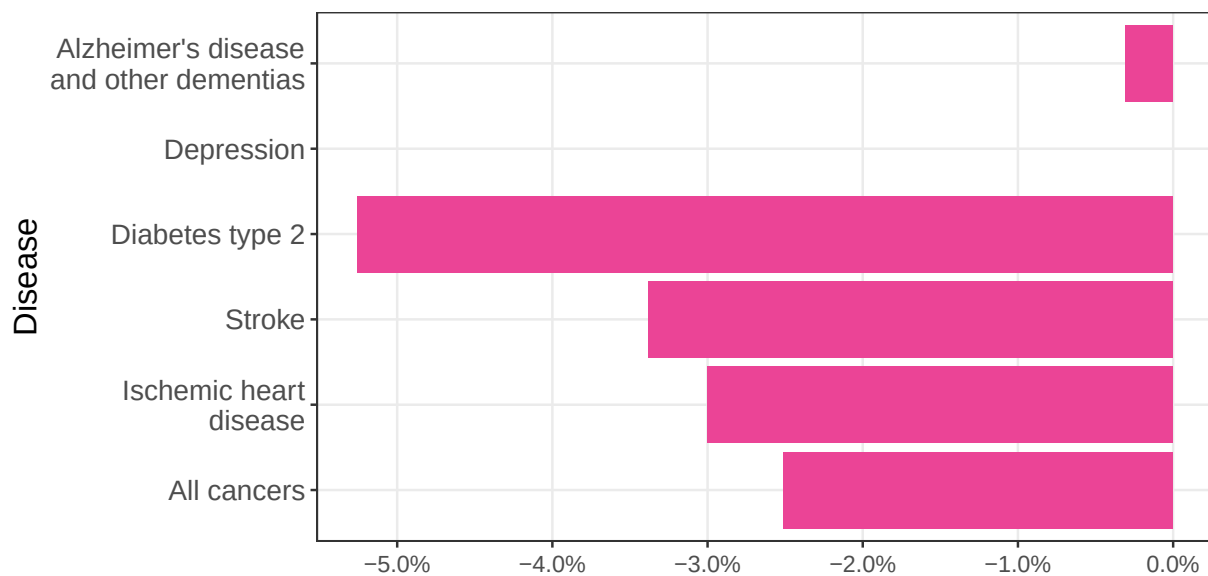
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

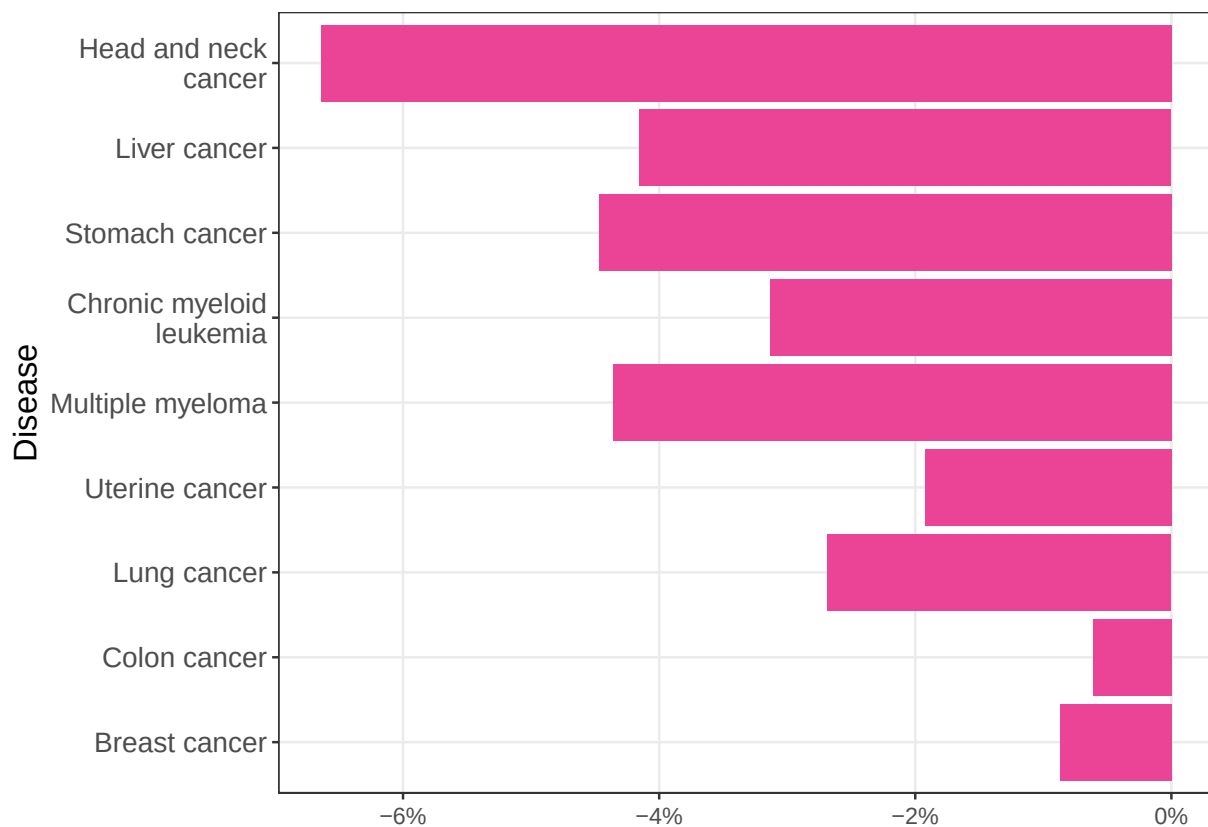
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.31%                   | 736                                                               |
| Breast cancer                           | 0.87%                   | 281                                                               |
| All cancers                             | 2.51%                   | 6,357                                                             |
| Colon cancer                            | 0.61%                   | 265                                                               |
| Chronic myeloid leukemia                | 3.13%                   | 64                                                                |
| Diabetes type 2                         | 5.25%                   | 2,597                                                             |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 6.64%                   | 264                                                               |
| Ischemic heart disease                  | 3.00%                   | 11,260                                                            |
| Liver cancer                            | 4.15%                   | 709                                                               |
| Multiple myeloma                        | 4.36%                   | 759                                                               |
| Stomach cancer                          | 4.47%                   | 946                                                               |
| Stroke                                  | 3.38%                   | 5,912                                                             |
| Lung cancer                             | 2.69%                   | 2,929                                                             |
| Uterine cancer                          | 1.92%                   | 139                                                               |

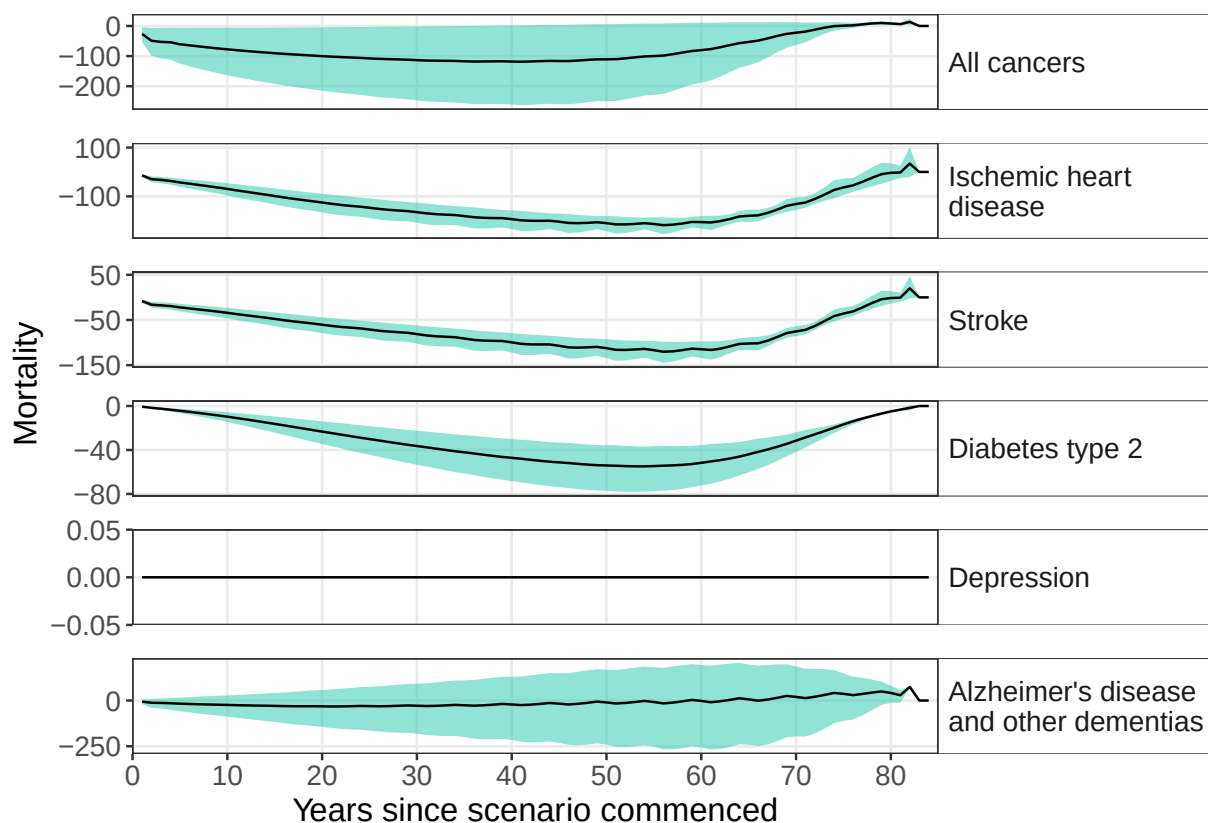
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

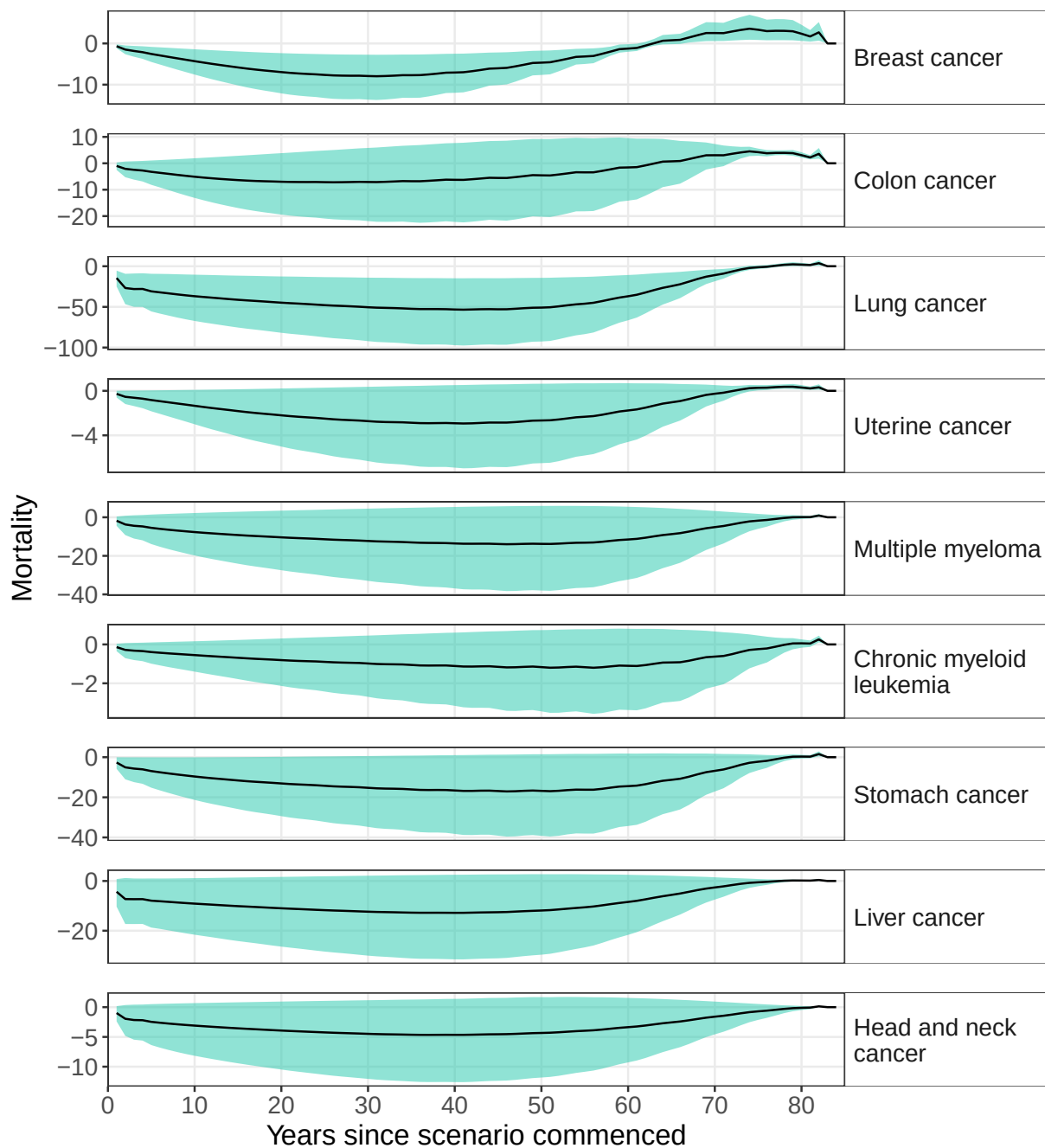


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

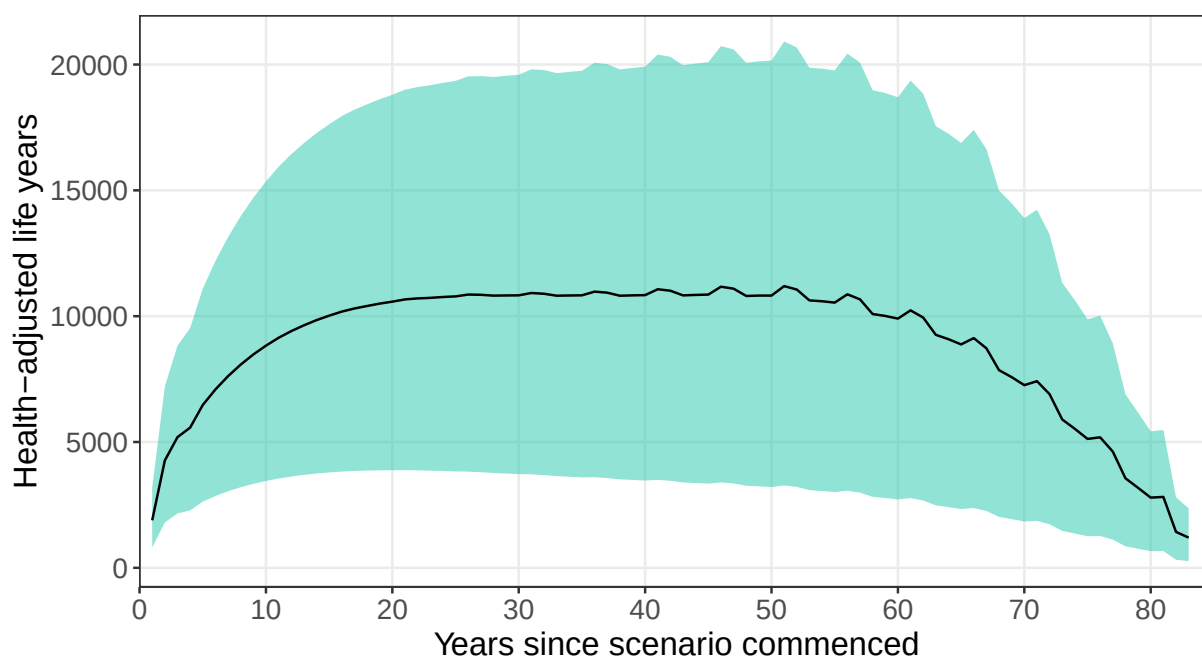
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **744,756** HALYs for the scenario population, which is **370** HALYs per 1,000 members of the population.



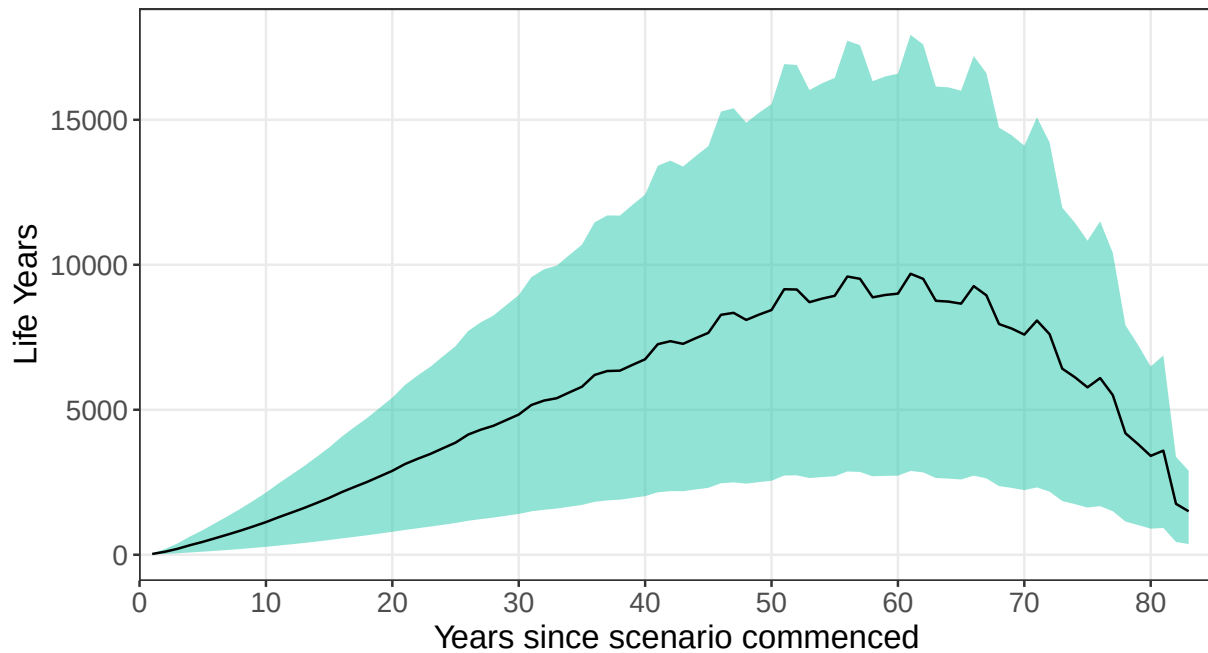
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **451,638** Life Years for the scenario population, which is **224** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **370** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **29,225,040** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **18,098,410** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **12,431,757** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$23,299         | \$14,261         | \$9,544          |
| Breast cancer                           | \$230,508        | \$146,734        | \$99,765         |
| All cancers                             | \$518,134        | \$331,358        | \$227,485        |
| Colon cancer                            | \$113,191        | \$74,875         | \$52,337         |
| Chronic myeloid leukemia                | \$26,565         | \$15,985         | \$10,555         |
| Diabetes type 2                         | \$139,694        | \$82,951         | \$53,688         |
| Depression                              | \$744,918        | \$569,478        | \$446,745        |
| Head and neck cancer                    | \$4,296          | \$2,793          | \$1,975          |
| Ischemic heart disease                  | \$283,628        | \$168,495        | \$109,662        |
| Liver cancer                            | \$4,254          | \$2,801          | \$2,001          |
| Multiple myeloma                        | \$63,233         | \$39,690         | \$27,187         |
| Stomach cancer                          | \$26,868         | \$16,794         | \$11,475         |
| Stroke                                  | \$43,552         | \$25,667         | \$16,639         |
| Lung cancer                             | \$37,917         | \$24,596         | \$17,358         |
| Uterine cancer                          | \$11,339         | \$7,144          | \$4,844          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

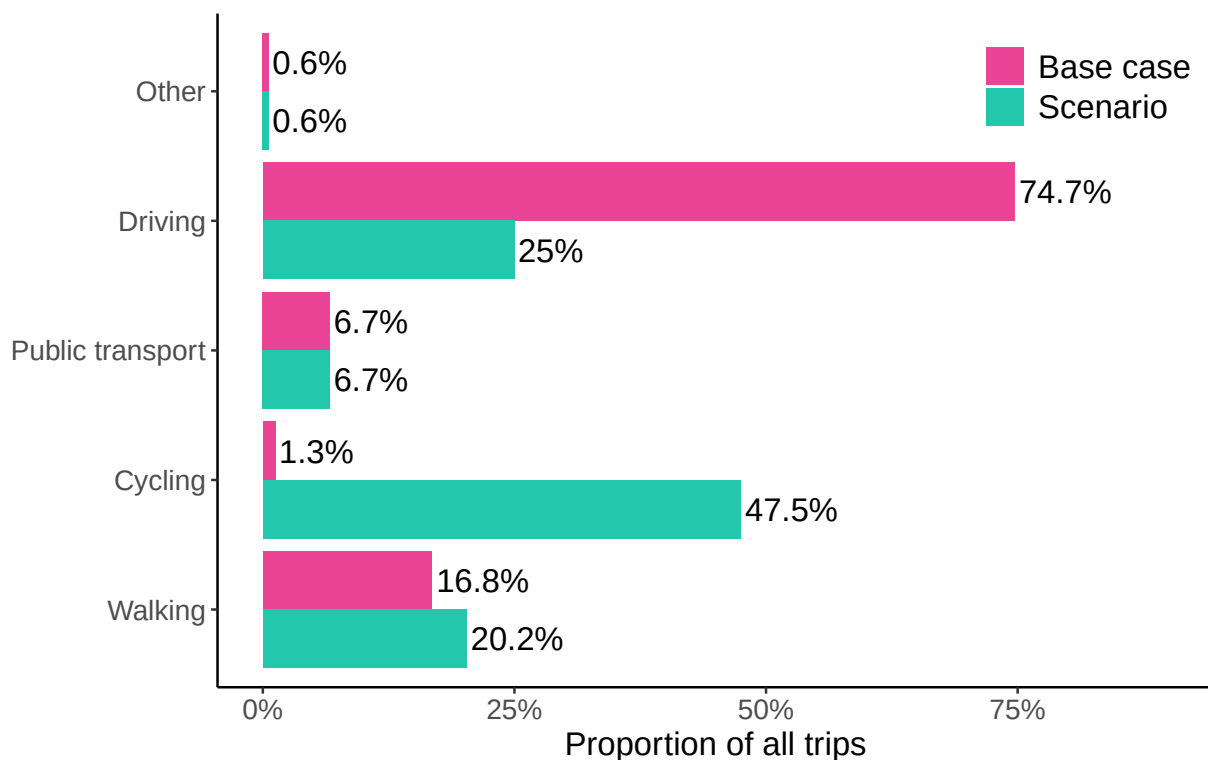
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## Scenario: replacing car trips under 1km with walking, and car trips between 1 and 10km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 1km with walking and replacing car trips between 1km and 10km with cycling for leisure, shopping, work, education or other purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 20.2%; cycling from 1.3% to 47.5%; and, from 74.7% to 25.0% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 75.2% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

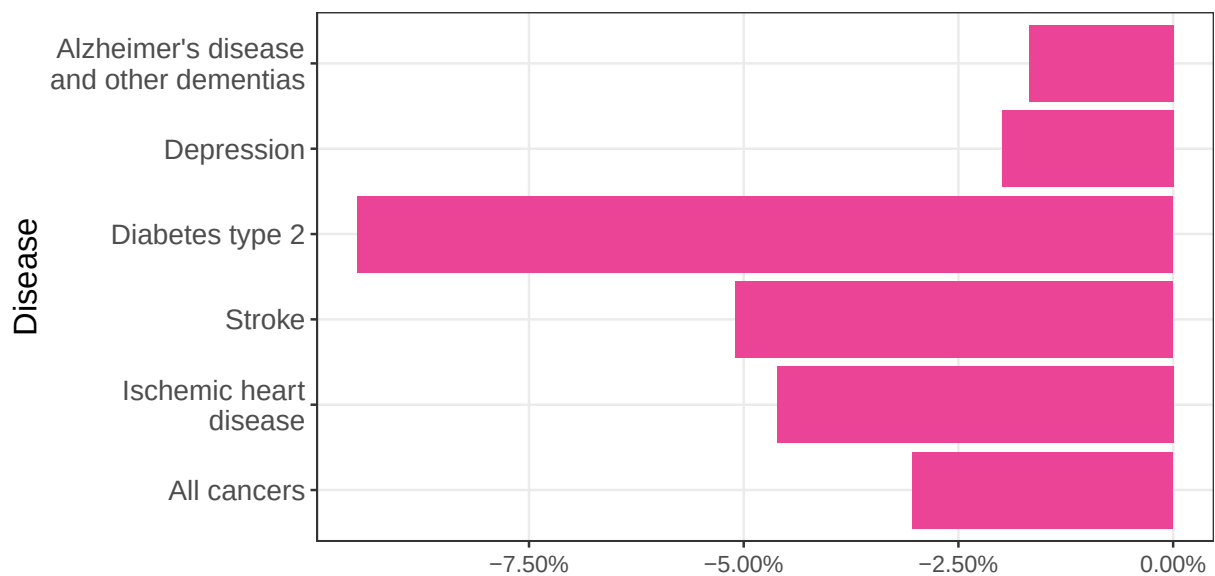
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

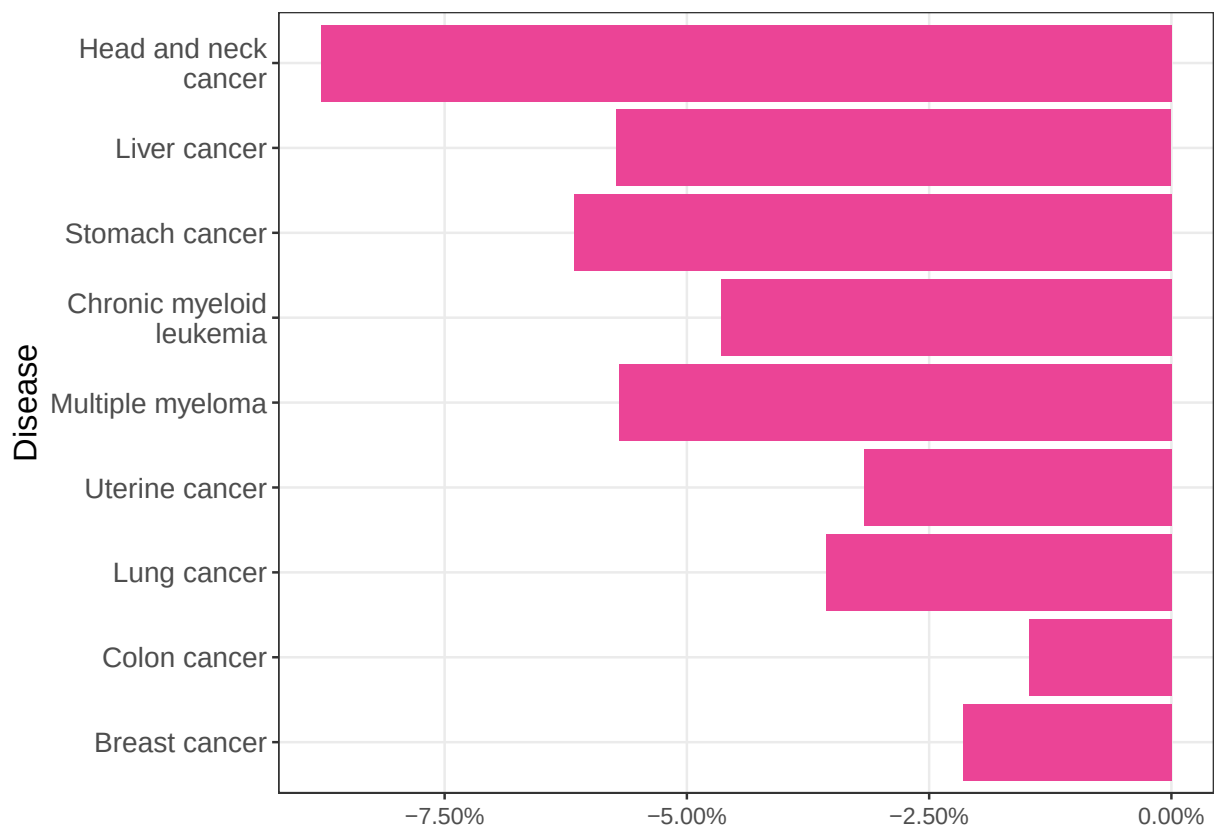
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 1.68%                              | 10,685                                                                      |
| Breast cancer                           | 2.15%                              | 1,578                                                                       |
| All cancers                             | 3.03%                              | 11,956                                                                      |
| Colon cancer                            | 1.47%                              | 1,906                                                                       |
| Chronic myeloid leukemia                | 4.65%                              | 127                                                                         |
| Diabetes type 2                         | 9.49%                              | 31,342                                                                      |
| Depression                              | 1.99%                              | 31,936                                                                      |
| Head and neck cancer                    | 8.78%                              | 387                                                                         |
| Ischemic heart disease                  | 4.61%                              | 40,600                                                                      |
| Liver cancer                            | 5.73%                              | 1,008                                                                       |
| Multiple myeloma                        | 5.70%                              | 1,106                                                                       |
| Stomach cancer                          | 6.16%                              | 1,472                                                                       |
| Stroke                                  | 5.09%                              | 13,255                                                                      |
| Lung cancer                             | 3.57%                              | 4,101                                                                       |
| Uterine cancer                          | 3.17%                              | 269                                                                         |

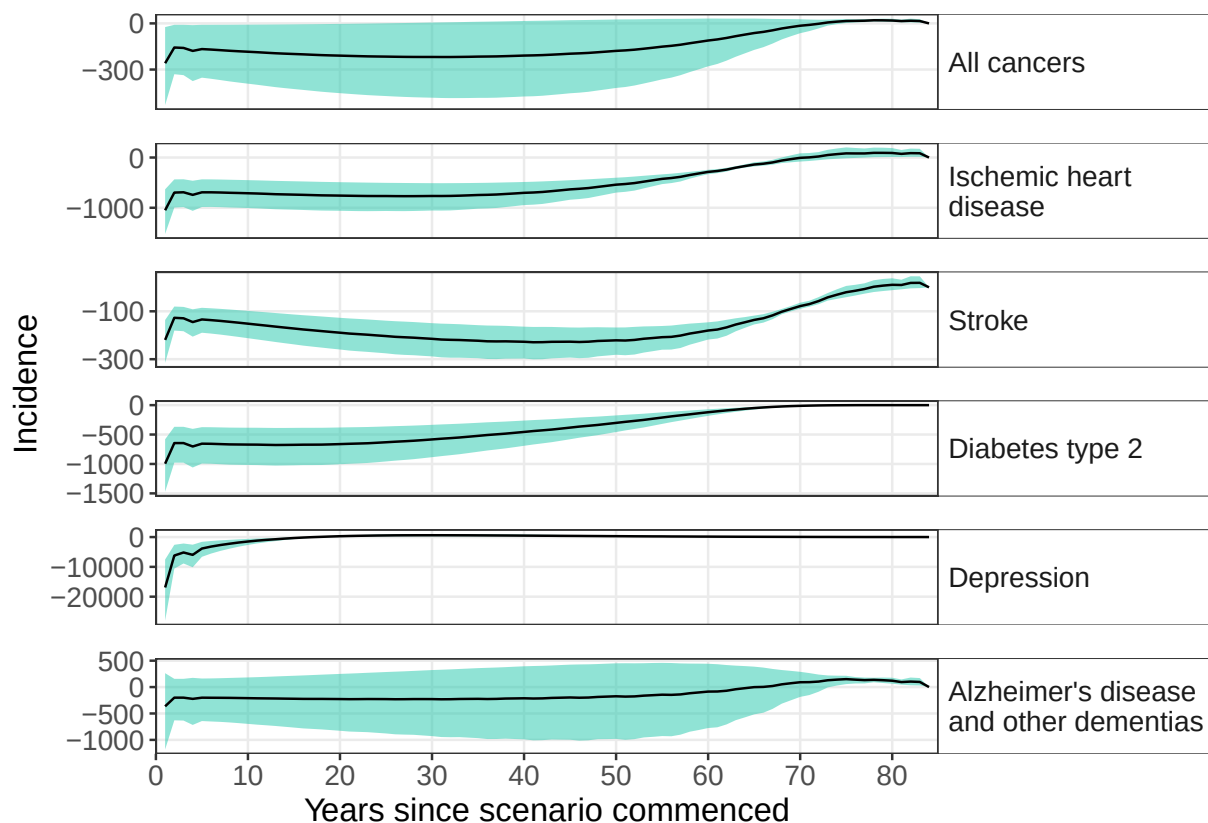
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



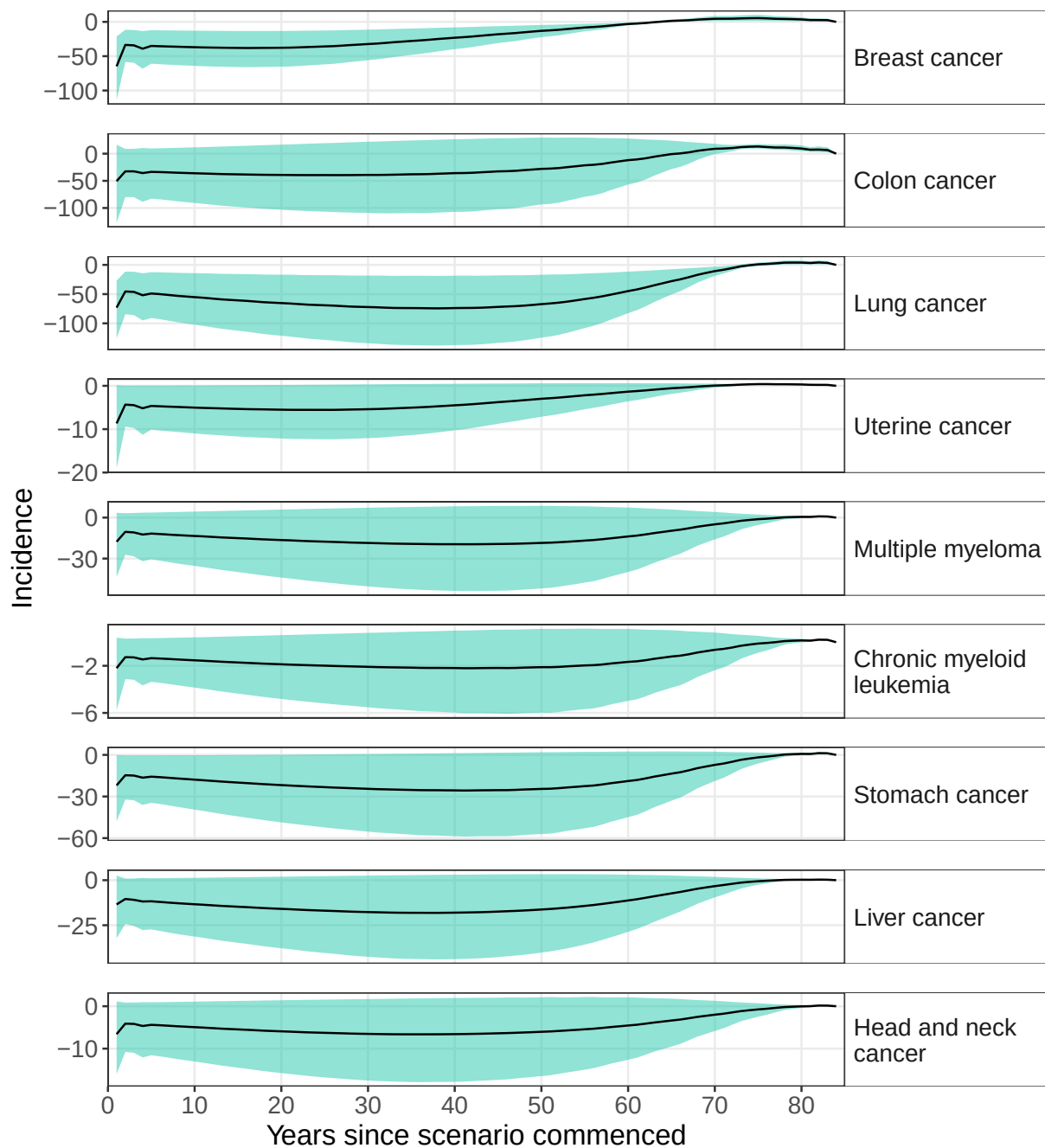
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

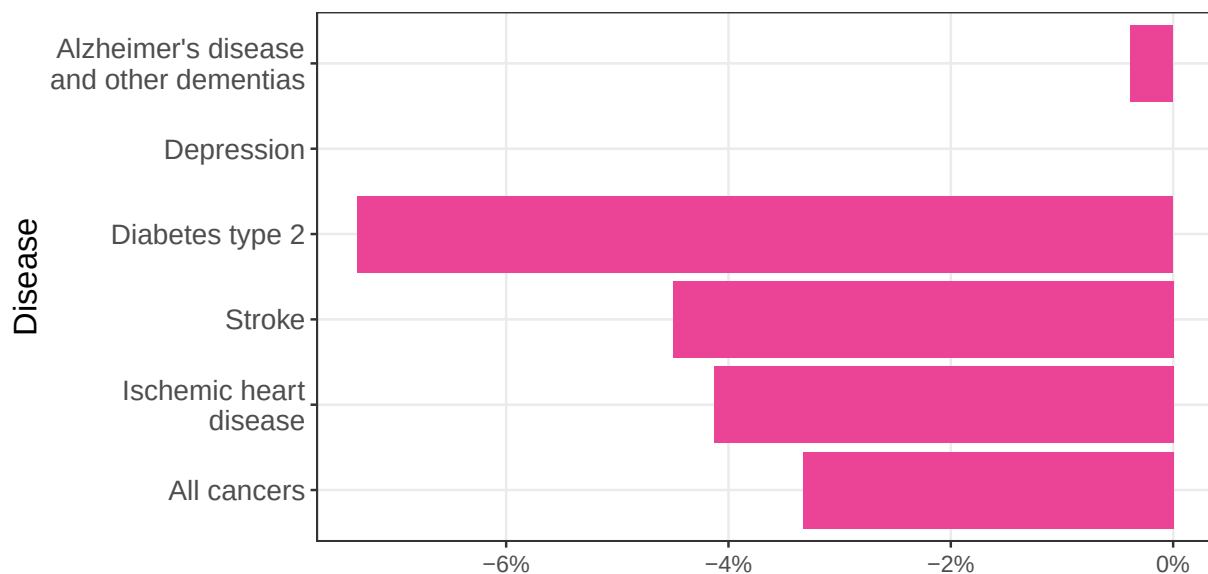
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

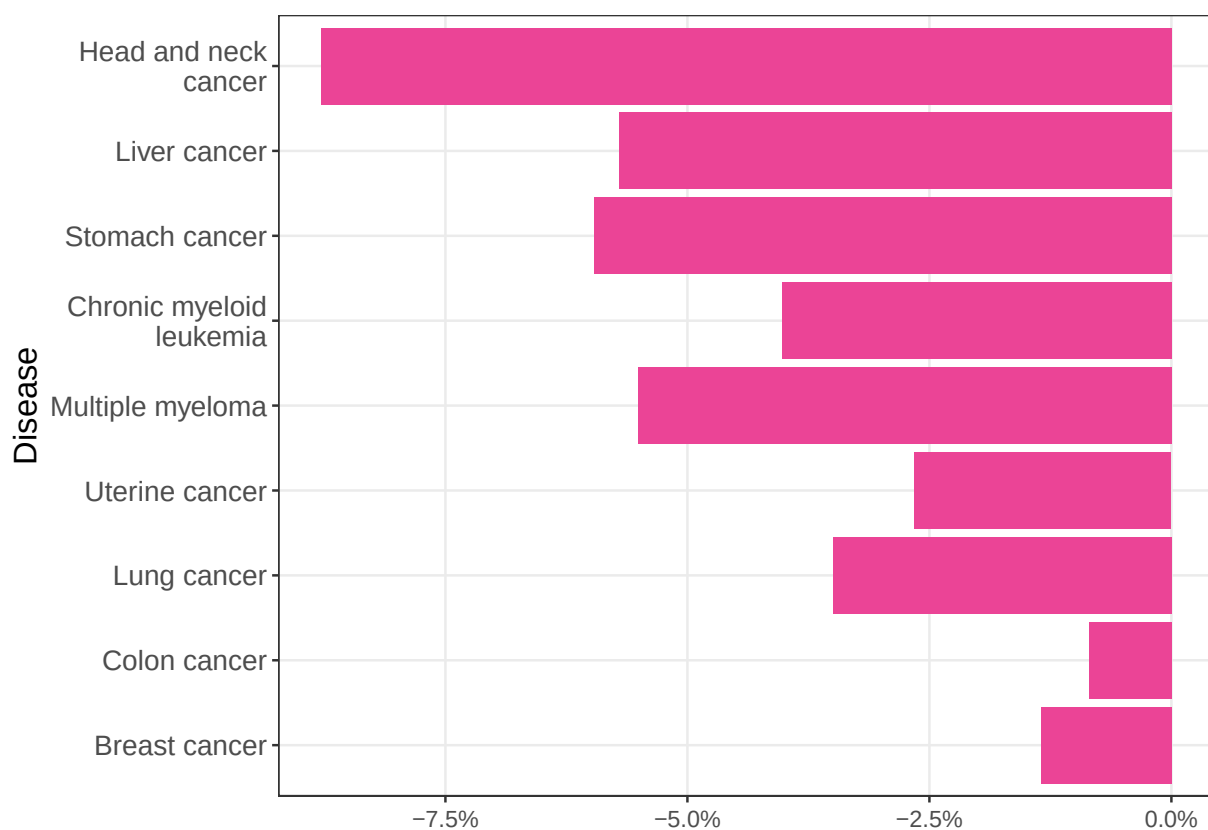
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.39%                   | 925                                                               |
| Breast cancer                           | 1.35%                   | 436                                                               |
| All cancers                             | 3.33%                   | 8,434                                                             |
| Colon cancer                            | 0.85%                   | 366                                                               |
| Chronic myeloid leukemia                | 4.03%                   | 83                                                                |
| Diabetes type 2                         | 7.34%                   | 3,625                                                             |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 8.78%                   | 350                                                               |
| Ischemic heart disease                  | 4.13%                   | 15,496                                                            |
| Liver cancer                            | 5.71%                   | 974                                                               |
| Multiple myeloma                        | 5.51%                   | 960                                                               |
| Stomach cancer                          | 5.97%                   | 1,262                                                             |
| Stroke                                  | 4.50%                   | 7,872                                                             |
| Lung cancer                             | 3.50%                   | 3,812                                                             |
| Uterine cancer                          | 2.65%                   | 191                                                               |

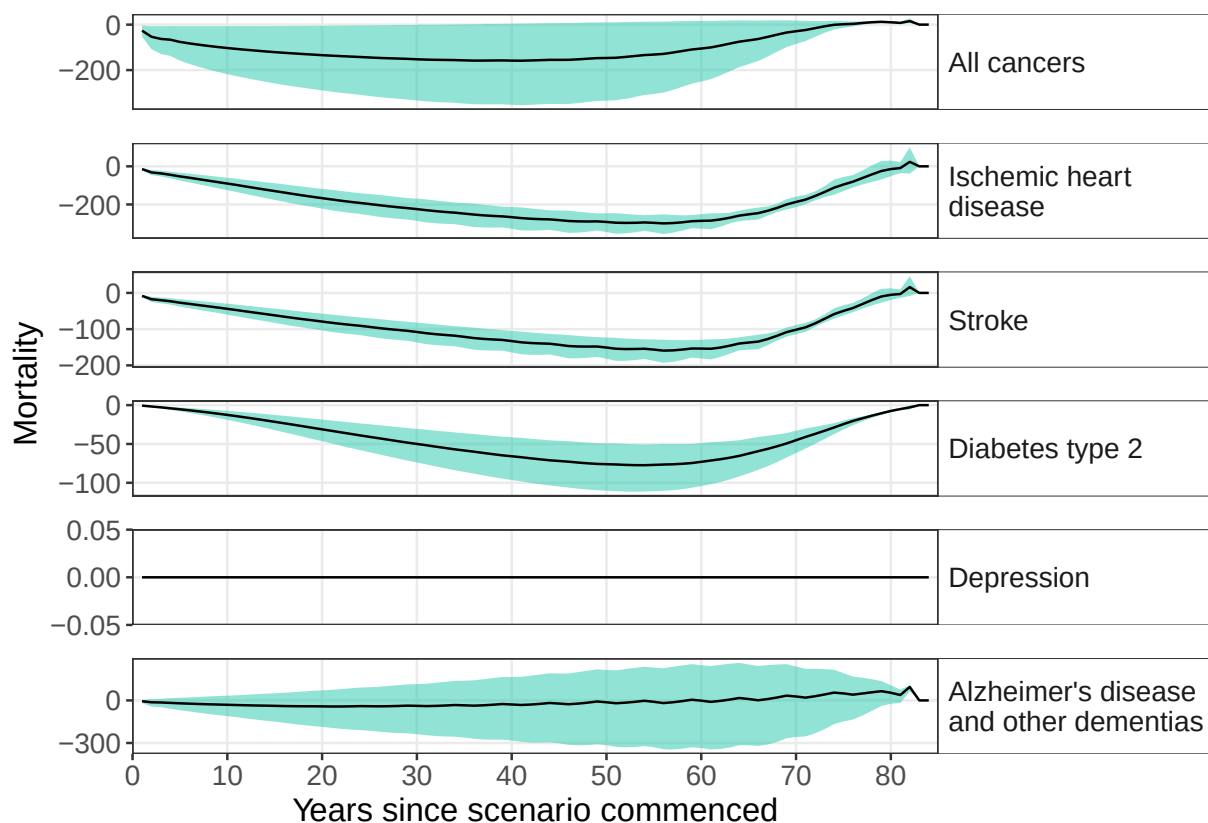
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

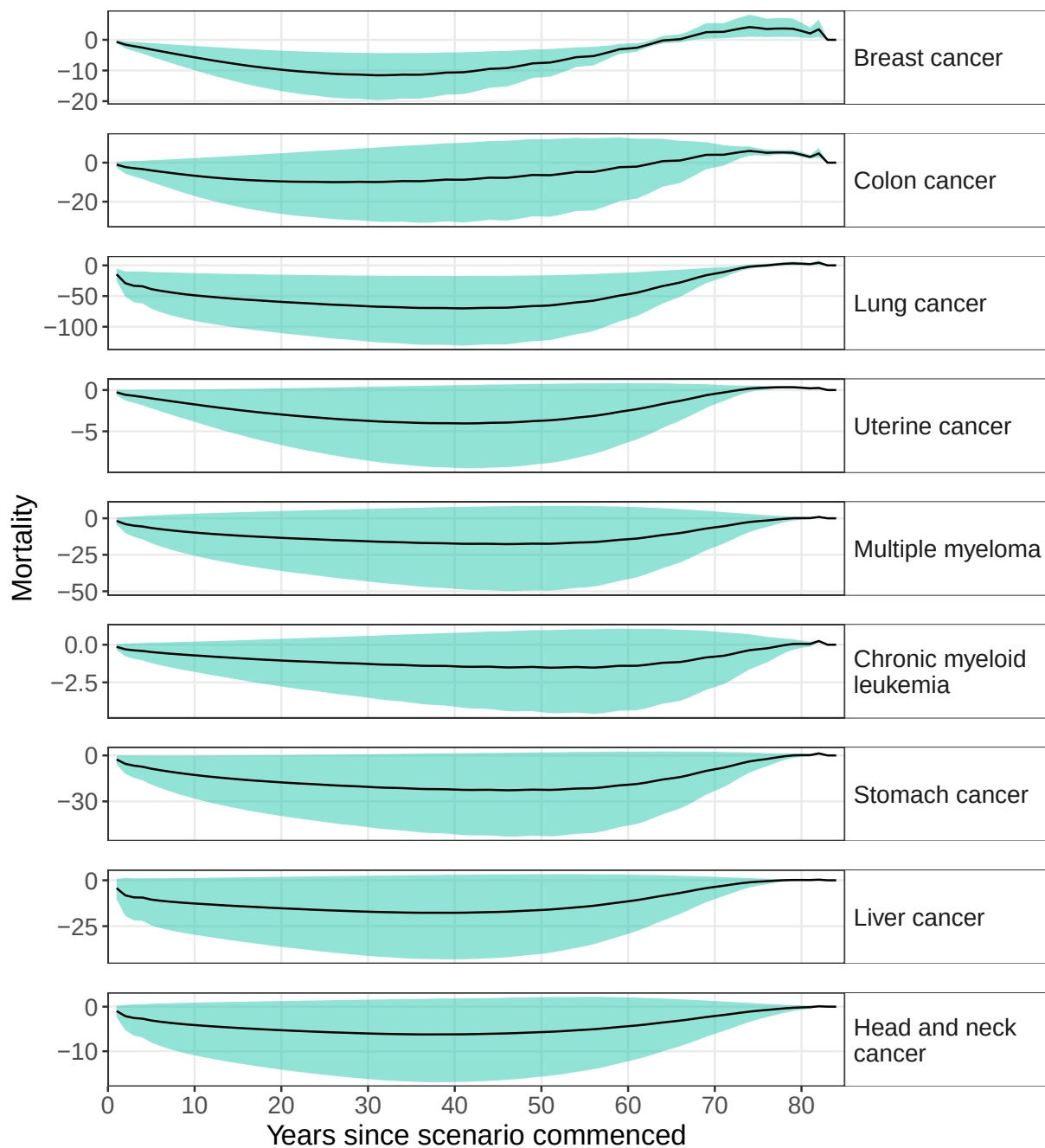


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

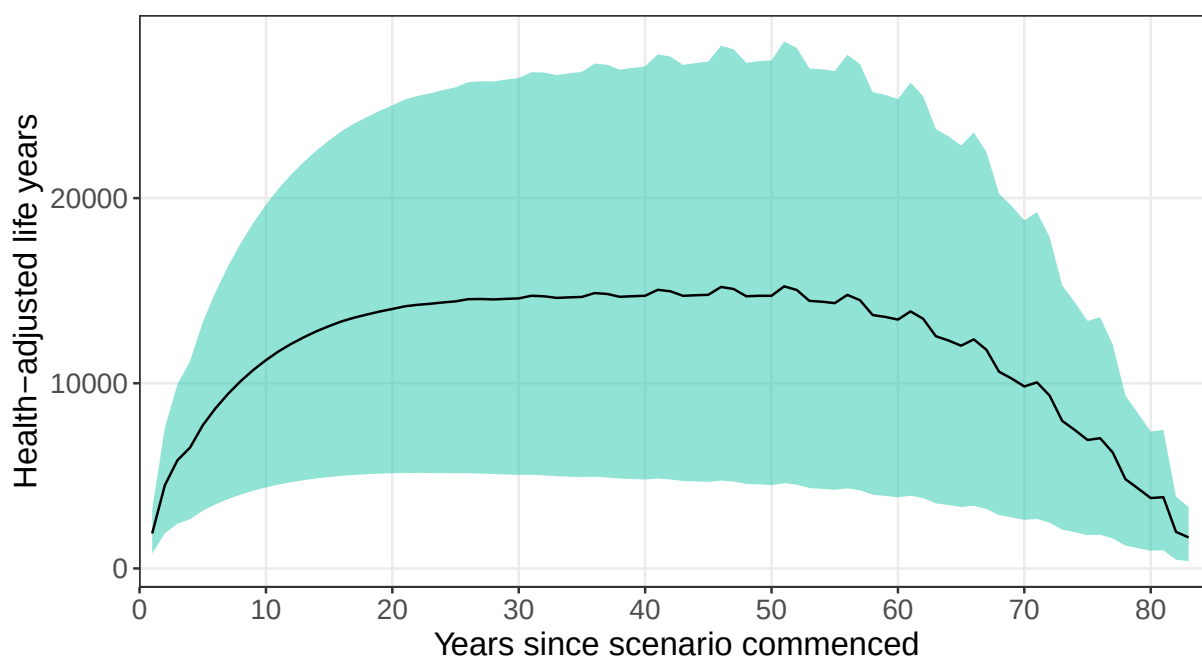
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **994,246** HALYs for the scenario population, which is **494** HALYs per 1,000 members of the population.



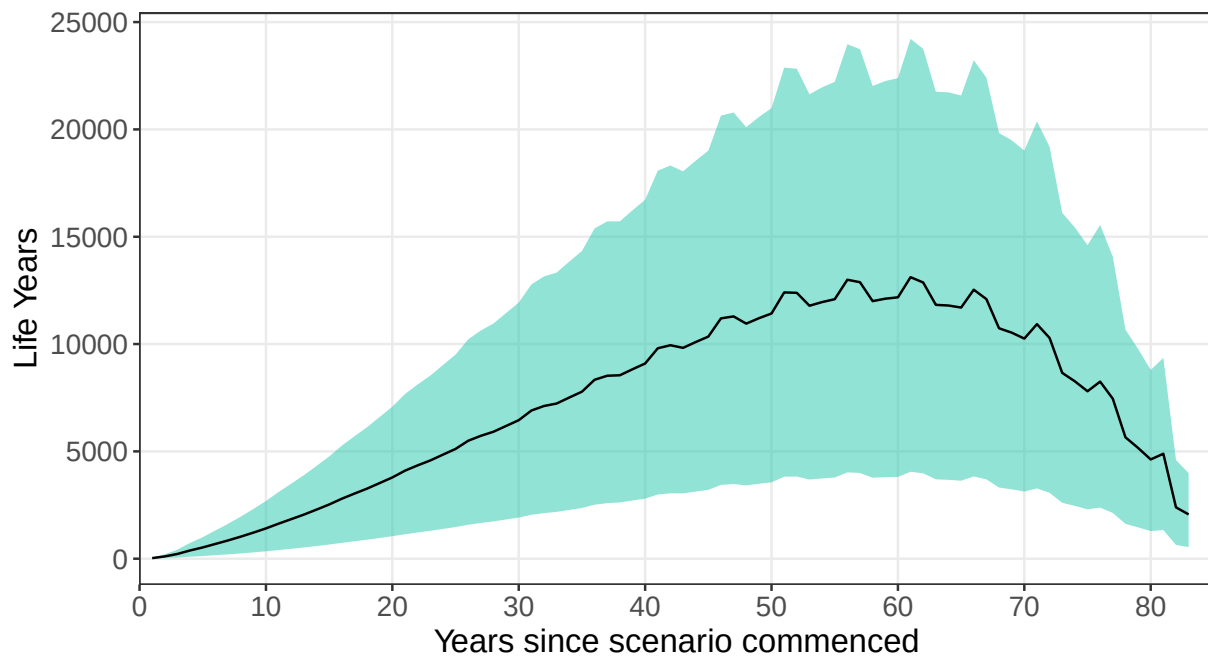
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **607,251** Life Years for the scenario population, which is **302** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated<sup>[a]</sup> by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year<sup>[b]</sup>. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting<sup>[c]</sup> to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).<sup>[2]</sup> Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations<sup>[3, 4]</sup> and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **494** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes<sup>[d]</sup>.

The HALYs gained in this scenario have a statistical value of:

- **38,342,345** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **23,442,857** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **15,906,876** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year<sup>[b]</sup>.



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$30,726         | \$18,694         | \$12,420         |
| Breast cancer                           | \$322,704        | \$202,287        | \$135,691        |
| All cancers                             | \$703,846        | \$445,529        | \$302,802        |
| Colon cancer                            | \$153,698        | \$100,767        | \$69,801         |
| Chronic myeloid leukemia                | \$33,845         | \$20,263         | \$13,299         |
| Diabetes type 2                         | \$191,878        | \$113,017        | \$72,533         |
| Depression                              | \$948,278        | \$718,246        | \$558,443        |
| Head and neck cancer                    | \$5,660          | \$3,663          | \$2,574          |
| Ischemic heart disease                  | \$387,398        | \$228,427        | \$147,428        |
| Liver cancer                            | \$5,899          | \$3,868          | \$2,748          |
| Multiple myeloma                        | \$81,171         | \$50,759         | \$34,585         |
| Stomach cancer                          | \$35,845         | \$22,297         | \$15,141         |
| Stroke                                  | \$57,185         | \$33,458         | \$21,519         |
| Lung cancer                             | \$49,909         | \$32,238         | \$22,619         |
| Uterine cancer                          | \$15,163         | \$9,470          | \$6,363          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

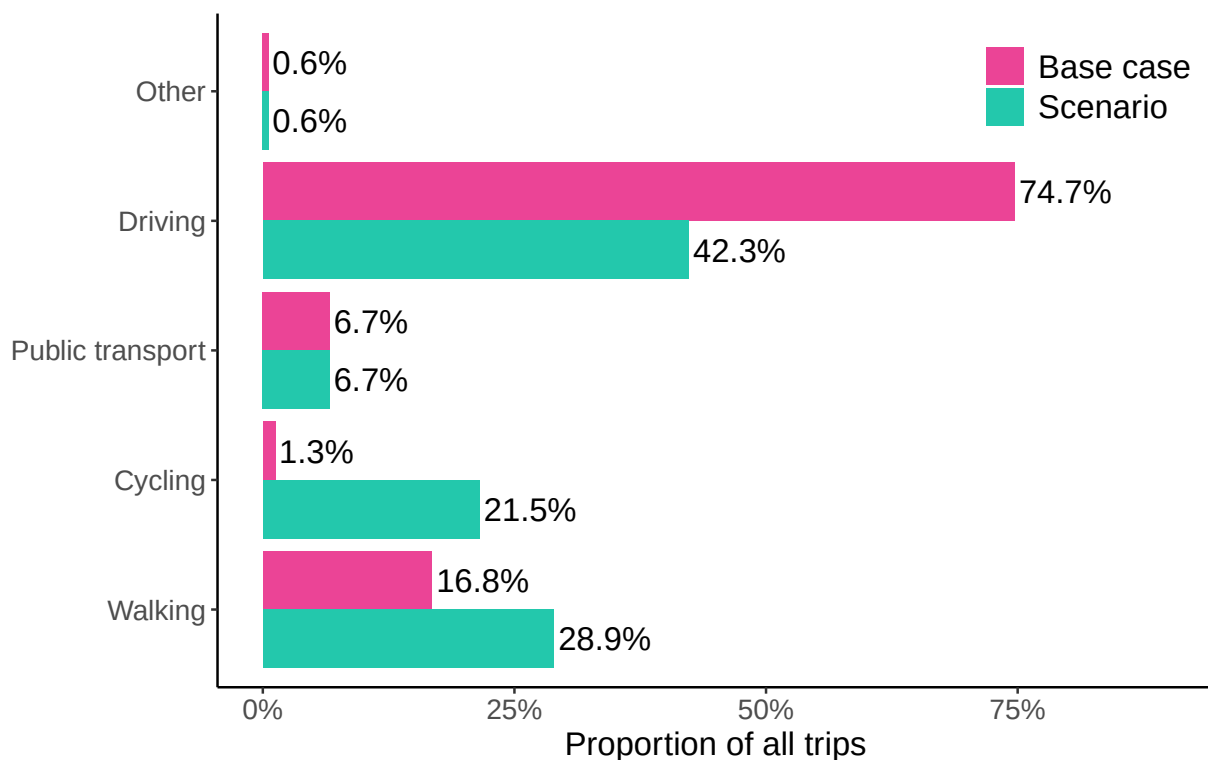
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
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## Scenario: replacing car trips under 2km with walking, and car trips between 2 and 5km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 2km with walking and replacing car trips between 2km and 5km with cycling for leisure, shopping, work, education or other purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 28.9%; cycling from 1.3% to 21.5%; and, from 74.7% to 42.3% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 67.9% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

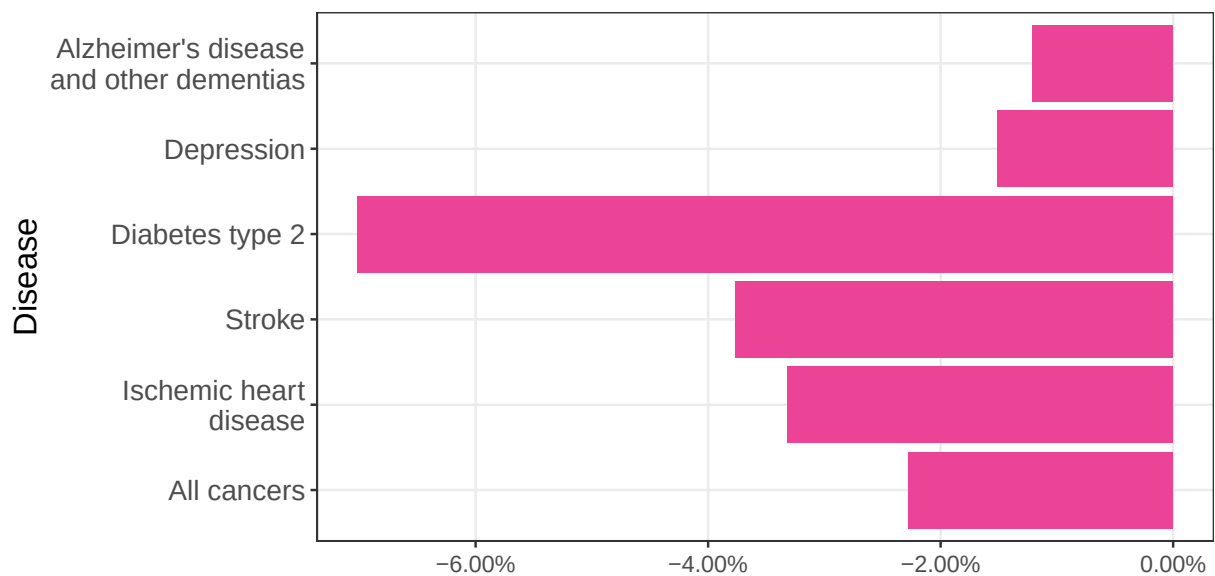
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

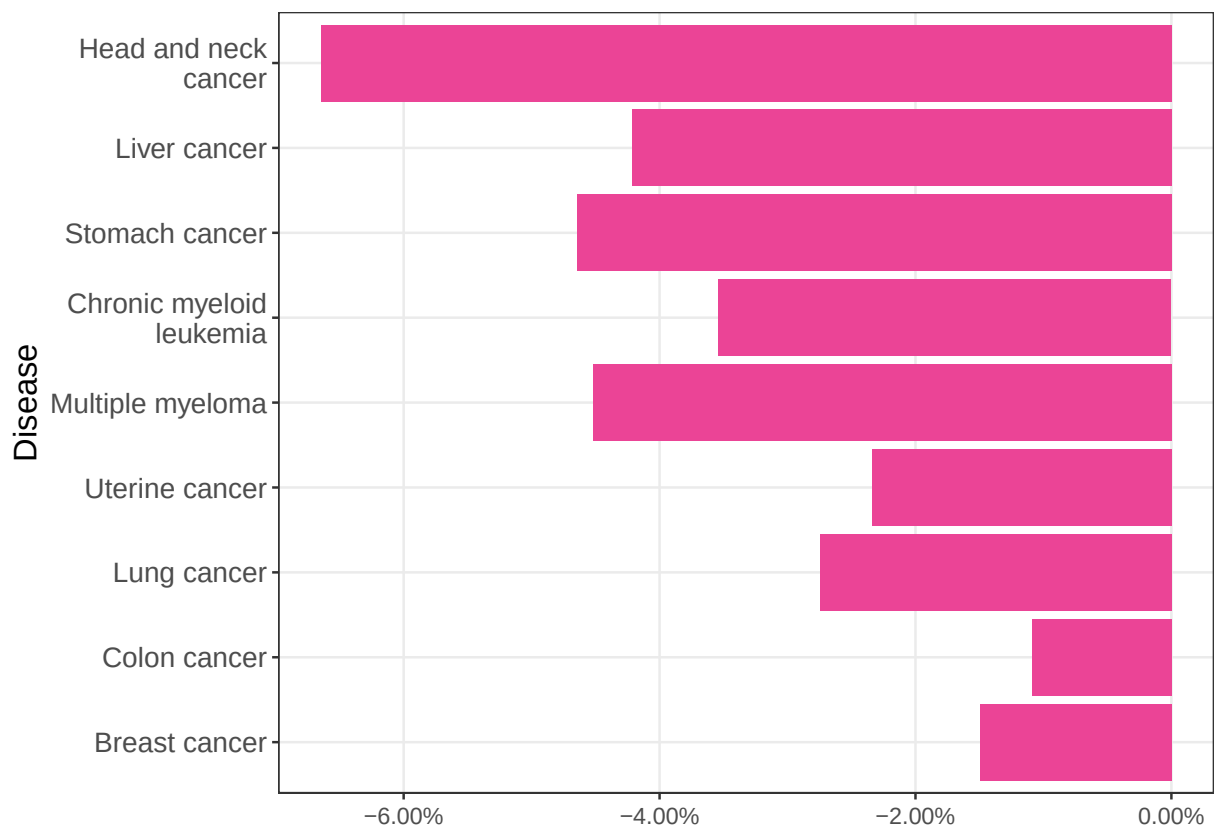
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 1.21%                              | 7,702                                                                       |
| Breast cancer                           | 1.50%                              | 1,095                                                                       |
| All cancers                             | 2.28%                              | 8,981                                                                       |
| Colon cancer                            | 1.09%                              | 1,413                                                                       |
| Chronic myeloid leukemia                | 3.54%                              | 97                                                                          |
| Diabetes type 2                         | 7.01%                              | 23,154                                                                      |
| Depression                              | 1.51%                              | 24,246                                                                      |
| Head and neck cancer                    | 6.65%                              | 293                                                                         |
| Ischemic heart disease                  | 3.32%                              | 29,208                                                                      |
| Liver cancer                            | 4.22%                              | 742                                                                         |
| Multiple myeloma                        | 4.52%                              | 876                                                                         |
| Stomach cancer                          | 4.65%                              | 1,109                                                                       |
| Stroke                                  | 3.76%                              | 9,791                                                                       |
| Lung cancer                             | 2.74%                              | 3,156                                                                       |
| Uterine cancer                          | 2.34%                              | 198                                                                         |

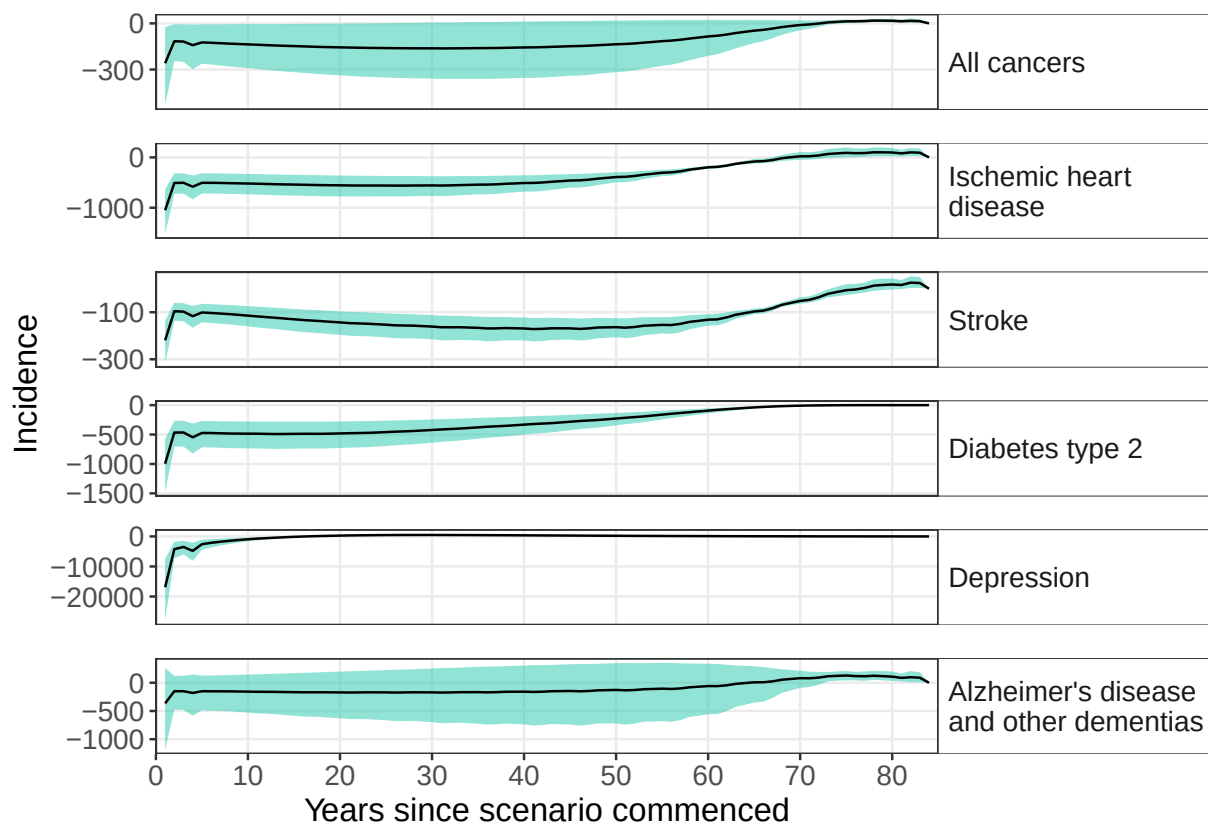
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



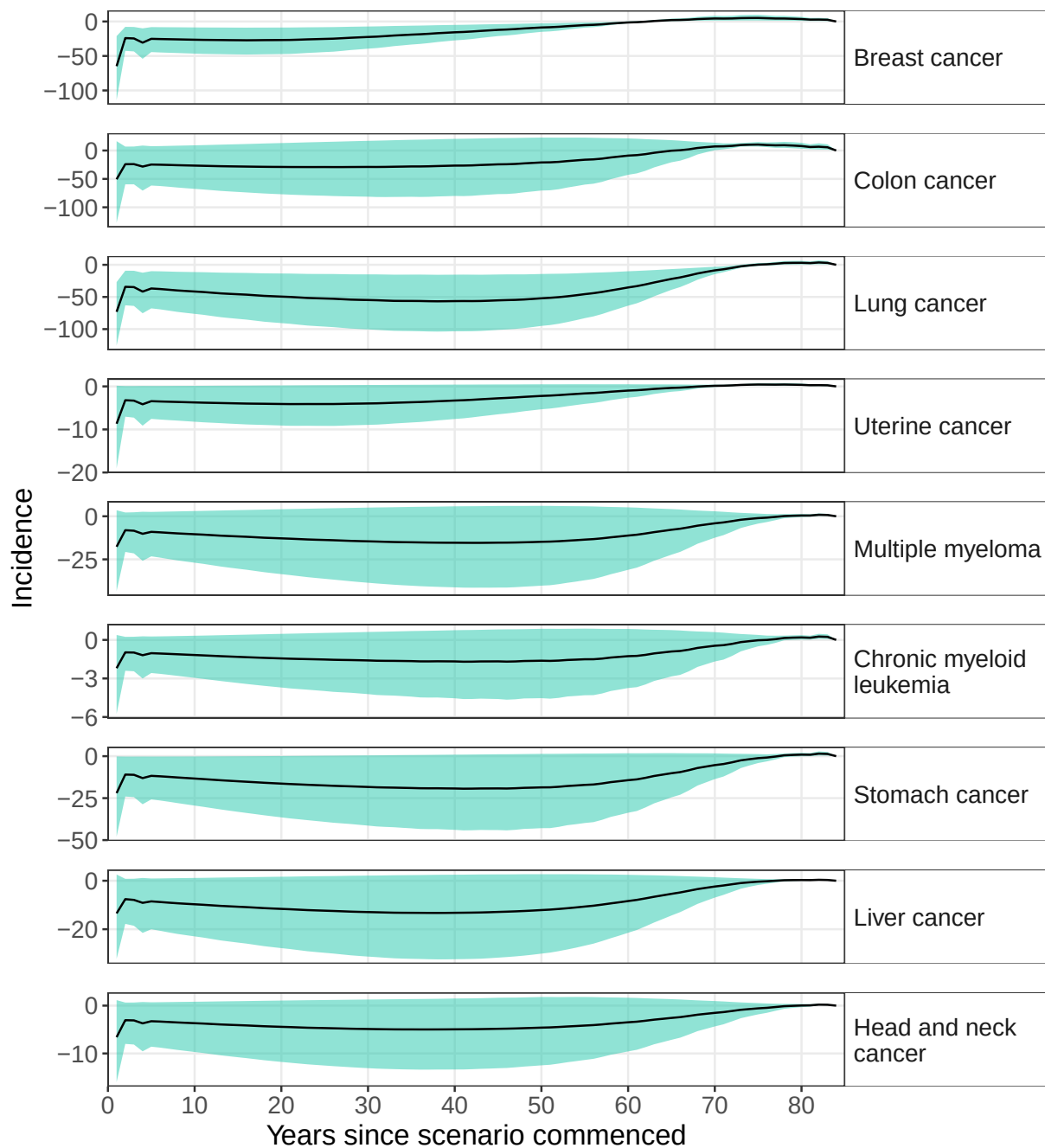
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

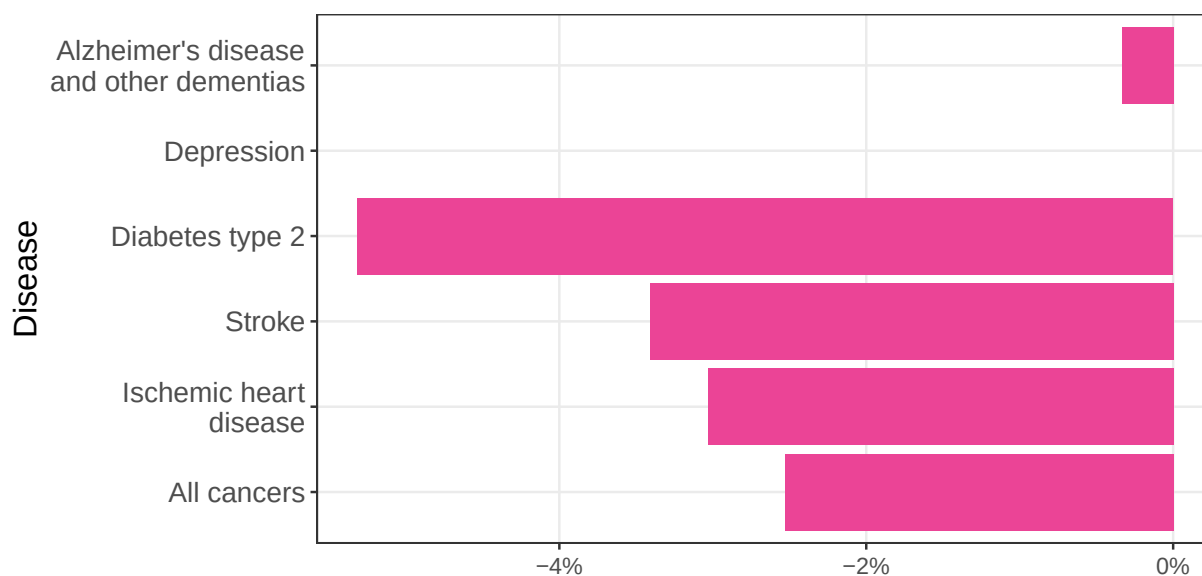
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

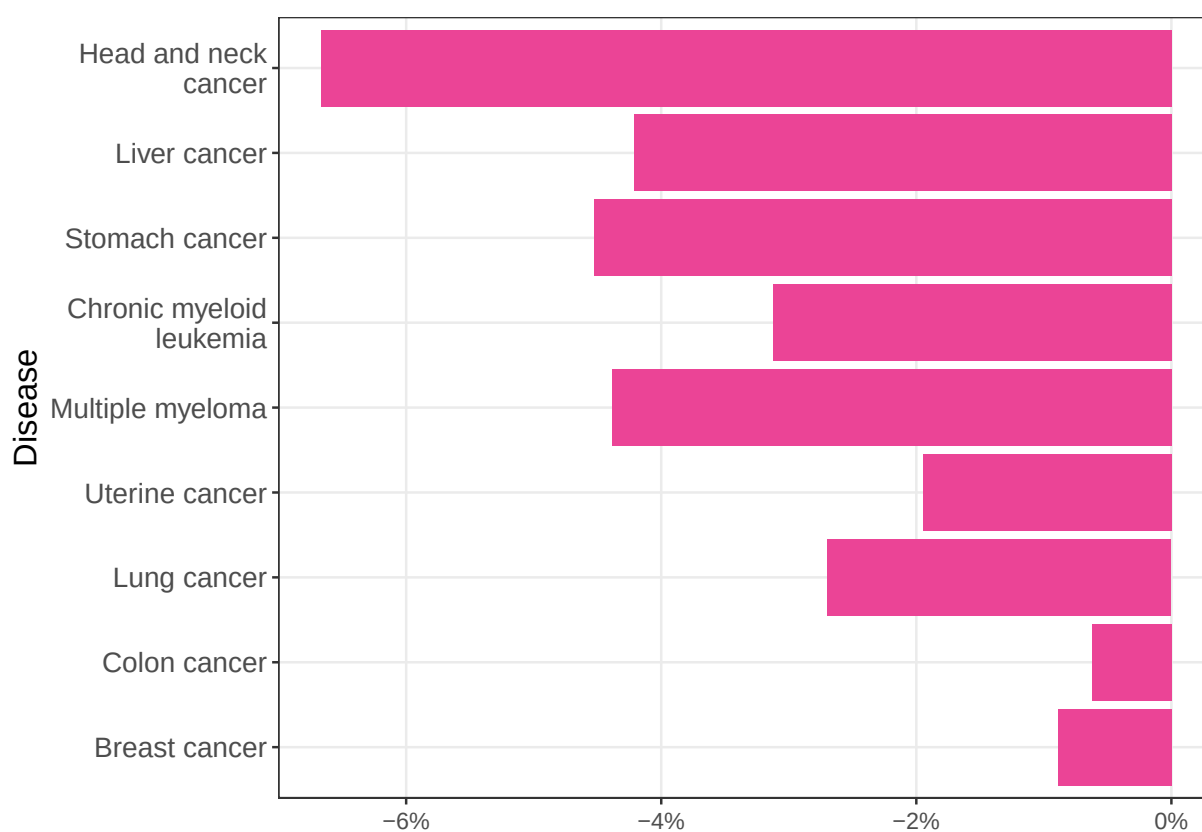
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.33%                   | 800                                                               |
| Breast cancer                           | 0.89%                   | 287                                                               |
| All cancers                             | 2.53%                   | 6,407                                                             |
| Colon cancer                            | 0.63%                   | 269                                                               |
| Chronic myeloid leukemia                | 3.12%                   | 64                                                                |
| Diabetes type 2                         | 5.31%                   | 2,626                                                             |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 6.67%                   | 265                                                               |
| Ischemic heart disease                  | 3.03%                   | 11,370                                                            |
| Liver cancer                            | 4.21%                   | 719                                                               |
| Multiple myeloma                        | 4.38%                   | 763                                                               |
| Stomach cancer                          | 4.53%                   | 957                                                               |
| Stroke                                  | 3.41%                   | 5,964                                                             |
| Lung cancer                             | 2.70%                   | 2,940                                                             |
| Uterine cancer                          | 1.95%                   | 140                                                               |

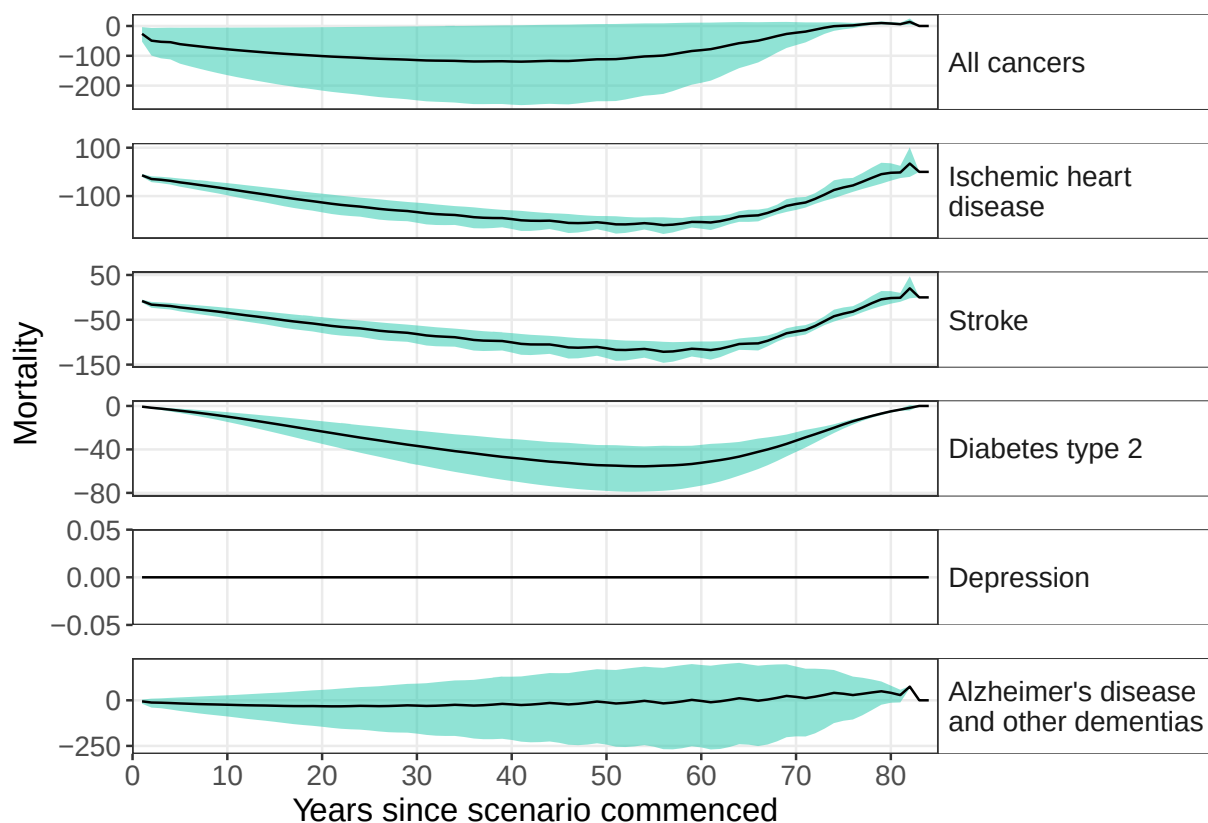
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

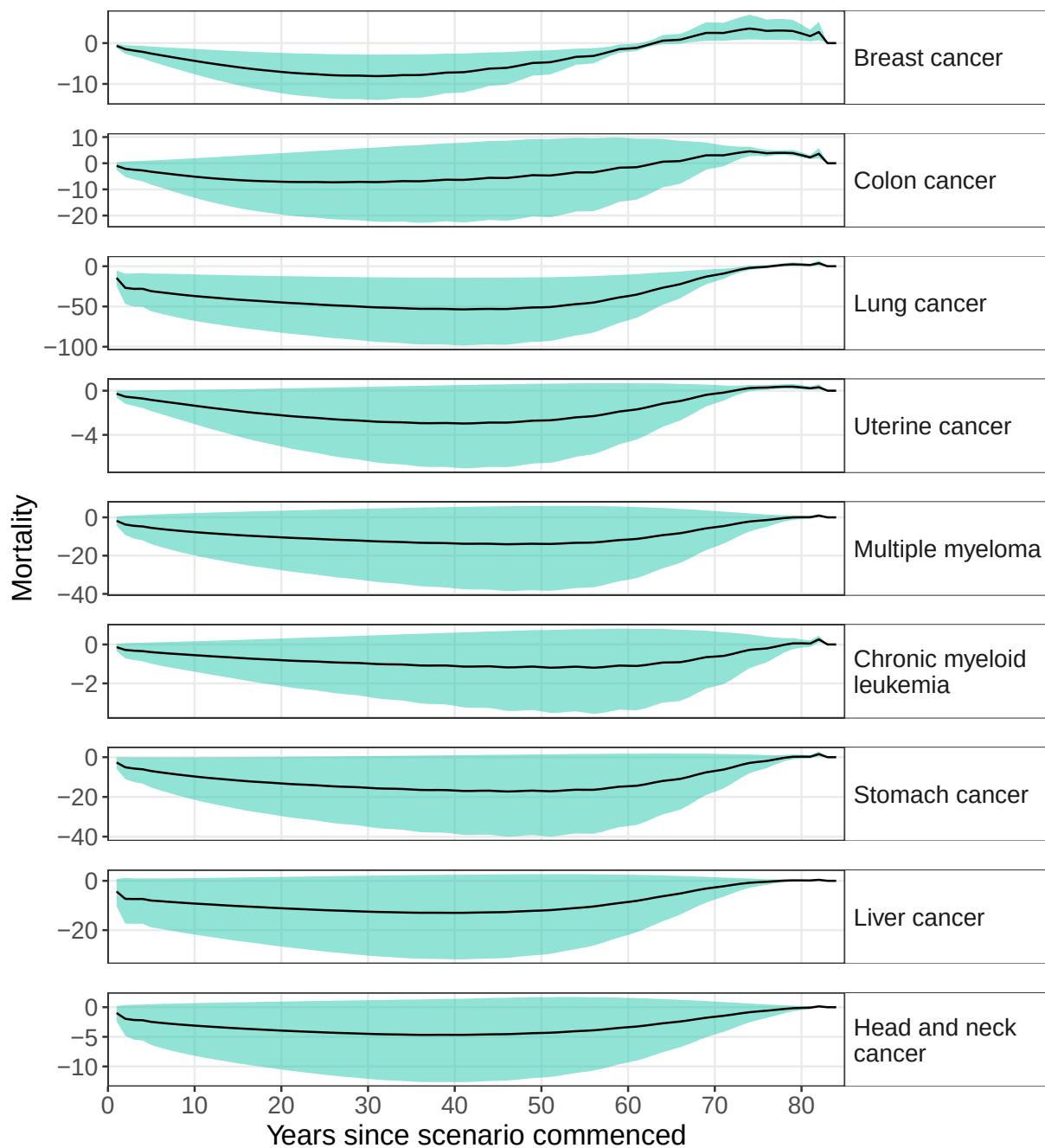


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

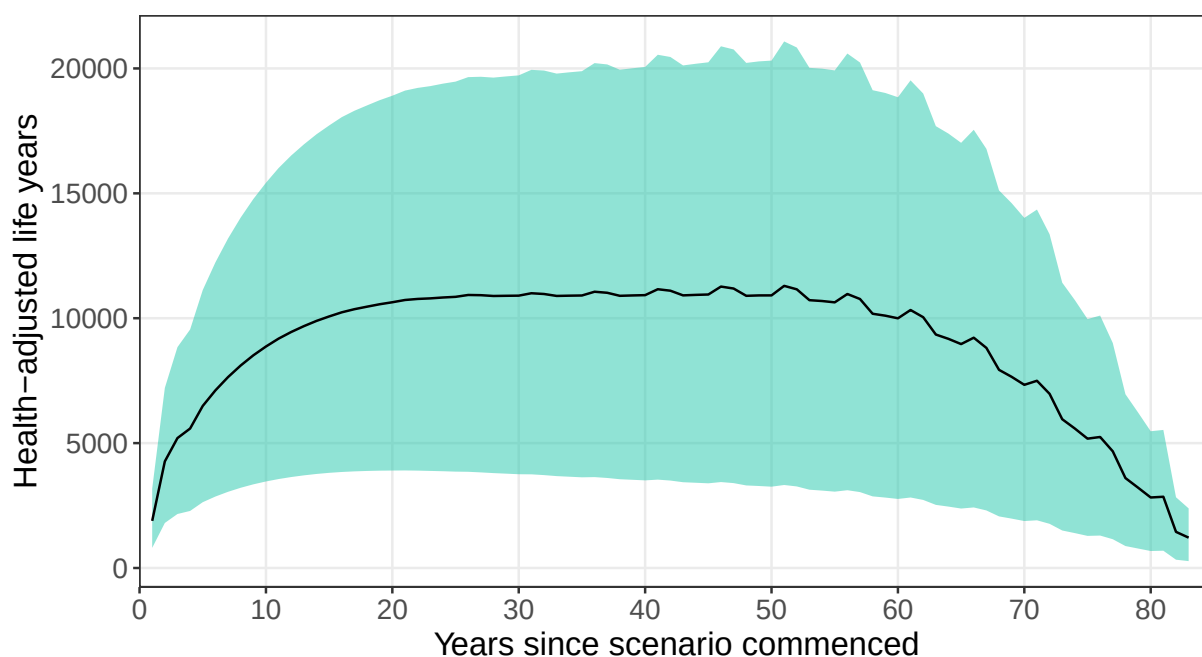
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **750,457** HALYs for the scenario population, which is **373** HALYs per 1,000 members of the population.



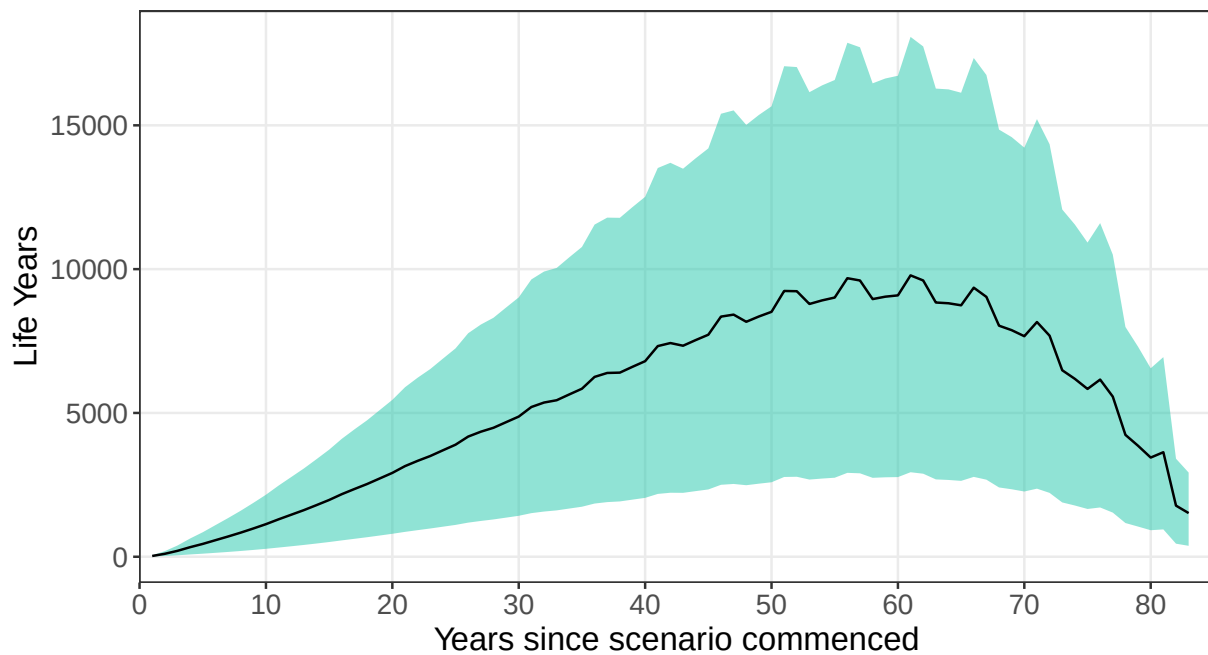
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **455,788** Life Years for the scenario population, which is **226** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **373** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **29,410,801** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **18,200,631** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **12,495,037** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$23,613         | \$14,435         | \$9,653          |
| Breast cancer                           | \$233,904        | \$148,757        | \$101,064        |
| All cancers                             | \$523,661        | \$334,677        | \$229,638        |
| Colon cancer                            | \$114,571        | \$75,712         | \$52,884         |
| Chronic myeloid leukemia                | \$26,587         | \$16,001         | \$10,566         |
| Diabetes type 2                         | \$141,126        | \$83,775         | \$54,203         |
| Depression                              | \$747,255        | \$571,301        | \$448,157        |
| Head and neck cancer                    | \$4,311          | \$2,803          | \$1,982          |
| Ischemic heart disease                  | \$285,840        | \$169,764        | \$110,457        |
| Liver cancer                            | \$4,303          | \$2,832          | \$2,022          |
| Multiple myeloma                        | \$63,430         | \$39,808         | \$27,264         |
| Stomach cancer                          | \$27,088         | \$16,925         | \$11,561         |
| Stroke                                  | \$43,820         | \$25,817         | \$16,732         |
| Lung cancer                             | \$38,057         | \$24,684         | \$17,419         |
| Uterine cancer                          | \$11,449         | \$7,210          | \$4,887          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

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## References

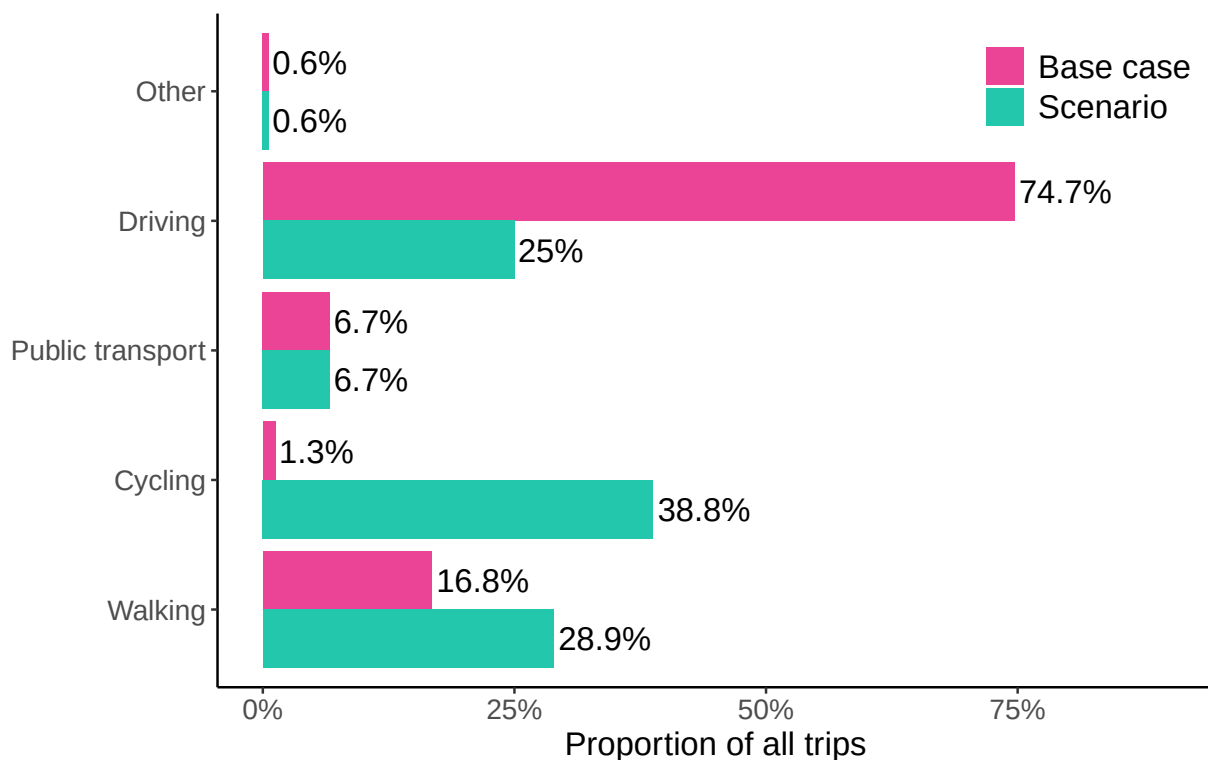
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12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577-1590.

## Scenario: replacing car trips under 2km with walking, and car trips between 2 and 10km with cycling for all trip purposes

This scenario shows the results of replacing car trips under 2km with walking and replacing car trips between 2km and 10km with cycling for leisure, shopping, work, education or other purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 28.9%; cycling from 1.3% to 38.8%; and, from 74.7% to 25.0% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 75.3% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

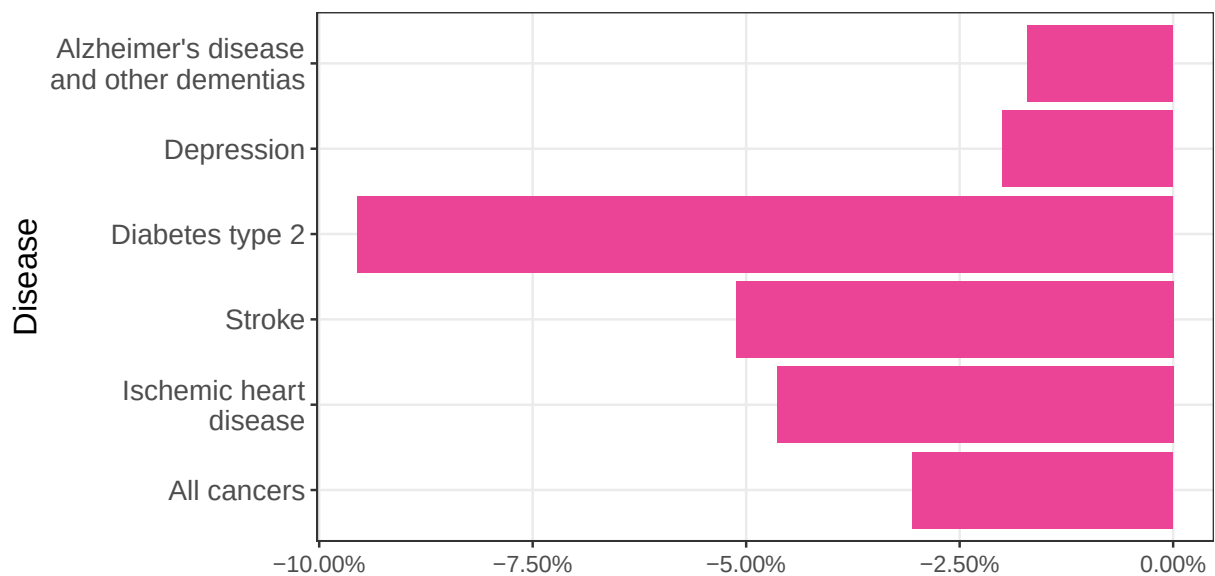
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

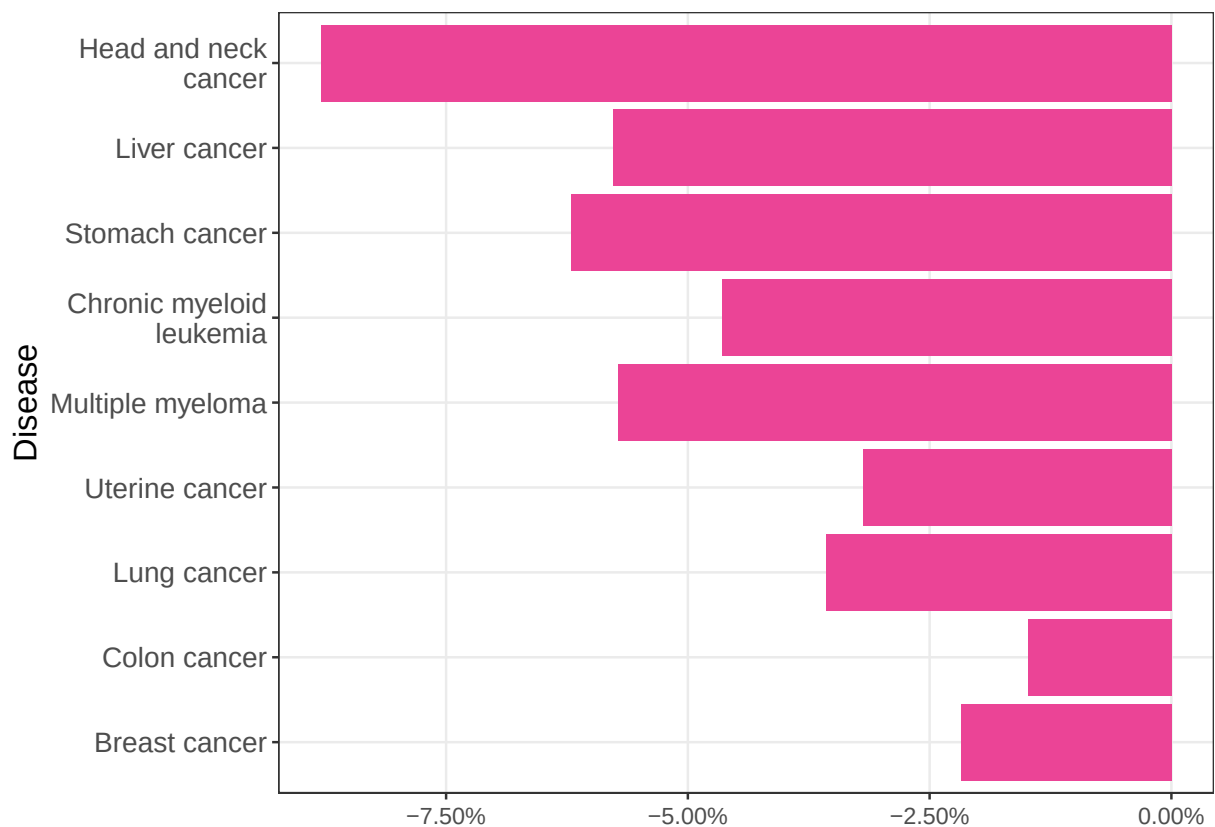
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 1.71%                              | 10,863                                                                      |
| Breast cancer                           | 2.17%                              | 1,592                                                                       |
| All cancers                             | 3.05%                              | 12,020                                                                      |
| Colon cancer                            | 1.49%                              | 1,925                                                                       |
| Chronic myeloid leukemia                | 4.64%                              | 127                                                                         |
| Diabetes type 2                         | 9.55%                              | 31,522                                                                      |
| Depression                              | 2.00%                              | 32,012                                                                      |
| Head and neck cancer                    | 8.79%                              | 388                                                                         |
| Ischemic heart disease                  | 4.64%                              | 40,838                                                                      |
| Liver cancer                            | 5.77%                              | 1,015                                                                       |
| Multiple myeloma                        | 5.72%                              | 1,110                                                                       |
| Stomach cancer                          | 6.21%                              | 1,483                                                                       |
| Stroke                                  | 5.12%                              | 13,320                                                                      |
| Lung cancer                             | 3.57%                              | 4,108                                                                       |
| Uterine cancer                          | 3.19%                              | 271                                                                         |

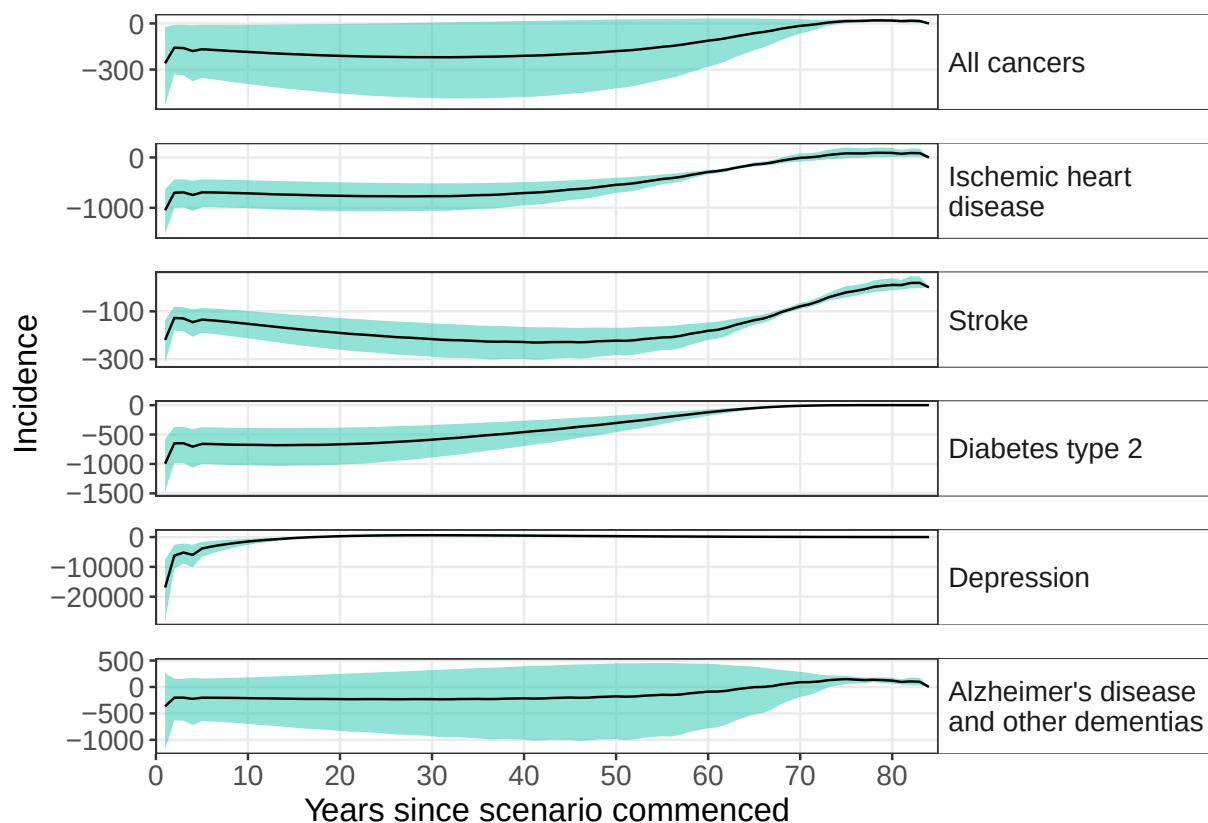
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



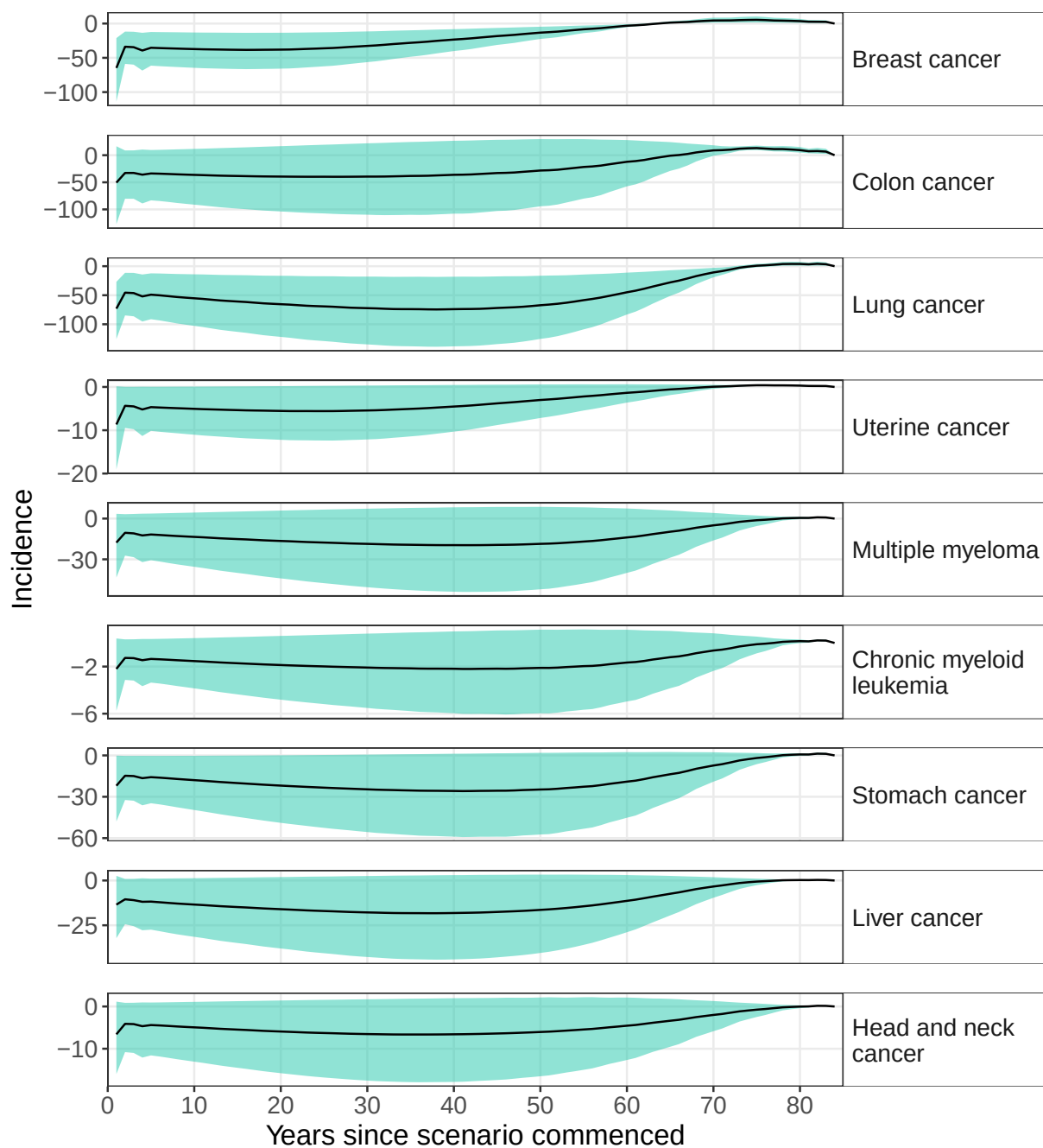
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

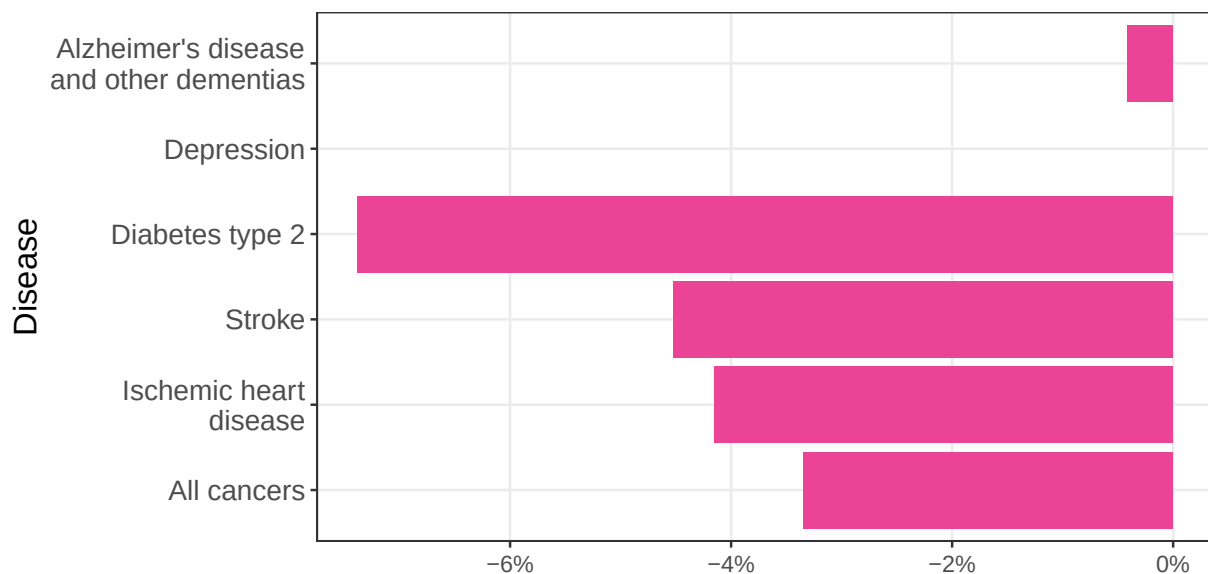
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

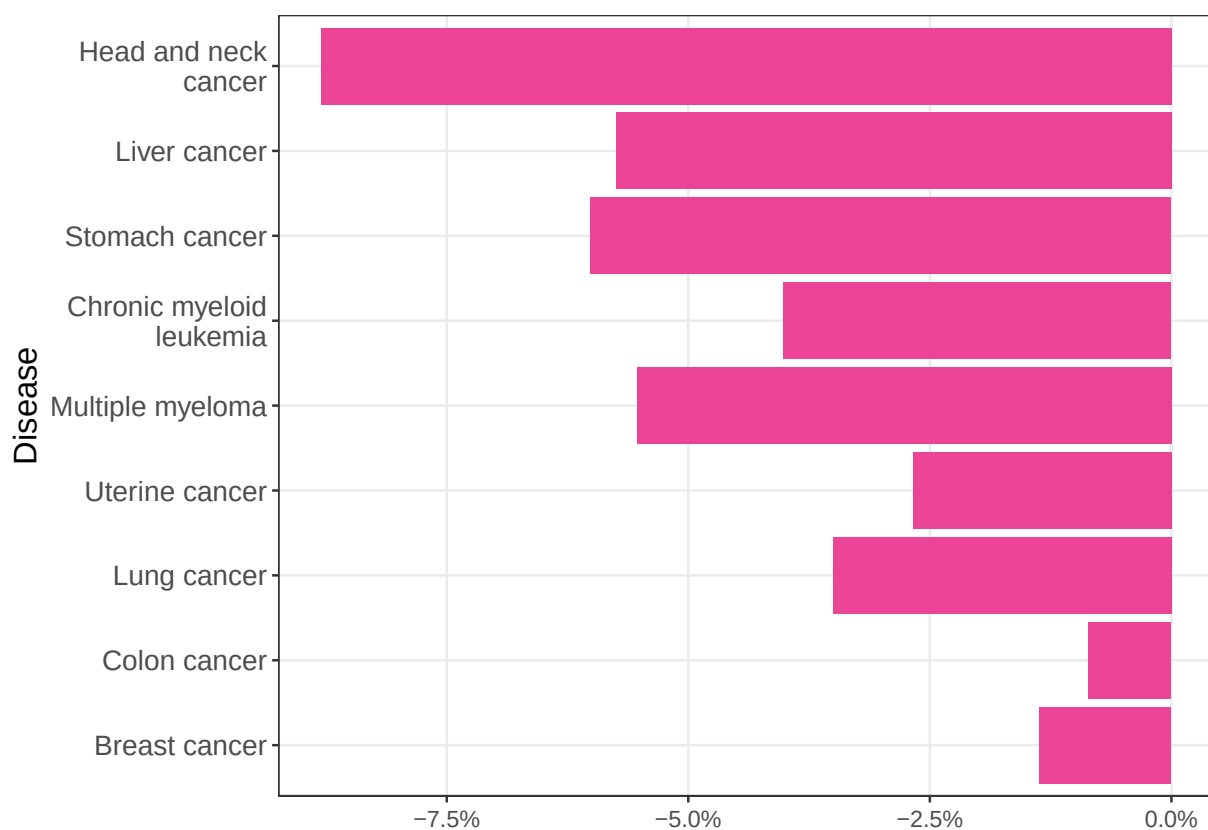
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | 0.41%                   | 986                                                               |
| Breast cancer                           | 1.37%                   | 441                                                               |
| All cancers                             | 3.34%                   | 8,470                                                             |
| Colon cancer                            | 0.86%                   | 369                                                               |
| Chronic myeloid leukemia                | 4.02%                   | 83                                                                |
| Diabetes type 2                         | 7.38%                   | 3,647                                                             |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 8.80%                   | 350                                                               |
| Ischemic heart disease                  | 4.15%                   | 15,580                                                            |
| Liver cancer                            | 5.75%                   | 981                                                               |
| Multiple myeloma                        | 5.53%                   | 963                                                               |
| Stomach cancer                          | 6.01%                   | 1,271                                                             |
| Stroke                                  | 4.52%                   | 7,913                                                             |
| Lung cancer                             | 3.50%                   | 3,818                                                             |
| Uterine cancer                          | 2.67%                   | 193                                                               |

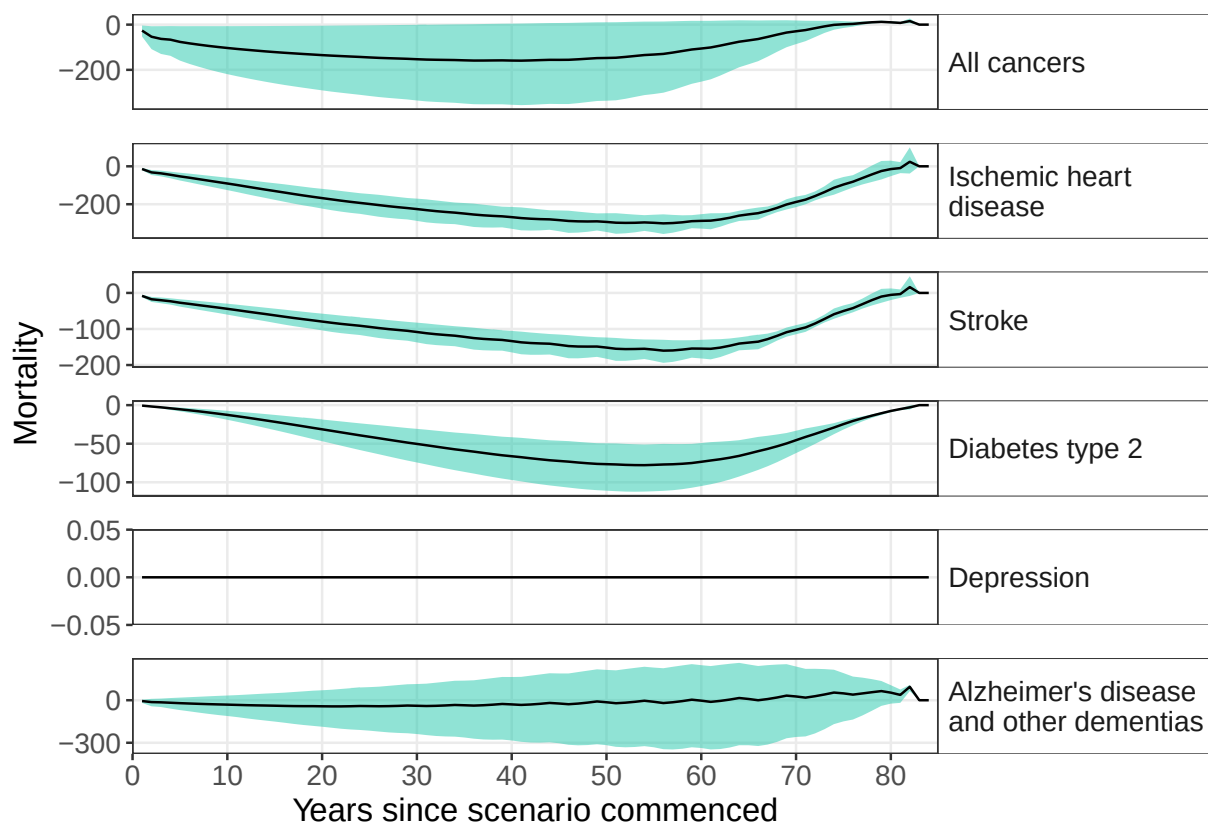
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

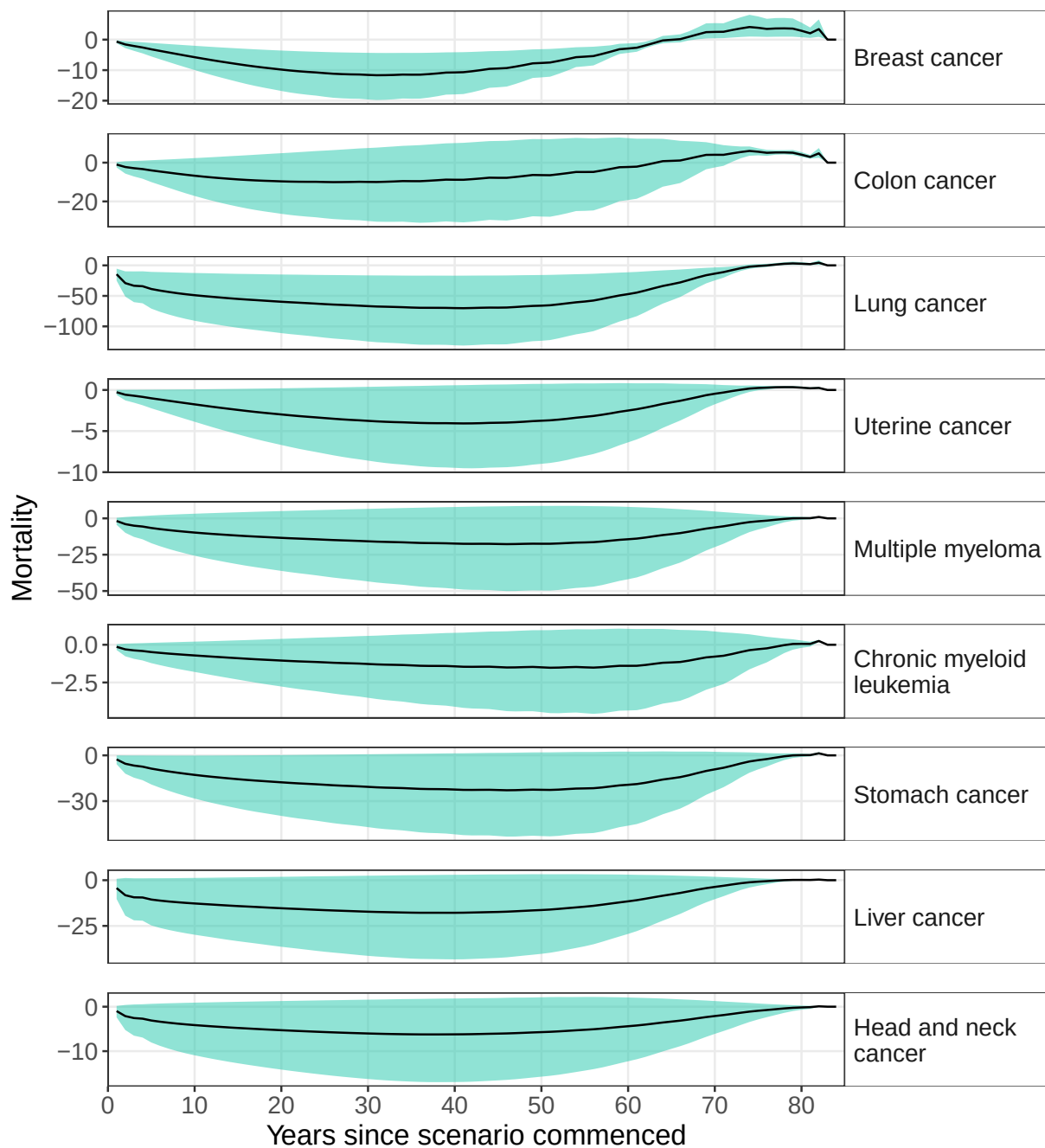


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

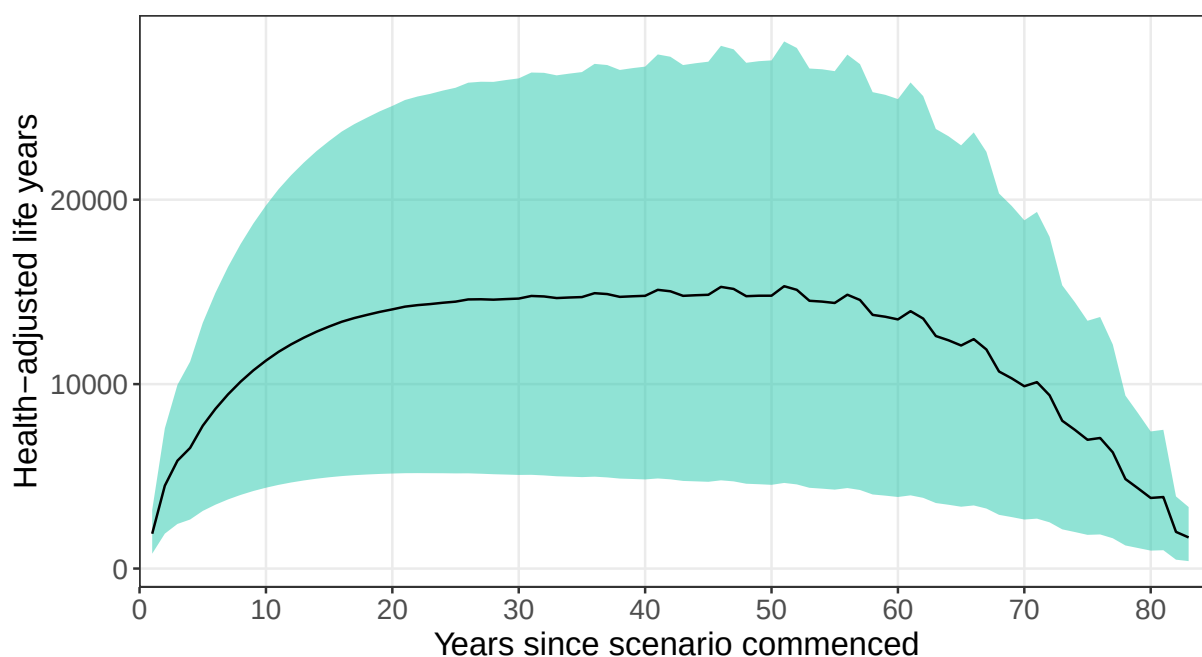
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **998,331** HALYs for the scenario population, which is **496** HALYs per 1,000 members of the population.



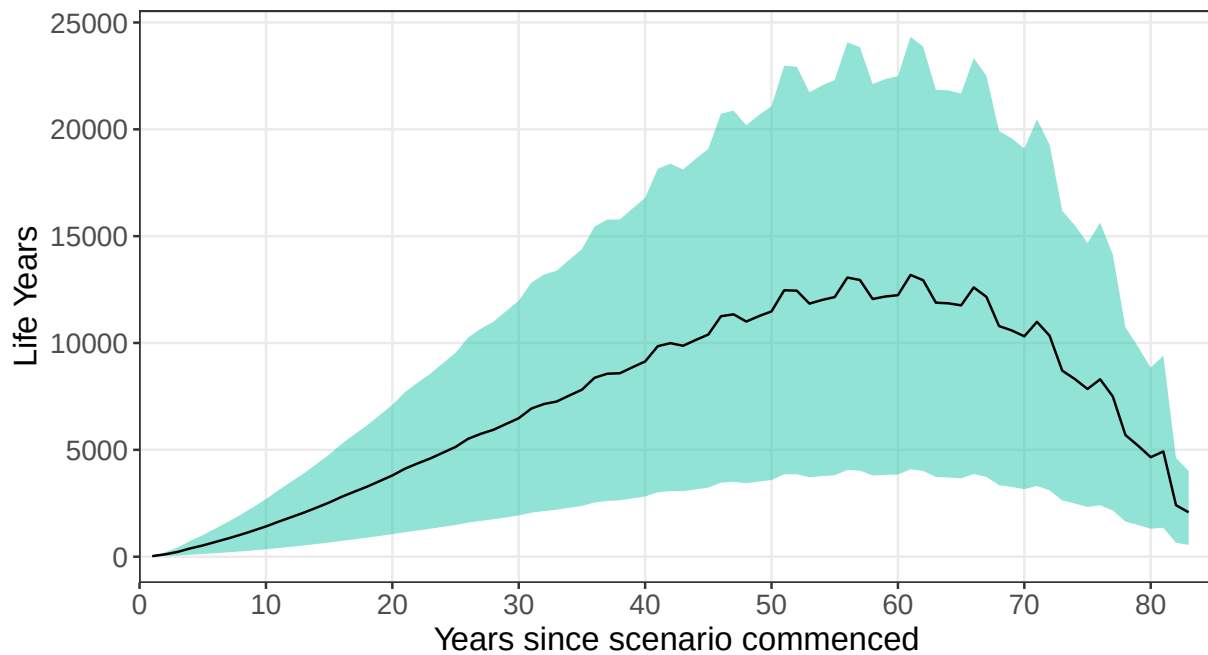
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **610,316** Life Years for the scenario population, which is **303** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **496** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **38,471,515** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **23,512,769** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **15,949,603** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$30,975         | \$18,830         | \$12,503         |
| Breast cancer                           | \$325,174        | \$203,753        | \$136,630        |
| All cancers                             | \$707,775        | \$447,877        | \$304,320        |
| Colon cancer                            | \$154,702        | \$101,373        | \$70,196         |
| Chronic myeloid leukemia                | \$33,842         | \$20,263         | \$13,300         |
| Diabetes type 2                         | \$192,892        | \$113,599        | \$72,897         |
| Depression                              | \$949,655        | \$719,344        | \$559,301        |
| Head and neck cancer                    | \$5,669          | \$3,668          | \$2,578          |
| Ischemic heart disease                  | \$388,941        | \$229,306        | \$147,977        |
| Liver cancer                            | \$5,934          | \$3,890          | \$2,764          |
| Multiple myeloma                        | \$81,278         | \$50,820         | \$34,624         |
| Stomach cancer                          | \$36,000         | \$22,389         | \$15,201         |
| Stroke                                  | \$57,365         | \$33,558         | \$21,581         |
| Lung cancer                             | \$49,985         | \$32,286         | \$22,653         |
| Uterine cancer                          | \$15,239         | \$9,516          | \$6,393          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

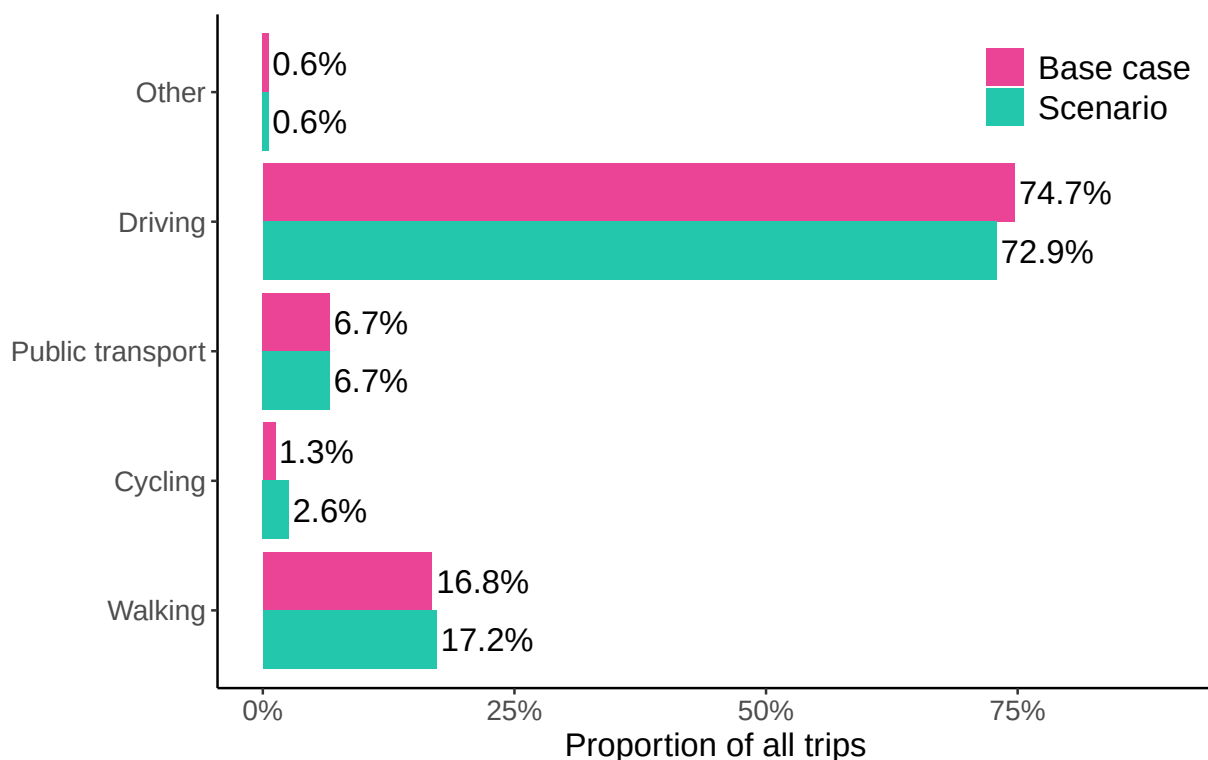
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2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
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9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
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## Scenario: replacing car trips under 1km with walking, and car trips between 1 and 2km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 1km with walking and replacing car trips between 1km and 2km with cycling for work related or education purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 17.2%; cycling from 1.3% to 2.6%; and, from 74.7% to 72.9% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 49.0% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

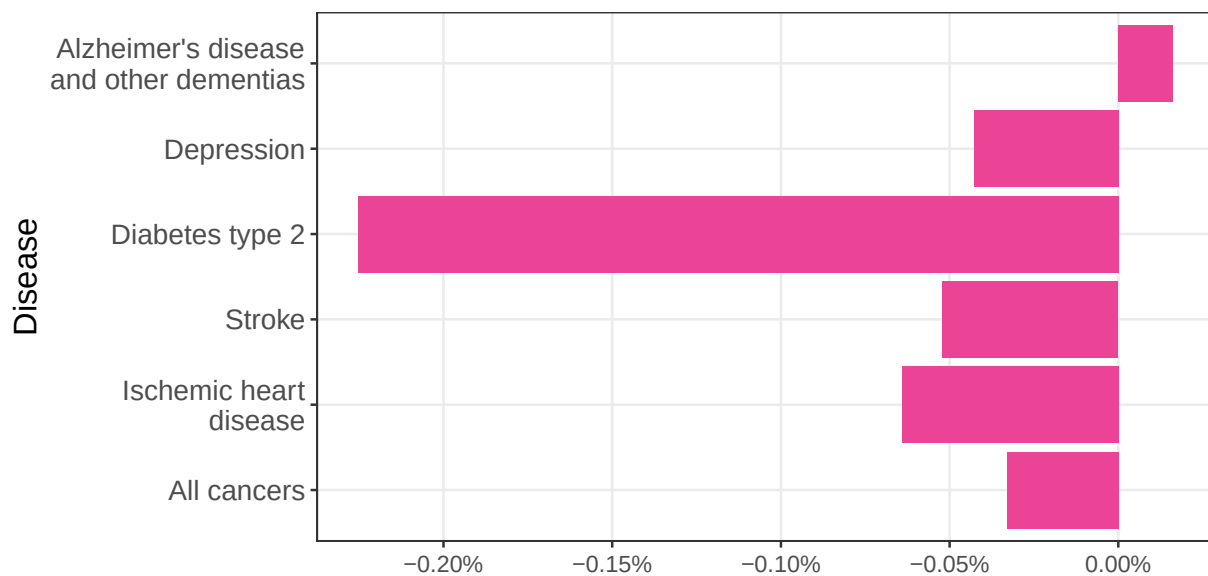
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

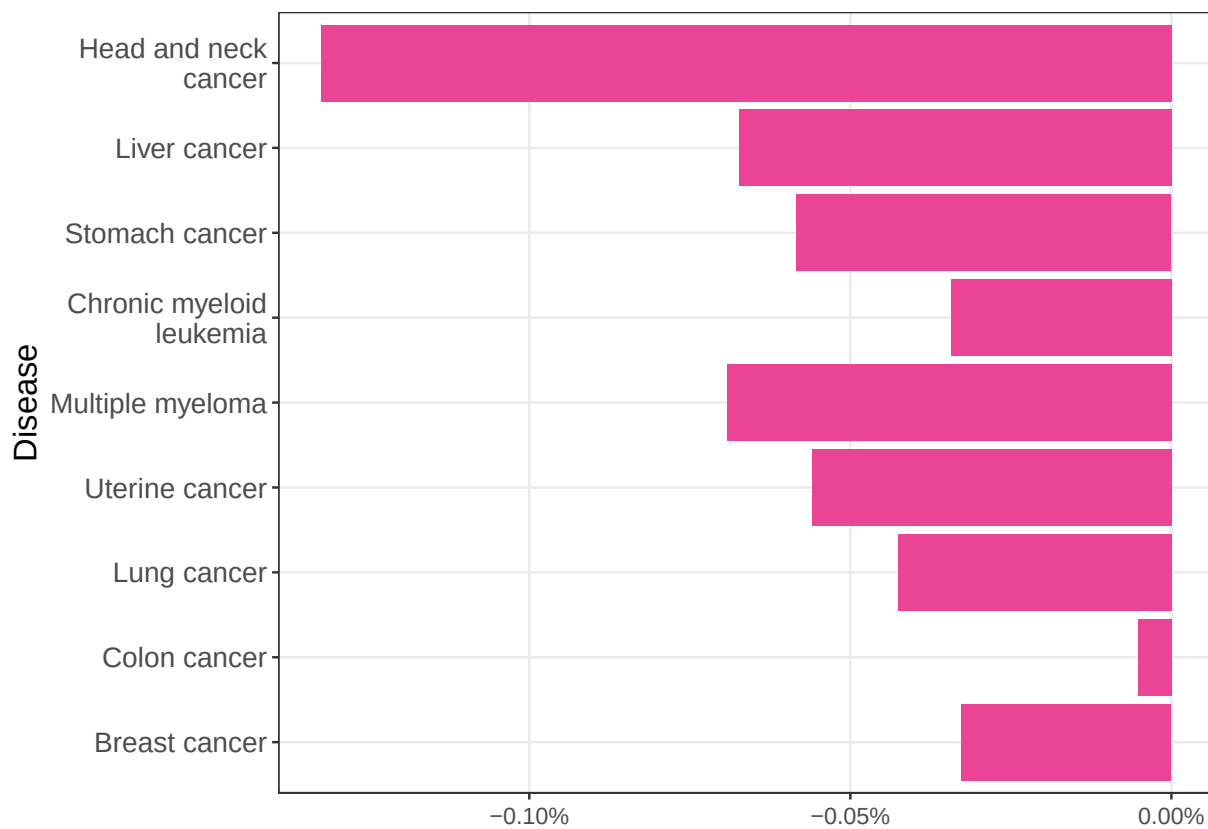
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.02%                             | -103                                                                        |
| Breast cancer                           | 0.03%                              | 24                                                                          |
| All cancers                             | 0.03%                              | 130                                                                         |
| Colon cancer                            | 0.01%                              | 7                                                                           |
| Chronic myeloid leukemia                | 0.03%                              | 1                                                                           |
| Diabetes type 2                         | 0.23%                              | 744                                                                         |
| Depression                              | 0.04%                              | 686                                                                         |
| Head and neck cancer                    | 0.13%                              | 6                                                                           |
| Ischemic heart disease                  | 0.06%                              | 565                                                                         |
| Liver cancer                            | 0.07%                              | 12                                                                          |
| Multiple myeloma                        | 0.07%                              | 13                                                                          |
| Stomach cancer                          | 0.06%                              | 14                                                                          |
| Stroke                                  | 0.05%                              | 136                                                                         |
| Lung cancer                             | 0.04%                              | 49                                                                          |
| Uterine cancer                          | 0.06%                              | 5                                                                           |

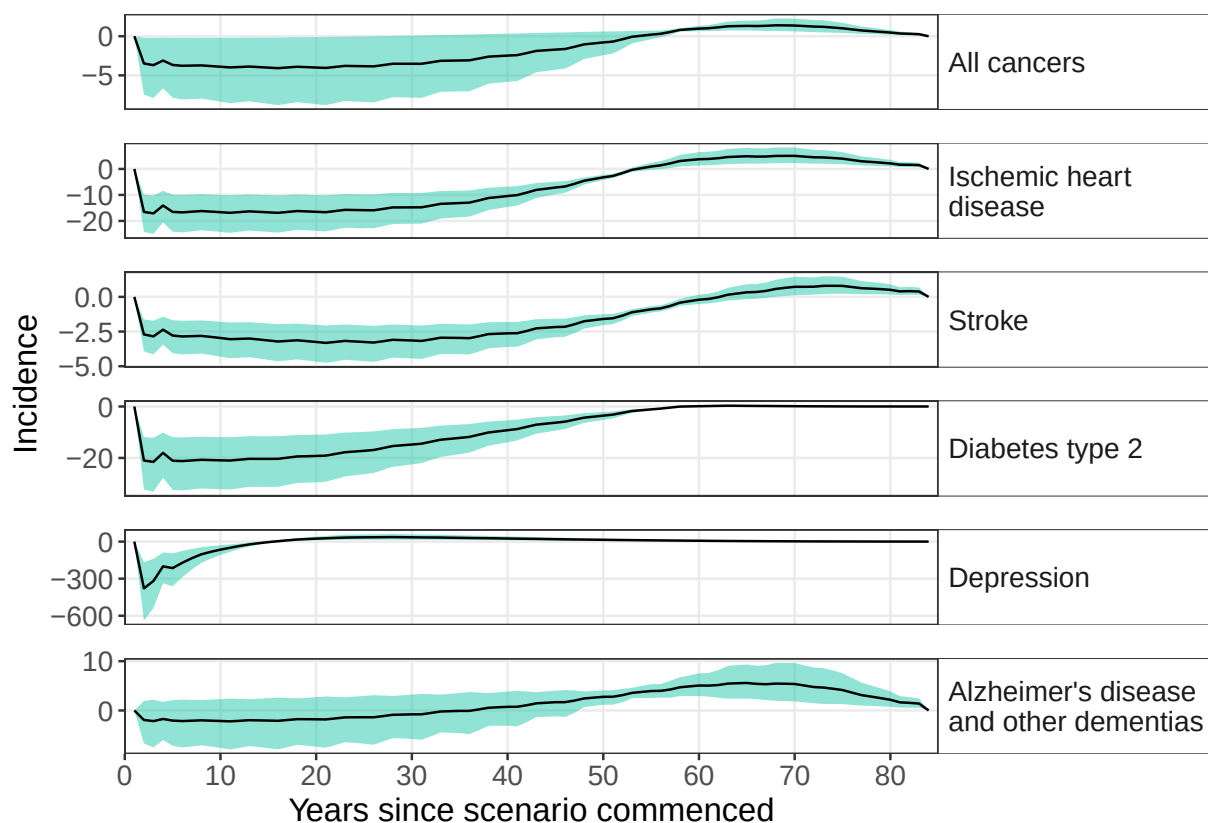
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



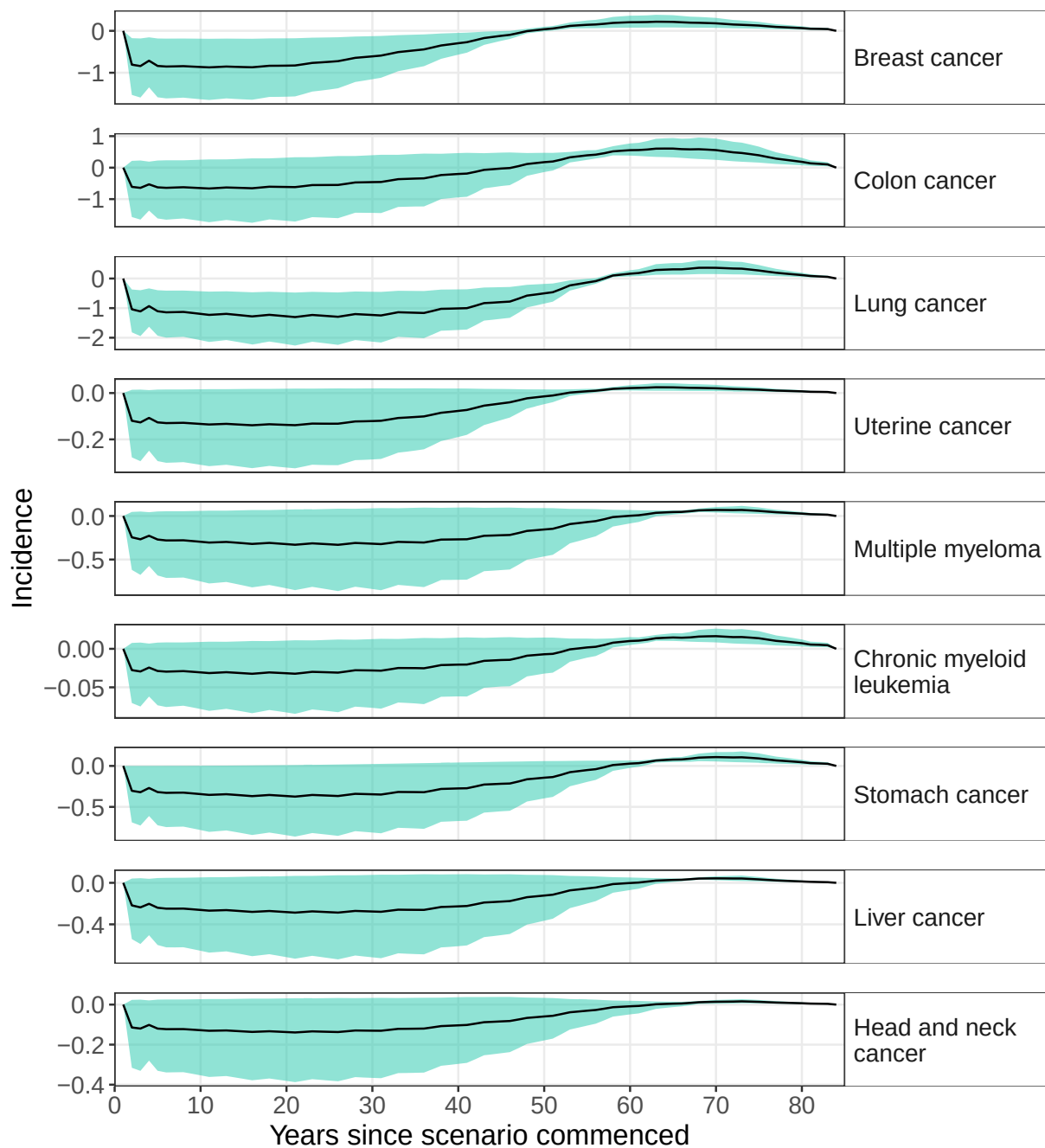
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

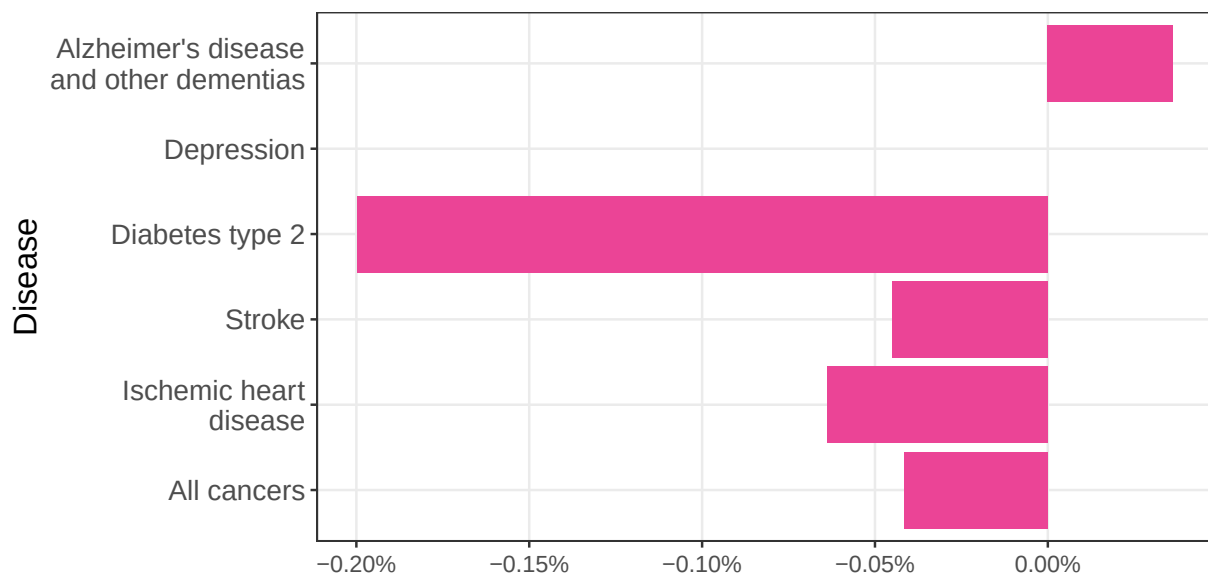
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

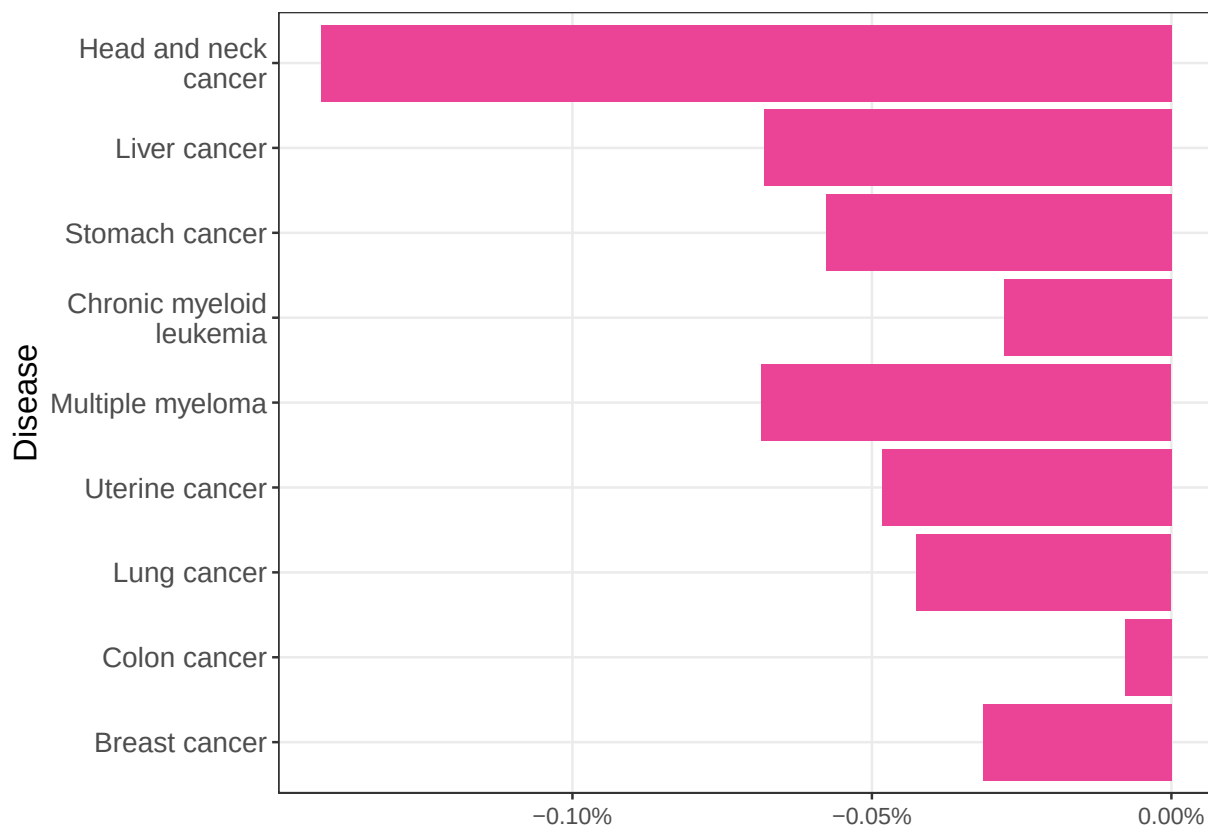
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.04%                  | -87                                                               |
| Breast cancer                           | 0.03%                   | 10                                                                |
| All cancers                             | 0.04%                   | 105                                                               |
| Colon cancer                            | 0.01%                   | 3                                                                 |
| Chronic myeloid leukemia                | 0.03%                   | 1                                                                 |
| Diabetes type 2                         | 0.20%                   | 99                                                                |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.14%                   | 6                                                                 |
| Ischemic heart disease                  | 0.06%                   | 239                                                               |
| Liver cancer                            | 0.07%                   | 12                                                                |
| Multiple myeloma                        | 0.07%                   | 12                                                                |
| Stomach cancer                          | 0.06%                   | 12                                                                |
| Stroke                                  | 0.05%                   | 79                                                                |
| Lung cancer                             | 0.04%                   | 46                                                                |
| Uterine cancer                          | 0.05%                   | 3                                                                 |

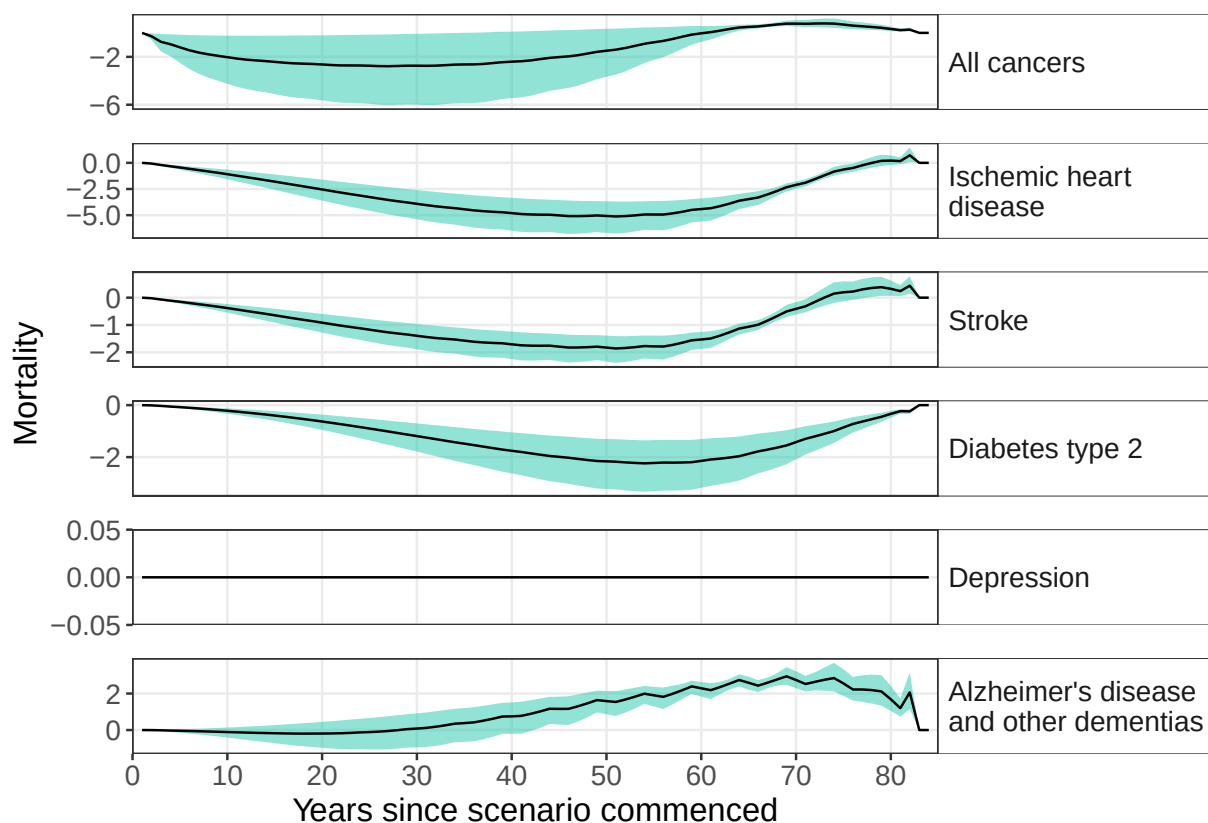
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

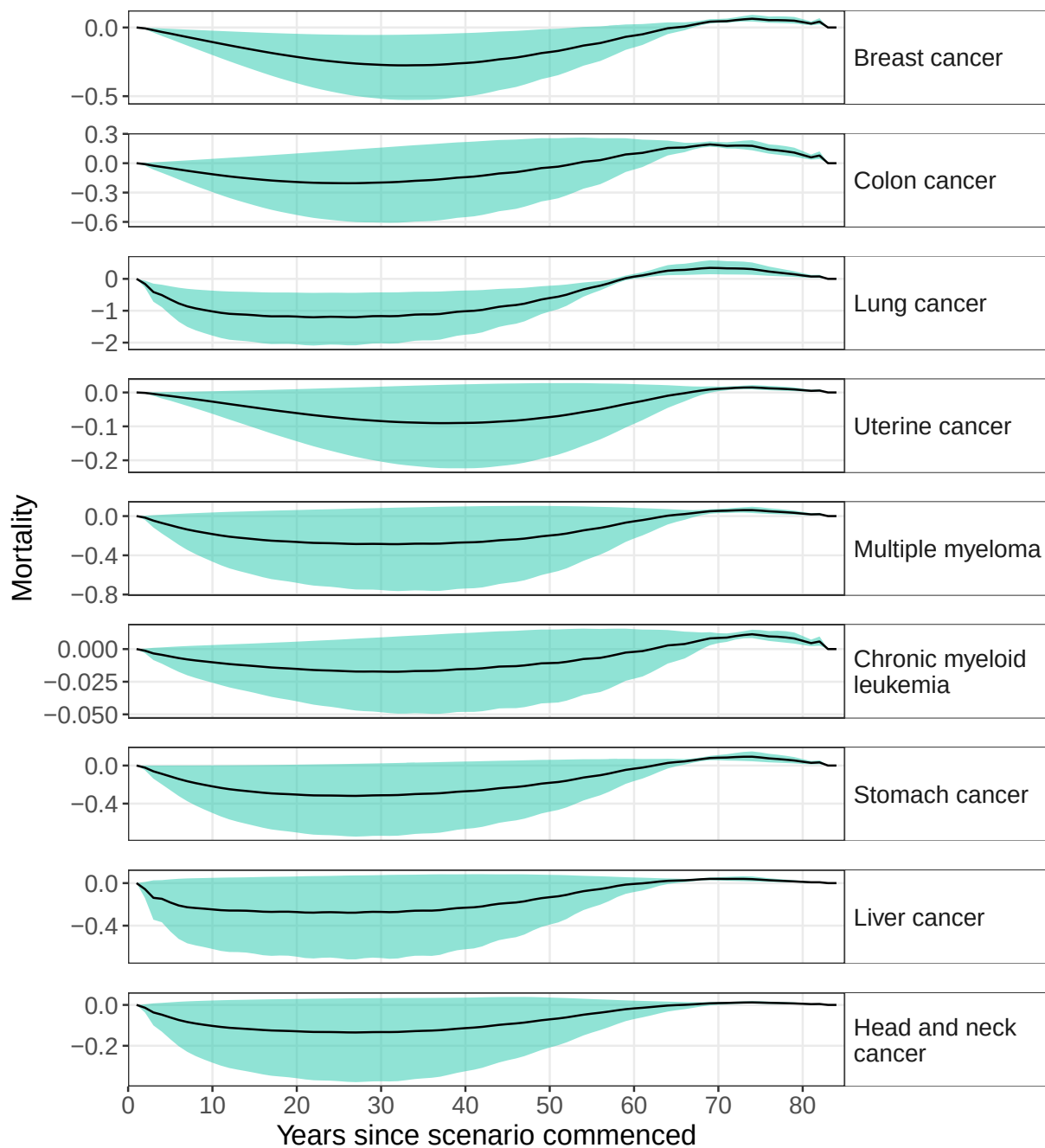


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **24,282** HALYs for the scenario population, which is **12** HALYs per 1,000 members of the population.



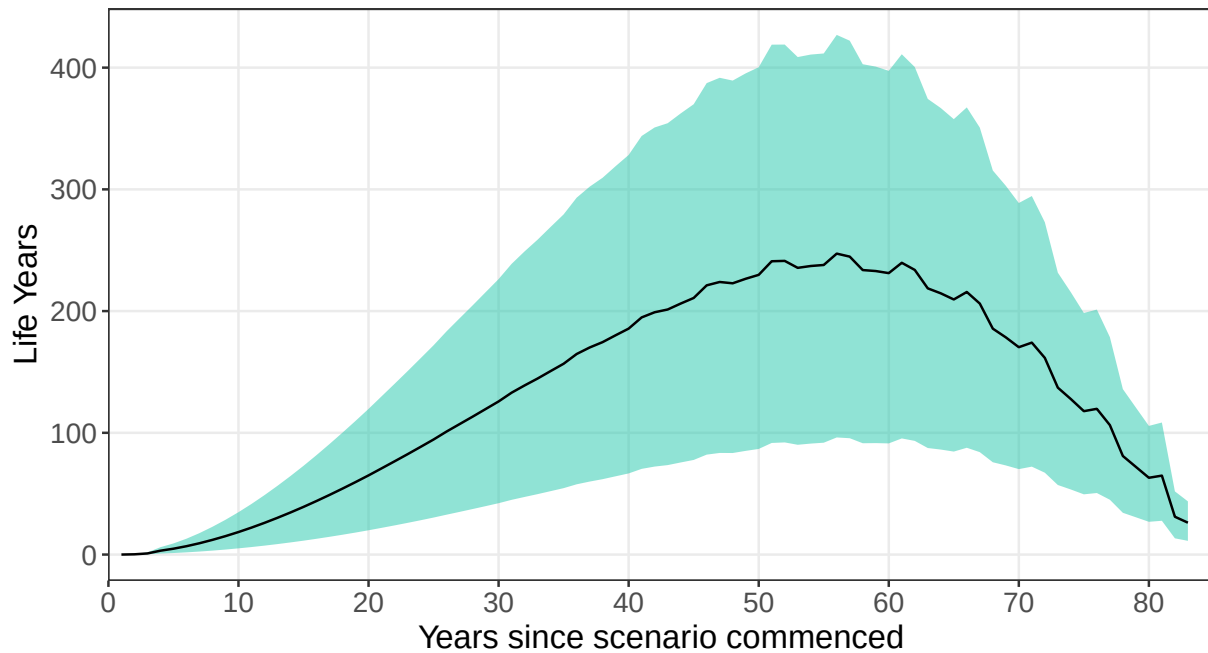
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **11,058** Life Years for the scenario population, which is **5.5** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **12** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **1,028,474** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **649,894** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **447,444** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$277            | \$194            | \$137            |
| Breast cancer                           | \$7,542          | \$4,629          | \$3,027          |
| All cancers                             | \$15,672         | \$9,955          | \$6,696          |
| Colon cancer                            | \$2,763          | \$1,889          | \$1,318          |
| Chronic myeloid leukemia                | \$826            | \$499            | \$324            |
| Diabetes type 2                         | \$5,820          | \$3,379          | \$2,129          |
| Depression                              | \$33,823         | \$24,626         | \$18,571         |
| Head and neck cancer                    | \$142            | \$97             | \$69             |
| Ischemic heart disease                  | \$10,062         | \$5,880          | \$3,719          |
| Liver cancer                            | \$119            | \$82             | \$59             |
| Multiple myeloma                        | \$2,001          | \$1,280          | \$874            |
| Stomach cancer                          | \$731            | \$471            | \$324            |
| Stroke                                  | \$1,385          | \$810            | \$513            |
| Lung cancer                             | \$1,117          | \$746            | \$529            |
| Uterine cancer                          | \$429            | \$264            | \$174            |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

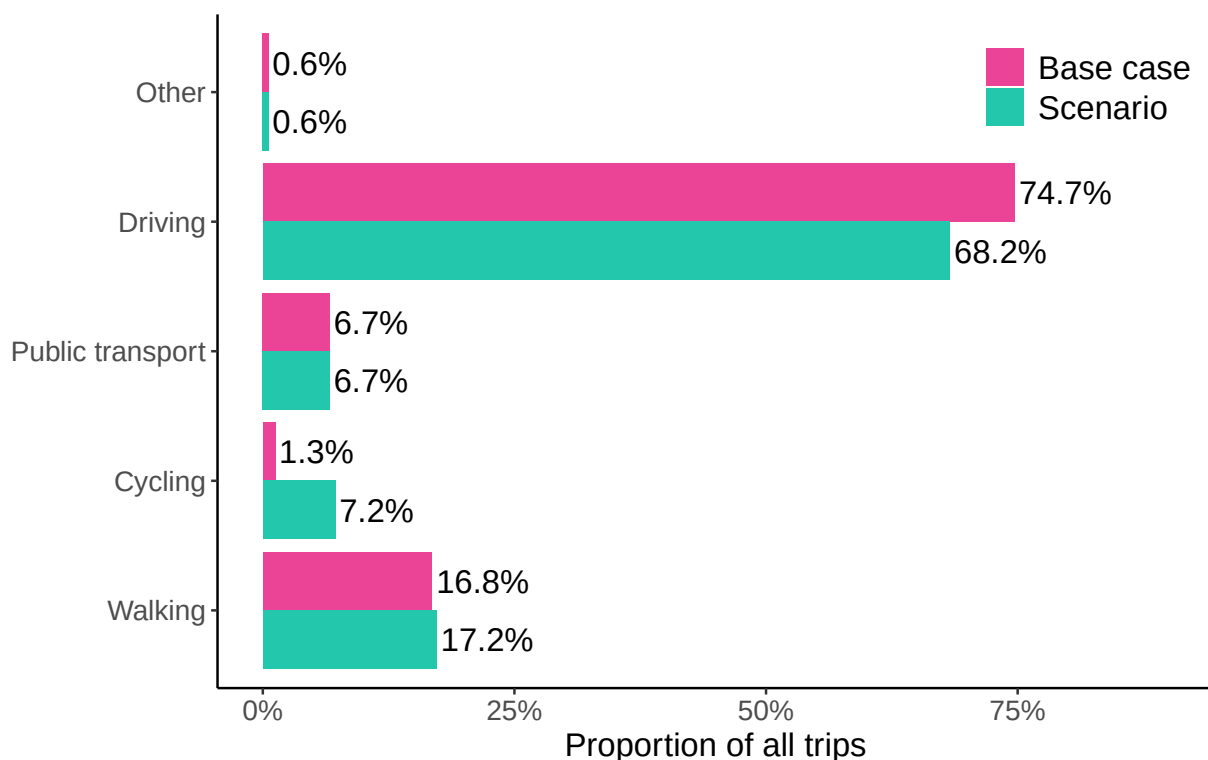
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
3. Ananthapavan, J., Moodie, M., Milat, A.J., & Carter, R. (2021). *Systematic review to update 'value of a statistical life' estimates for Australia*. International journal of environmental research and public health, 2021. 18(11): p. 6168.
4. Terrill, M. & Batrouney, H. (2018). *Unfreezing discount rates: Transport infrastructure for tomorrow*. Grattan Institute.
5. Abelson, P. (2008). *Establishing a monetary value for lives saved: issues and controversies*. Canberra: Office of Best Practice Regulation, Department of Finance and Deregulation.
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7. Haacker, M., Hallett, T.B. & Atun, R. (2020). *On discount rates for economic evaluations in global health*. Health Policy and Planning, 2020. 35(1): p. 107-114.
8. Australian Institute of Health and Welfare (2023). Technical Notes: Estimating Spending per prevalent case of disease. Health system spending per case of disease and for certain risk factors, Estimating the spending per prevalent case of disease - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
11. Zapata-Diomedí, B., Boulangé, C., Giles-Corti, B., Phelan, K., Washington, S., Veerman, L.J., & Gunn, L. (2019). Physical activity-related health and economic benefits of building walkable neighbourhoods: A modelled comparison between brownfield and greenfield developments. International Journal of Behavioural Nutrition and Physical Activity.
12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577-1590.

## Scenario: replacing car trips under 1km with walking, and car trips between 1 and 5km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 1km with walking and replacing car trips between 1km and 5km with cycling for work related or education purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 17.2%; cycling from 1.3% to 7.2%; and, from 74.7% to 68.2% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 53.2% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

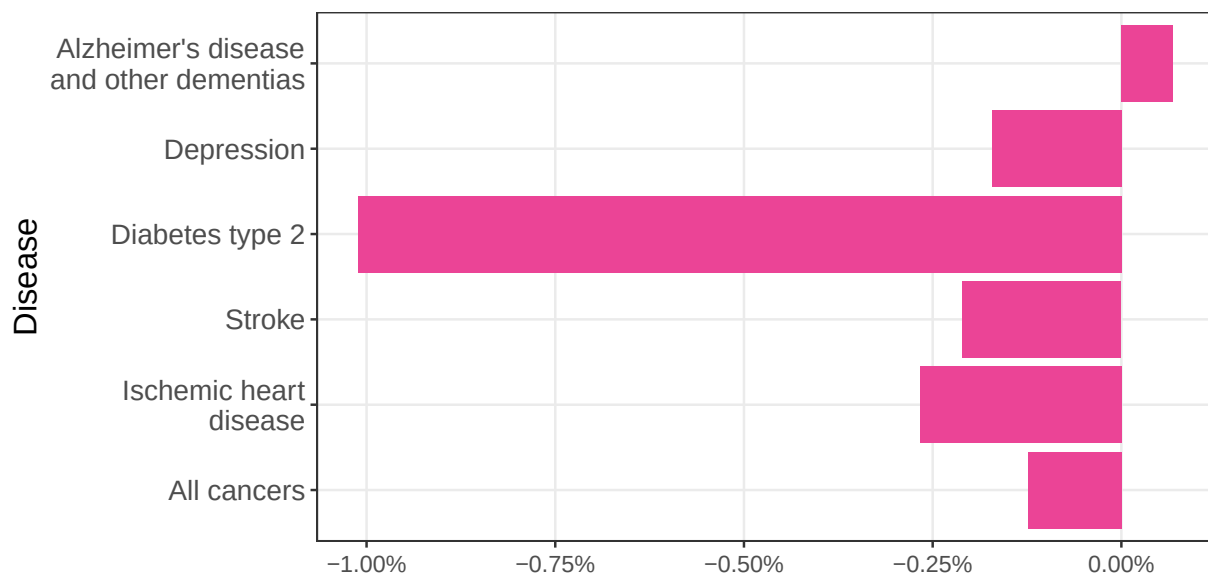
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

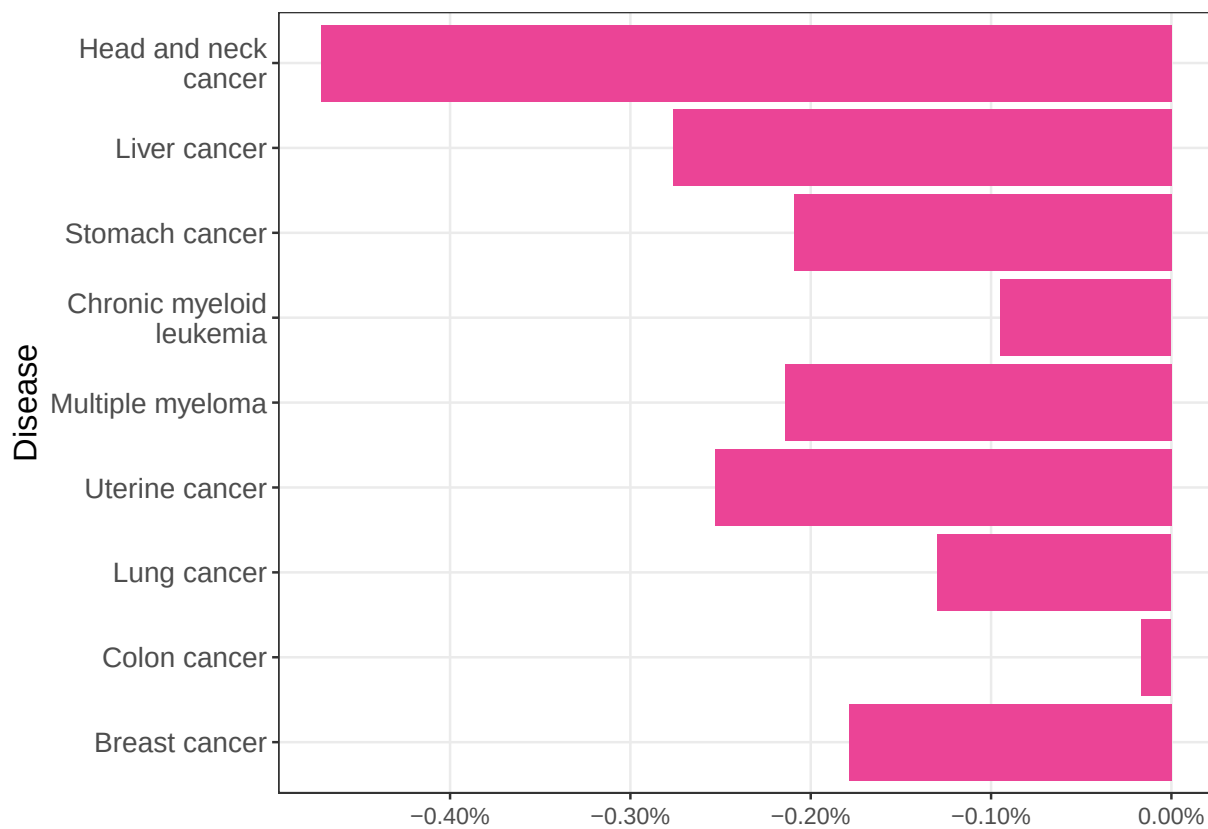
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.07%                             | -437                                                                        |
| Breast cancer                           | 0.18%                              | 131                                                                         |
| All cancers                             | 0.12%                              | 486                                                                         |
| Colon cancer                            | 0.02%                              | 22                                                                          |
| Chronic myeloid leukemia                | 0.09%                              | 3                                                                           |
| Diabetes type 2                         | 1.01%                              | 3,340                                                                       |
| Depression                              | 0.17%                              | 2,747                                                                       |
| Head and neck cancer                    | 0.47%                              | 21                                                                          |
| Ischemic heart disease                  | 0.27%                              | 2,347                                                                       |
| Liver cancer                            | 0.28%                              | 49                                                                          |
| Multiple myeloma                        | 0.21%                              | 42                                                                          |
| Stomach cancer                          | 0.21%                              | 50                                                                          |
| Stroke                                  | 0.21%                              | 548                                                                         |
| Lung cancer                             | 0.13%                              | 149                                                                         |
| Uterine cancer                          | 0.25%                              | 21                                                                          |

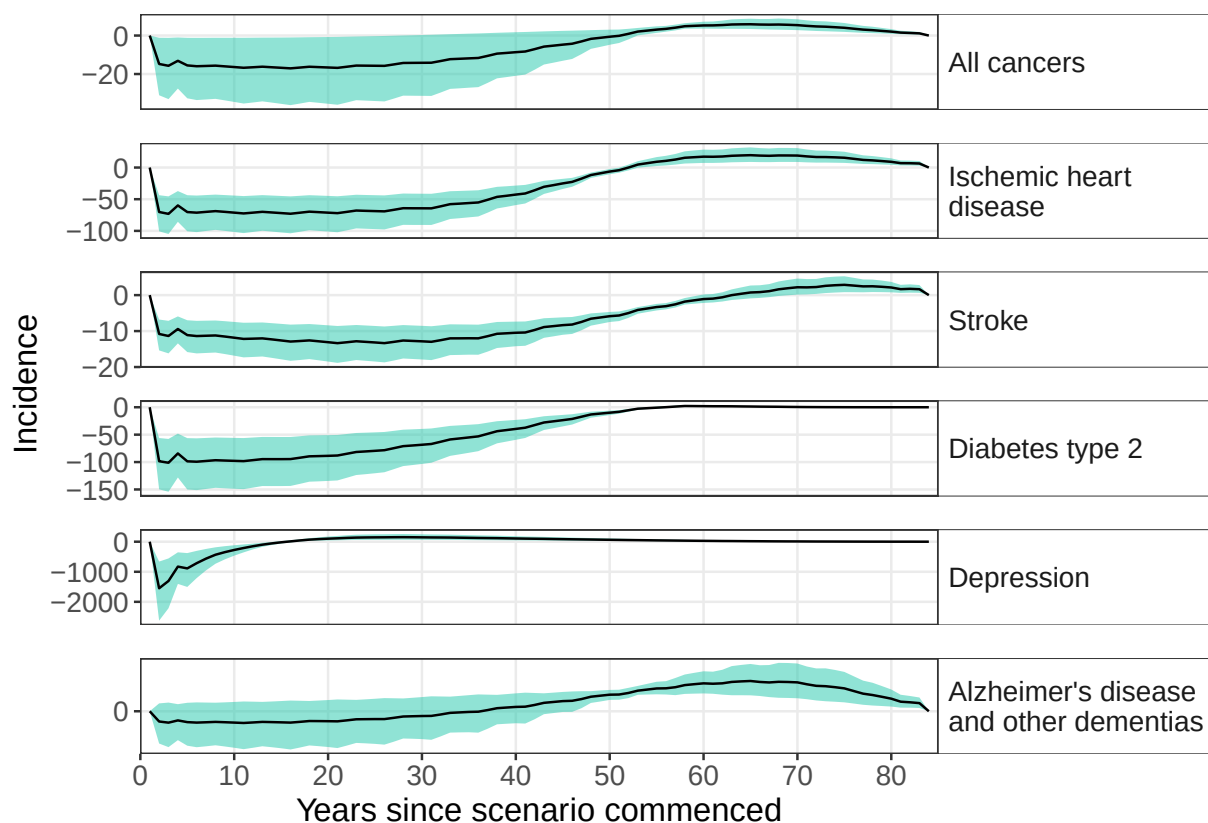
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



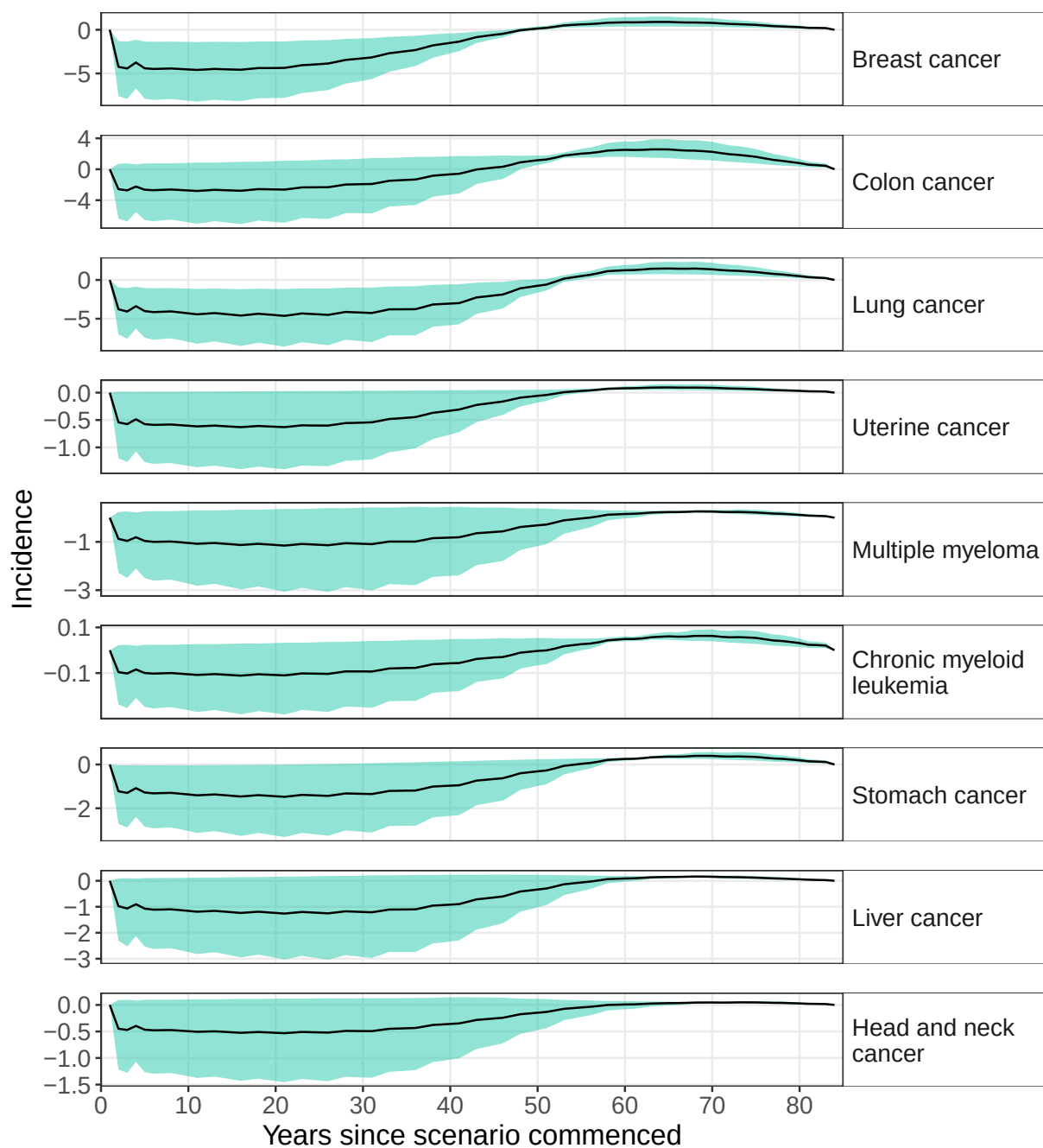
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

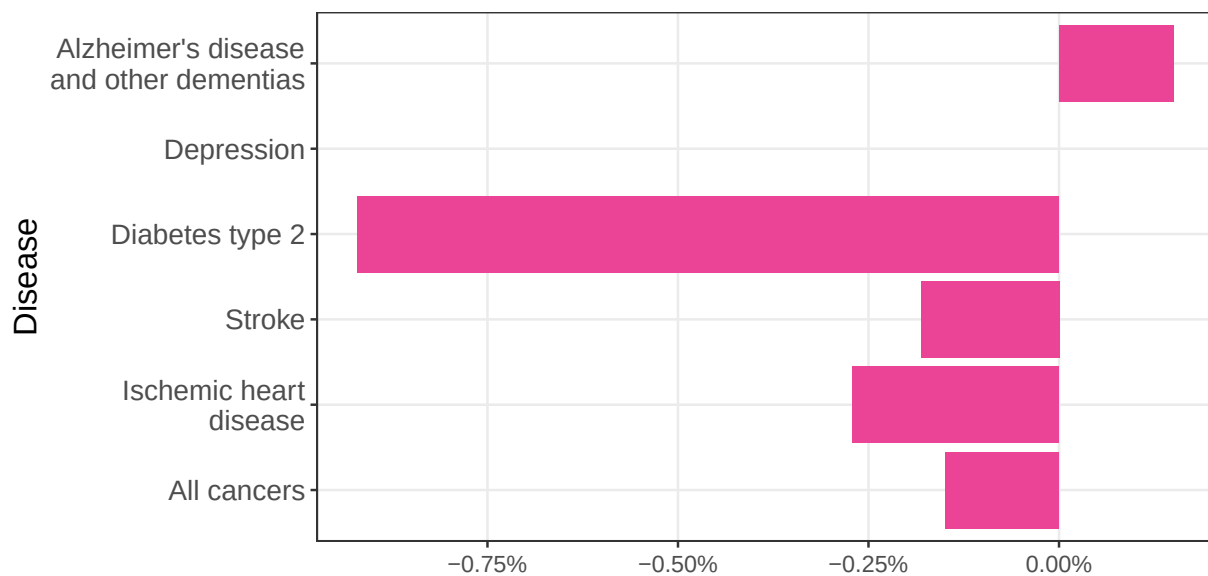
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

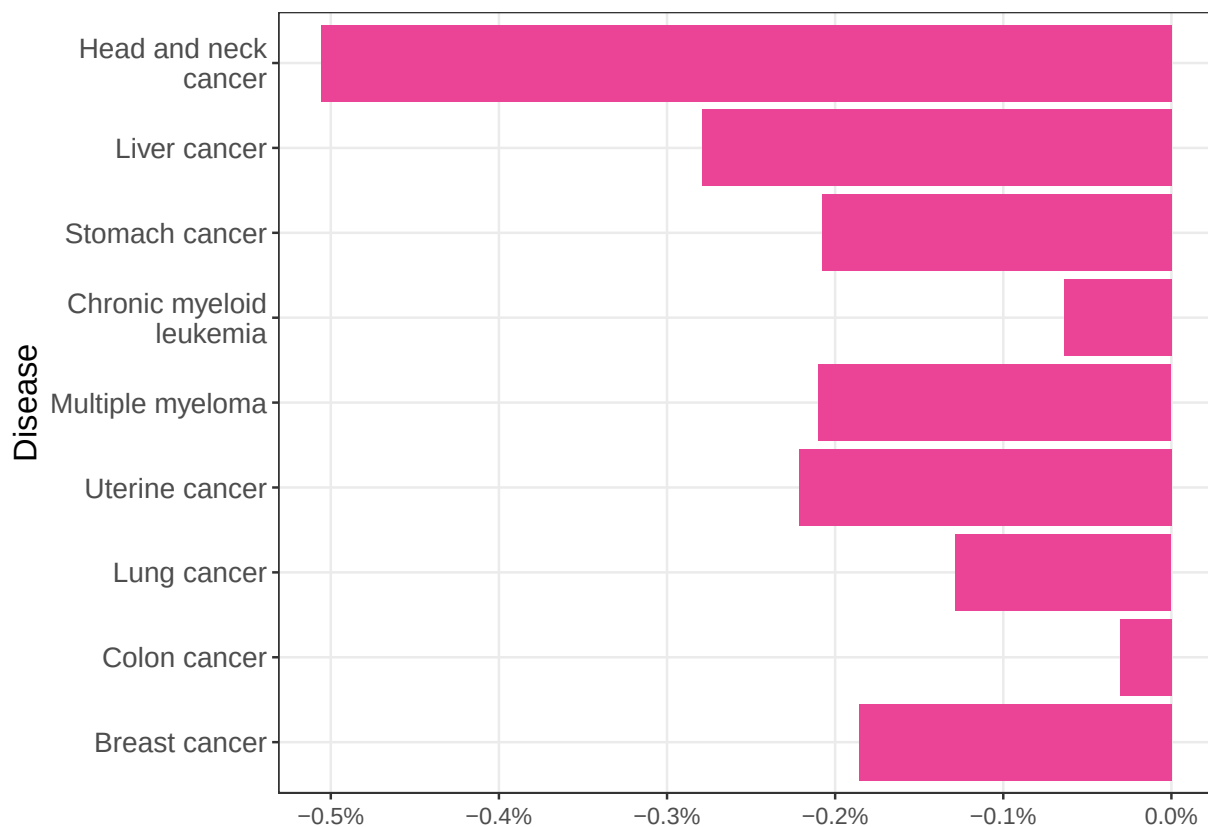
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.15%                  | -358                                                              |
| Breast cancer                           | 0.19%                   | 60                                                                |
| All cancers                             | 0.15%                   | 378                                                               |
| Colon cancer                            | 0.03%                   | 13                                                                |
| Chronic myeloid leukemia                | 0.06%                   | 1                                                                 |
| Diabetes type 2                         | 0.92%                   | 455                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.51%                   | 20                                                                |
| Ischemic heart disease                  | 0.27%                   | 1,017                                                             |
| Liver cancer                            | 0.28%                   | 48                                                                |
| Multiple myeloma                        | 0.21%                   | 37                                                                |
| Stomach cancer                          | 0.21%                   | 44                                                                |
| Stroke                                  | 0.18%                   | 317                                                               |
| Lung cancer                             | 0.13%                   | 140                                                               |
| Uterine cancer                          | 0.22%                   | 16                                                                |

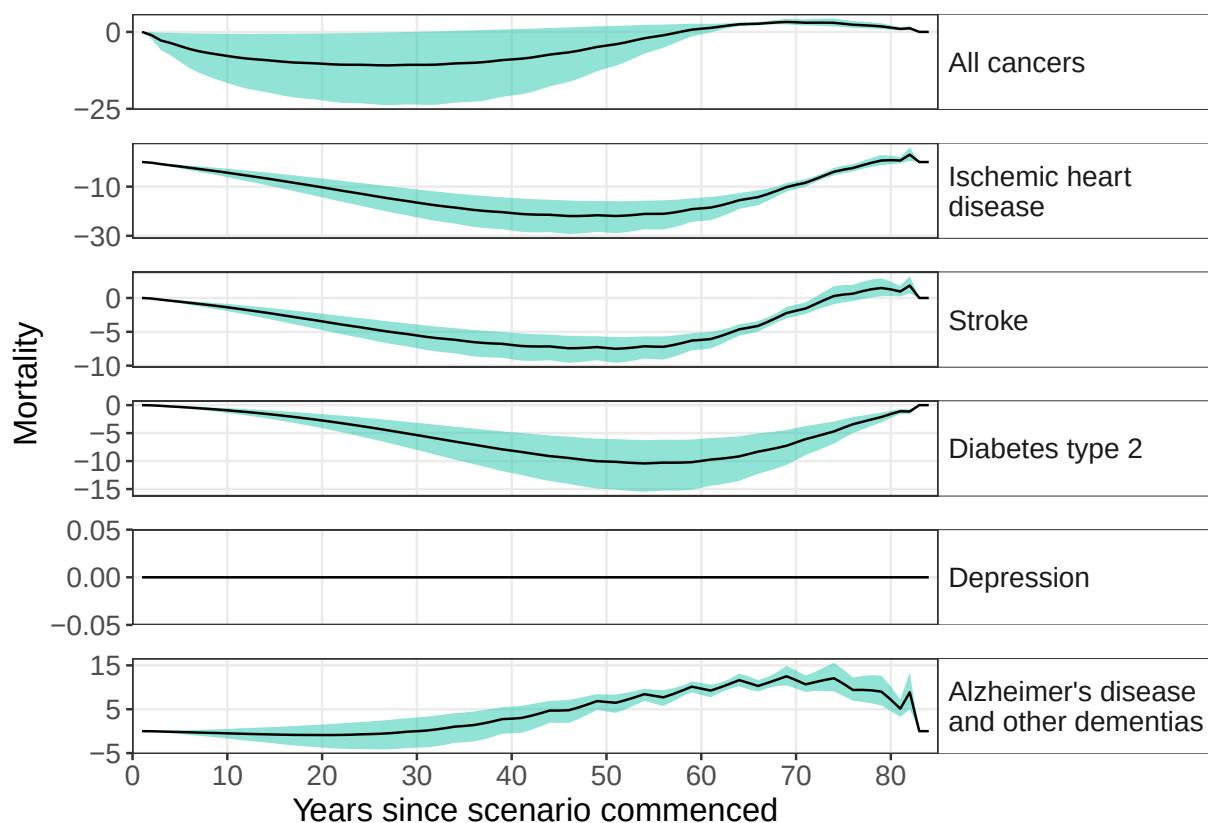
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

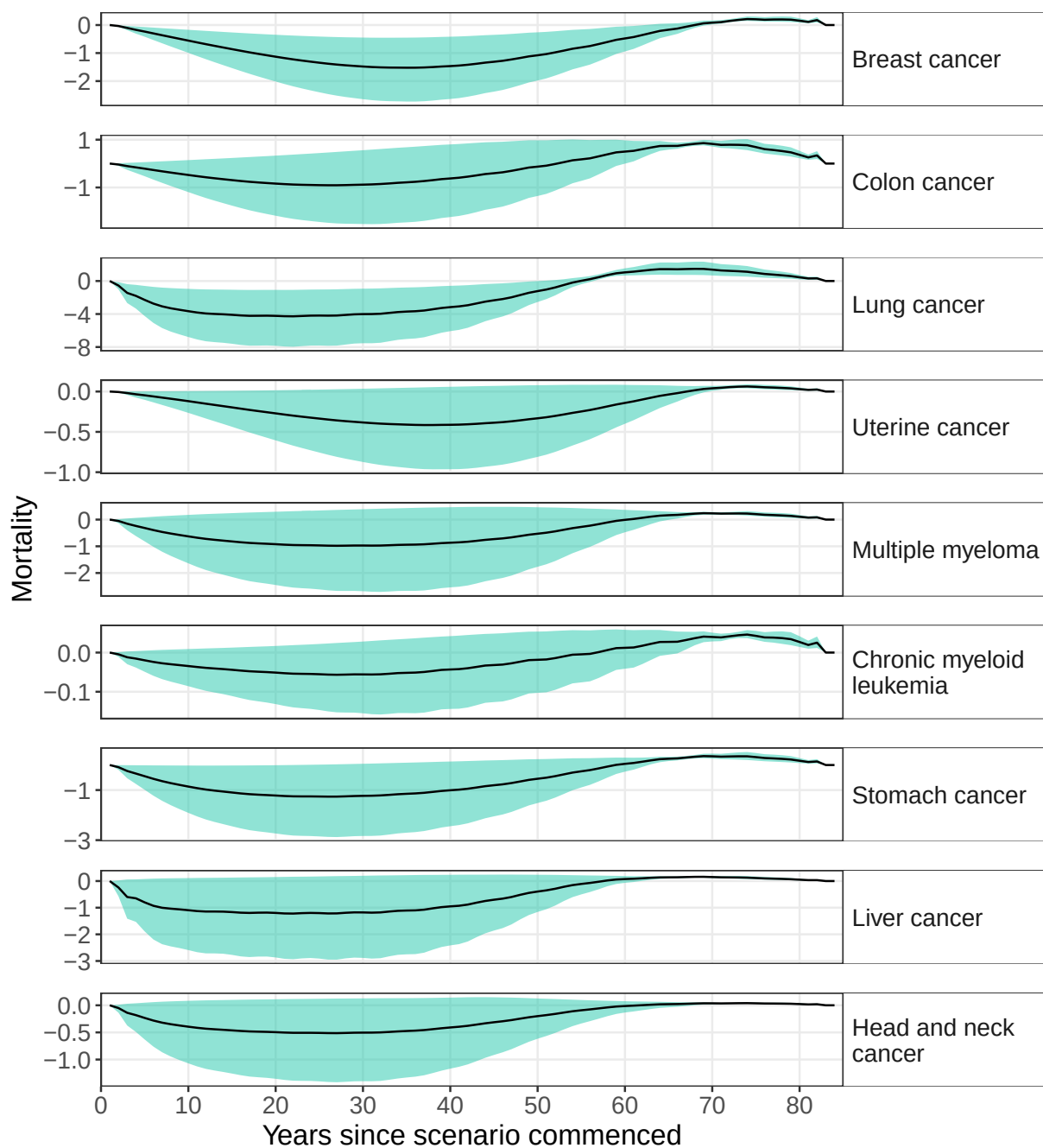


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **102,986** HALYs for the scenario population, which is **51** HALYs per 1,000 members of the population.



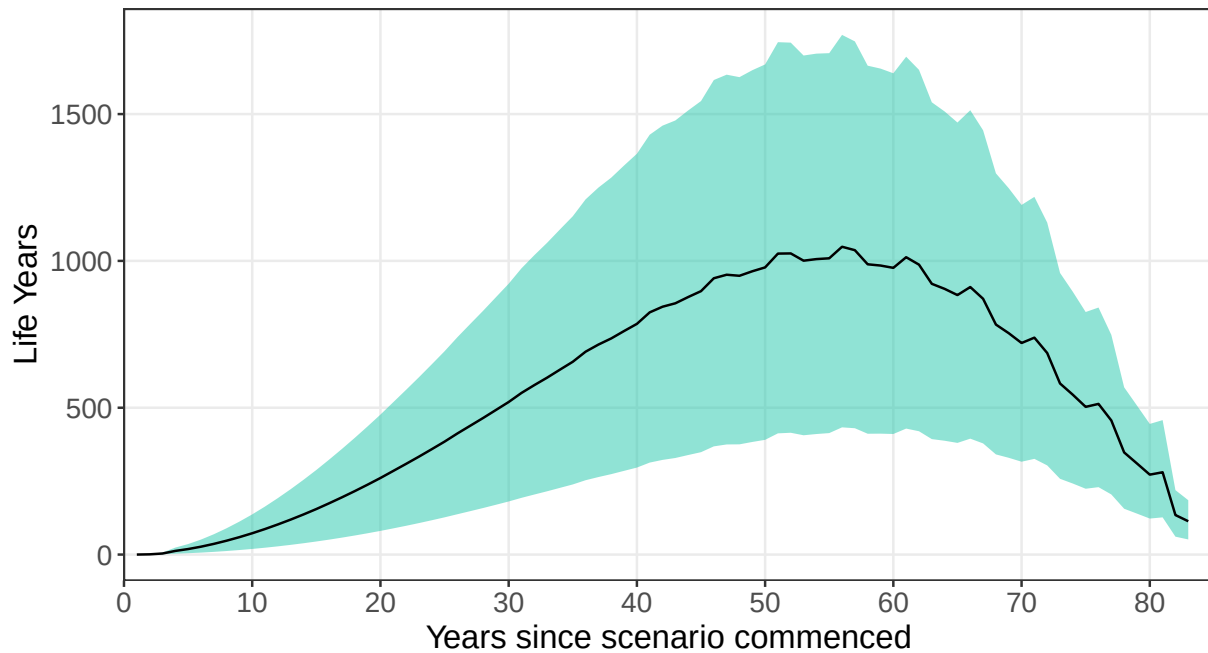
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **46,550** Life Years for the scenario population, which is **23** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **51** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **4,336,658** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **2,730,485** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **1,874,086** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$1,044          | \$754            | \$537            |
| Breast cancer                           | \$41,036         | \$24,985         | \$16,250         |
| All cancers                             | \$73,001         | \$46,168         | \$30,922         |
| Colon cancer                            | \$11,935         | \$8,210          | \$5,738          |
| Chronic myeloid leukemia                | \$2,796          | \$1,702          | \$1,111          |
| Diabetes type 2                         | \$27,564         | \$16,001         | \$10,076         |
| Depression                              | \$140,499        | \$102,147        | \$76,932         |
| Head and neck cancer                    | \$555            | \$380            | \$273            |
| Ischemic heart disease                  | \$44,540         | \$26,000         | \$16,433         |
| Liver cancer                            | \$543            | \$377            | \$275            |
| Multiple myeloma                        | \$7,203          | \$4,661          | \$3,203          |
| Stomach cancer                          | \$2,934          | \$1,911          | \$1,322          |
| Stroke                                  | \$5,681          | \$3,321          | \$2,099          |
| Lung cancer                             | \$4,023          | \$2,735          | \$1,959          |
| Uterine cancer                          | \$1,966          | \$1,211          | \$795            |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

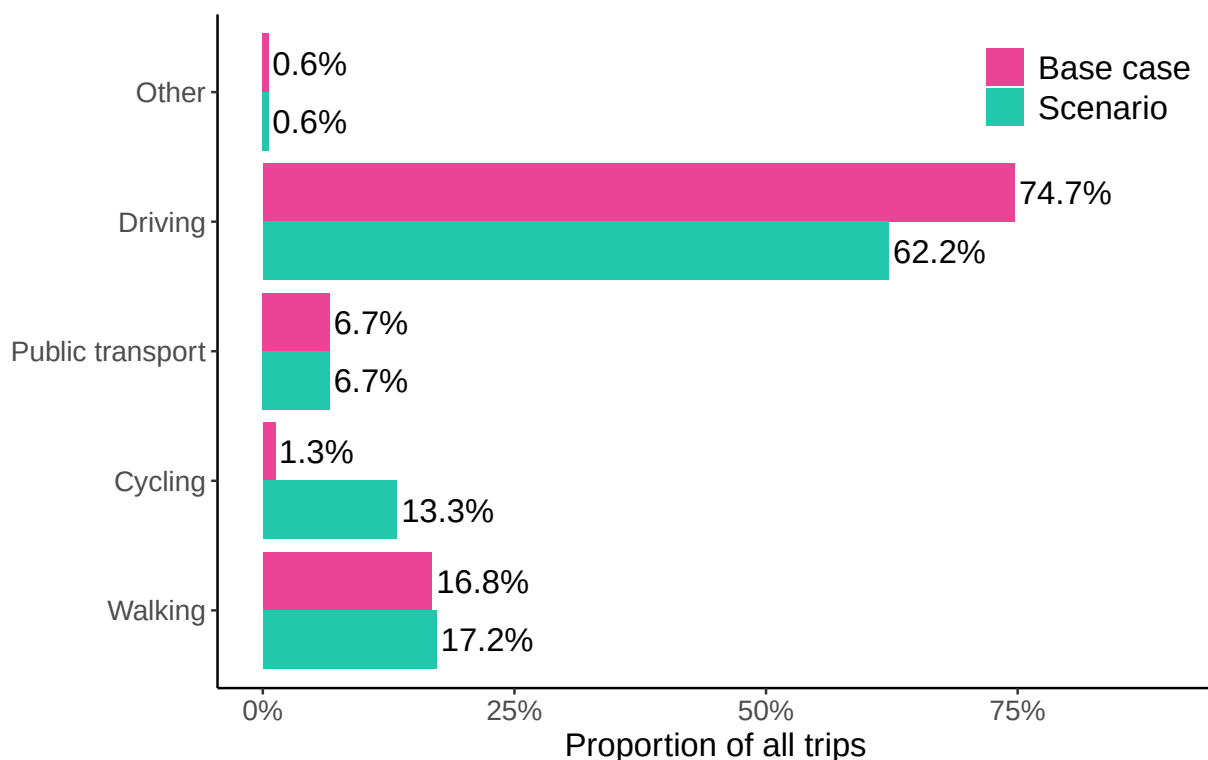
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2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
3. Ananthapavan, J., Moodie, M., Milat, A.J., & Carter, R. (2021). *Systematic review to update 'value of a statistical life' estimates for Australia*. International journal of environmental research and public health, 2021. 18(11): p. 6168.
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8. Australian Institute of Health and Welfare (2023). Technical Notes: Estimating Spending per prevalent case of disease. Health system spending per case of disease and for certain risk factors, Estimating the spending per prevalent case of disease - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
11. Zapata-Diomed, B., Boulangé, C., Giles-Corti, B., Phelan, K., Washington, S., Veerman, L.J., & Gunn, L. (2019). Physical activity-related health and economic benefits of building walkable neighbourhoods: A modelled comparison between brownfield and greenfield developments. International Journal of Behavioural Nutrition and Physical Activity.
12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577-1590.

## Scenario: replacing car trips under 1km with walking, and car trips between 1 and 10km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 1km with walking and replacing car trips between 1km and 10km with cycling for work related or education purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 17.2%; cycling from 1.3% to 13.3%; and, from 74.7% to 62.2% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 58.1% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

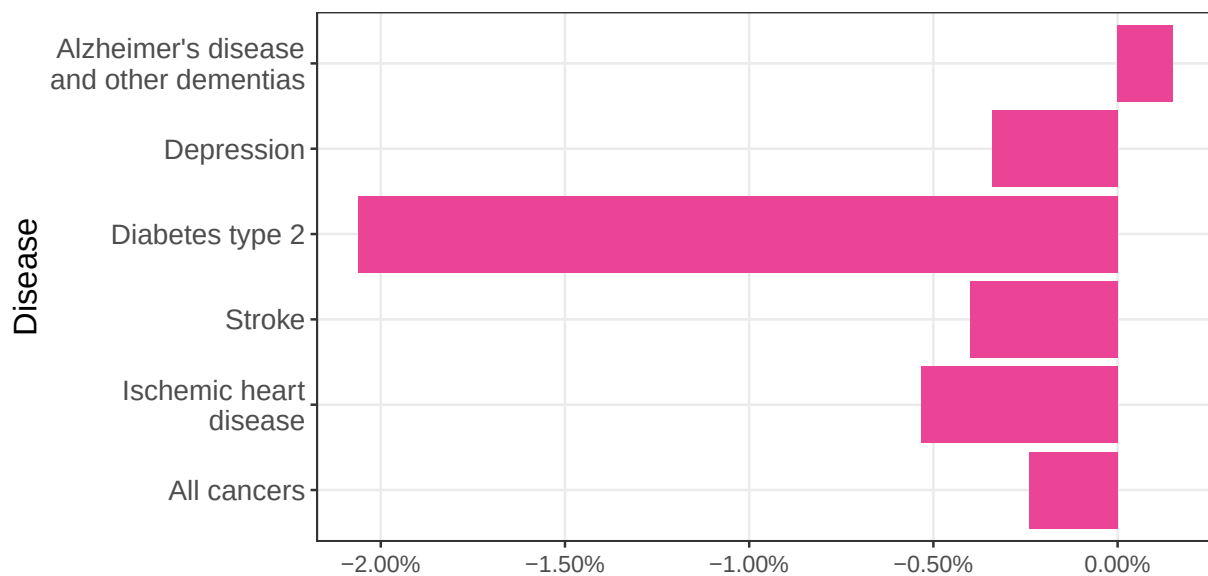
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

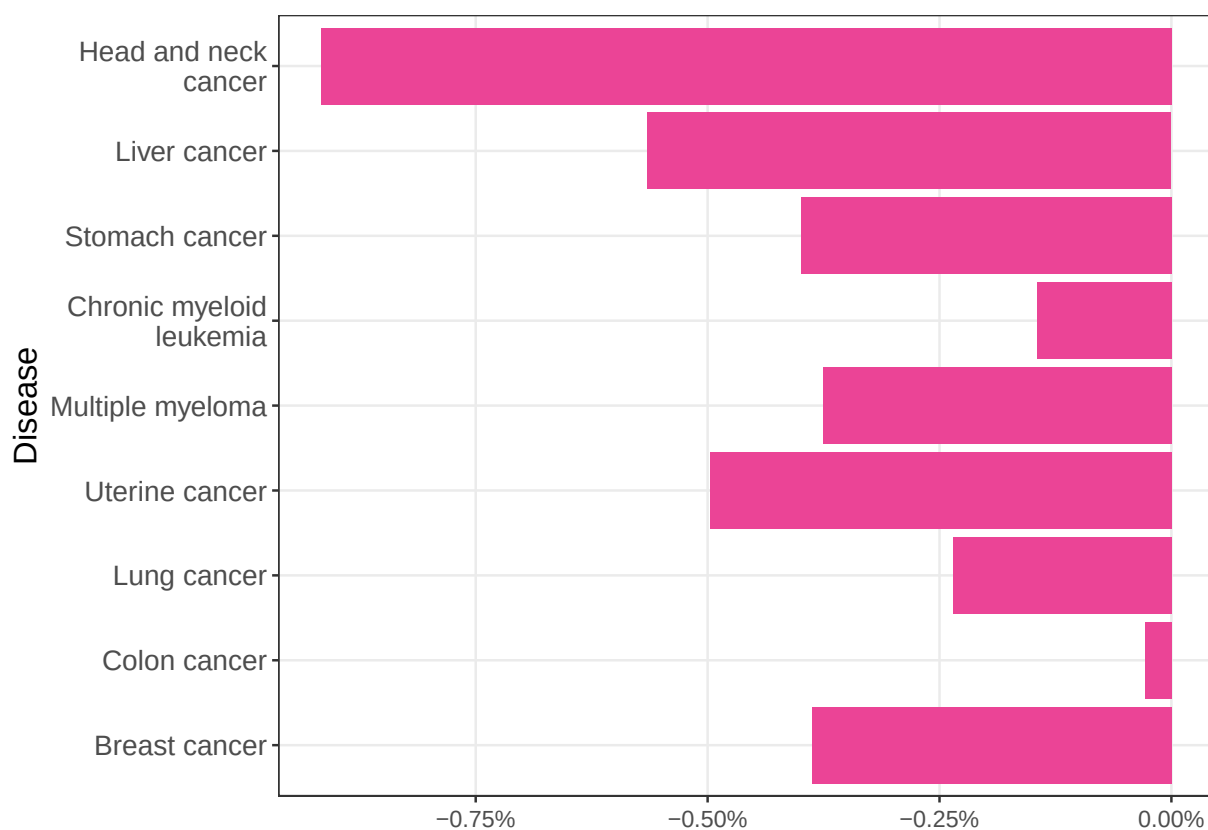
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.15%                             | -960                                                                        |
| Breast cancer                           | 0.39%                              | 284                                                                         |
| All cancers                             | 0.24%                              | 945                                                                         |
| Colon cancer                            | 0.03%                              | 37                                                                          |
| Chronic myeloid leukemia                | 0.15%                              | 4                                                                           |
| Diabetes type 2                         | 2.06%                              | 6,809                                                                       |
| Depression                              | 0.34%                              | 5,477                                                                       |
| Head and neck cancer                    | 0.92%                              | 40                                                                          |
| Ischemic heart disease                  | 0.53%                              | 4,685                                                                       |
| Liver cancer                            | 0.56%                              | 99                                                                          |
| Multiple myeloma                        | 0.38%                              | 73                                                                          |
| Stomach cancer                          | 0.40%                              | 95                                                                          |
| Stroke                                  | 0.40%                              | 1,042                                                                       |
| Lung cancer                             | 0.24%                              | 271                                                                         |
| Uterine cancer                          | 0.50%                              | 42                                                                          |

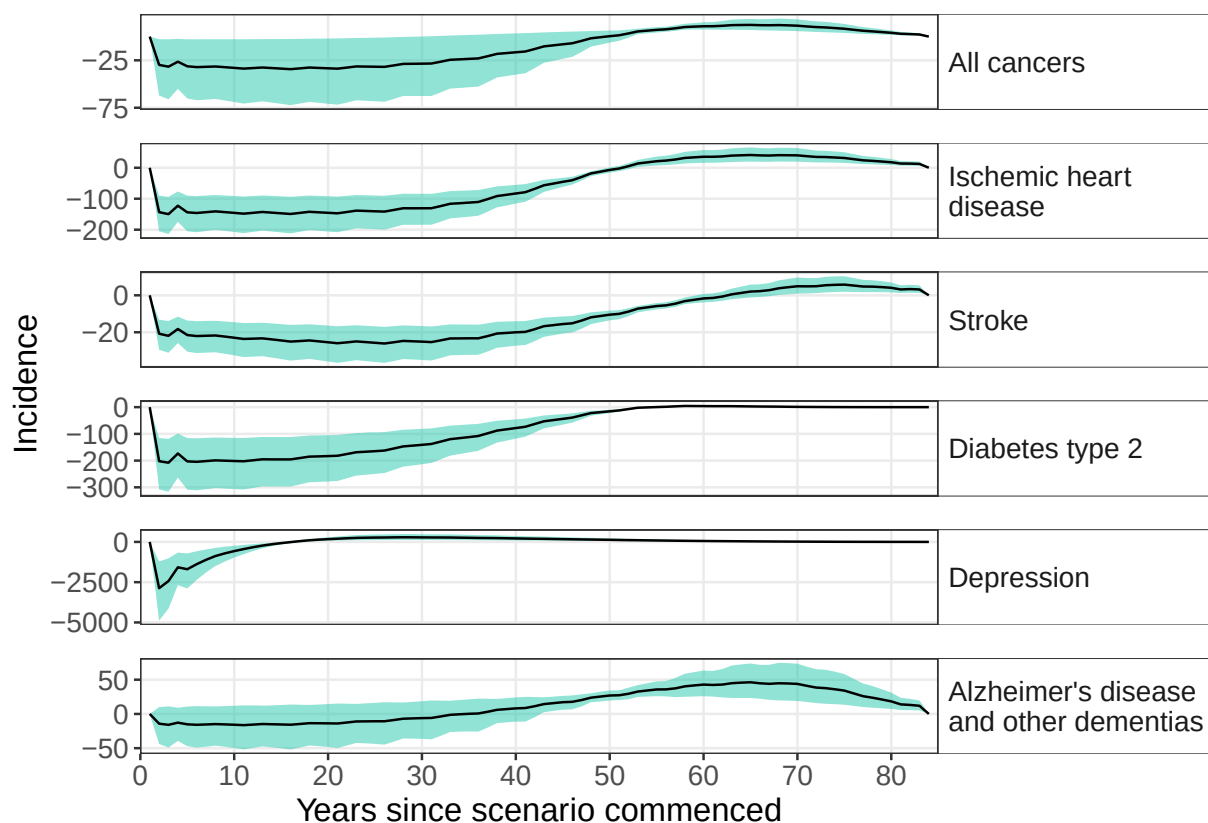
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



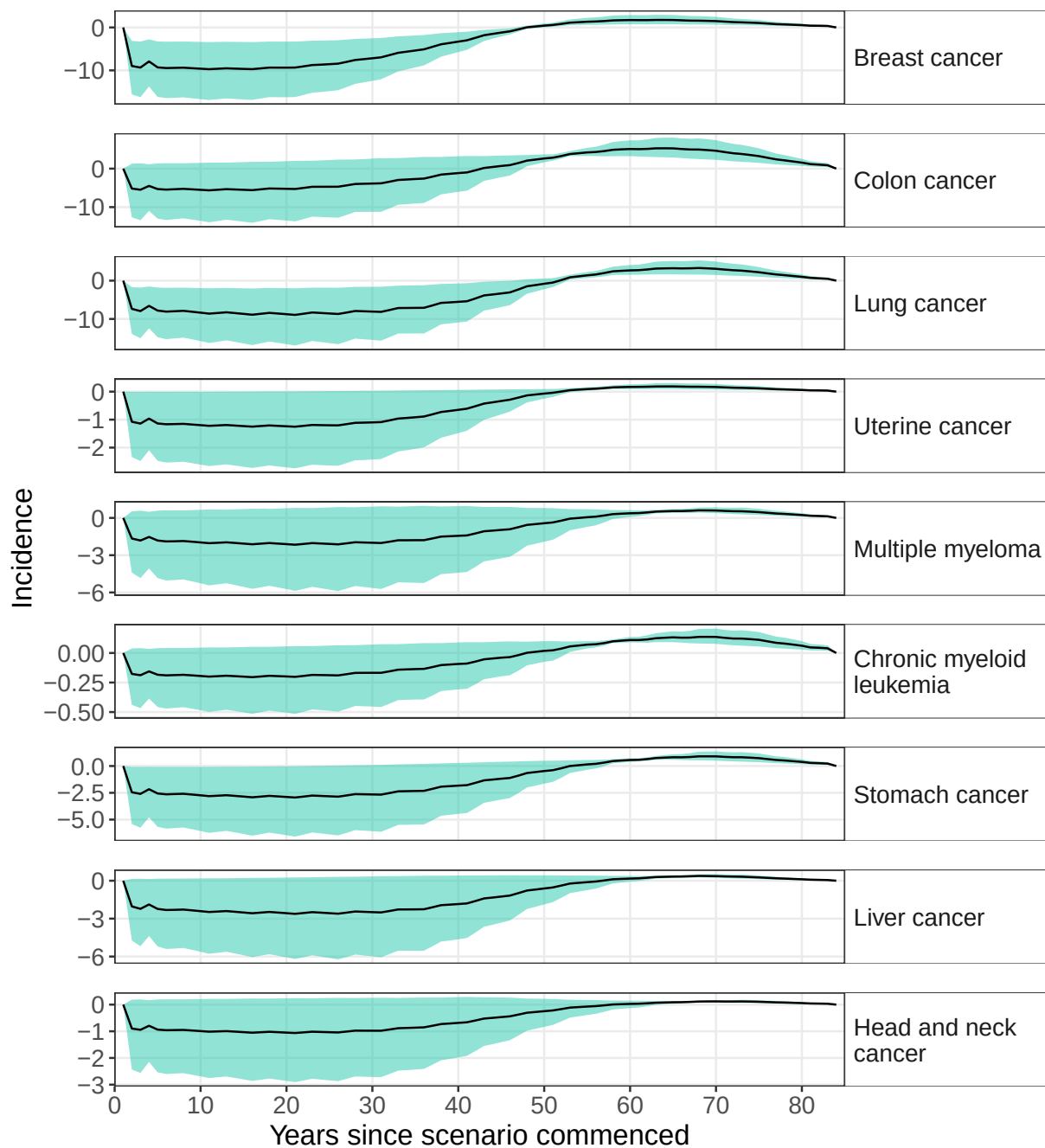
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

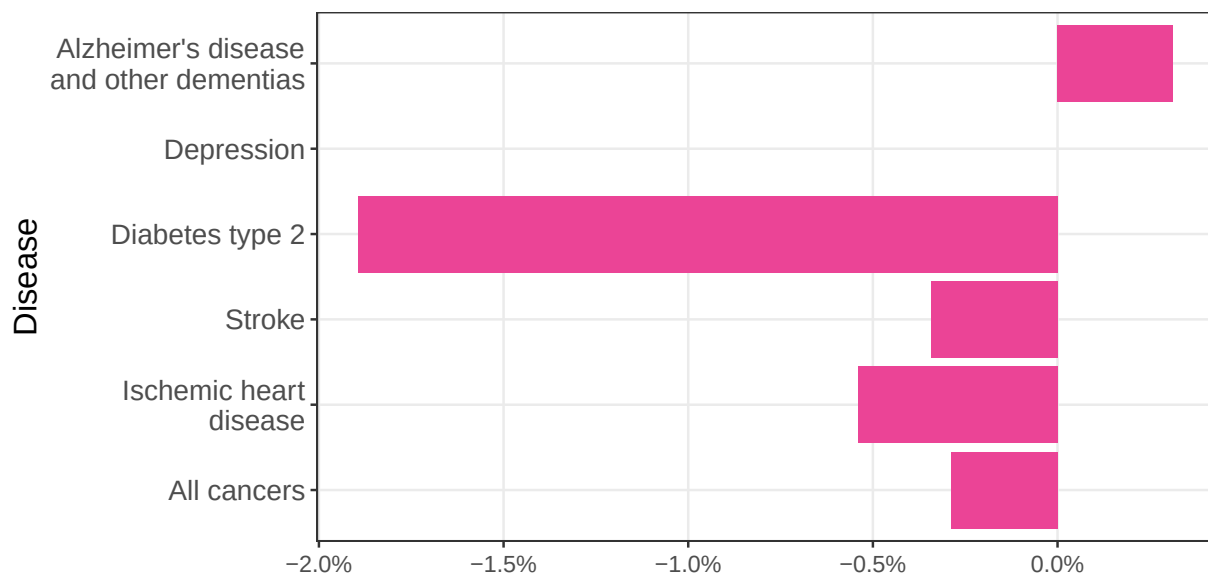
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

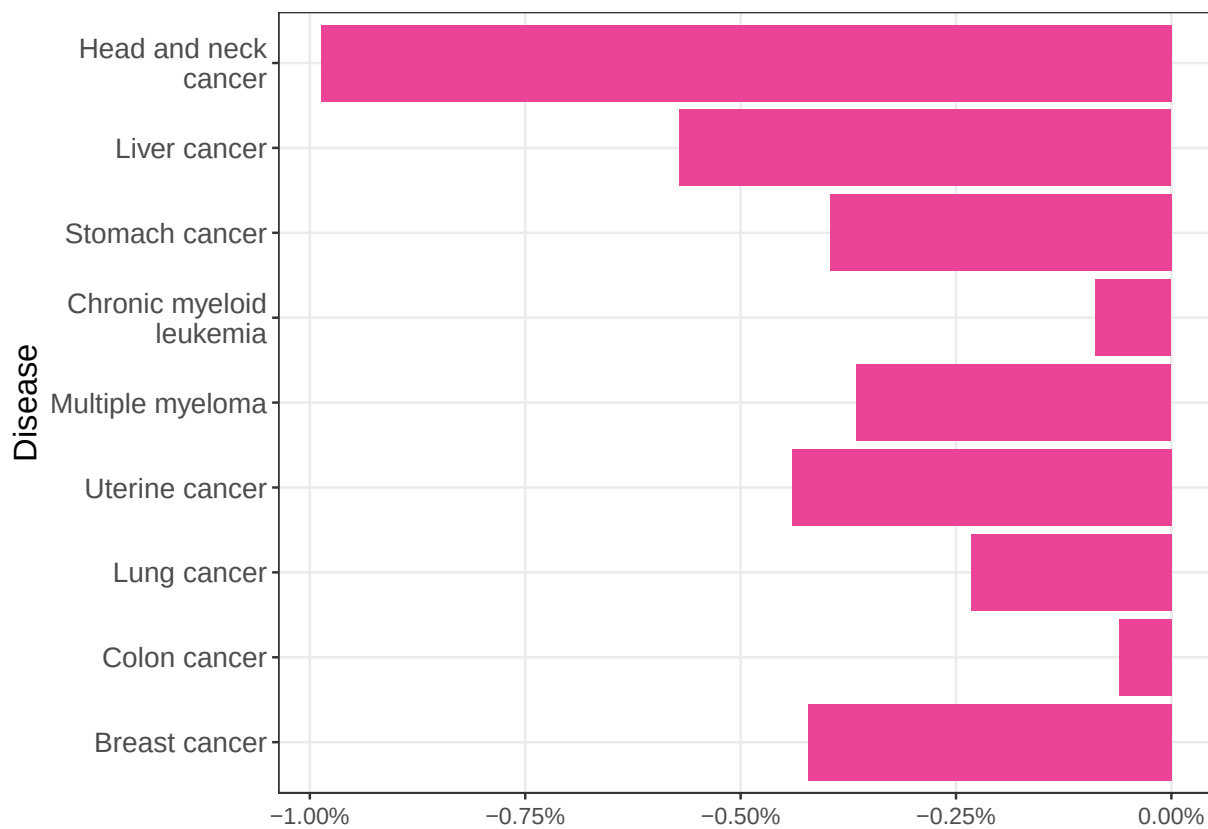
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.31%                  | -749                                                              |
| Breast cancer                           | 0.42%                   | 136                                                               |
| All cancers                             | 0.29%                   | 733                                                               |
| Colon cancer                            | 0.06%                   | 26                                                                |
| Chronic myeloid leukemia                | 0.09%                   | 2                                                                 |
| Diabetes type 2                         | 1.89%                   | 936                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.99%                   | 39                                                                |
| Ischemic heart disease                  | 0.54%                   | 2,026                                                             |
| Liver cancer                            | 0.57%                   | 97                                                                |
| Multiple myeloma                        | 0.37%                   | 64                                                                |
| Stomach cancer                          | 0.40%                   | 84                                                                |
| Stroke                                  | 0.34%                   | 600                                                               |
| Lung cancer                             | 0.23%                   | 254                                                               |
| Uterine cancer                          | 0.44%                   | 32                                                                |

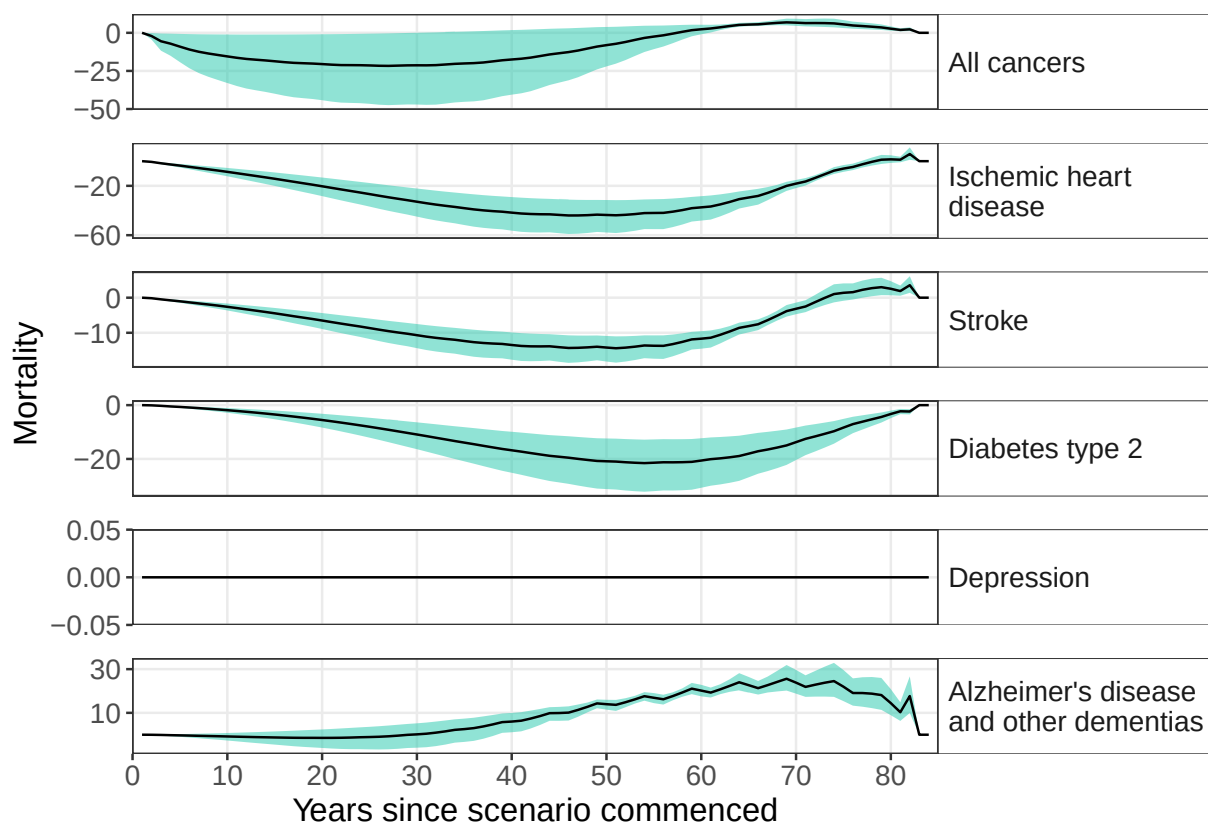
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

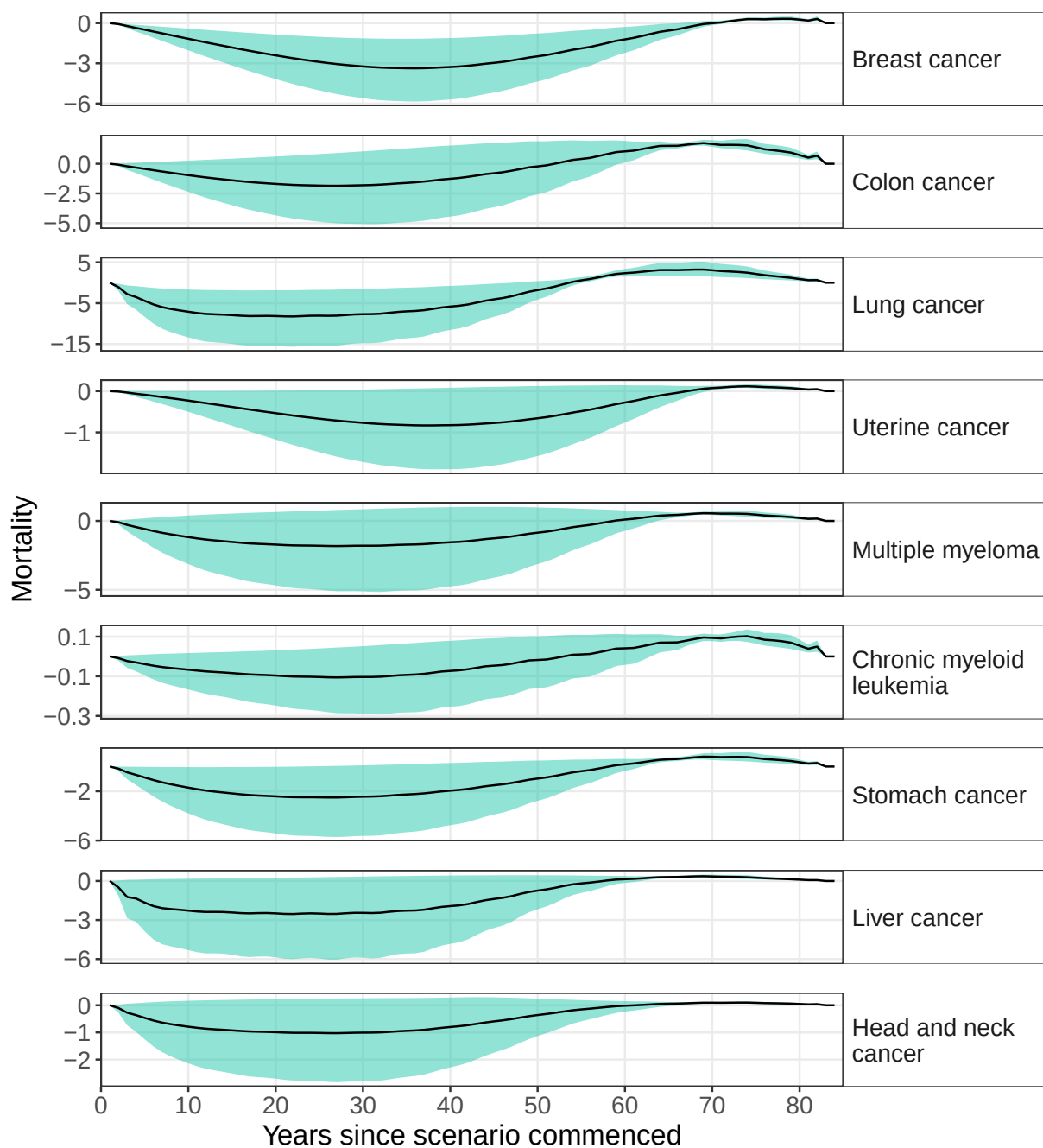


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

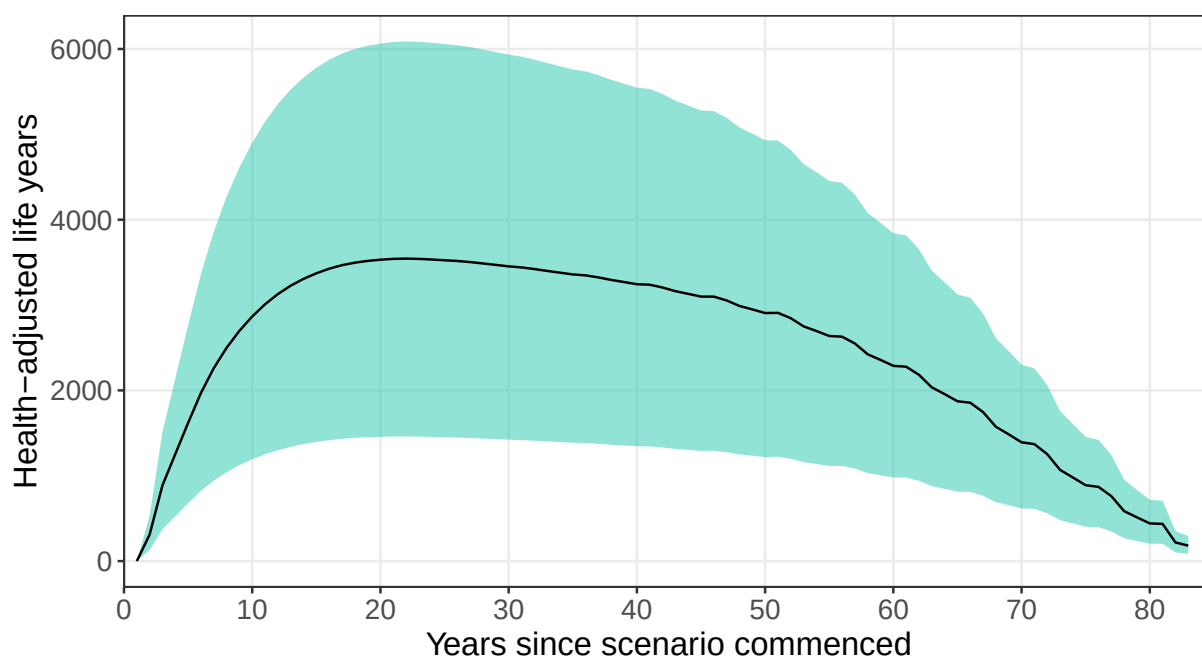
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **204,976** HALYs for the scenario population, which is **102** HALYs per 1,000 members of the population.



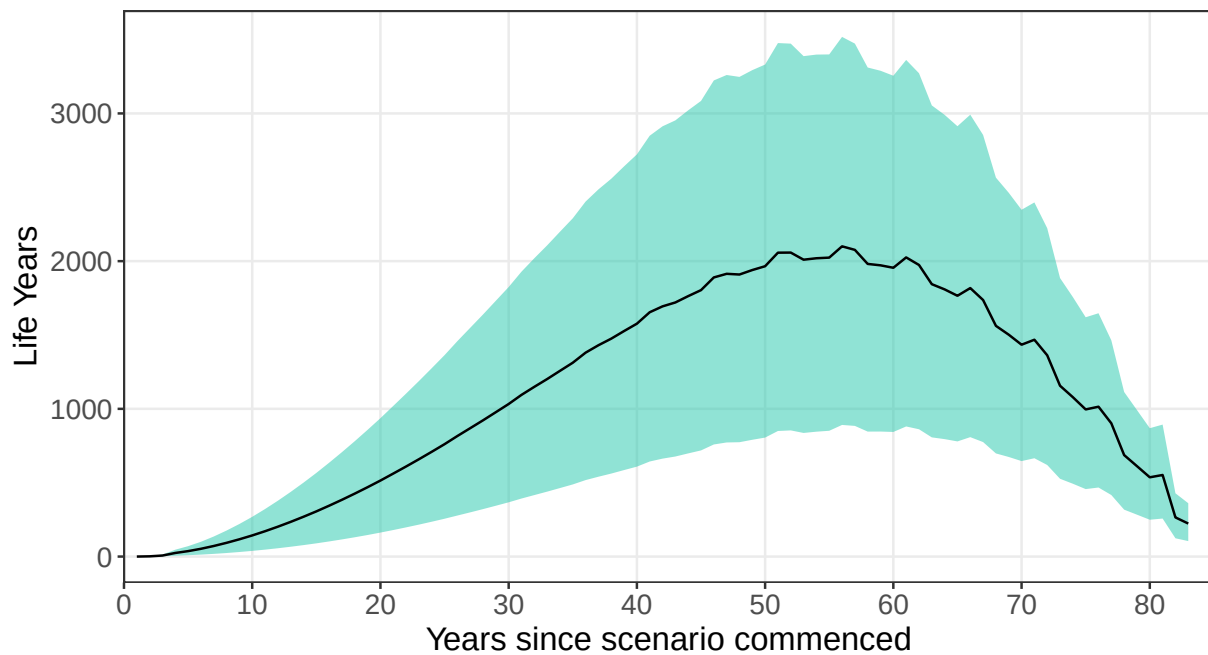
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **92,934** Life Years for the scenario population, which is **46** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **102** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **8,565,317** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **5,363,149** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **3,662,033** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$1,918          | \$1,416          | \$1,018          |
| Breast cancer                           | \$88,970         | \$53,868         | \$34,891         |
| All cancers                             | \$151,599        | \$95,525         | \$63,794         |
| Colon cancer                            | \$24,133         | \$16,618         | \$11,617         |
| Chronic myeloid leukemia                | \$5,018          | \$3,073          | \$2,013          |
| Diabetes type 2                         | \$57,194         | \$33,194         | \$20,895         |
| Depression                              | \$271,655        | \$196,606        | \$147,485        |
| Head and neck cancer                    | \$1,101          | \$756            | \$544            |
| Ischemic heart disease                  | \$92,173         | \$53,847         | \$34,034         |
| Liver cancer                            | \$1,147          | \$798            | \$583            |
| Multiple myeloma                        | \$13,598         | \$8,833          | \$6,083          |
| Stomach cancer                          | \$5,861          | \$3,829          | \$2,652          |
| Stroke                                  | \$11,067         | \$6,471          | \$4,085          |
| Lung cancer                             | \$7,805          | \$5,330          | \$3,828          |
| Uterine cancer                          | \$3,940          | \$2,425          | \$1,591          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

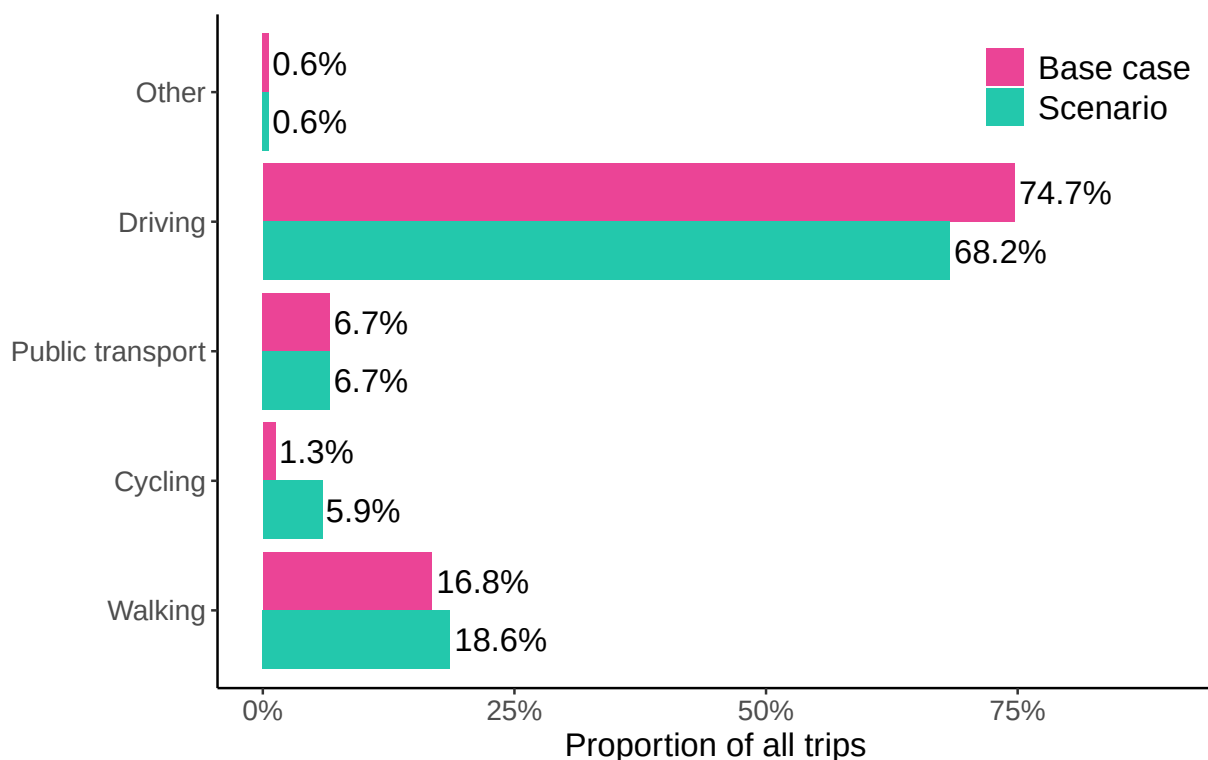
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745-758.
3. Ananthapavan, J., Moodie, M., Milat, A.J., & Carter, R. (2021). *Systematic review to update 'value of a statistical life' estimates for Australia*. International journal of environmental research and public health, 2021. 18(11): p. 6168.
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9. Australian Institute of Health and Welfare (2023). Health system spending per case of disease and for certain risk factors. Health system spending per case of disease and for certain risk factors, Data - Australian Institute of Health and Welfare (aihw.gov.au). Accessed September 20, 2023.
10. Global Burden of Disease (2019). Global Health Data Exchange. <https://vizhub.healthdata.org/gbd-results>. Accessed September 20, 2023.
11. Zapata-Diomed, B., Boulangé, C., Giles-Corti, B., Phelan, K., Washington, S., Veerman, L.J., & Gunn, L. (2019). Physical activity-related health and economic benefits of building walkable neighbourhoods: A modelled comparison between brownfield and greenfield developments. International Journal of Behavioural Nutrition and Physical Activity.
12. Khorasani, E., Davari, M., Kebriaeezadeh, A., Fatemi, F., Akbari Sari, A., & Varahrami, V. (2022). A comprehensive review of official discount rates in guidelines of health economic evaluations over time: the trends and roots. The European Journal of Health Economics, 23(9), 1577-1590.

## Scenario: replacing car trips under 2km with walking, and car trips between 2 and 5km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 2km with walking and replacing car trips between 2km and 5km with cycling for work related or education purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 18.6%; cycling from 1.3% to 5.9%; and, from 74.7% to 68.2% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 53.4% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

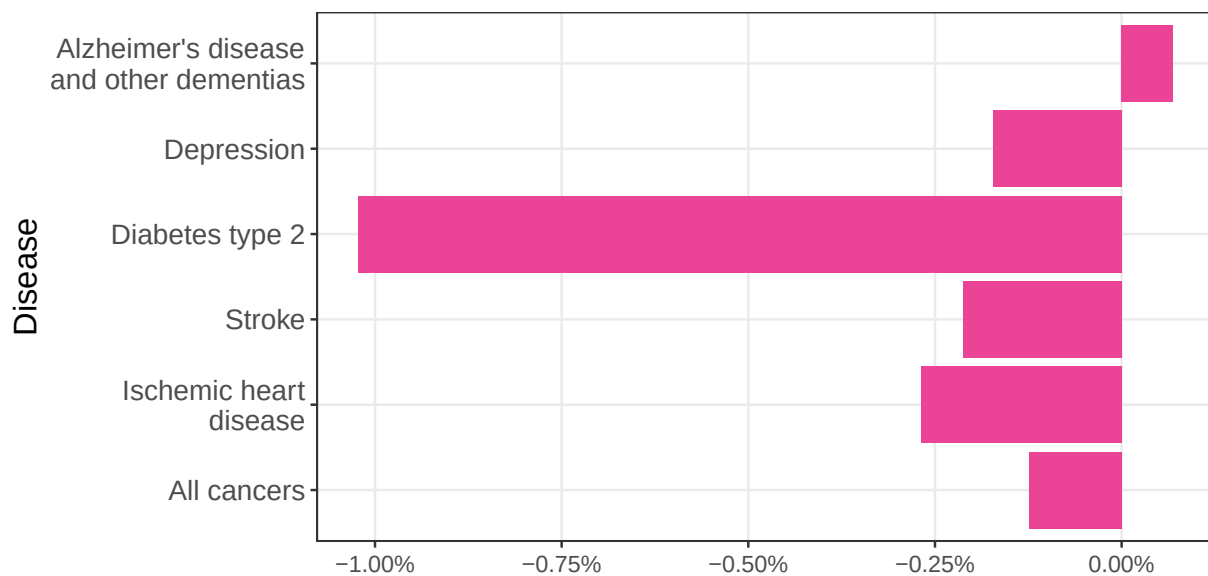
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

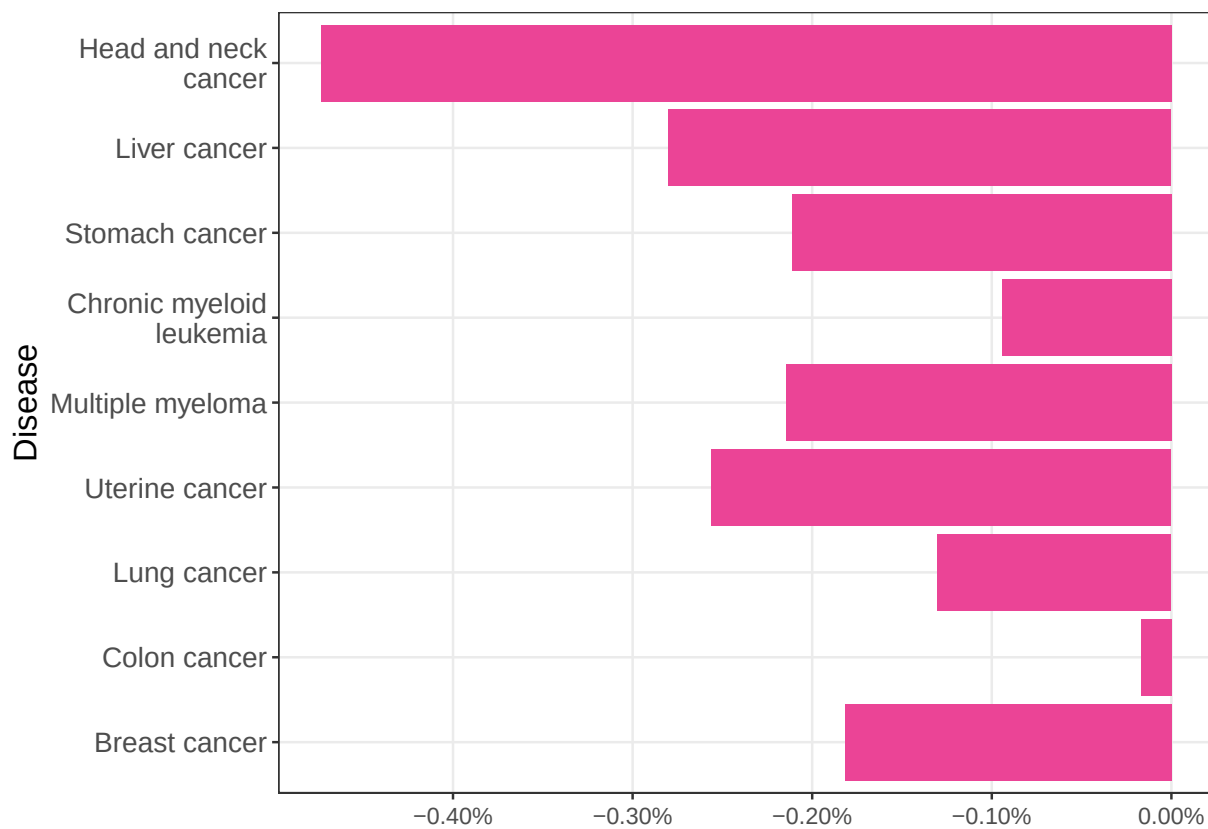
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.07%                             | -439                                                                        |
| Breast cancer                           | 0.18%                              | 133                                                                         |
| All cancers                             | 0.12%                              | 491                                                                         |
| Colon cancer                            | 0.02%                              | 22                                                                          |
| Chronic myeloid leukemia                | 0.09%                              | 3                                                                           |
| Diabetes type 2                         | 1.02%                              | 3,378                                                                       |
| Depression                              | 0.17%                              | 2,762                                                                       |
| Head and neck cancer                    | 0.47%                              | 21                                                                          |
| Ischemic heart disease                  | 0.27%                              | 2,370                                                                       |
| Liver cancer                            | 0.28%                              | 49                                                                          |
| Multiple myeloma                        | 0.21%                              | 42                                                                          |
| Stomach cancer                          | 0.21%                              | 50                                                                          |
| Stroke                                  | 0.21%                              | 553                                                                         |
| Lung cancer                             | 0.13%                              | 150                                                                         |
| Uterine cancer                          | 0.26%                              | 22                                                                          |

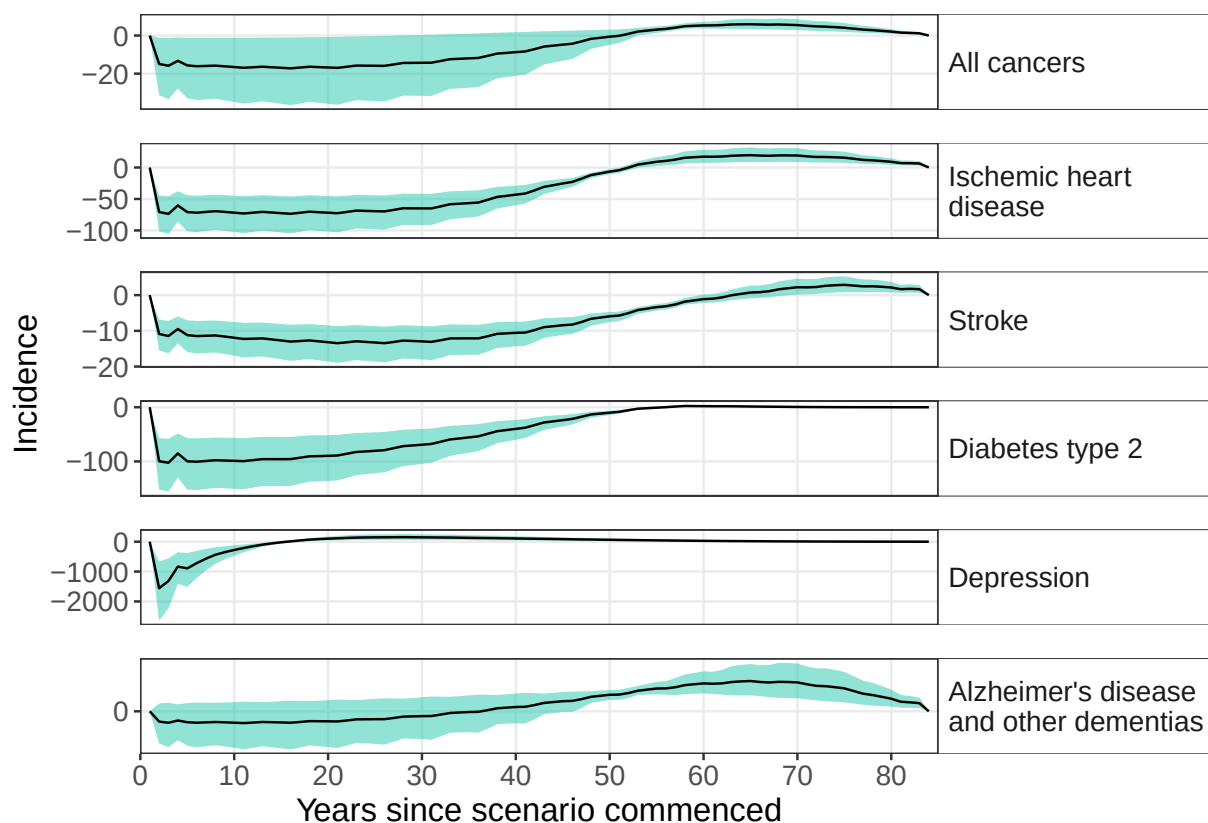
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



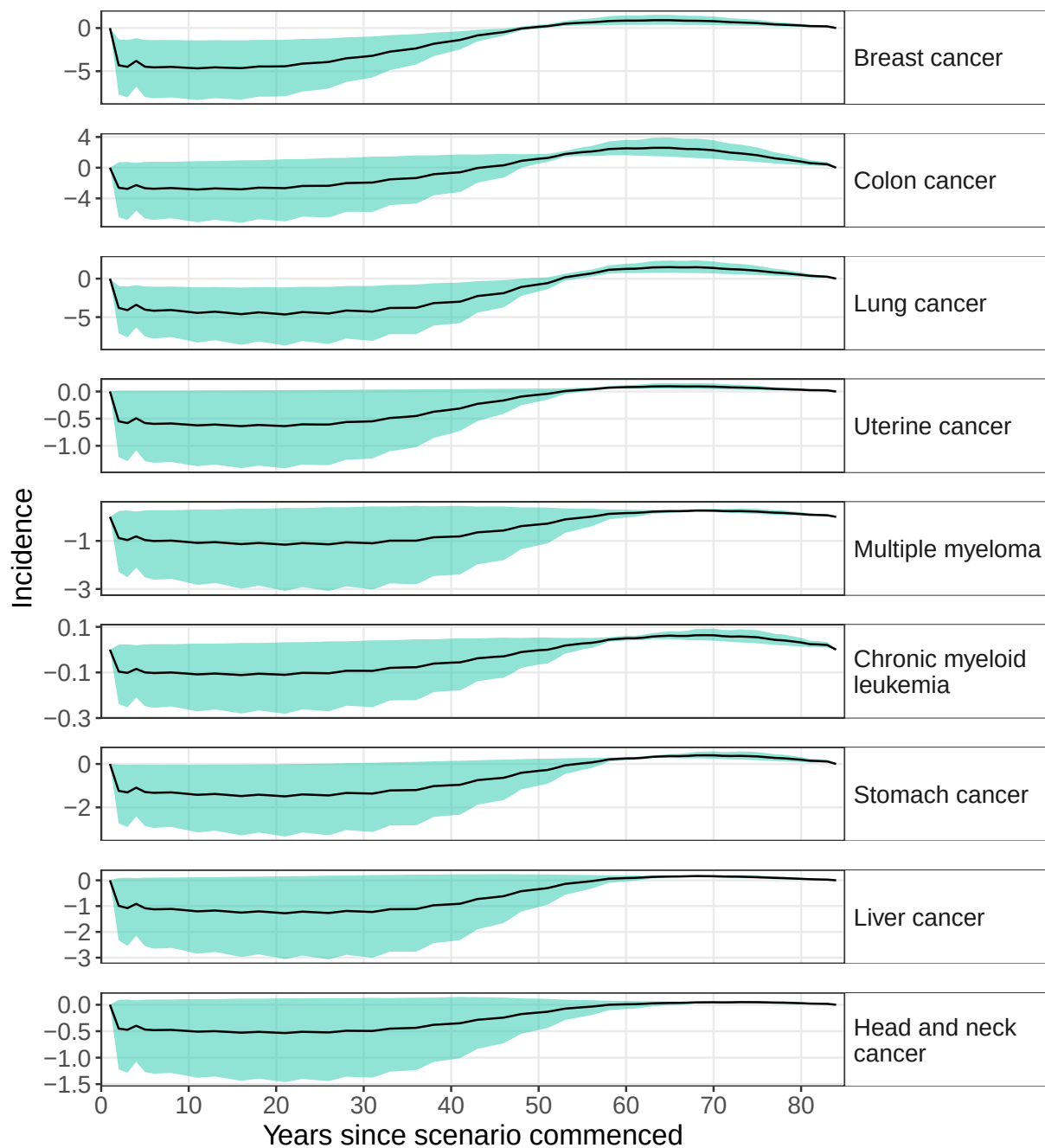
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

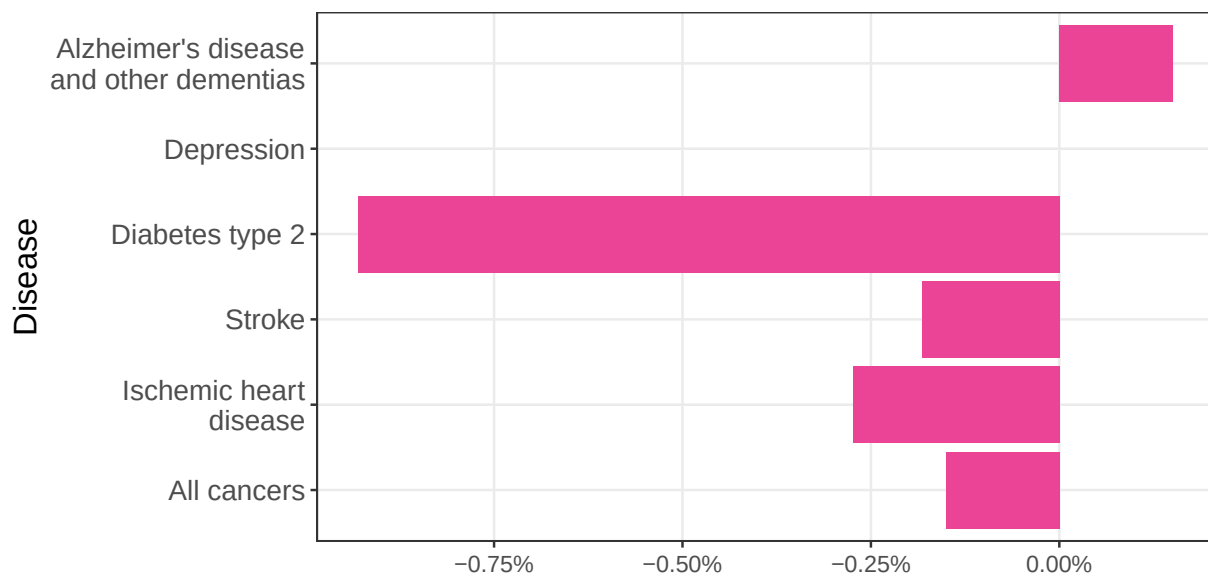
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

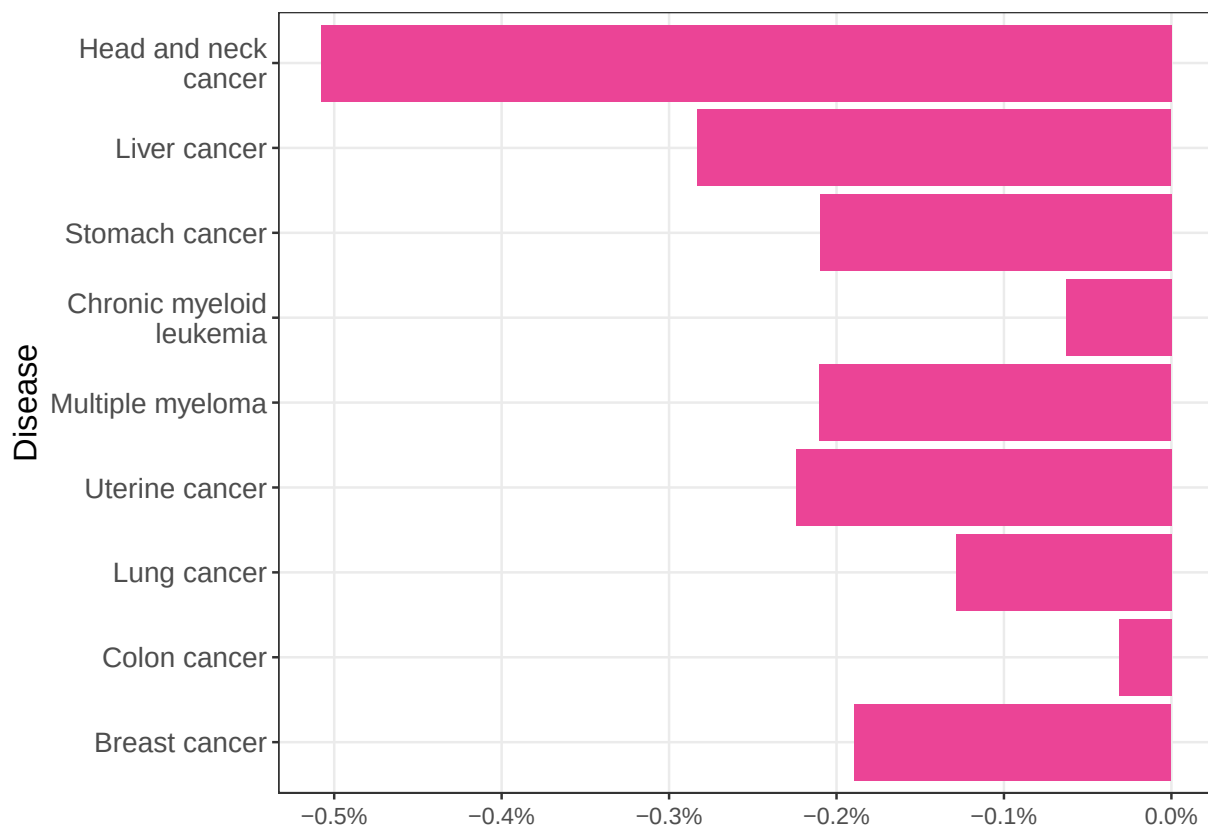
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.15%                  | -361                                                              |
| Breast cancer                           | 0.19%                   | 61                                                                |
| All cancers                             | 0.15%                   | 382                                                               |
| Colon cancer                            | 0.03%                   | 13                                                                |
| Chronic myeloid leukemia                | 0.06%                   | 1                                                                 |
| Diabetes type 2                         | 0.93%                   | 460                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.51%                   | 20                                                                |
| Ischemic heart disease                  | 0.27%                   | 1,026                                                             |
| Liver cancer                            | 0.28%                   | 48                                                                |
| Multiple myeloma                        | 0.21%                   | 37                                                                |
| Stomach cancer                          | 0.21%                   | 44                                                                |
| Stroke                                  | 0.18%                   | 319                                                               |
| Lung cancer                             | 0.13%                   | 141                                                               |
| Uterine cancer                          | 0.22%                   | 16                                                                |

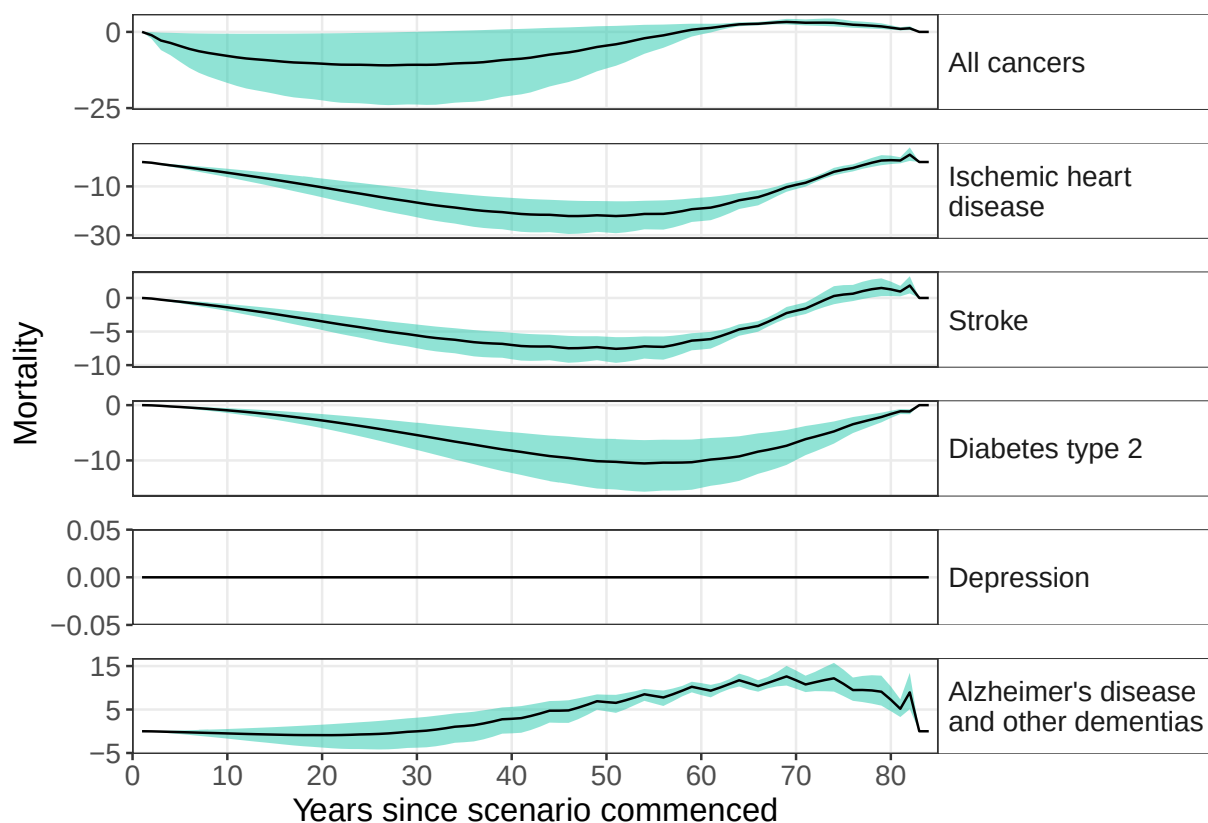
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

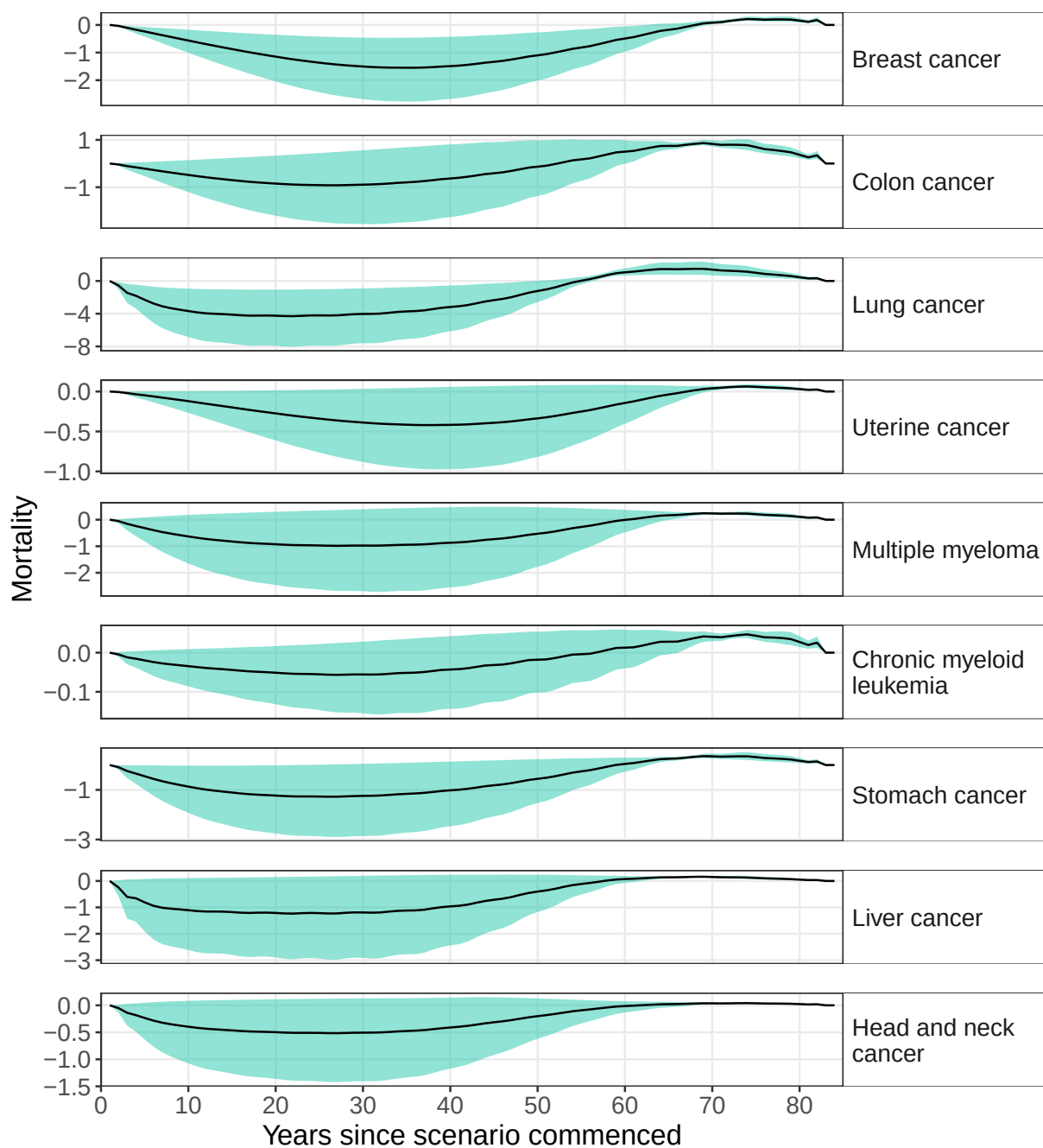


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **103,790** HALYs for the scenario population, which is **52** HALYs per 1,000 members of the population.



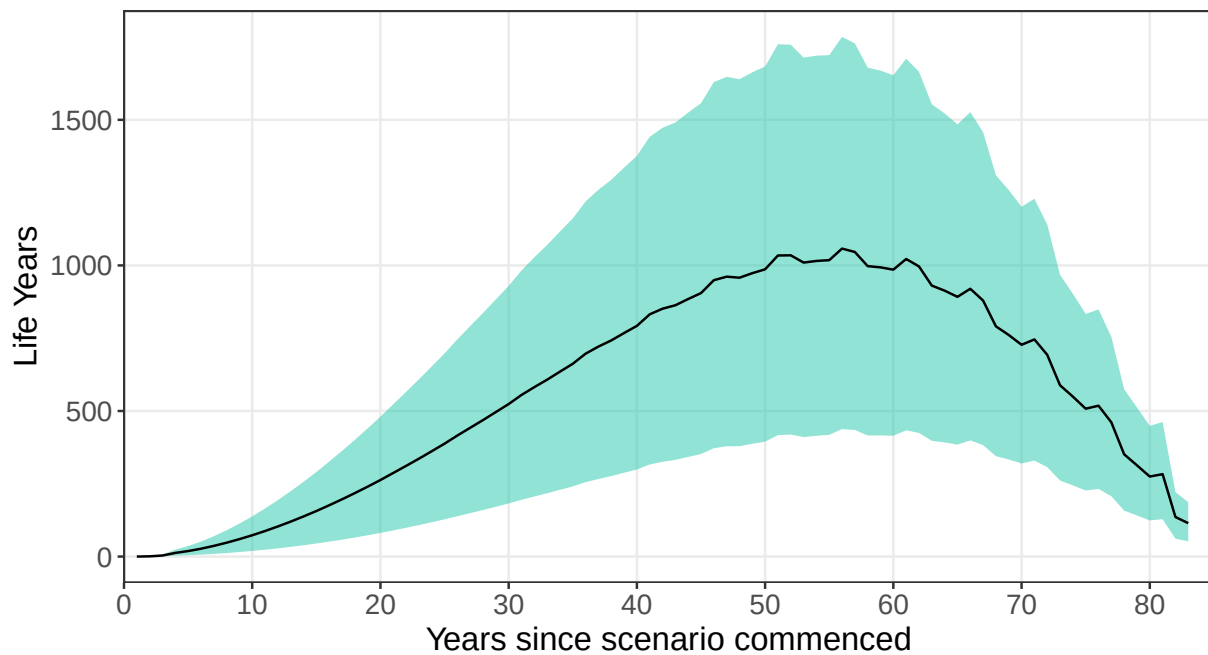
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **46,971** Life Years for the scenario population, which is **23** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **52 HALYs** per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **4,367,947** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **2,749,415** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **1,886,696** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$1,059          | \$764            | \$543            |
| Breast cancer                           | \$41,722         | \$25,396         | \$16,515         |
| All cancers                             | \$73,944         | \$46,748         | \$31,303         |
| Colon cancer                            | \$12,086         | \$8,308          | \$5,805          |
| Chronic myeloid leukemia                | \$2,801          | \$1,706          | \$1,114          |
| Diabetes type 2                         | \$27,876         | \$16,181         | \$10,190         |
| Depression                              | \$141,294        | \$102,734        | \$77,378         |
| Head and neck cancer                    | \$557            | \$381            | \$274            |
| Ischemic heart disease                  | \$44,907         | \$26,214         | \$16,568         |
| Liver cancer                            | \$549            | \$381            | \$278            |
| Multiple myeloma                        | \$7,229          | \$4,678          | \$3,215          |
| Stomach cancer                          | \$2,959          | \$1,928          | \$1,333          |
| Stroke                                  | \$5,723          | \$3,346          | \$2,114          |
| Lung cancer                             | \$4,042          | \$2,748          | \$1,969          |
| Uterine cancer                          | \$1,990          | \$1,225          | \$805            |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

These figures use AIHW estimates of the amounts spent through the health system in 2018–19 for each case of disease. This is extracted from Health system spending per case of disease and for certain risk factors, Table 1 – Estimates of health system spending per case, by burden of disease group, condition and sex, Australia 2018–2019.[\[9\]](#) For head and neck cancers, supplementary figures were obtained from the Global Burden of Disease incidence data.[\[10\]](#)

## b. What are discount rates and how are they relevant here?

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## References

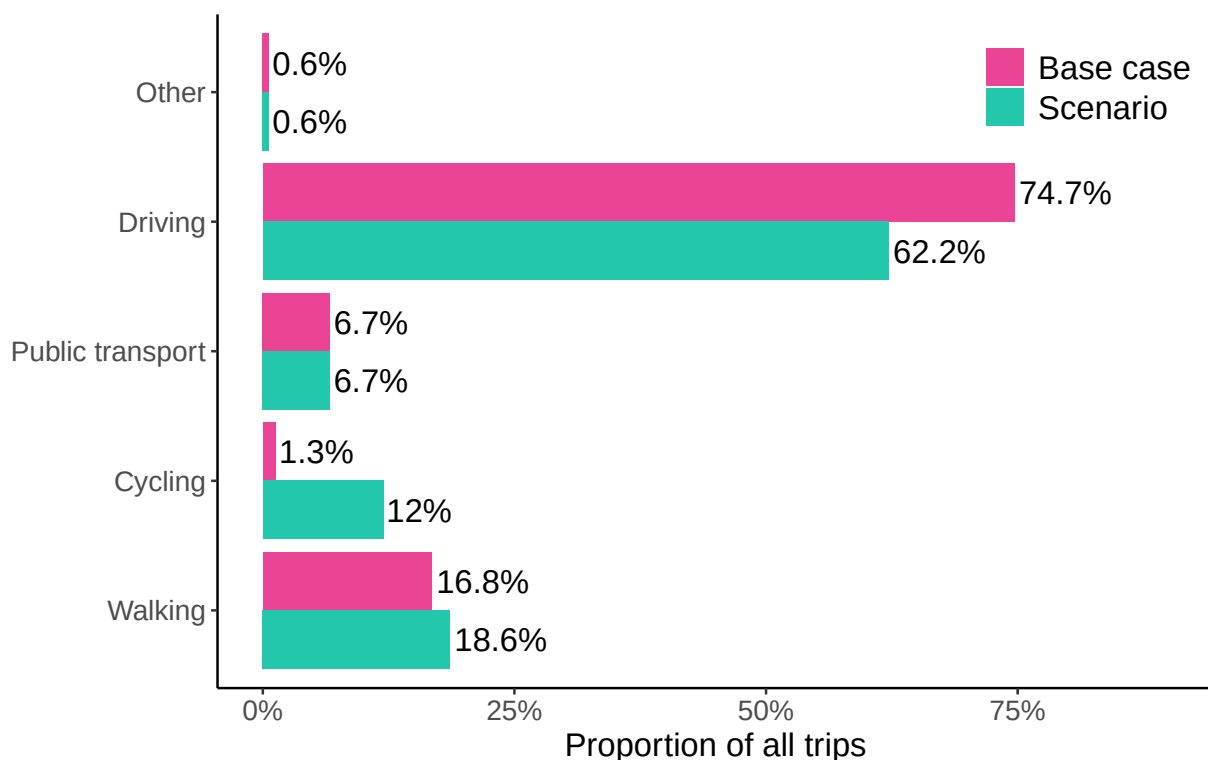
1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115–134.
2. Attema, A.E., Brouwer, W.B. & Claxton, K. (2018). *Discounting in economic evaluations*. Pharmacoeconomics. 36: p. 745–758.
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## Scenario: replacing car trips under 2km with walking, and car trips between 2 and 10km with cycling for commuting trip purposes

This scenario shows the results of replacing car trips under 2km with walking and replacing car trips between 2km and 10km with cycling for work related or education purposes for all adults of all ages.

This implies that the selected scenario results in a mode shift in walking from 16.8% to 18.6%; cycling from 1.3% to 12.0%; and, from 74.7% to 62.2% for car trips taken as either a driver or passenger.

Increases in walking and cycling translate into a shift from 47.9% to 58.2% of the population accumulating the required minutes spent being moderately (150 – 300 mins) or vigorously physically active (75 – 150 mins) or an equivalent combination of both contributing to recommended levels as detailed in the Physical Activity Guidelines.



**Figure 1:** Distribution of base case and scenario trips.

## Incidence

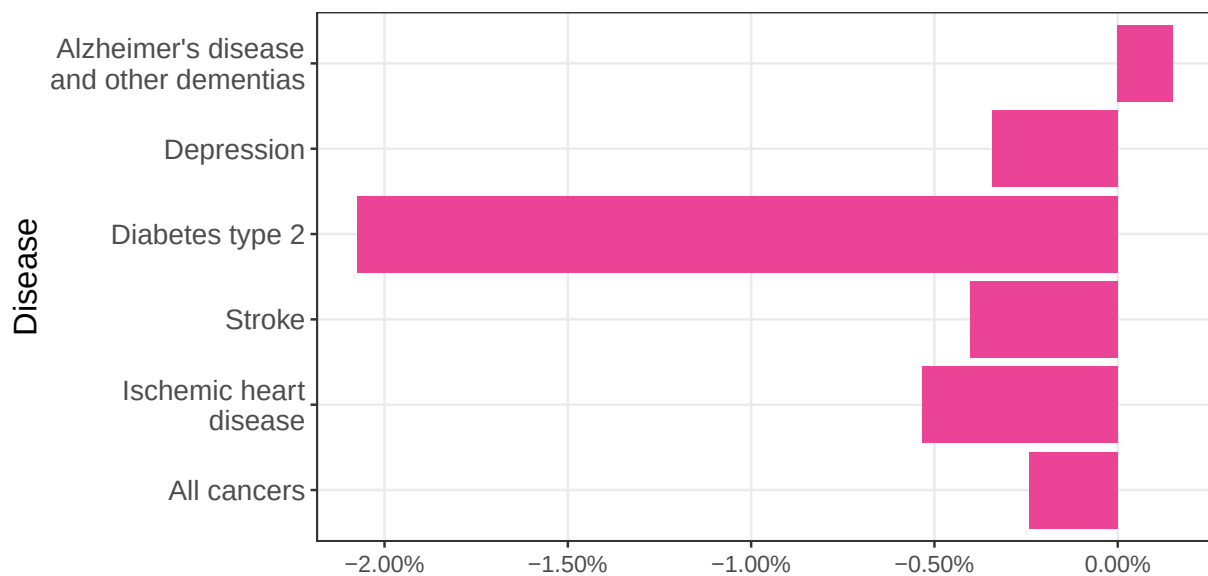
Incidence describes the rate of occurrence of new cases of a disease over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of disease prevented, due to increases in physical activity associated with the scenario. Figure 2 presents the change (%) in the disease incidence across the life course. Figure 3 presents how the difference in disease incidence changes over time, by year, using a snapshot of the population from 2019.

Table 1 shows how the scenario impacts the incidence of chronic diseases as both as a percentage and total number of prevented cases.

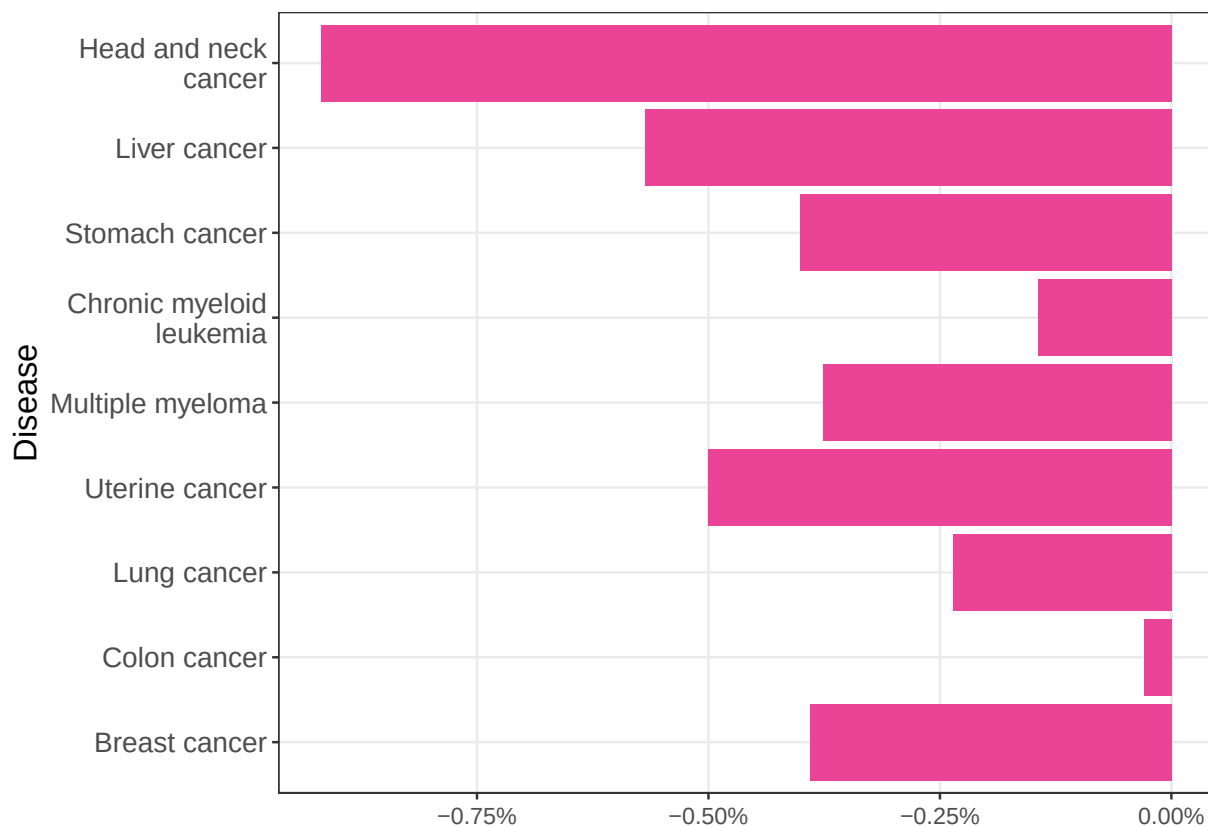
**Table 1.** Chronic disease incidence reduction and total number of prevented cases of disease measured across the years of the simulation

| Disease*                                | Incidence of disease is reduced by | Total number of prevented cases of disease aggregated across the simulation |
|-----------------------------------------|------------------------------------|-----------------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.15%                             | -962                                                                        |
| Breast cancer                           | 0.39%                              | 286                                                                         |
| All cancers                             | 0.24%                              | 950                                                                         |
| Colon cancer                            | 0.03%                              | 38                                                                          |
| Chronic myeloid leukemia                | 0.14%                              | 4                                                                           |
| Diabetes type 2                         | 2.07%                              | 6,844                                                                       |
| Depression                              | 0.34%                              | 5,491                                                                       |
| Head and neck cancer                    | 0.92%                              | 40                                                                          |
| Ischemic heart disease                  | 0.53%                              | 4,706                                                                       |
| Liver cancer                            | 0.57%                              | 100                                                                         |
| Multiple myeloma                        | 0.38%                              | 73                                                                          |
| Stomach cancer                          | 0.40%                              | 96                                                                          |
| Stroke                                  | 0.40%                              | 1,047                                                                       |
| Lung cancer                             | 0.24%                              | 272                                                                         |
| Uterine cancer                          | 0.50%                              | 42                                                                          |

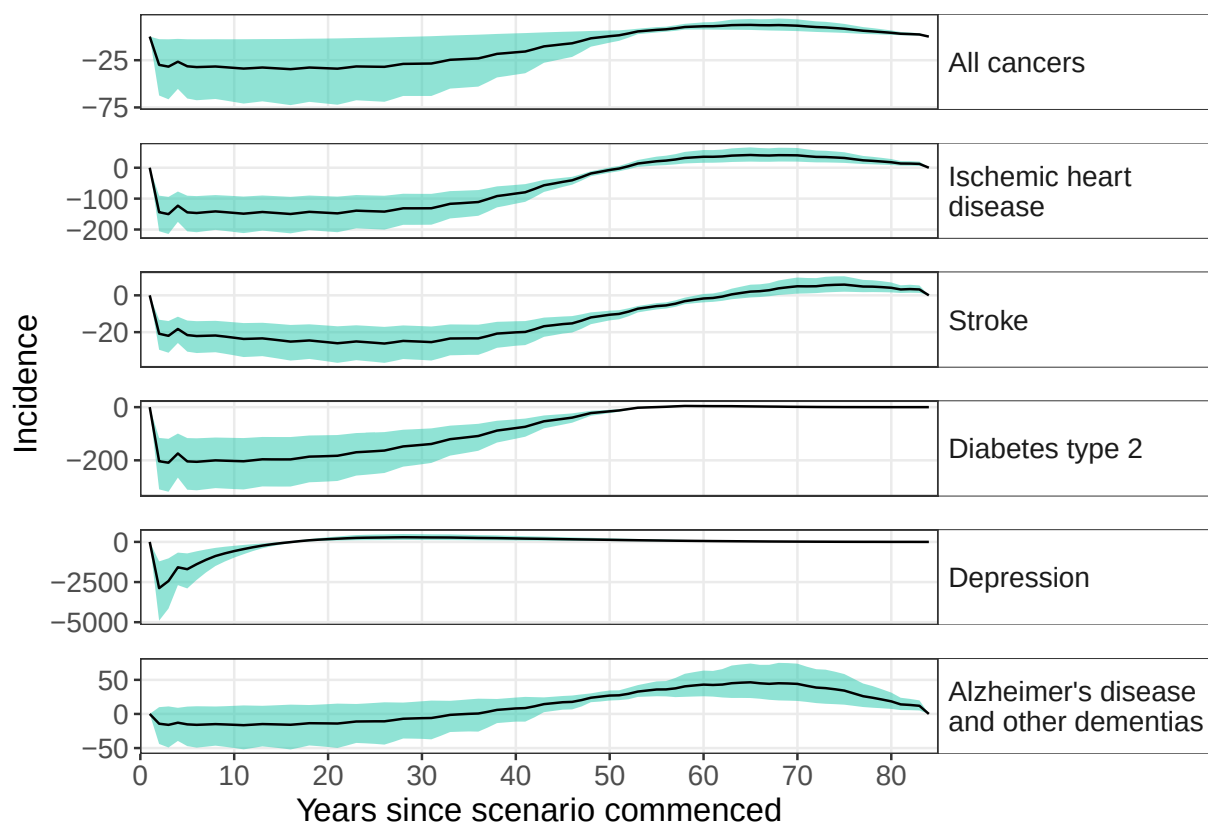
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 2.a:** Percentage change in incidence by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.



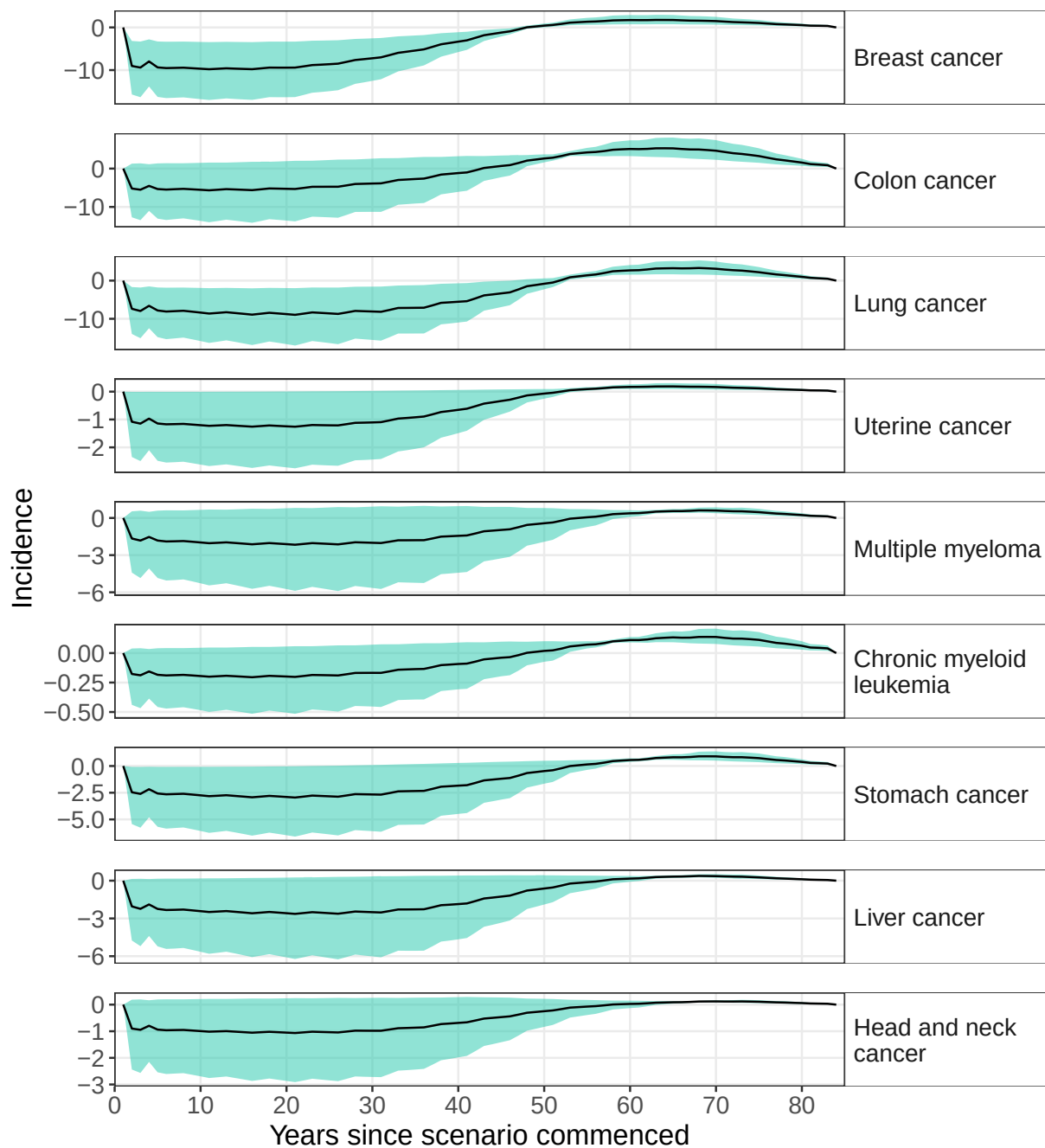
**Figure 2.b:** Percentage change in incidence for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 3.a:** The change in incidence over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.





**Figure 3.b:** Percentage change in incidence for specific cancers.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in incidence returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Mortality

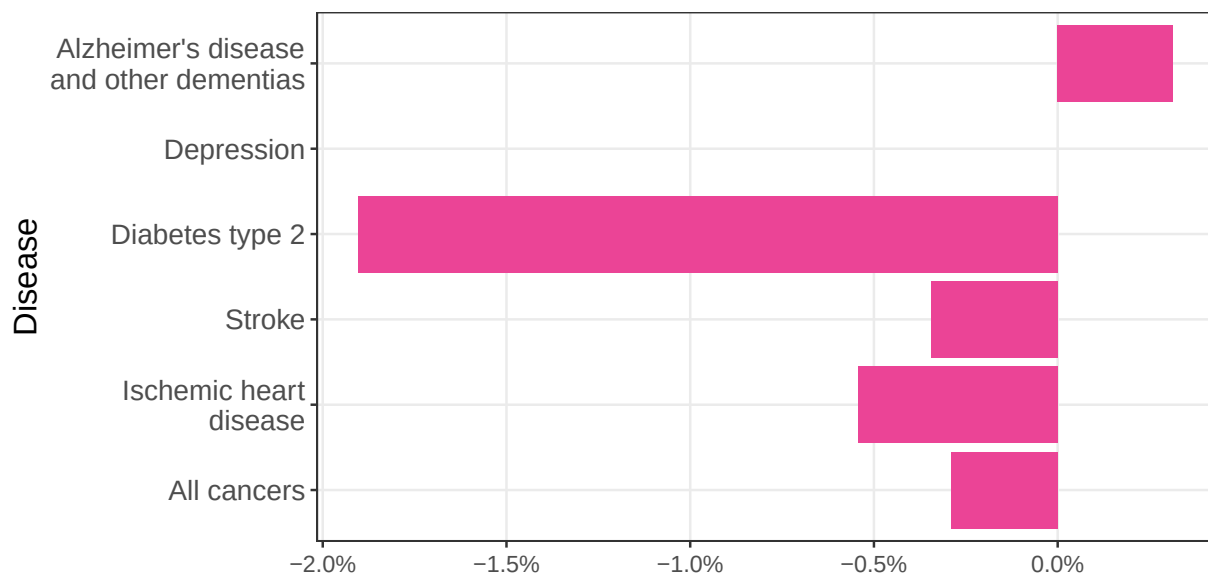
Mortality is the number of deaths due to a given disease over over a time period. In this example, results for females/males of all ages (n= 2,013,587) are presented as cases of prevented deaths due to increases in physical activity associated with the scenario. Figure 4 presents the total change in mortality over the life course. Figure 5 presents the difference in the number of deaths by year using a snapshot of the population from 2019.

Table 2 shows how the scenario impacts reductions in mortality presented as a percentage and total number of prevented deaths caused by chronic diseases.

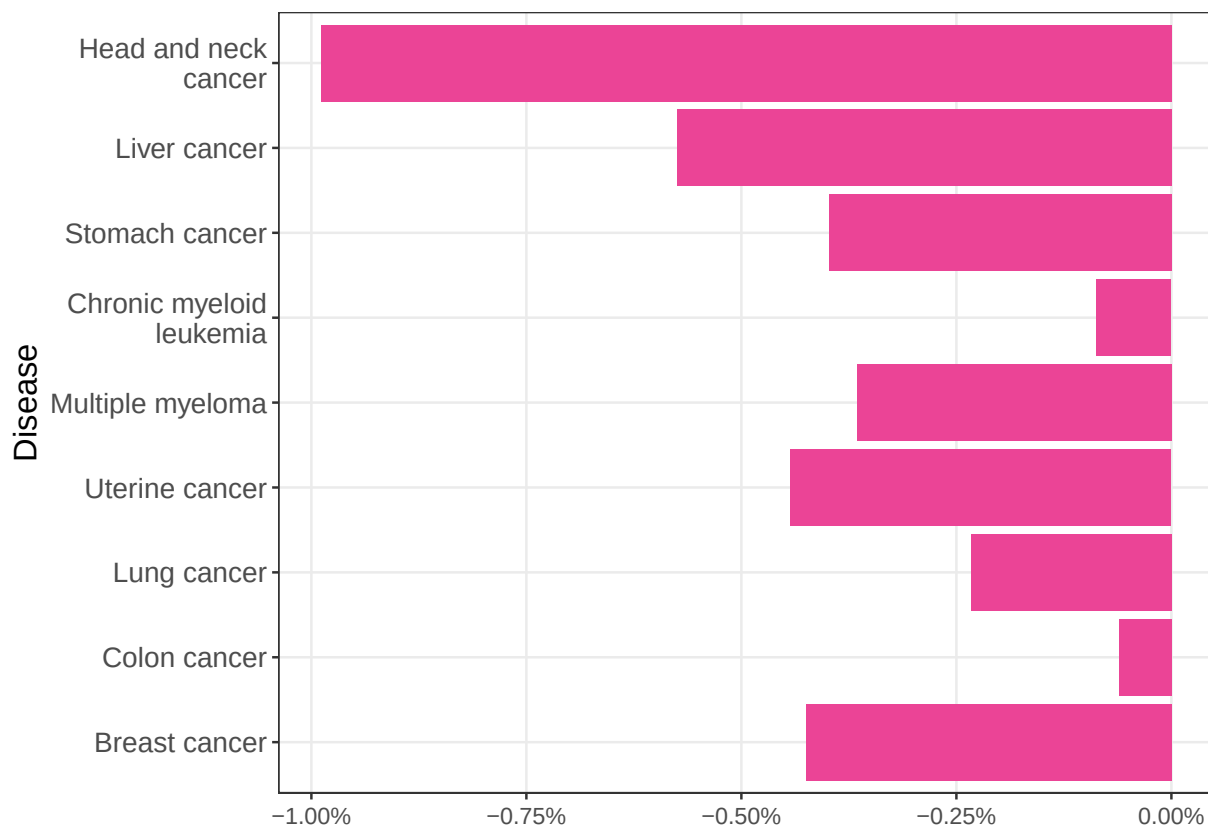
**Table 2.** Percentage reduction in mortality and total number of prevented deaths by chronic disease measured across the years of the simulation.

| Disease*                                | Mortality is reduced by | Total number of prevented deaths aggregated across the simulation |
|-----------------------------------------|-------------------------|-------------------------------------------------------------------|
| Alzheimer's disease and other dementias | -0.31%                  | -752                                                              |
| Breast cancer                           | 0.43%                   | 137                                                               |
| All cancers                             | 0.29%                   | 737                                                               |
| Colon cancer                            | 0.06%                   | 26                                                                |
| Chronic myeloid leukemia                | 0.09%                   | 2                                                                 |
| Diabetes type 2                         | 1.90%                   | 941                                                               |
| Depression                              | 0.00%                   | 0                                                                 |
| Head and neck cancer                    | 0.99%                   | 39                                                                |
| Ischemic heart disease                  | 0.54%                   | 2,035                                                             |
| Liver cancer                            | 0.57%                   | 98                                                                |
| Multiple myeloma                        | 0.37%                   | 64                                                                |
| Stomach cancer                          | 0.40%                   | 84                                                                |
| Stroke                                  | 0.34%                   | 603                                                               |
| Lung cancer                             | 0.23%                   | 254                                                               |
| Uterine cancer                          | 0.44%                   | 32                                                                |

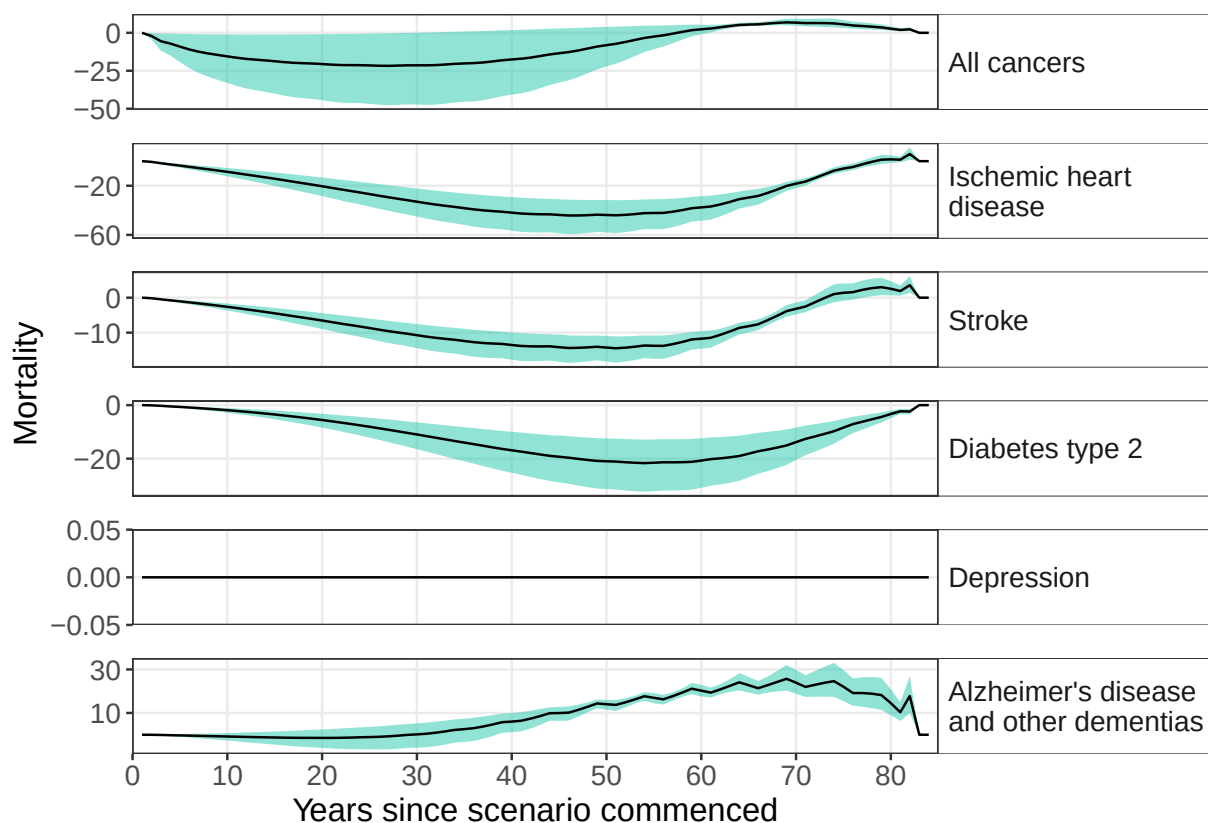
\* Negative figures indicate an increase in disease. This can occur because the scenarios increase physical activity improving population and physical health allowing the population to live longer but making them susceptible to other degenerative and age related diseases such as Alzheimer's and other dementias. Some scenarios result in minor shifts in chronic disease reduction as shown by zeros for incidence and disease. This is more common for scenarios involving older age groups who undertake less commuting trips.



**Figure 4.a:** Percentage change in mortality by disease. Notes: In this model, breast and uterine cancers only apply to populations including females.

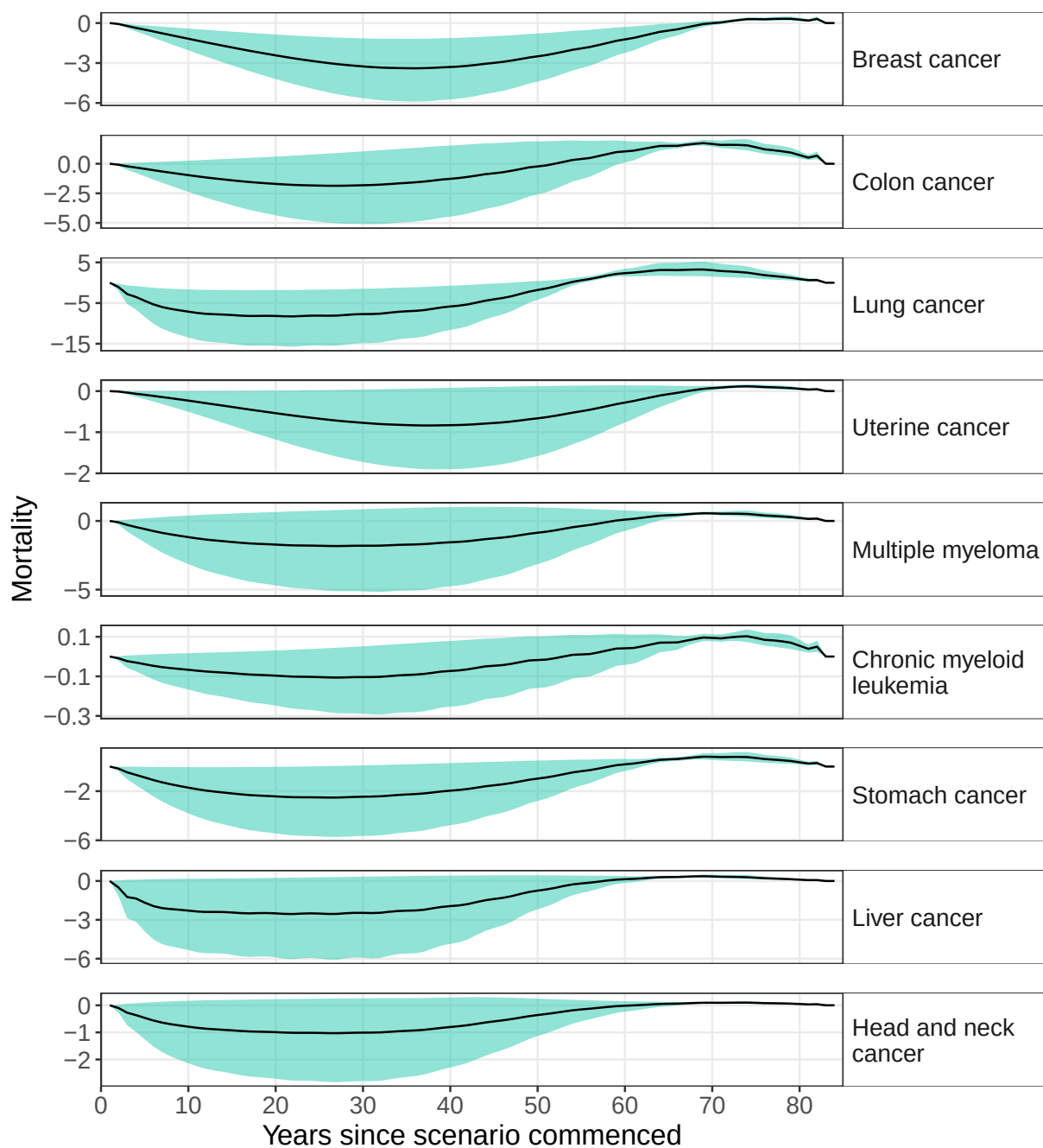


**Figure 4.b:** Percentage change in mortality for specific cancers. Notes: In this model, breast and uterine cancers only apply to populations including females.



**Figure 5.a:** The change in mortality over time by disease.

Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.



**Figure 5.b:** Percentage change in mortality for specific cancers.

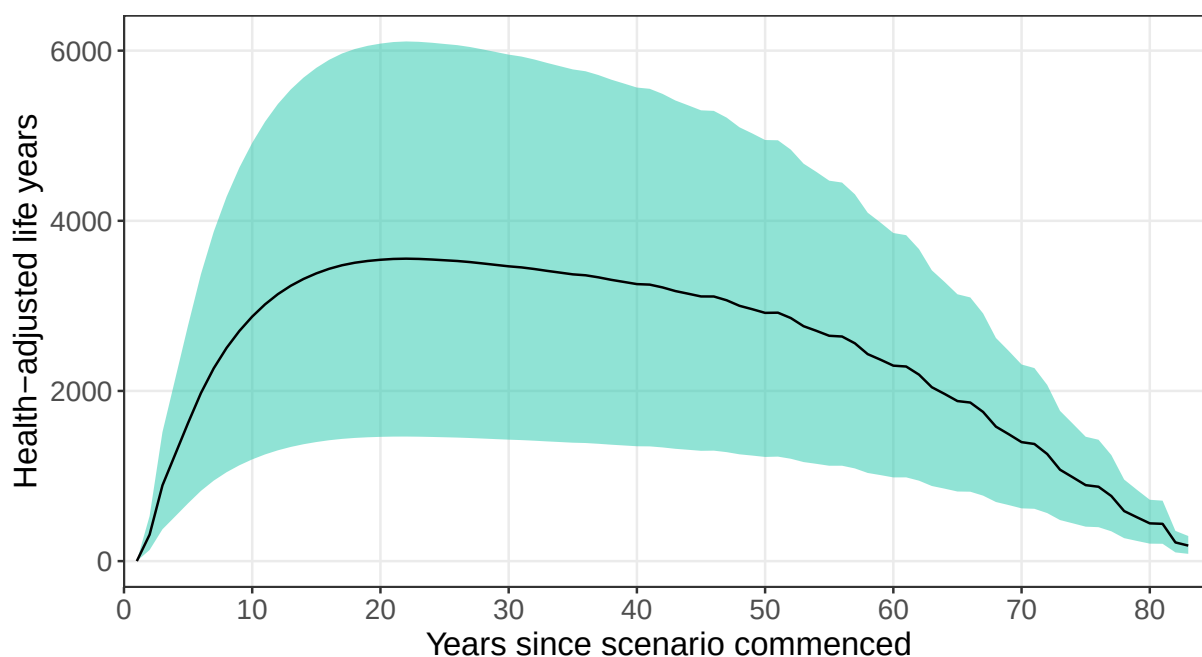
Notes: In this model, breast and uterine cancer only apply to populations including females. Time=0 at baseline year 2019. In the above figure, the change in mortality returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. For example, older age groups included in the original simulated model will not reach the maximum simulation range of 80 years because the time period extends beyond a lifetime. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Health

Figures 6 and 7 below show the change in Health Adjusted Life Years (HALYs)<sup>1</sup> and Life Years<sup>2</sup> for a snapshot of the population from 2019 for the scenario. Both figures show that the greatest gains from increasing physical activity occur midway through the life cycle with most of the gains occurring cumulatively in the long term. The decline from the mid-point onwards is due to individuals dying from natural causes within the model.

### HALYS

The model estimates a total of **205,707** HALYs for the scenario population, which is **102** HALYs per 1,000 members of the population.



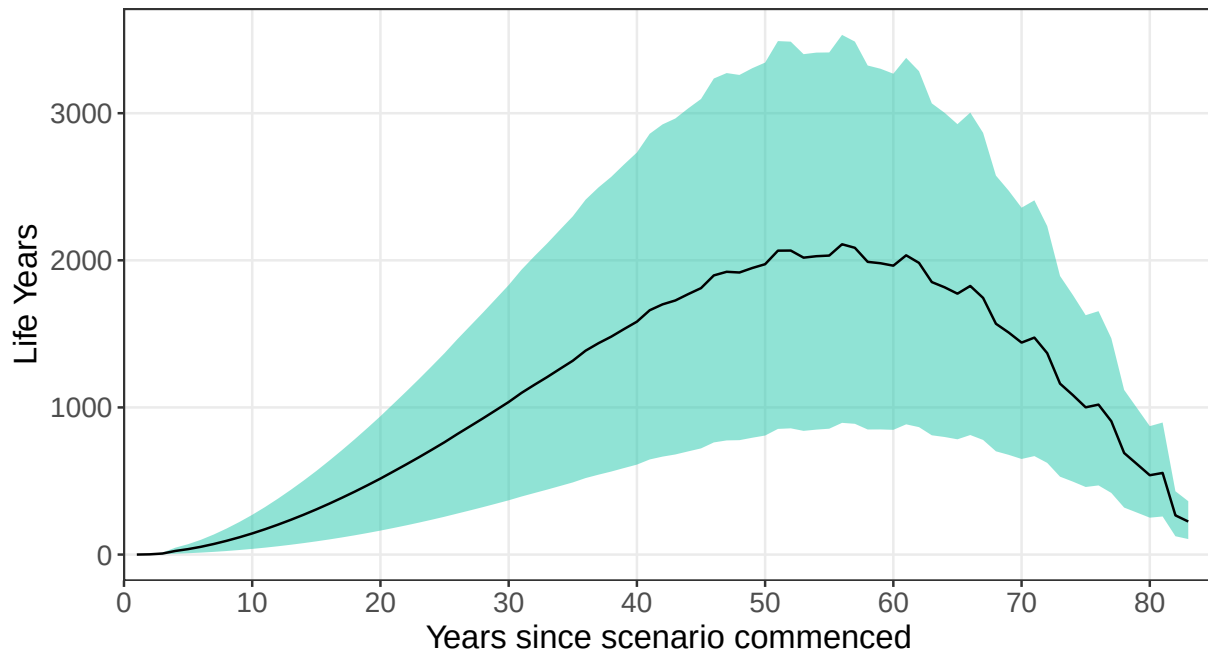
**Figure 6.** Total health-adjusted life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

<sup>1</sup>Health Adjusted Life Years are holistic measures of health that account for morbidity, mortality and quality of life.

<sup>2</sup>Life Years are similar to a HALYs however they exclude the quality of life component.

## Life Years

The model estimates a total of **93,322** Life Years for the scenario population, which is **46** Life Years per 1,000 members of the population.



**Figure 7.** Total life years gained due to the scenario. Notes: Time = 0 at baseline year 2019. In the above figure, the change in Life Years returns towards zero across time as the modelled population is fixed and no new people are allowed into the simulation model. The aqua shading represents uncertainty in the results as modelling can only represent indicative values. Please refer to the technical documentation for more detail.

## Value

The value of improvements to community health can be calculated[**a**] by translating the Health Adjusted Life Years (HALYs) from each scenario into dollar terms using the value of a statistical life year[**b**]. The value of a statistical life year is an estimate of the amount a society is willing to trade to reduce the risk of death for one year.

In the simulation model, HALYs are generated across time and are cumulative. Thus, to help us understand the value of HALYs across time in present day terms, it is necessary to use discounting[**c**] to reduce HALYs generated at the future point in time. Discounted HALYs from these future points can be added up to give the aggregate value of HALYs in today's terms as a measure of the value of improvements to community health arising from the chosen scenario.

The size of the discount rate can impact the aggregated value of HALYs and there is considerable debate on what discount rates should be used (with some arguing that health should not be discounted at all).[2] Hence, it is common to use a variety of discount rates to allow for differing risks, preferences and sensitivity when valuing health. The figures presented below were calculated using discount rates of 3%, 5% and 7% based on recent recommendations [3, 4] and represent the value of HALYs in present day terms resulting from an increase in physical activity from the chosen scenario.

### The value of improvements to community health

The model estimates a total of **HALYs**, Health Adjusted Life Years (HALYs) gained for the scenario population, which is **102** HALYs per 1,000 members of the scenario population. The figures below represent the value of improvements to community health from the chosen scenario. These figures can be used in summary reports and for advocacy purposes[**d**].

The HALYs gained in this scenario have a statistical value of:

- **8,593,572** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **5,380,176** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **3,673,341** per 1,000 members of the population, when calculated using a discount rate of 7%.

#### a. What is meant by value and how can it be measured?

Value is conceptual and measures a sense of worth or usefulness of something to individuals or to a society. Measuring the value of something, such as health, enables it to be included in assessments or analyses such as cost-benefit analyses to recognise its relative importance.

Value can be derived in many ways and a common approach is to use monetary terms, such as dollars. Valuing something using dollars is not the same as equating it with its price. Prices represent the amount at which something can be traded, prices therefore represent the amount of money for buying or selling something such as food, clothing or to pay bills. One way to evaluate health in dollar terms is to use the Value of a Statistical Life and Value of a Statistical Life Year[**b**].



## **b. What is the Value of a Statistical Life and Value of a Statistical Life Year?**

The value of a **statistical life** is the estimated amount that a society is willing to trade to reduce the risk of death. The word 'Statistical' refers to the average value for life and therefore means the value of a statistical life doesn't relate to any specific individual. This value can change across risk factors and different societies who may value life differently. There are various ways of measuring the value of a statistical life with most approaches using revealed or stated preference approaches.[3] In Australia, the Office of Best Practice Regulation estimates a statistical life at \$5.3M in 2022 dollar terms, and assumes that the life is of a young person with at least another 40 years to live.[5, 6]

### **Value of a Statistical Life Year**

The value of a statistical life year is the estimated amount that a society is willing to trade to reduce the risk of death over **one year**. It can be derived from the value of a statistical life or measured directly using surveys or willingness to pay techniques.[5] The current value of a **statistical life year** is \$227,000 in 2022 dollars based on current estimates from the Office of Best Practice Regulation.[6] The value of a statistical life year is useful for evaluating small increases in life years instead of evaluating full life expectancy. It is appropriate for valuing the Health Adjusted Life Years estimated from the scenarios and modelling presented in this tool. For the modelling and results presented here, the value of \$227,000 was converted to 2019 dollars based on the Wage Price Index for Brisbane.

## **c. What are discount rates and how are they relevant here?**

Discount rates are used to translate future amounts into an equivalent amount in today's terms or present value. This process is also known as a Net Present Value calculation (NPV). NPV calculations are useful for evaluating competing projects which may have different monetary costs or benefits that occur at different times in the future. Discount rates also represent risk, the time value of money and opportunity costs.[2]

In practice, a variety of discount rates are often used for estimating the value of costs and benefits in projects. For health, there remains considerable debate on which discount rates should be applied. Many argue that the value of health should not be discounted. Considering current discourse and following best practice approaches, three discount rates were used for discounting future Health Adjusted Life Years arising from the scenarios modelled within this tool. Discount rates of 3%, 5% and 7% were used. [3, 4, 7]

## **d. Application in advocacy and reporting**

This section uses figures to show how the value of community health (estimated from HALYs and the value of statistical life year) can be used for reporting and advocacy purposes.

The simulation model uses **population-based estimates** for disease morbidity and mortality and is best applied to larger groups of people. It also assumes that the people of interest have similar characteristics and behaviours to the population data used in the simulation model and scenarios. The **example** below shows results from a scenario that replaces car trips with walking trips for distances of 0–2 km for All age groups.

Example:

The HALYs gained in this scenario have a statistical value of:

- **\$10,859,605** per 1,000 members of the population, when calculated using a discount rate of 3%,
- **\$6,662,541** per 1,000 members of the population, when calculated using a discount rate of 5%,
- **\$4,533,392** per 1,000 members of the population, when calculated using a discount rate of 7%.

This **example** shows that the HALYs gained in this scenario have a statistical value of \$10,859,605 per 1,000 members of the population using a discount of 3%.

This figure can be divided by 1,000 to give a per person figure. Once a per person figure is established, it can be multiplied by the number of people in any population size of interest for use in reports or as evidence to advocate for benefits associated with shifts to active transport modes.

$$\$10,859,605 / 1,000 = \$10,859.61 \text{ per person value}$$

A good example of how this model can be applied links to previous research that investigated the impact of new more walkable development in Altona North on a population of 21,000 people [11]. If we assume that these people have similar characteristics to the underlying population based estimates and behaviours based on the travel survey data in the simulation model underlying this tool, then the value of community health according to the chosen scenario can be calculated as:

$$21,000 \text{ (people)} \times \$10,859 \text{ (statistical value from HALYs gained)} = \$228 \text{ M.}$$

## Savings

An increase in physical activity due to the chosen scenario reduces chronic disease cases across a lifetime and reduces spending for each disease within the health care system resulting in overall health care cost savings[\[a\]](#).

Table 3 provides estimated health care cost savings associated with the prevented cases of chronic diseases per 1,000 members of the population according to the selected scenario. These figures are based on applying average health care system costs per prevalent case of disease and using three alternative discount rates**[b]**:

**Table 3.** Total health care cost savings by disease per 1,000 members of the population.

| Disease                                 | 3% discount rate | 5% discount rate | 7% discount rate |
|-----------------------------------------|------------------|------------------|------------------|
| Alzheimer's disease and other dementias | \$1,932          | \$1,425          | \$1,024          |
| Breast cancer                           | \$89,612         | \$54,253         | \$35,139         |
| All cancers                             | \$152,478        | \$96,064         | \$64,146         |
| Colon cancer                            | \$24,274         | \$16,709         | \$11,679         |
| Chronic myeloid leukemia                | \$5,023          | \$3,076          | \$2,016          |
| Diabetes type 2                         | \$57,479         | \$33,359         | \$20,999         |
| Depression                              | \$272,358        | \$197,126        | \$147,880        |
| Head and neck cancer                    | \$1,103          | \$757            | \$545            |
| Ischemic heart disease                  | \$92,501         | \$54,037         | \$34,153         |
| Liver cancer                            | \$1,153          | \$802            | \$586            |
| Multiple myeloma                        | \$13,619         | \$8,847          | \$6,093          |
| Stomach cancer                          | \$5,884          | \$3,844          | \$2,663          |
| Stroke                                  | \$11,105         | \$6,493          | \$4,099          |
| Lung cancer                             | \$7,822          | \$5,342          | \$3,836          |
| Uterine cancer                          | \$3,962          | \$2,438          | \$1,600          |

### a. What do we mean by health care cost savings?

To calculate health care cost savings for each disease, the annual costs for each disease in each year is multiplied by the number of prevented cases of each disease for each scenario. This results in a total saving in spending for each disease by year. The savings in spending for future years are discounted**[b]** with annual savings aggregated to give a total amount saved for each disease. Total savings are presented as the amount saved per 1,000 members of the population to enable comparisons against populations of different sizes.

We use the term **health care cost saving** because it represents a reduction in health spending. However, the Australian Institute of Health and Welfare (AIHW) stress that the term cost is broad and not representative of the full cost experienced by individuals, families, or the health system, consequently AIHW use the term spending.[\[8\]](#)

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## References

1. Gold, M. R., Stevenson, D., & Fryback, D. G. (2002). *HALYS and QALYS and DALYS, Oh My: similarities and differences in summary measures of population Health*. Annual review of public health, 23(1), 115-134.
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